

**Potential Theory on the Berkovich
Projective Line (DRAFT
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Matthew Baker

Robert Rumely

E-mail address: mbaker@math.gatech.edu, rr@math.uga.edu

SCHOOL OF MATHEMATICS, GEORGIA INSTITUTE OF TECHNOLOGY,
ATLANTA, GA 30332-0140, USA

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF GEORGIA, ATHENS,
GA 30602-7403, USA

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Preface

Summary

This book is a revised and expanded version of the lecture notes from the Berkovich Space seminar held at the University of Georgia during Spring, 2004. Those notes ([6], 150 pages), entitled “Analysis and Dynamics on the Berkovich Projective Line”, are available online at <http://arxiv.org/abs/math/0407426>.

In addition to providing a concrete introduction to Berkovich’s theory of analytic spaces, we develop the foundations of potential theory on the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$. This has applications in arithmetic geometry to the study canonical heights, arithmetic intersection theory, and arithmetic dynamics, among other things. After describing in detail the topological structure of the Berkovich projective line, we introduce the Hsia kernel, the fundamental kernel for potential theory, and define a Laplacian operator on $\mathbb{P}_{\text{Berk}}^1$. We then develop a theory of capacities, harmonic, and subharmonic functions, all of which are strikingly similar to their classical counterparts over $\mathbb{C}$. Finally, we give some applications to non-archimedean dynamics, including the construction of a canonical probability measure on $\mathbb{P}_{\text{Berk}}^1$ attached to a rational function of degree at least 2.

Some related work

In his recent doctoral thesis [72], Amaury Thuillier has independently proved a number of results concerning potential theory on general Berkovich curves, including the construction of a Laplacian operator and a theory of harmonic and subharmonic functions. Thuillier also gives interesting applications of his work to Arakelov intersection theory. Because of the great generality and scope of Thuillier’s work, the present monograph may be more accessible to the novice reader than Thuillier’s thesis. In addition, we prove some results here which are not contained in [72], but are needed for applications to the theory of dynamical heights (see [5] and [1]). Charles Favre and Juan Rivera-Letelier [28, 29], as well as Antoine Chambert-Loir [22], have independently given constructions of the canonical measure on $\mathbb{P}_{\text{Berk}}^1$ attached

to a rational function. The material in Chapter 8 on the dynamics of rational maps utilizes earlier work of Rivera-Letelier ([53, 54, 55, 56], see also [58]). Our discussion of multiplicities in Chapter 7 has also been influenced by Rivera-Letelier's work.

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Differences from previous version

There are a number of differences between the present manuscript and the preliminary version [6] which was posted to the arXiv preprint server in July 2004. For one thing, we have corrected a number of errors in the previous version.

In addition, we have revised all of the statements and proofs so that they work over an arbitrary complete and algebraically closed field K endowed with a non-trivial non-archimedean absolute value, rather than just over the field $\mathbb{C}_p$. The main difference is that the Berkovich projective line over $\mathbb{C}_p$ is *metrizable* and has countable *branching* at every point, whereas in general the Berkovich projective line over K is non-metrizable and has uncountable branching. This required a significant reworking of many of our original proofs, since [6] relies in several places on arguments which are valid for metric spaces but not for an arbitrary compact Hausdorff space. Consequently, the present book makes more demands on the reader in terms of topological prerequisites, since for example we now make heavy use of nets rather than sequences. In some sense, this works against the concrete and “elementary” exposition that we have strived for. However, the changes seem desirable for at least two reasons. First of all, some of the proofs become more natural once the crutch of metrizability is removed. Second, and perhaps more importantly, we expect that the theory for more

general fields will have significant applications. The first author’s paper [1] is one example of this: a Northcott-type theorem for heights over a general field k endowed with a product formula is proved by working locally at each place v on the Berkovich projective line over $\mathbb{C}_v$ (the smallest complete and algebraically closed valued containing k and possessing an absolute value extending the given one $||_v$ on k). As another example, Kontsevich and Soibelman [44] have recently used Berkovich’s theory over fields such as the completion of the algebraic closure of $\mathbb{C}[[T]]$ to study homological mirror symmetry. We mention also the work of Favre and Jonsson [27] on the valuative tree, as well as the related work of Jan Kiwi [42], which have applications to complex dynamics and complex pluripotential theory.

Here is a brief summary of some of the other major changes which have been made:

- We have added a detailed Introduction giving a summary of our work and providing an overview of the historical context for it.
- Several appendices have been added in order to make the book more self-contained.
- We give a different construction of $\mathbb{P}_{\text{Berk}}^1$ (analogous to the “Proj” construction in algebraic geometry) which makes it easier to understand the action of a rational function.
- We have included a discussion of $\mathbb{R}$ -trees, and in particular of $\mathbb{P}_{\text{Berk}}^1$ as a “profinite $\mathbb{R}$ -tree”.
- We have simplified the discussion in Chapter 7 on analytic multiplicities.
- We have greatly expanded the material on dynamics of rational maps, including a Montel-type theorem and incorporating some earlier work of Rivera-Letelier.
- As an application of capacity theory, we state and prove a Berkovich space version of the Fekete-Szegö theorem.

Introduction

Overview

This research monograph has several goals. The first is to provide the reader with an introduction to Berkovich's theory of analytic spaces by focusing on the special case of the Berkovich projective line. The second, and primary, goal is to develop the foundations of potential theory on $\mathbb{P}_{\text{Berk}}^1$, including the definition of a measure-valued Laplacian operator, capacity theory, and a theory of harmonic and subharmonic functions. A third goal is to give some applications of potential theory on $\mathbb{P}_{\text{Berk}}^1$, especially to the dynamics of rational maps defined over $\mathbb{C}_p$.

Contents

We now give a more detailed overview of the contents of this book.

The Berkovich affine and projective lines. Let K be an algebraically closed field which is complete with respect to a nontrivial nonarchimedean absolute value. The topology on K induced by the given absolute value is Hausdorff, but it is also totally disconnected and not locally compact. This makes it very difficult to define in a direct way a good local notion of an analytic function on K . Tate dealt with this problem by developing the subject which is now known as *rigid analysis*, in which one works with a certain Grothendieck topology on K . This gives a satisfactory theory of analytic functions on K , but the underlying topological space is unchanged, so a number of problems remain for other applications. For example, the topology on K makes it difficult to define a Laplacian operator on K analogous to the classical Laplacian on $\mathbb{C}$, or to work sensibly with probability measures on K . Rigid analysis does not provide any apparent help with these difficulties.

However, these difficulties, and many more, can be resolved in an extremely satisfactory way using Berkovich's theory. The Berkovich affine line $\mathbb{A}_{\text{Berk}}^1$ over K is a locally compact, Hausdorff, and path-connected topological space which contains K (with the topology induced by the given absolute value) as a dense subspace. One obtains

the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$ by adjoining to $\mathbb{A}_{\text{Berk}}^1$ in a suitable manner a point at infinity; the resulting space $\mathbb{P}_{\text{Berk}}^1$ is a compact, Hausdorff, and path-connected topological space which contains $\mathbb{P}^1(K)$ (with its natural topology) as a dense subspace. In fact, $\mathbb{A}_{\text{Berk}}^1$ and $\mathbb{P}_{\text{Berk}}^1$ are better than just path-connected: they are *uniquely* path-connected, in the sense that any two distinct points can be joined by a unique arc. The unique path-connectedness is closely related to the fact that $\mathbb{A}_{\text{Berk}}^1$ and $\mathbb{P}_{\text{Berk}}^1$ are endowed with a natural tree structure. (More specifically, they are $\mathbb{R}$ -trees, as defined in §4.) The tree structure on $\mathbb{A}_{\text{Berk}}^1$ (resp. $\mathbb{P}_{\text{Berk}}^1$) can be used to define a *Laplacian operator* in terms of the classical Laplacian on a finite graph. This in turn leads to a good theory of harmonic and subharmonic functions which closely parallels the classical theory over $\mathbb{C}$.

The definition of $\mathbb{A}_{\text{Berk}}^1$ is quite simple, and makes sense with K replaced by an arbitrary field k endowed with a (possibly archimedean or even trivial) absolute value. As a set, $\mathbb{A}_{\text{Berk},k}^1$ consists of all multiplicative seminorms on the polynomial ring $k[T]$ which extend the usual absolute value on k . (A *multiplicative seminorm* on a ring A is a function $[\]_x : A \rightarrow \mathbb{R}_{\geq 0}$ satisfying $[0]_x = 0$, $[1]_x = 1$, $[fg]_x = [f]_x \cdot [g]_x$, and $[f + g]_x \leq [f]_x + [g]_x$ for all $f, g \in A$.) By an aesthetically desirable abuse of notation, we will identify seminorms $[\]_x$ with points $x \in \mathbb{A}_{\text{Berk},k}^1$, and we will usually omit explicit reference to the field k , writing $\mathbb{A}_{\text{Berk}}^1$ and assuming that we are working over a given complete and algebraically closed non-archimedean field K . The topology on $\mathbb{A}_{\text{Berk},k}^1$ is defined to be the weakest one for which $x \mapsto [f]_x$ is continuous for every $f \in A$.

To motivate this definition, we observe that in the classical setting, every multiplicative seminorm on $\mathbb{C}[T]$ which extends the usual absolute value on $\mathbb{C}$ is of the form $f \mapsto |f(z)|$ for some $z \in \mathbb{C}$. (This is a consequence of the well-known Gelfand-Mazur theorem from functional analysis.) It is then easy to see that $\mathbb{A}_{\text{Berk},\mathbb{C}}^1$ is homeomorphic to $\mathbb{C}$ itself, and also to the Gelfand spectrum (i.e., the space of all maximal ideals) in $\mathbb{C}[T]$.

In the non-archimedean world, K can once again be identified with the Gelfand space of maximal ideals in $K[T]$, but now there are many more multiplicative seminorms on $K[T]$ than just the ones given by evaluation at a point of K . The prototypical example is to fix a closed ball $B(a, r) = \{z \in K : |z - a| \leq r\}$ in K , and to define $[\]_{B(a,r)}$ by

$$[f]_{B(a,r)} = \sup_{z \in B(a,r)} |f(z)|.$$

It is an elementary consequence of Gauss' lemma that $[\cdot, \cdot]_{B(a,r)}$ is *multiplicative*, and the other axioms for a seminorm are trivially satisfied. Thus each ball $B(a, r)$ gives rise to a point of $\mathbb{A}_{\text{Berk}}^1$. Note that this includes balls for which $r \notin |K^*|$, i.e., “irrational balls” for which the set $\{z \in K : |z - a| = r\}$ is empty. We may consider the point a as a “degenerate” ball of radius zero. It is not hard to see that distinct balls $B(a, r)$ with $r \geq 0$ give rise to distinct multiplicative seminorms on $K[T]$, and therefore the set of all such balls embeds naturally into $\mathbb{A}_{\text{Berk}}^1$.

Suppose $x, x' \in \mathbb{A}_{\text{Berk}}^1$ are distinct points corresponding to the (possibly degenerate) balls $B(a, r), B(a', r')$, respectively. The unique path in $\mathbb{A}_{\text{Berk}}^1$ between x and x' has a very intuitive description. If $B(a, r) \subset B(a', r')$, it consists of all points of $\mathbb{A}_{\text{Berk}}^1$ corresponding to balls containing $B(a, r)$ and contained in $B(a', r')$. The collection of such “intermediate balls” is totally ordered by containment, and if $a = a'$ it is just $\{B(a, t) : r \leq t \leq r'\}$. If $B(a, r)$ and $B(a', r')$ are distinct, the unique path between x and x' consists of all points of $\mathbb{A}_{\text{Berk}}^1$ corresponding to balls of the form $B(a, t)$ with $r \leq t \leq |a - a'|$. The ball $B(a, |a - a'|)$ is the smallest one containing both $B(a, r)$ and $B(a', r')$, and if $x \vee x'$ denotes the point of $\mathbb{A}_{\text{Berk}}^1$ corresponding to $B(a, |a - a'|)$, then the path from x to x' is just the path from x to $x \vee x'$ followed by the path from $x \vee x'$ to x' .

In particular, if a, a' are distinct points of K , then one can visualize the path in $\mathbb{A}_{\text{Berk}}^1$ from a to a' as follows: Start increasing the “radius” of the degenerate ball $B(a, r)$ until you have a ball which also contains a' . This ball can be written as $B(a', s)$ with $s = |a - a'|$. Now decrease s until the radius reaches zero. In this way we have “connected” the totally disconnected space K by adding points corresponding to closed balls in K . In order to obtain a *compact* space, however, it is necessary in general to add even more points. For K may not be *spherically complete* (this happens, e.g., when $K = \mathbb{C}_p$): there may be decreasing sequences of closed balls with empty intersection. Intuitively, we need to add in points corresponding to such sequences in order to obtain a space which has a chance of being compact. More precisely, if we return to the definition of $\mathbb{A}_{\text{Berk}}^1$ in terms of multiplicative seminorms, it is easy to see that if $\{B(a_i, r_i)\}$ is any decreasing nested sequence of closed balls, then the map

$$f \mapsto \lim_{i \rightarrow \infty} [f]_{B(a_i, r_i)}$$

defines a multiplicative seminorm on $K[T]$ extending the usual absolute value on K . One can show that two such sequences of balls with empty

intersection define the same seminorm if and only if the sequences are *cofinal*. This yields a large number additional points of $\mathbb{A}_{\text{Berk}}^1$ which we haven't yet considered. According to *Berkovich's classification theorem*, we have now described all points of $\mathbb{A}_{\text{Berk}}^1$: every point $x \in \mathbb{A}_{\text{Berk}}^1$ corresponds to a decreasing sequence $\{B(a_i, r_i)\}$ of closed balls, and we can categorize the points of $\mathbb{A}_{\text{Berk}}^1$ into four types according to the nature of $B = \cap B(a_i, r_i)$:

- (Type I) B is a point of K .
- (Type II) B is a closed ball with radius belonging to $|K^*|$.
- (Type III) B is a closed ball with radius *not* belonging to $|K^*|$.
- (Type IV) $B = \emptyset$.

As a set, $\mathbb{P}_{\text{Berk}}^1$ can be obtained from $\mathbb{A}_{\text{Berk}}^1$ by adding a type I point denoted ∞ . The topology on $\mathbb{P}_{\text{Berk}}^1$ is that of the one-point compactification.

Following Rivera-Letelier, we write $\mathbb{H}_{\text{Berk}}$ for the subset of $\mathbb{P}_{\text{Berk}}^1$ consisting of all points of type II, III, or IV (Berkovich “hyperbolic space”). We also write $\mathbb{H}_{\text{Berk}}^{\mathbb{Q}}$ for the set of type II points, and $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ for the set of points of type II or III.

The description of points of $\mathbb{A}_{\text{Berk}}^1$ in terms of closed balls is very useful, because it allows one to visualize quite concretely the abstract space of multiplicative seminorms which we started with. It also shows us that $\mathbb{A}_{\text{Berk}}^1$ is endowed with a natural partial order, defined by saying that $x \leq y$ if and only if $[f]_x \leq [f]_y$ for all $f \in K[T]$. In terms of balls, if x, y are points of Type I, II, or III, one can show that $x \leq y$ if and only if the ball corresponding to x is contained in the ball corresponding to y . (We leave it as an exercise to the reader to extend this description of the partial order to points of Type IV.) For any pair of points $x, y \in \mathbb{A}_{\text{Berk}}^1$, there is a unique least upper bound $x \vee y \in \mathbb{A}_{\text{Berk}}^1$ with respect to this partial order. We can extend the partial order to $\mathbb{P}_{\text{Berk}}^1$ by declaring that $x \leq \infty$ for all $x \in \mathbb{A}_{\text{Berk}}^1$.

Writing

$$[x, x'] = \{z \in \mathbb{P}_{\text{Berk}}^1 : x \leq z \leq x'\} \cup \{z \in \mathbb{P}_{\text{Berk}}^1 : x' \leq z \leq x\} ,$$

it is easy to see that the unique path between $x, y \in \mathbb{P}_{\text{Berk}}^1$ is just

$$[x, x \vee y] \cup [x \vee y, y] .$$

There is a canonical *metric* ρ on $\mathbb{H}_{\text{Berk}}$ which is of great importance for potential theory. First, define a function $\text{diam} : \mathbb{A}_{\text{Berk}}^1 \rightarrow \mathbb{R}_{\geq 0}$ by setting $\text{diam}(x) = \lim r_i$ if x corresponds to the nested sequence $\{B(a_i, r_i)\}$. This is well-defined independent of the choice of nested sequence. If $x \in \mathbb{H}_{\text{Berk}}^{\mathbb{R}}$, then $\text{diam}(x)$ is just the diameter (= radius) of the corresponding closed ball. Because K is complete, if x is of type

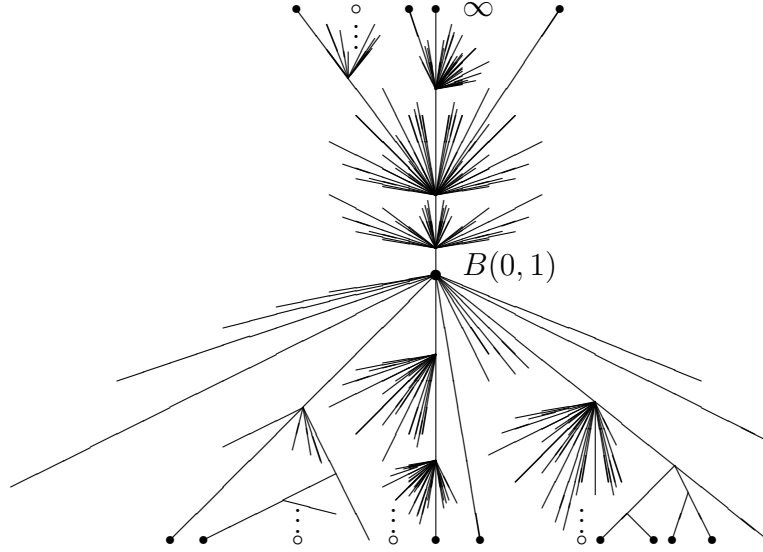


FIGURE 1. The Berkovich projective line (adapted from an illustration of Joe Silverman)

IV then $\text{diam}(x) > 0$. Thus $\text{diam}(x) = 0$ for $x \in \mathbb{A}_{\text{Berk}}^1$ of Type I, and $\text{diam}(x) > 0$ for $x \in \mathbb{H}_{\text{Berk}}$.

If $x, y \in \mathbb{H}_{\text{Berk}}$ with $x \leq y$, we define

$$\rho(x, y) = \log_v \frac{\text{diam}(y)}{\text{diam}(x)} ,$$

where $\log_v$ denotes the logarithm to the base q_v , with $q_v > 1$ a fixed real number chosen so that $-\log_v$ is a suitably normalized valuation on K .

More generally, for $x, y \in \mathbb{H}_{\text{Berk}}$ arbitrary, we define

$$\rho(x, y) = \rho(x, x \vee y) + \rho(x \vee y, y) .$$

It is not hard to verify that ρ defines a metric on $\mathbb{H}_{\text{Berk}}$ satisfying the ultrametric inequality. One can extend ρ to a singular metric on $\mathbb{P}_{\text{Berk}}^1$ by declaring that if $x \in \mathbb{P}^1(K)$ and $y \in \mathbb{P}_{\text{Berk}}^1$, we have $\rho(x, y) = +\infty$ if $x \neq y$ and 0 if $x = y$. However, we will usually only consider ρ as being defined on $\mathbb{H}_{\text{Berk}}$.

It is important to note that the topology on $\mathbb{H}_{\text{Berk}}$ defined by the metric ρ is *not* the subspace topology induced from the Berkovich (or Gelfand) topology on $\mathbb{P}_{\text{Berk}}^1 \supset \mathbb{H}_{\text{Berk}}$.

The group $\text{PGL}(2, K)$ of *Möbius transformations* acts continuously on $\mathbb{P}_{\text{Berk}}^1$ in a natural way compatible with the usual action on $\mathbb{P}^1(K)$, and this action preserves $\mathbb{H}_{\text{Berk}}$, $\mathbb{H}_{\text{Berk}}^{\mathbb{Q}}$, and $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$. Using the definition of $\mathbb{P}_{\text{Berk}}^1$ in terms of multiplicative seminorms, we have $[f]_M(x) = [f \circ$

$M]_x$ for $M \in \mathrm{PGL}(2, K)$. The action of $\mathrm{PGL}(2, K)$ on $\mathbb{P}_{\mathrm{Berk}}^1$ can also be described concretely in terms of Berkovich's classification, using the fact that each $M \in \mathrm{PGL}(2, K)$ takes closed balls to closed balls. An important observation is that $\mathrm{PGL}(2, K)$ acts *isometrically* on $\mathbb{H}_{\mathrm{Berk}}$, i.e.,

$$\rho(M(x), M(y)) = \rho(x, y)$$

for all $x, y \in \mathbb{H}_{\mathrm{Berk}}$ and all $M \in \mathrm{PGL}(2, K)$. This shows that the metric ρ is “coordinate-free”.

The diameter function diam can also be used to extend the usual distance function $|x - y|$ on K to $\mathbb{A}_{\mathrm{Berk}}^1$. We call this extension the *Hsia kernel*, and denote it by $\delta(x, y)_\infty$. Formally, for $x, y \in \mathbb{A}_{\mathrm{Berk}}^1$ we have

$$\delta(x, y)_\infty = \mathrm{diam}(x \vee y) .$$

It is easy to see that if $x, y \in K$ then $\delta(x, y)_\infty = |x - y|$. More generally, one has the formula

$$\delta(x, y)_\infty = \limsup_{(x_0, y_0) \rightarrow (x, y)} |x_0 - y_0| ,$$

where $(x_0, y_0) \in K \times K$ and the convergence implicit in the $\lim \sup$ is with respect to the (product) Berkovich topology on $\mathbb{P}_{\mathrm{Berk}}^1 \times \mathbb{P}_{\mathrm{Berk}}^1$. The Hsia kernel satisfies all of the axioms for an ultrametric with one exception: we have $\delta(x, x)_\infty > 0$ for $x \in \mathbb{H}_{\mathrm{Berk}}$.

The function $-\log_v \delta(x, y)_\infty$, which generalizes the usual potential theory kernel $-\log |x - y|$, leads to a theory of capacities on $\mathbb{P}_{\mathrm{Berk}}^1$ which generalizes that of [62], and which has many features in common with classical capacity theory over $\mathbb{C}$.

There is also a *generalized Hsia kernel* $\delta(x, y)_\zeta$ with respect to an arbitrary point $\zeta \in \mathbb{P}_{\mathrm{Berk}}^1$; we refer the reader to §4 for details.

We now come to an important description of $\mathbb{P}_{\mathrm{Berk}}^1$ as a *profinite* $\mathbb{R}$ -tree. An $\mathbb{R}$ -tree is a metric space (T, d) such that for each distinct pair of points $x, y \in T$, there is a unique path in T from x to y . A *branch point* is a point $x \in T$ for which $T \setminus \{x\}$ has either one or more than two connected components. A *finite* $\mathbb{R}$ -tree is an $\mathbb{R}$ -tree which has only finitely many branch points. Intuitively, a finite $\mathbb{R}$ -tree is just a tree in the usual graph-theoretic sense, but where the edges are thought of as line segments having definite lengths. Finally, a *profinite* $\mathbb{R}$ -tree is an inverse limit of finite $\mathbb{R}$ -trees.

Here's how these definitions play out in the case of $\mathbb{P}_{\mathrm{Berk}}^1$. If $S \subset \mathbb{P}_{\mathrm{Berk}}^1$, define the *convex hull* of S to be the smallest path-connected subset of $\mathbb{P}_{\mathrm{Berk}}^1$ containing S . (This is the same as the set of all paths between points of S .) By abuse of terminology, a *subgraph* of $\mathbb{P}_{\mathrm{Berk}}^1$ will mean the convex hull of a finite subset $S \subset \mathbb{H}_{\mathrm{Berk}}^{\mathbb{R}}$. Every subgraph Γ is

a finite $\mathbb{R}$ -tree, and the collection of all subgraphs of $\mathbb{P}_{\text{Berk}}^1$ is a directed set under inclusion. Moreover, if $\Gamma \leq \Gamma'$, then by a basis property of $\mathbb{R}$ -trees, there is a continuous *retraction map* $r_{\Gamma', \Gamma} : \Gamma' \rightarrow \Gamma$. In §4, we will show that $\mathbb{P}_{\text{Berk}}^1$ is homeomorphic to the inverse limit $\varprojlim \Gamma$ over all subgraphs $\Gamma \subset \mathbb{P}_{\text{Berk}}^1$. (Intuitively, this is just a topological formulation of Berkovich's classification theorem.) This description of $\mathbb{P}_{\text{Berk}}^1$ as a profinite $\mathbb{R}$ -tree provides a convenient way to visualize the topology on $\mathbb{P}_{\text{Berk}}^1$: two points are “close” if they retract to the same point of a “large” subgraph.

We remark that the *direct limit* of the Γ 's (with respect to inclusion) is precisely $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ endowed with the metric topology coming from ρ .

A base of open sets for the topology on $\mathbb{P}_{\text{Berk}}^1$ is given by the *open affinoid subsets*, which are the sets of the form $r_{\mathbb{P}_{\text{Berk}}^1, \Gamma}^{-1}(V)$ for Γ a subgraph of $\mathbb{P}_{\text{Berk}}^1$ and V an open subset of Γ . (Here $r_{\mathbb{P}_{\text{Berk}}^1, \Gamma}$ denotes the natural map from $\mathbb{P}_{\text{Berk}}^1$ to Γ coming from the universal property of the inverse limit.) We will refer to a connected open affinoid subset of $\mathbb{P}_{\text{Berk}}^1$ as *simple domain*. Simple domains can be completely characterized as the connected open subsets of $\mathbb{P}_{\text{Berk}}^1$ having only finitely many boundary points. If U is an open subset of $\mathbb{P}_{\text{Berk}}^1$, a *simple subdomain* of U is defined to be an open affinoid set whose closure is contained in U .

Laplacians. The profinite $\mathbb{R}$ -tree structure on $\mathbb{P}_{\text{Berk}}^1$ also leads directly to the construction of a Laplacian operator. On a finite $\mathbb{R}$ -tree (or more generally any metrized graph) Γ , there is a natural Laplacian operator Δ generalizing the well-known combinatorial Laplacian on a weighted graph. If $f : \Gamma \rightarrow \mathbb{R}$ is sufficiently smooth, then there is a unique Borel measure $\Delta_{\Gamma}(f)$ of total mass zero on Γ such that

$$(0.1) \quad \int_{\Gamma} \psi \Delta_{\Gamma}(f) := \int_{\Gamma} f'(x) \psi'(x)$$

for all suitable test functions ψ . We call the measure $\Delta_{\Gamma}(f)$ the Laplacian of f on Γ . We denote by $\text{BDV}(\Gamma)$ the largest class of continuous functions f on Γ for which the distribution defined by

$$(0.2) \quad \psi \mapsto \int_{\Gamma} f(x) \Delta \psi$$

is represented by a Borel measure; a simple integration by parts argument shows that the resulting measure coincides with the one defined by (0.1) when f is sufficiently smooth.

The Laplacian operator satisfies an important compatibility property with respect to the partially ordered set of subgraphs of $\mathbb{P}_{\text{Berk}}^1$. If

$\Gamma \leq \Gamma'$ and $f \in \text{BDV}(\Gamma')$, then we have

$$(0.3) \quad \Delta_\Gamma(f|_\Gamma) = (r_{\Gamma',\Gamma})_* \Delta_{\Gamma'}(f) .$$

We define $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ to be the collection of all functions $f : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R} \cup \{\pm\infty\}$ such that:

- $f|_\Gamma \in \text{BDV}(\Gamma)$ for each subgraph Γ .
- The measures $|\Delta_\Gamma(f)|$ have uniformly bounded total mass.

Note that belonging to $\text{BDV}(\Gamma)$ imposes no condition on the values of f at points of $\mathbb{P}^1(K)$.

Using the compatibility property (0.3), one shows that if $f \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, then the collection of measures $\{\Delta_\Gamma\}$ “cohere” to give a unique Borel measure $\Delta(f)$ of total mass zero on the inverse limit space $\mathbb{P}_{\text{Berk}}^1$ for which

$$(r_{\mathbb{P}_{\text{Berk}}^1,\Gamma})_* \Delta(f) = \Delta_\Gamma(f)$$

for all subgraphs Γ of $\mathbb{P}_{\text{Berk}}^1$. We call $\Delta(f)$ the *Laplacian* of f on $\mathbb{P}_{\text{Berk}}^1$.

Similarly, if U is a domain (i.e., connected open subset) in $\mathbb{P}_{\text{Berk}}^1$, one defines a class $\text{BDV}(U)$ of functions $f : U \rightarrow \mathbb{R} \cup \{\pm\infty\}$ for which the Laplacian $\Delta_{\overline{U}}(f)$ is a bounded Borel measure of total mass zero supported on the closure of U . The measure $\Delta_{\overline{U}}(f)$ has the property that

$$(r_{\overline{U},\Gamma})_* \Delta(f) = \Delta_\Gamma(f)$$

for all subgraphs Γ of $\mathbb{P}_{\text{Berk}}^1$ contained in U .

As a concrete example, fix $y \in \mathbb{A}_{\text{Berk},\mathbb{C}_p}^1$ and let $f : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R} \cup \{\pm\infty\}$ be defined by $f(\infty) = -\infty$ and

$$f(x) = -\log_p \delta(x, y)_\infty$$

for $x \in \mathbb{A}_{\text{Berk}}^1$. Then $f \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and

$$(0.4) \quad \Delta f = \delta_y - \delta_\infty$$

is a discrete measure on $\mathbb{P}_{\text{Berk}}^1$ supported on $\{y, \infty\}$. Intuitively, the explanation for the formula (0.4) is as follows. The function f is locally constant away from the path $\Lambda = [y, \infty]$ from y to ∞ ; more precisely, we have $f(x) = r_{\mathbb{P}_{\text{Berk}}^1,\Lambda}(x)$. Moreover, the restriction of f to Λ is linear (with respect to the distance function ρ) with slope -1 . For any test function ψ , we therefore have the “heuristic” calculation

$$\int_{\mathbb{P}_{\text{Berk}}^1} \psi \Delta f = \int_\Lambda \psi \Delta f = \int_\Lambda f'(x) \psi'(x) = - \int_y^\infty \psi'(x) dx = \psi(y) - \psi(\infty) .$$

(To make this calculation rigorous, one needs to exhaust $\Lambda = [y, \infty]$ by an increasing sequence of line segments $\Gamma \subset \mathbb{H}_{\text{Berk}}^{\mathbb{R}}$, and then observe that the corresponding measures $\Delta_\Gamma f$ converge weakly to $\delta_y - \delta_\infty$.)

Equation (0.4) shows that $-\log_v \delta(x, y)_\infty$, like its classical counterpart $-\log |x-y|$ over $\mathbb{C}$, is a fundamental solution (in the sense of distributions) to the Laplace equation. This “explains” why $-\log_v \delta(x, y)_\infty$ is the correct kernel for doing capacity theory.

More generally, let $\varphi \in K(T)$ be a nonzero rational function with zeros and poles given by the divisor $\text{div}(\varphi)$ on $\mathbb{P}^1(K)$. The usual action of φ on $\mathbb{P}^1(K)$ extends naturally to an action of φ on $\mathbb{P}_{\text{Berk}}^1$, and there is a continuous function $-\log_v[\varphi]_x : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}^1 \cup \{\pm\infty\}$ extending the usual map $x \mapsto -\log_v |\varphi(x)|$ on $\mathbb{P}^1(K)$. One derives from (0.4) the following version of the *Poincaré-Lelong formula*:

$$\Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v[\varphi]_x) = \delta_{\text{div}(\varphi)} .$$

Harmonic functions. If U is a domain in $\mathbb{P}_{\text{Berk}}^1$, a real-valued function $f : U \rightarrow \mathbb{R}$ is called *strongly harmonic* on U if it is continuous, belongs to $\text{BDV}(U)$, and if $\Delta_{\bar{v}}(f)$ is supported on ∂U . f is *harmonic* on U if every point $x \in U$ has a connected open neighborhood on which f is strongly harmonic.

Harmonic functions on domains $U \subseteq \mathbb{P}_{\text{Berk}}^1$ satisfy many properties analogous to their classical counterparts over $\mathbb{C}$. For example, a harmonic function which attains its maximum or minimum value on U must be constant. There is also an analogue of the Poisson Formula: if f is a harmonic function on an open affinoid U , then f extends uniquely to the boundary ∂U , and the values of f on U can be computed explicitly in terms of $f|_{\partial U}$. A version of Harnack’s principle holds as well: the limit of a monotonically increasing sequence of nonnegative harmonic functions on U is either harmonic or identically $+\infty$. Unlike the classical case (where a hypothesis of uniform convergence is required), a *pointwise* limit of harmonic functions is automatically harmonic. As is the case over $\mathbb{C}$, harmonicity is preserved under pullbacks by meromorphic functions.

Capacities. Fix $\zeta \in \mathbb{P}_{\text{Berk}}^1$, and let E be a compact subset of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. (For concreteness, the reader may wish to imagine that $\zeta = \infty$.) By analogy with the classical theory over $\mathbb{C}$, and also with the non-archimedean theory developed in [62], one can define the *logarithmic capacity* of E with respect to ζ . This is done as follows.

Given a probability measure ν on $\mathbb{P}_{\text{Berk}}^1$ with support contained in E , we define the energy integral

$$I_\zeta(\nu) = \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) .$$

Letting ν vary over the collection $\mathbb{P}(E)$ of all probability measures supported on E , one defines the *Robin constant*

$$V_\zeta(E) = \inf_{\nu \in \mathbb{P}(E)} I_\zeta(\nu) .$$

The *logarithmic capacity* of E relative to ζ is then defined to be

$$\gamma_\zeta(E) = q_v^{-V_\zeta(E)} .$$

Any countable subset of $\mathbb{P}^1(K)$ has capacity zero. On the other hand, if $E \cap \mathbb{H}_{\text{Berk}} \neq \emptyset$, then $\gamma_\zeta(E) > 0$. In particular, a singleton set can have positive capacity, a phenomenon which has no classical analogue. The property of having capacity 0 relative to ζ is independent of the chosen point ζ .

As a more elaborate example, if $K = \mathbb{C}_p$ and $E = \mathbb{Z}_p \subset \mathbb{A}^1(\mathbb{C}_p) \subset \mathbb{A}_{\text{Berk}, \mathbb{C}_p}^1$, then $\gamma_\infty(E) = p^{-1/(p-1)}$. Since $\delta(x, y)_\infty = |x - y|$ for $x, y \in K$, this follows from the same computation as in [62], Example 5.2.13.

If E is compact, and if $\gamma_\zeta(E) > 0$, we show that there is a *unique* probability measure $\mu_{E, \zeta}$ on E , called the *equilibrium measure* of E with respect to ζ , which minimizes energy (i.e., for which $I_\zeta(\mu_{E, \zeta}) = V_\zeta(E)$). As in the classical case, $\mu_{E, \zeta}$ is always supported on the boundary of E .

Closely linked to the theory of capacities is the theory of *potential functions*. For each probability measure ν supported on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, one defines the potential function $u_\nu(z, \zeta)$ by

$$u_\nu(z, \zeta) = \int -\log_v \delta(z, w)_\zeta d\nu(w) .$$

As in classical potential theory, potential functions need not be continuous, but they do share several of the distinguishing features of continuous functions. For example, $u_\nu(z, \zeta)$ is lower semi-continuous, and is continuous at each $z \notin \text{supp}(\nu)$. Potential functions on $\mathbb{P}_{\text{Berk}}^1$ also satisfy the following analogues of Maria's theorem and Frostman's theorem from complex potential theory:

THEOREM (Maria). If $u_\nu(z, \zeta) \leq M$ on $\text{supp}(\nu)$, then $u_\nu(z, \zeta) \leq M$ for all $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

THEOREM (Frostman). If E has positive capacity, then the equilibrium potential $u_E(z, \zeta)$ satisfies $u_E(z, \zeta) \leq V_\zeta(E)$ for all $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, and $u_E(z, \zeta) = V_\zeta(E)$ for all $z \in E$ outside a set of capacity zero.

As in capacity theory over $\mathbb{C}$, one can also define the *transfinite diameter* and the *Chebyshev constant* of E , and they turn out to both be equal to the logarithmic capacity of E . In fact, we define three

different variants of the Chebyshev constant and prove that they are all equal.

We define the *Green's function of E relative to ζ* to be

$$G(z, \zeta; E) = V_\zeta(E) - u_E(z, \zeta)$$

for all $z \in \mathbb{P}_{\text{Berk}}^1$. We show that the Green's function is everywhere nonnegative, and is strictly positive on the connected component U_ζ of $\mathbb{P}_{\text{Berk}}^1 \setminus E$ containing ζ . Also, $G(z, \zeta; E)$ is finite on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, with a logarithmic singularity at ζ , and it is harmonic in $U_\zeta \setminus \{\zeta\}$. Additionally, $G(z, \zeta; E)$ is identically zero on the complement of U_ζ outside a set of capacity zero. The Laplacian of $G(z, \zeta; E)$ on $\mathbb{P}_{\text{Berk}}^1$ is equal to $\delta_\zeta - \mu_{E, \zeta}$. Finally, as in the classical case, the Green's function is symmetric as a function of z and ζ : we have

$$G(z_1, z_2; E) = G(z_2, z_1; E)$$

for all $z_1, z_2 \notin E$.

As an arithmetic application of the theory of capacities on $\mathbb{P}_{\text{Berk}}^1$, we formulate a generalization to $\mathbb{P}_{\text{Berk}}^1$ of the generalized Fekete and Fekete-Szegö theorems from [62]. The proof is rather easy, since it goes by reducing the general case to the special case of *RL*-domains, which has already been proved in [62]. Nonetheless, our result is aesthetically pleasing because the statement is very simple, and unlike the treatment in [62], there are no thorny issues of “algebraic capacitability” to deal with. The existence of this kind of simpler reformulation is directly related to the fact that $\mathbb{P}_{\text{Berk}}^1$ is locally compact while $\mathbb{P}^1(K)$ is not.

Subharmonic functions. We give two equivalent characterizations of what it means for a function on a domain $U \subseteq \mathbb{P}_{\text{Berk}}^1$ to be subharmonic. The first, which we take as the definition, is as follows. We say that a function $f : U \rightarrow \mathbb{R} \cup \{-\infty\}$ is *strongly subharmonic* if it is upper semicontinuous, satisfies an additional technical semicontinuity hypothesis at points of $\mathbb{P}^1(K)$, and if the positive part of $\Delta_U(f)$ is supported on ∂U . f is *subharmonic* on U if every point of U has a connected open neighborhood on which f is strongly subharmonic. We also say that f is *superharmonic* on U if $-f$ is subharmonic on U . As an example, if ν is a probability measure on $\mathbb{P}_{\text{Berk}}^1$ and $\zeta \notin \text{supp}(\nu)$, the potential function $u_\nu(x, \zeta)$ is strongly superharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ and is strongly subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \text{supp}(\nu)$. A function f is harmonic on U if and only if it is both subharmonic and superharmonic on U .

As a second characterization of subharmonic functions, we say that $f : U \rightarrow \mathbb{R} \cup \{-\infty\}$ (not identically $-\infty$) is *domination subharmonic* on the domain U if it is upper semicontinuous, and if for each simple subdomain V of U and each harmonic function f on V for which $f \leq$

h on ∂V , we have $f \leq h$ on V . A difficult theorem, proved in §2 of Chapter 6, states that f is subharmonic on U if and only if it is domination subharmonic on U .

Subharmonic functions also satisfy the Maximum Principle: if U is a domain in $\mathbb{P}_{\text{Berk}}^1$ and f is a subharmonic function which attains its maximum value on U , then f is constant. In addition, subharmonic functions on domains in $\mathbb{P}_{\text{Berk}}^1$ are stable under many of the same operations (e.g., convex combinations, maximum, monotone convergence, uniform convergence) as their classical counterparts. There is also an analogue of the Riesz Decomposition Theorem, according to which a subharmonic function on a simple subdomain $V \subset U$ can be written as the difference of a harmonic function and a potential function. We also show that subharmonic functions can be well-approximated by continuous functions of a special form, which we call *smooth* functions.

In §8, we define the notion of an Arakelov-Green's function on $\mathbb{P}_{\text{Berk}}^1$, and prove an *energy minimization principle* which is used in the proof of the main result in [5].

Multiplicities. If $\varphi \in K(T)$ is a non-constant rational function, then as mentioned above, the action of φ on $\mathbb{P}^1(K)$ extends naturally to an action of φ on $\mathbb{P}_{\text{Berk}}^1$. We use the theory of Laplacians to define *analytic multiplicities* which generalize the usual multiplicity of φ at a point $a \in \mathbb{P}^1(K)$ (i.e., the multiplicity of a as a preimage of $b = \varphi(a) \in \mathbb{P}^1(K)$). As a consequence, we are able to prove that the extended map $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is a surjective open mapping. We also obtain a purely topological interpretation of the analytic multiplicity, which shows that our multiplicities coincide with those originally defined by Rivera-Letelier. For each $a \in \mathbb{P}_{\text{Berk}}^1$, the multiplicity of φ at a is a positive integer, and it is equal to 1 if and only if φ is locally injective at a . For each $b \in \mathbb{P}_{\text{Berk}}^1$, the sum of the multiplicities of φ over all preimages of b is equal to the degree of φ .

Using these analytic multiplicities, we define the pushforward and pullback under φ of a bounded Borel measure on $\mathbb{P}_{\text{Berk}}^1$. The pushforward and pullback operations satisfy the expected functoriality properties, including the fact that if f is subharmonic on U , then $f \circ \varphi$ is subharmonic on $\varphi^{-1}(U)$ and the Laplacian of $f \circ \varphi$ is the pullback under φ of the Laplacian of f .

Applications to the dynamics of rational maps. Suppose $\varphi \in K(T)$ is a rational function of degree $d \geq 2$. In §1, we construct a *canonical probability measure* μ_φ on $\mathbb{P}_{\text{Berk}}^1$ attached to φ , whose properties are analogous to the well-known measure on $\mathbb{P}^1(\mathbb{C})$ first

defined by Lyubich and Freire-Lopes-Mañé. The measure μ_φ is φ -invariant (i.e., satisfies $\varphi_*(\mu_\varphi) = \mu_\varphi$), and satisfies the functional equation $\varphi^*(\mu_\varphi) = d \cdot \mu_\varphi$. In §2, we prove an explicit formula and functional equation for the Arakelov-Green's function $g_{\mu_\varphi}(x, y)$ associated to μ_φ . These results, along with the energy minimization principle mentioned earlier, play a key role in some of the applications of our theory to arithmetic dynamics over global fields (see [1] and [5]).

We then discuss various results analogous to the Fatou-Julia theory of iteration of rational maps on $\mathbb{P}^1(\mathbb{C})$. In particular, we define the Berkovich Fatou and Julia sets of φ , and prove that the Berkovich Julia set (like its complex counterpart, but unlike its counterpart in $\mathbb{P}^1(K)$) is always non-empty. By making use of the classification theorem for Fatou components due to Rivera-Letelier, we prove a Berkovich space analogue of Montel's theorem for the iterates of a rational map.

We note that although Berkovich introduced his theory of analytic spaces with rather different goals in mind, Berkovich spaces are very well adapted to the study of non-archimedean dynamics. The fact that the topological space $\mathbb{P}_{\text{Berk}}^1$ is both compact and connected means in practice that many of the difficulties encountered in “classical” non-archimedean dynamics disappear when one defines the Fatou and Julia sets as subsets of $\mathbb{P}_{\text{Berk}}^1$. For example, there is essentially no ambiguity as to what the correct definitions of the Berkovich Fatou and Julia sets should be, since the notions of normality and equicontinuity coincide on $\mathbb{P}_{\text{Berk}}^1$. Also, the notion of a connected component becomes completely straightforward in the Berkovich setting, so that one avoids having to discuss various possible definitions of a Fatou component (e.g. the D -components versus analytic components found in R. Benedetto's paper [7], or the definition given by J. Rivera-Letelier in [55]).

Appendices. In Appendix A, we review a number of facts from real analysis and point-set topology which are used throughout the text. Some of these (e.g., the Riesz Representation Theorem) are quite well-known, while others (e.g., the Portmanteau Theorem) are hard to find precise references for. In the latter cases, we have provided self-contained proofs. We include a detailed discussion of nets in topological spaces: since the space $\mathbb{P}_{\text{Berk}, K}^1$ is not in general metrizable, sequences do not suffice for our purposes when discussing notions like continuity.

In Appendix B, we discuss $\mathbb{R}$ -trees and their relation to Gromov's theory of hyperbolic spaces.

Appendix C provides an almost entirely self-contained account of the necessary background from the theory of metrized graphs, weaving together various facts from [4] and [23], as well as proving some facts

which are used throughout the book but for which we could not find any references.

Appendix D gives a brief overview of some basic definitions and results from Berkovich’s theory of non-archimedean analytic spaces. This material is included in order to give the reader some perspective on the relationship between the special cases dealt with in this book (the Berkovich unit disc, affine line, and projective line) and the general setting of Berkovich’s theory. For the most part, we do not provide proofs, but we do include a proof, different from the one given in Berkovich’s book, that the spectrum of a Banach ring is compact.

Rumely’s “Capacity Theory on Algebraic Curves”. To some extent, the present work can be seen as a “modern” reworking, for the special case of $\mathbb{P}^1$, of the non-archimedean potential theory on curves developed in Rumely’s 1989 book [62]. We recall briefly some of the goals and achievements of that book.

Capacity is a measurement of the size of subsets of $\mathbb{C}$ which arises naturally in physics when considering electrostatic potential functions. It turns out to play an important role in complex analysis, as well; for example, there are results which more or less assert that sets of capacity zero are “invisible” to holomorphic functions. The first significant application of capacity theory to arithmetic was given by Fekete and Szegő [30] who proved the following result.

THEOREM. Suppose E is a subset of $\mathbb{C}$ which is stable under complex conjugation. Let $\gamma(E)$ be its logarithmic capacity. Then:

- A) (Fekete’s theorem) If $\gamma(E) < 1$, then there is some neighborhood of E which contains only finitely many complete conjugate sets of algebraic integers.
- B) (Fekete-Szegő theorem) If $\gamma(E) \geq 1$, then every neighborhood of E contains infinitely many such sets.

For simplicity, we will refer to parts (A) and (B) together as the Fekete-Szegő theorem. The nontrivial aspect of this result is that it deals with algebraic *integers*. It is quite easy to find algebraic *numbers* all of whose conjugates lie near a specified set E , but being an algebraic integer is a much more global restriction: it means that conditions are being imposed at every place of some number field k , not just the archimedean places. Indeed, an algebraic number $\alpha \in k$ is an algebraic integer if and only if it lies in the closed unit disc $\mathcal{O}_v$ of the completion k_v for all places v . Thus the Fekete-Szegő theorem can be viewed as an *adelic* statement about algebraic numbers whose conjugates satisfy some geometric constraints at every place v of k .

From this point of view, one is naturally led to try to formulate an adelic version of the Fekete-Szegö theorem, in which one may, at a finite set of non-archimedean places, replace the v -adic unit disc with a more general subset of $\bar{k}_v$. Progress was first made in this direction in the 1960's by Bertrandias, and then around 1980 Cantor's new ideas resulted in a general adelic version of the Fekete-Szegö theorem on $\mathbb{P}^1$.

The purpose of Rumely's book [62] was to extend Cantor's theory, especially the adelic Fekete-Szegö theorem, to arbitrary curves. In Rumely's theory, one has not only a good notion of capacity for suitable non-archimedean sets, but also of Green's functions, equilibrium measures, and other classical notions from complex potential theory.

A related theory, based on slightly different foundations, was developed shortly thereafter by Kani in [40].

As a powerful application of the adelic Fekete-Szegö theorem for curves proved in [62], Rumely proved in [63] that there is a decision procedure for resolving diophantine equations over $\bar{\mathbb{Z}}$ (the ring of all algebraic integers), thus providing a positive answer to a variant of Hilbert's 10th problem. Moret-Bailly [48] was subsequently able to generalize the decision procedure over $\bar{\mathbb{Z}}$ without using capacity theory, although Rumely's original approach does yield additional quantitative information not obtainable by Moret-Bailly's methods. (As a side note, Richard Taylor has used Moret-Bailly's work in studying Artin's conjecture concerning L -functions of Galois representations, see e.g. [71].)

One problem with non-archimedean potential theory, as developed in [62], [40], and other works, is that some of the constructions feel a bit artificial, and there are some technical notions (e.g., "algebraic capacitability") which one must come to terms with that do not play a prominent role the complex case. In addition, certain fundamental constructions from the complex case do not carry over. For example, a basic fact from complex potential theory is that the closed unit disc in $\mathbb{C}$ has capacity 1, and its unique "equilibrium measure" – the probability measure of minimum energy supported on the unit disc – is normalized Haar measure on the unit circle. Over the field $\mathbb{C}_p$ of " p -adic complex numbers", closed balls are not compact, and there is no longer an obvious candidate for the equilibrium measure of the unit disc. This is unfortunate, since morally speaking one feels that such a measure "ought to exist". For example, the function $\log^+ |z|_p$ has all the properties required to be considered as a *Green's function* for the complement of the unit disc in $\mathbb{C}_p$, just like the function $\log^+ |z|$ over $\mathbb{C}$. In the complex case, the equilibrium measure and Green's function

determine each other: the Green's function can be obtained by integrating the equilibrium measure against the “fundamental potential kernel” $-\log|x-y|$, and the equilibrium measure can be obtained by taking the (distributional) Laplacian of the Green's function. Unfortunately, in the version of capacity theory developed in [62] and [40], there is no Laplacian operator to help make sense of a p -adic version of this correspondence.

As explained in this book, the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$ offers a beautiful solution to these conundrums. Any compact subset E of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$ possesses an “equilibrium measure”, defined as in the classical case as a probability measure of minimum energy supported on E , and when the capacity of E is positive, the equilibrium measure is unique. For example, the equilibrium measure for the unit disc $B(0, 1)$ is a point mass supported at the point of $\mathbb{P}_{\text{Berk}}^1$ corresponding to $B(0, 1)$ (the “Gauss point”). The fact that the equilibrium measure is supported entirely on the set $\mathbb{H}_{\text{Berk}} = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$ illustrates why it is difficult to understand concepts like the equilibrium measure without using Berkovich's theory.

The inclusion relations between closed discs in $\mathbb{C}_p$ give rise to a natural “profinite $\mathbb{R}$ -tree structure” on $\mathbb{P}_{\text{Berk}}^1$, and there is a natural Laplacian operator on $\mathbb{P}_{\text{Berk}}^1$ defined in terms of the branching of this tree. And just as in the classical case, the equilibrium measure and Green's function for a compact set $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$ of positive capacity determine one another. As in the complex case, the Green's function is obtained by integrating the equilibrium measure against a fundamental potential kernel (the *Hsia kernel*) which is a natural extension of $-\log|x-y|_p$ to the Berkovich projective line. Conversely, the equilibrium measure can be obtained as the Laplacian of the corresponding Green's function.

Notation

We set the following notation, which will be used throughout unless otherwise noted.

K	a complete and algebraically closed nonarchimedean field.
$ \cdot $	the absolute value on K , which we assume to be non-trivial.
$ K^* $	the value group of K , i.e., $\{ \alpha : \alpha \in K^*\}$.
$\mathcal{O}$	the valuation ring of K .
$\mathfrak{m}$	the maximal ideal of $\mathcal{O}$.
$\tilde{K}$	the residue field of K .
q_v	a fixed real number greater than 1, chosen so that $-\log_{q_v} : K^* \rightarrow \mathbb{R}$ is a suitably normalized valuation v on K .
$\log_v$	shorthand for $\log_{q_v}$
$B(a, r)$	the closed disc $\{z \in K : z - a \leq r\}$ of radius r about a in K . Here r is any positive real number, and sometimes we allow the degenerate case $r = 0$ as well. If $r \in K^* $ we call the disc <i>rational</i> , and if $r \notin K^* $ we call it <i>irrational</i> .
$B(a, r)^-$	the open disc $\{z \in K : z - a < r\}$ of radius r about a in K .
$\mathbb{P}_{\text{Berk}}^1$	the Berkovich projective line over K .
$\mathbb{H}_{\text{Berk}}$	the “hyperbolic space” $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$.
$\mathbb{C}_p$	the completion of a fixed algebraic closure of $\mathbb{Q}_p$ for some prime number p .

CHAPTER 1

Topological description of the Berkovich unit disc

In this chapter, we recall Berkovich's theorem that points of the Berkovich unit disc $\mathcal{B}(0, 1)$ over K can be identified with equivalence classes of sequences of nested closed discs $\{B(a_i, r_i)\}_{i=1,2,\dots}$ contained in the unit disc $B(0, 1)$ of K . This leads to an explicit description of the Berkovich unit disc as a sort of “infinitely branched tree”; more precisely, we show that $\mathcal{B}(0, 1)$ is an inverse limit of finite $\mathbb{R}$ -trees.

1. Definition of $\mathcal{B}(0, 1)$

Let $\mathcal{A} = K\langle T \rangle$ be the ring of all formal power series with coefficients in K , converging on the unit disc $B(0, 1)$. That is, $\mathcal{A}$ is the ring of all power series $f(T) = \sum_{i=0}^{\infty} a_i T^i \in K[[T]]$ such that $\lim_{i \rightarrow \infty} |a_i| = 0$. Equipped with the *Gauss norm* $\|\cdot\|$ defined by $\|f\| = \max_i(|a_i|)$, $\mathcal{A}$ becomes a Banach algebra over K .

A *multiplicative seminorm* on $\mathcal{A}$ is a function $[\cdot]_x : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ such that $[0]_x = 0$, $[1]_x = 1$, $[f \cdot g]_x = [f]_x \cdot [g]_x$ and $[f + g]_x \leq [f]_x + [g]_x$ for all $f, g \in \mathcal{A}$. It is a *norm* if $[f]_x = 0$ if and only if $f = 0$.

A multiplicative seminorm $[\cdot]_x$ is called *bounded* if there is a constant C_x such that $[f]_x \leq C_x \|f\|$ for all $f \in \mathcal{A}$. It is well-known (see [31, Proposition 5.2]) that boundedness is equivalent to continuity with respect to the Banach norm topology on $\mathcal{A}$.

The reason for writing the x in $[\cdot]_x$ is that we will be considering the space of all bounded multiplicative seminorms on $\mathcal{A}$, and we will identify the seminorm $[\cdot]_x$ with a point x in this space.

It can be deduced from the definition that a bounded multiplicative seminorm $[\cdot]_x$ on $\mathcal{A}$ behaves just like a nonarchimedean absolute value, except that its kernel may be nontrivial. For example, $[\cdot]_x$ satisfies the following properties:

LEMMA 1.1. *Let $[\cdot]_x$ be a bounded multiplicative seminorm on $\mathcal{A}$. Then:*

- (1) $[f]_x \leq \|f\|$ for all $f \in \mathcal{A}$.
- (2) $[c]_x = |c|$ for all $c \in K$.
- (3) $[f + g]_x \leq \max([f]_x, [g]_x)$ for all $f, g \in \mathcal{A}$, with equality if $[f]_x \neq [g]_x$.

PROOF. (1) For each n , $([f]_x)^n = [f^n]_x \leq C_x \|f^n\| = C_x \|f\|^n$, so $[f]_x \leq C_x^{1/n} \|f\|$, and letting $n \rightarrow \infty$ gives the desired inequality.

(2) By the definition of the Gauss norm, $\|c\| = |c|$. If $c = 0$ then trivially $[c]_x = 0$; otherwise, $[c]_x \leq \|c\| = |c|$ and $[c^{-1}]_x \leq \|c^{-1}\| = |c^{-1}|$, while multiplicativity gives $[c]_x \cdot [c^{-1}]_x = [c \cdot c^{-1}]_x = 1$. Combining these gives $[c]_x = |c|$.

(3) The binomial theorem shows that for each n ,

$$\begin{aligned} ([f + g]_x)^n &= [(f + g)^n]_x = \left[\sum_{k=0}^n \binom{n}{k} f^k g^{n-k} \right]_x \\ &\leq \sum_{k=0}^n \left| \binom{n}{k} \right| \cdot [f]_x^k [g]_x^{n-k} \leq \sum_{k=0}^n [f]_x^k [g]_x^{n-k} \\ &\leq (n+1) \cdot \max([f]_x, [g]_x)^n. \end{aligned}$$

Taking n^{th} roots and passing to a limit gives the desired inequality. If in addition $[f]_x < [g]_x$, then $[g]_x \leq \max([f + g]_x, [-f]_x)$ implies that $[f + g]_x = [g]_x$. $\square$

As a set, the Berkovich unit disc $\mathcal{B}(0, 1)$ is defined to be the functional analytic *spectrum* of $\mathcal{A}$, i.e., the set of all bounded multiplicative seminorms $[\cdot]_x$ on $K\langle T \rangle$. The set $\mathcal{B}(0, 1)$ is clearly non-empty, since it contains the Gauss norm. By abuse of notation, we will often denote by just x the seminorm $[\cdot]_x \in \mathcal{B}(0, 1)$. To make this set into a topological space, the Berkovich unit disc is equipped with the *Gelfand topology*, which is the weakest topology such that for all $f \in A$ and all $\alpha \in \mathbb{R}$, the sets

$$\begin{aligned} U(f, \alpha) &= \{x \in \mathcal{B}(0, 1) : [f]_x < \alpha\} \\ V(f, \alpha) &= \{x \in \mathcal{B}(0, 1) : [f]_x > \alpha\} \end{aligned}$$

are open. This topology makes $\mathcal{B}(0, 1)$ into a compact Hausdorff space, see Theorem D.2. It is also connected, and even path-connected; this will emerge as a simple consequence of our description in §4 below of $\mathcal{B}(0, 1)$ as a profinite $\mathbb{R}$ -tree. See also [9, Corollary 3.2.3] for a generalization to higher dimensions.

2. Berkovich's classification

A useful observation is that each $x \in \mathcal{B}(0, 1)$ is determined by its values on the linear polynomials $T - a$, $a \in B(0, 1)$. Indeed, fix $x \in \mathcal{B}(0, 1)$. By the Weierstrass Preparation Theorem ([17, Theorem

5.2.2/1]), each $f \in K\langle T \rangle$ can be uniquely written as

$$f = c \cdot \prod_{j=1}^m (T - a_j) \cdot u(T)$$

where $c \in K$, $a_j \in B(0, 1)$ for each j , and $u(T)$ is a unit power series, that is $u(T) = 1 + \sum_{i=1}^{\infty} a_i T^i \in K\langle T \rangle$ with $|a_i| < 1$ for all $i \geq 1$ and $\lim_{i \rightarrow \infty} |a_i| = 0$. It is easy to see that $[u]_x = 1$. Indeed, $u(T)$ has a multiplicative inverse $u^{-1}(T)$ of the same form, and by the definition of the Gauss norm $\|u\| = \|u^{-1}\| = 1$. Since $[u]_x \leq \|u\| = 1$, $[u^{-1}]_x \leq \|u^{-1}\| \leq 1$, and $[u]_x \cdot [u^{-1}]_x = [u \cdot u^{-1}]_x = 1$, we must have $[u]_x = 1$. It follows that

$$[f]_x = |c| \cdot \prod_{j=1}^m [T - a_j]_x.$$

Thus x is determined by the values of $[\]_x$ on the linear polynomials $T - a_j$.

Before proceeding further, we give some examples of elements of $\mathcal{B}(0, 1)$. For each $a \in B(0, 1)$ we have the *evaluation seminorm*

$$[f]_a = |f(a)|.$$

The boundedness of $[f]_a$ follows easily from the ultrametric inequality, and it is obvious that $[\]_a$ is multiplicative.

Also, for each subdisc $B(a, r) \subset B(0, 1)$, we have the *supremum norm*

$$[f]_{B(a, r)} = \sup_{z \in B(a, r)} |f(z)|.$$

One of the miracles of the non-archimedean universe is that this norm is *multiplicative*. This is a consequence of the Maximum Modulus Principle in non-archimedean analysis (see [17, Propositions 5.1.4/2 and 5.1.4/3]), which tells us that if $g(T) = \sum_{i=0}^{\infty} a_i (T - a)^i \in K[[T]]$ converges on $B(a, r)$ (i.e., if $\lim |a_i| r^i = 0$), then

$$[f]_{B(a, r)} = \sup |a_i| r^i.$$

More generally, for any decreasing sequence of discs $x = \{B(a_i, r_i)\}_{i \geq 1}$, one can consider the limit seminorm

$$[f]_x = \lim_{i \rightarrow \infty} [f]_{B(a_i, r_i)}.$$

Berkovich's classification theorem asserts that every point $x \in \mathcal{B}(0, 1)$ arises in this way:

THEOREM 1.2. (Berkovich [9], p.18) *Every $x \in \mathcal{B}(0, 1)$ can be realized as*

$$(1.1) \quad [f]_x = \lim_{i \rightarrow \infty} [f]_{B(a_i, r_i)}$$

for some sequence of nested discs $B(a_1, r_1) \supseteq B(a_2, r_2) \supseteq \dots$. If this sequence has a nonempty intersection, then either

- 1) the intersection is a single point a , in which case $[f]_x = |f(a)|$,*
- or*
- 2) the intersection is a closed disc $B(a, r)$ (where r may or may not belong to the value group of K), in which case $[f]_x = [f]_{B(a, r)}$.*

PROOF. Fix $x \in \mathcal{B}(0, 1)$, and consider the family of (possibly degenerate) discs

$$\mathcal{F} = \{B(a, [T - a]_x) : a \in B(0, 1)\} .$$

We claim that the family $\mathcal{F}$ is totally ordered by containment. Indeed, if $a, b \in B(0, 1)$ and $[T - a]_x \geq [T - b]_x$, then

$$(1.2) \quad \begin{aligned} |a - b| &= [a - b]_x = [(T - b) - (T - a)]_x \\ &\leq \max([T - a]_x, [T - b]_x) = [T - a]_x , \end{aligned}$$

with equality if $[T - a]_x > [T - b]_x$. In particular $b \in B(a, [T - a]_x)$, and

$$B(a, [T - a]_x) \supseteq B(b, [T - b]_x) .$$

Put $r = \inf_{a \in B(0, 1)} [T - a]_x$, and choose a sequence of points $a_i \in B(0, 1)$ such that the numbers $r_i = [T - a_i]_x$ satisfy $\lim_{i \rightarrow \infty} r_i = r$.

We claim that for each polynomial $T - a$ with $a \in B(0, 1)$, we have

$$(1.3) \quad [T - a]_x = \lim_{i \rightarrow \infty} [T - a]_{B(a_i, r_i)} .$$

By definition of r , for each $a \in B(0, 1)$ we have $[T - a]_x \geq r$. If $[T - a]_x = r$, then for each a_i we have $r_i = [T - a_i]_x \geq |a_i - a|$ (by (1.2) applied to $a = a_i$ and $b = a$), so $a \in B(a_i, r_i)$. Hence

$$[T - a]_{B(a_i, r_i)} = \sup_{z \in B(a_i, r_i)} |z - a| = r_i .$$

Since $\lim_{i \rightarrow \infty} r_i = r$, (1.3) holds in this case.

If $[T - a]_x > r$, then for each a_i with $[T - a]_x > [T - a_i]_x$ (which will be all but finitely many a_i), we have $[T - a]_x = |a - a_i|$ by the strict case in (1.2), which means that $|a - a_i| > [T - a_i]_x = r_i$. Hence

$$[T - a]_{B(a_i, r_i)} = \sup_{z \in B(a_i, r_i)} |z - a| = |a - a_i| = [T - a]_x .$$

Thus the limit on the right side of (1.3) stabilizes at $[T - a]_x$, and (1.3) holds in this case as well. As noted previously, $[\]_x$ is determined by its values on the polynomials $T - a$, so for all $f \in K\langle T \rangle$,

$$(1.4) \quad [f]_x = \lim_{i \rightarrow \infty} [f]_{B(a_i, r_i)} .$$

Now suppose the intersection of the family $\mathcal{F}$ is non-empty, and contains the point a . Formula (1.3) gives

$$[T - a]_x = \lim_{i \rightarrow \infty} [T - a]_{B(a_i, r_i)} \leq \lim_{i \rightarrow \infty} r_i = r ,$$

while the definition of r shows that $[T - a]_x \geq r$. Thus $[T - a]_x = r$. Hence the disc $B(a, r)$ (which may consist of a single point if $r = 0$) is a minimal element of $\mathcal{F}$. The arguments above show that formula (1.4) holds for *any* sequence of discs $B(a_i, r_i)$ such that $r_i = [T - a_i]_x$ satisfies $\lim r_i = r$. If we take $a_i = a$ for each i , then $r_i = [T - a]_x = r$, and (1.4) gives $[f]_x = [f]_{B(a, r)}$. If $r = 0$, it gives $[f]_x = |f(a)|$. $\square$

When $K = \mathbb{C}_p$ for some prime p , for example, it is important to note that there *are* sequences of nested discs $\{B(a_i, r_i)\}$ with empty intersection. Such sequences necessarily satisfy $r = \lim r_i > 0$, since if $r = 0$, the completeness of K shows the intersection is a point $a \in K$.

To construct an example of such a sequence, fix $0 < r < 1$, and choose a sequence $\{r_i\}$ which decreases monotonically to r , with $r < r_i \leq 1$ for each i . The algebraic closure $\tilde{K}$ is countable, and is dense in K . Enumerate the elements of $\tilde{K} \cap B(0, 1)$ as $\{\alpha_j\}_{j \geq 1}$. Define a sequence of discs $B(a_i, r_i)$ as follows. Take $B(a_1, r_1) = B(\alpha_1, r_1)$. Suppose $B(a_i, r_i)$ has been constructed. Let j_i be the least index of an element with $\alpha_{j_i} \in B(a_i, r_i)$. Since $r_i > r_{i+1}$, $B(a_i, r_i) \setminus B(\alpha_{j_i}, r_{i+1})$ is nonempty; let a_{i+1} be any element of it. Then $\alpha_{j_i} \notin B(a_{i+1}, r_{i+1})$.

Clearly the sequence j_i increases to ∞ , indeed one sees inductively that $j_i \geq i$. For each i , the construction has arranged that the α_j with $j \leq j_i$ do not belong to $B(a_{i+1}, r_{i+1})$. It follows that $\bigcap_{i=1}^{\infty} B(a_i, r_i)$ contains no elements of $\tilde{K}$. If it were nonempty, it would be a disc $B(a, r)$. However, every such disc contains elements of $\tilde{K}$. Hence $\bigcap_{i=1}^{\infty} B(a_i, r_i)$ must be empty.

A nonarchimedean field in which every nested sequence of closed discs has nonempty intersection is called *spherically complete*. For example, finite extensions of $\mathbb{Q}_p$ are spherically complete, while as we have just seen $\mathbb{C}_p$ is not spherically complete. It is easy to see that every spherically complete field is complete, and it is known (see [26],

Chapter 7) that every nonarchimedean field L is contained in an algebraically closed and spherically complete field whose absolute value extends the one on L .

This brings us to Berkovich's classification of elements of $\mathcal{B}(0, 1)$:

1. Points corresponding to sequences $\{B(a_i, r_i)\}$ with $\lim r_i = 0$ are said to be of *Type I*. As noted above, the completeness of K assures that the intersection of such a sequence is a point $a \in K$, and the corresponding seminorm is $[\]_a$. We will also call these points the “classical points”.
2. Points corresponding to sequences $\{B(a_i, r_i)\}$ with nonempty intersection, for which $r = \lim r_i > 0$ belongs to the value group $|K^*|$ of K , are said to be of *Type II*. These correspond to a supremum norm $[\]_{B(a, r)}$.
3. Points corresponding to sequences $\{B(a_i, r_i)\}$ with nonempty intersection, but for which $r = \lim r_i > 0$ does not belong to the value group of K , are said to be of *Type III*. These also correspond to a supremum norm $[\]_{B(a, r)}$; however the disc $B(a, r)$ is not an “affinoid domain” in the sense of classical rigid analysis. (Or in Berkovich's terminology, $B(a, r)$ is not a *strict affinoid*.)
4. Points corresponding to sequences $\{B(a_i, r_i)\}$ with empty intersection are said to be of *Type IV*. As noted before, necessarily $\lim r_i > 0$.

We let $\zeta_{a, r}$ denote the point of $\mathcal{B}(0, 1)$ of type II or III corresponding to the closed or irrational disc $B(a, r)$. Allowing degenerate discs (i.e., $r = 0$), we can extend this notation to points of type I: we let $\zeta_{a, 0}$ be the point of $\mathcal{B}(0, 1)$ corresponding to $a \in B(0, 1)$.

Following the terminology introduced by Chambert-Loir in [22], the distinguished point $\zeta_{0, 1}$ in $\mathcal{B}(0, 1)$, corresponding to the Gauss norm $\|f\| = [\]_{B(0, 1)}$, will be called the *Gauss point*. We will sometimes write ζ_{Gauss} instead of $\zeta_{0, 1}$.

We will call two sequences of nested discs *equivalent* if they define the same point in $\mathcal{B}(0, 1)$.

LEMMA 1.3. *Two nested sequences $\{B(a_i, r_i)\}$ and $\{B(a'_i, r'_i)\}$ of closed discs are equivalent if and only if*

- (a) *each has a nonempty intersection, and their intersections are the same; or*
- (b) *both have empty intersection, and each sequence is cofinal in the other.*

(Cofinal means that for each i , there is a j such that $B(a_i, r_i) \supseteq B(a'_j, r'_j)$, and for each j there is an i such that $B(a'_j, r'_j) \supseteq B(a_i, r_i)$.)

PROOF. Since the limit (2.4) is a decreasing one, it is clear that two sequences $\{B(a_i, r_i)\}$ and $\{B(a'_i, r'_i)\}$ are equivalent if either condition (a) or (b) is satisfied.

These conditions are also necessary. Indeed, from Theorem 1.2 it is clear that two sequences with nonempty intersection are equivalent if and only if they have the same intersection. A sequence x with nonempty intersection $B(a, r)$ (possibly $r = 0$) cannot be equivalent to any sequence $y = \{B(a_i, r_i)\}$ with empty intersection, since for any i with $a \notin B(a_i, r_i)$ we have

$$[T - a_i]_x = [T - a_i]_{B(a, r)} = |a - a_i| > r_i,$$

whereas

$$[T - a_i]_y \leq [T - a_i]_{B(a_i, r_i)} = r_i.$$

Finally, two sequences $x = \{B(a_i, r_i)\}$ and $y = \{B(a'_i, r'_i)\}$ with empty intersection which are not cofinal cannot be equivalent. Since x is not cofinal in y , after removing some initial terms of x we can assume that $B(a_1, r_1) \not\supseteq B(a'_i, r'_i)$ for any i . Since $\bigcap_{i=1}^{\infty} B(a'_i, r'_i) = \emptyset$, after deleting some initial terms of y we can assume that $a_1 \notin B(a'_1, r'_1)$. As any two discs are either disjoint, or one contains the other, it must be that $B(a_1, r_1)$ and $B(a'_1, r'_1)$ are disjoint. Since $B(a'_i, r'_i) \subseteq B(a'_1, r'_1)$ for all i , we have $[T - a_1]_{B(a'_i, r'_i)} = |a_1 - a'_1| > r_1$ for each i . Hence

$$[T - a_1]_x \leq [T - a_1]_{B(a_1, r_1)} = r_1,$$

while

$$[T - a_1]_y = \lim_{i \rightarrow \infty} [T - a_1]_{B(a'_i, r'_i)} = |a_1 - a'_1| > r_1.$$

□

REMARK 1.4. More generally, every nested *family* (or *net*) of closed discs in $\mathcal{B}(0, 1)$ defines a point of $\mathcal{B}(0, 1)$, and we call two families of nested discs equivalent if they define the same point in $\mathcal{B}(0, 1)$. The proof of Theorem 1.2 shows that every nested family of closed discs is equivalent to a sequence of closed discs, so that equivalence classes of nested families and nested sequences coincide.

3. Topology on $\mathcal{B}(0, 1)$

By definition, the Gelfand topology on $\mathcal{B}(0, 1)$ is generated by the open sets $U(f, \alpha) = \{x \in \mathcal{B}(0, 1) : [f]_x < \alpha\}$ and $V(f, \alpha) = \{x \in \mathcal{B}(0, 1) : [f]_x > \alpha\}$, for $f \in K\langle T \rangle$ and $\alpha \in \mathbb{R}$. Since the value group of K is dense in $\mathbb{R}_{>0}$, it suffices to consider α belonging to the value

group of K . Also, by the Weierstrass Preparation Theorem and the fact that any unit power series $u(T)$ satisfies $[u]_x \equiv 1$, we can restrict to polynomials $f(T) \in K[T]$ with roots in $B(0, 1)$.

Given a nonconstant polynomial $f(T) = c \prod_{i=1}^n (T - a_i)^{m_i} \in K[T]$, and $\alpha > 0$ belonging to the value group of K , there is a well-known description of the set $\{z \in K : |f(z)| \leq \alpha\}$ as a finite union of closed discs $\bigcup_{i=1}^N B(a_i, r_i)$ (c.f. [21], Theorem 3.1.2, p.180). Here the centers can be taken to be roots of $f(T)$, and each r_i belongs to the value group of K . If desired, one can assume that the discs in the decomposition are pairwise disjoint. However, for us it will be more useful to assume that all the roots occur as centers, so $N = n$, and that if a_i and a_j are roots with $a_j \in B(a_i, r_i)$, then $B(a_i, r_i) = B(a_j, r_j)$.

Taking the union over an increasing sequence of α , we can lift the requirement that α belongs to the value group of K (note that if α is not in the value group, then $\{z \in K : |f(z)| = \alpha\}$ is empty.) Thus, any $\alpha > 0$ determines a collection of numbers $r_i > 0$, which belong to the value group of K if α does, such that

$$\{z \in K : |f(z)| \leq \alpha\} = \bigcup_{i=1}^n B(a_i, r_i) .$$

Here we assume as before that $a_1, \dots, a_n$ are the roots of $f(z)$, and if $a_j \in B(a_i, r_i)$ then $r_j = r_i$. Using this and the factorization of $f(T)$, it is easy to see that

$$\{z \in K : |f(z)| < \alpha\} = \bigcup_{i=1}^n B(a_i, r_i)^- ,$$

where $B(a_i, r_i)^- = \{z \in K : |z - a_i| < r_i\}$.

For any closed disc $B(b, t) \subset B(a_i, r_i)^-$ with t in the value group of K , one sees readily that

$$\sup_{z \in B(b, t)} |f(z)| < \alpha .$$

Likewise, on any $B(b, t) \subset B(a_i, r_i) \setminus (\bigcup_{a_j \in B(a_i, r_i)} B(a_j, r_i)^-)$ one has $|f(z)| \equiv \alpha$, and on any $B(b, t)$ disjoint from $\bigcup_{i=1}^n B(a_i, r_i)$ one has $|f(z)| \equiv \beta$ for some $\beta > \alpha$.

Suppose $x \in \mathcal{B}(0, 1)$ corresponds to a sequence of nested discs $\{B(b_j, t_j)\}$. Without loss, we can assume that each t_j belongs to the value group of K . We will say that x is associated to an open disc $B(a, r)^-$ if there is some j such that $B(b_j, t_j) \subset B(a, r)^-$. We say that x is associated to a closed disc $B(a, r)$ if there is some j such that $B(b_j, t_j) \subset B(a, r)$, or if $\bigcap_{j=1}^\infty B(b_j, t_j) = B(a, r)$. From the assertions in the previous paragraph, it follows that $[f]_x < \alpha$ if and only if x is

associated to some $B(a_i, r_i)^-$. Likewise, $[f]_x > \alpha$ if and only if x is not associated to any of the $B(a_i, r_i)$.

LEMMA 1.5. *Let $x \in \mathcal{B}(0, 1)$. Then:*

A) *x is associated to an open disc $B(a, r)^-$ if and only if $[T - a]_x < r$.*

B) *x is associated to a closed disc $B(a, r)$ if and only if $[T - a]_x \leq r$.*

PROOF. Suppose $x \in \mathcal{B}(0, 1)$ corresponds to a sequence of nested discs $\{B(b_j, t_j)\}$ with each t_j in the value group of K . We have $[T - a]_{B(b_j, t_j)} = \max(t_j, |b_j - a|)$. Thus, if $[T - a]_x < r$, then there is some j for which $\max(t_j, |b_j - a|) < r$ and this implies $B(b_j, t_j) \subset B(a, r)^-$. Conversely, if $B(b_j, t_j) \subset B(a, r)^-$ then $t_j < r$ and $|b_j - a| < r$ so $[T - a]_x \leq [T - a]_{B(b_j, t_j)} < r$. This proves (A).

For (B), first suppose x is associated to $B(a, r)$. If some $B(b_j, t_j) \subset B(a, r)$ then clearly $[T - a]_x \leq [T - a]_{B(b_j, t_j)} \leq r$. By Theorem 1.2 if $\cap_{j=1}^\infty B(b_j, t_j) = B(a, r)$ then $[T - a]_x = [T - a]_{B(a, r)} = r$. Conversely, suppose $[T - a]_x \leq r$. If $B(b_j, t_j) \subset B(a, r)$ for some j then x is certainly associated to $B(a, r)$. Otherwise, for each j either $B(b_j, t_j)$ is disjoint from $B(a, r)$ or $B(b_j, t_j) \supset B(a, r)$. If some $B(b_{j_0}, t_{j_0})$ is disjoint from $B(a, r)$ then by the ultrametric inequality $|z - a| = |b_{j_0} - a| > r$ for all $z \in B(b_{j_0}, t_{j_0})$. Since the discs $B(b_j, t_j)$ are nested, this means that $[T - a]_{B(b_j, t_j)} = |b_{j_0} - a|$ for all $j \geq j_0$, and hence that $[T - a]_x = |b_{j_0} - a|$. This contradicts $[T - a]_x \leq r$, so it must be that each $B(b_j, t_j)$ contains $B(a, r)$. Thus, $\cap_{j=1}^\infty B(b_j, t_j) = B(a, t)$ for some $t \geq r$. By Theorem 1.2, $[T - a]_x = t$. Since we have assumed $[T - a]_x \leq r$ this gives $t = r$, so x is associated to $B(a, r)$. $\square$

This leads us to define open and closed “Berkovich discs”, as follows. For $a \in B(0, 1)$ and $r > 0$, write

$$\begin{aligned} \mathcal{B}(a, r)^- &= \{x \in \mathcal{B}(0, 1) : [T - a]_x < r\} , \\ \mathcal{B}(a, r) &= \{x \in \mathcal{B}(0, 1) : [T - a]_x \leq r\} . \end{aligned}$$

With this notation, our discussion above shows that

$$U(f, \alpha) = \bigcup_{i=1}^N \mathcal{B}(a_i, r_i)^- , \quad V(f, \alpha) = \mathcal{B}(0, 1) \setminus \bigcup_{i=1}^N \mathcal{B}(a_i, r_i) .$$

Taking finite intersections of sets of the form $U(f, \alpha)$ and $V(f, \alpha)$ gives a basis for the open sets in the Gelfand topology. Since the value group $|K^*|$ is dense in $\mathbb{R}$, it clearly suffices to take $\alpha \in |K^*|$, in which case $r, r_i \in |K^*|$ as well. Thus:

PROPOSITION 1.6. *A basis for the open sets of $\mathcal{B}(0, 1)$ is given by the sets*

$$\mathcal{B}(a, r)^-, \quad \mathcal{B}(a, r)^- \setminus \bigcup_{i=1}^N \mathcal{B}(a_i, r_i), \quad \text{and} \quad \mathcal{B}(0, 1) \setminus \bigcup_{i=1}^N \mathcal{B}(a_i, r_i),$$

where a and the a_i range over $B(0, 1)$, and where r, r_i all belong to the value group $|K^*|$.

The special subsets appearing in Proposition 1.6 are called *connected strict open affinoids* in $\mathcal{B}(0, 1)$. If we lift the requirement that $r, r_i \in |K^*|$, we obtain the more general class of *connected open affinoids* in $\mathcal{B}(0, 1)$. We will also refer to these as *simple domains*. Equivalently, a connected open affinoid in $\mathcal{B}(0, 1)$ is any finite intersection of open Berkovich discs. An *open affinoid* in $\mathcal{B}(0, 1)$ is defined as a finite union of connected open affinoids.

We can define (connected, strict) *closed affinoids* in $\mathcal{B}(0, 1)$ similarly, using closed Berkovich discs instead of open ones.

4. Tree structure of $\mathcal{B}(0, 1)$

In this section, we will use Berkovich's classification theorem to prove that $\mathcal{B}(0, 1)$ is homeomorphic to an $\mathbb{R}$ -tree endowed with its weak topology, and to an inverse limit of finite $\mathbb{R}$ -trees. (Recall from Appendix B that an $\mathbb{R}$ -tree is a metric space (T, d) such that for any two points $x, y \in T$, there is a *unique* geodesic segment $[x, y]$ in T joining x to y .)

First, we define a partial order on $\mathcal{B}(0, 1)$ as follows. For $x, y \in \mathcal{B}(0, 1)$, define $x \leq y$ iff $[f]_x \leq [f]_y$ for all $f \in K\langle T \rangle$. Clearly, the Gauss point ζ_{Gauss} is the unique maximal point with respect to this partial order: by Lemma 1.1, we have $x \leq \zeta_{\text{Gauss}}$ for all $x \in \mathcal{B}(0, 1)$.

It is useful to interpret this partial order in terms of Berkovich's classification of points of $\mathcal{B}(0, 1)$. We begin with the following lemma (recall that $\zeta_{a,r}$ denotes the point of $\mathcal{B}(0, 1)$ of Type II or III corresponding to the closed disc $B(a, r)$):

LEMMA 1.7. *We have $\zeta_{a,r} \leq \zeta_{a',r'}$ if and only if $B(a, r) \subseteq B(a', r')$.*

PROOF. Let $x = \zeta_{a,r}$ and $x' = \zeta_{a',r'}$. Recall that for $f \in K\langle T \rangle$, we have $[f]_x = \sup_{z \in B(a,r)} |f(z)|$ and $[f]_{x'} = \sup_{z \in B(a',r')} |f(z)|$. If $B(a, r) \subseteq B(a', r')$, it is therefore clear that $x \leq x'$.

On the other hand, we see that

$$B(a, r) \subseteq B(a', r') \iff \sup_{z \in B(a,r)} |T - a'| \leq r' \iff [T - a']_x \leq r'.$$

So if $B(a, r) \not\subseteq B(a', r')$, then $[T - a']_x > r' = [T - a']_{x'}$, and thus we do not have $x \leq x'$. $\square$

Thus the partial order $\leq$ corresponds to containment of discs on the points of Type II and III. We can easily extend this observation to *all* points of $\mathcal{B}(0, 1)$. For this, note that every point of $\mathcal{B}(0, 1)$ can be represented by a *strictly decreasing* sequence of discs.

LEMMA 1.8. *Let $x, y \in \mathcal{B}(0, 1)$ be arbitrary, and choose strictly decreasing sequences of discs $\{B(a_i, r_i)\}, \{B(a'_i, r'_i)\}$ corresponding to x and y , respectively. Then $x \leq y$ if and only if for each natural number k , there exist $m, n \geq k$ such that $B(a_m, r_m) \subseteq B(a'_n, r'_n)$.*

PROOF. Write $B_i = B(a_i, r_i)$ and $B'_j = B(a'_j, r'_j)$. Recall that for every $f \in K\langle T \rangle$, we have

$$[f]_x = \inf_i [f]_{B_i}$$

and similarly

$$[f]_y = \inf_j [f]_{B'_j}.$$

If for each k there exist $m, n \geq k$ such that $B_m \subseteq B'_n$, then for k sufficiently large and any $\varepsilon > 0$, we have

$$[f]_x \leq [f]_{B_m} \leq [f]_{B'_n} \leq [f]_y + \varepsilon.$$

As f and ε are arbitrary, it follows that $x \leq y$.

Conversely, suppose that $x \leq y$, and fix $k \geq 1$. Consider the function $f' = T - a'_{k+1}$. Since the sequence $\{B'_j\}$ is strictly decreasing, we have $[f']_{B'_{k+1}} = r'_{k+1} < r'_k$, so that $[f']_{B'_{k+1}} \leq r'_k - \varepsilon$ for some $\varepsilon > 0$. On the other hand, for m sufficiently large we have

$$[f']_{B'_{k+1}} \geq [f']_y \geq [f']_x \geq [f']_{B_m} - \varepsilon.$$

It follows that $[T - a'_{k+1}]_{B_m} \leq r'_k$. Since $B'_k = B(a'_k, r'_k) = B(a'_{k+1}, r'_k)$, it follows that $B_m \subseteq B'_k$, and we may take $n = k$. $\square$

It follows easily from Lemma 1.8 that if $x \in \mathcal{B}(0, 1)$ is a point of Type I, then $y \leq x$ if and only if $y = x$, and $x \leq y$ if and only if $x \in B(a'_j, r'_j)$ for all j . Similarly, if $x \in \mathcal{B}(0, 1)$ is a point of Type IV, then $y \leq x$ if and only if $y = x$. Thus we have:

COROLLARY 1.9. *The Gauss point ζ_{Gauss} is the unique maximal point of $\mathcal{B}(0, 1)$, and the minimal points of $\mathcal{B}(0, 1)$ are precisely the points of Type I and Type IV.*

Let $\text{diam} : \mathcal{B}(0, 1) \rightarrow \mathbb{R}_{\geq 0}$ be the function sending $x \in \mathcal{B}(0, 1)$ to $[T]_x$. If x corresponds to a sequence of nested discs $\{B(a_i, r_i)\}$, then $r = \lim_{i \rightarrow \infty} r_i$. We call $r = \text{diam}(x)$ the *diameter* (or *radius*) of x .

We claim that the partially ordered set $\mathcal{B}(0, 1)$, together with the function $1 - \text{diam}$, is a parametrized rooted tree, i.e., that the pair $(\mathcal{B}(0, 1), 1 - \text{diam})$ satisfies axioms (P1)-(P5) of §4 of Appendix B. Indeed, axioms (P1), (P3), and (P4) are obvious, and axioms (P2) and (P5) follow easily from Lemma 1.8 and the fact that two discs in K are either disjoint or one contains the other. Using the correspondence discussed in §4 of Appendix B, we see that:

LEMMA 1.10. *The metric*

$$d(x, y) = \text{diam}(x) + \text{diam}(y) - 2 \text{diam}(x \vee y)$$

makes $\mathcal{B}(0, 1)$ into an $\mathbb{R}$ -tree.

We call d the *small metric* on $\mathcal{B}(0, 1)$. It is important to note that the Berkovich, or Gelfand, topology on $\mathcal{B}(0, 1)$ is *not* the same as the topology induced by d . In fact, as we now show, the Berkovich topology coincides with the *weak topology* on the $\mathbb{R}$ -tree $(\mathcal{B}(0, 1), d)$ (see Appendix B for a definition of the weak topology on an $\mathbb{R}$ -tree).

PROPOSITION 1.11. *$\mathcal{B}(0, 1)$ with its Berkovich topology is homeomorphic to the weak topology on the $\mathbb{R}$ -tree $(\mathcal{B}(0, 1), d)$.*

Before giving the proof, we make a few preliminary remarks.

Recall that in terms of the partial order on $\mathcal{B}(0, 1)$, a closed Berkovich disc $\mathcal{B}(a, r)$ consists of all points $z \in \mathcal{B}(0, 1)$ with $z \leq \zeta_{a,r}$. The open Berkovich disc $\mathcal{B}(a, r)^-$ is the connected component of $\mathcal{B}(0, 1) \setminus \zeta_{a,r}$ containing a , and the open Berkovich disc $\mathcal{B}(0, 1) \setminus \mathcal{B}(a, r)$ is the connected component of $\mathcal{B}(0, 1) \setminus \{\zeta_{a,r}\}$ containing ζ_{Gauss} . The point $\zeta_{a,r}$ is the unique boundary point of $\mathcal{B}(a, r)^-$ (resp. $\mathcal{B}(0, 1) \setminus \mathcal{B}(a, r)$), as one sees easily from the fact that the complement of $\mathcal{B}(a, r)^-$ (resp. $\mathcal{B}(0, 1) \setminus \mathcal{B}(a, r)$) is a union of open Berkovich discs.

In particular, *tangent vectors* at $\zeta_{a,r}$ in the $\mathbb{R}$ -tree $(\mathcal{B}(0, 1), d)$ can be identified with the open Berkovich discs having $\zeta_{a,r}$ as a boundary point. Recall from Appendix B that for $x \in \mathcal{B}(0, 1)$, the space T_x of tangent vectors at x is defined to be the set of equivalence classes of paths $[x, y]$ emanating from x , where y is any point of $\mathcal{B}(0, 1)$ not equal to x . Here we declare that two paths $[x, y_1], [x, y_2]$ are *equivalent* if they share a common initial segment. There is a natural bijection between elements $\vec{v}$ of T_x and connected components of $\mathcal{B}(0, 1) \setminus \{x\}$. We denote by $U(x; \vec{v})$ the connected component of $\mathcal{B}(0, 1) \setminus \{x\}$ corresponding to $\vec{v} \in T_x$. As x and $\vec{v}$ vary, the open sets $U(x; \vec{v})$ generate the weak topology

PROOF OF PROPOSITION 1.11. Let X denote $\mathcal{B}(0, 1)$ equipped with the weak $\mathbb{R}$ -tree topology. The natural map ι (which is the identity map

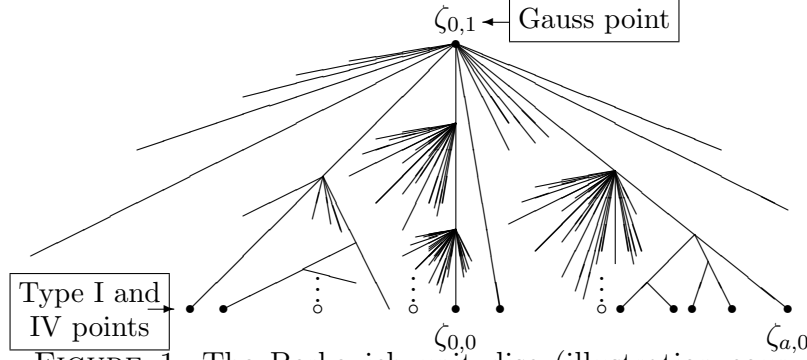


FIGURE 1. The Berkovich unit disc (illustration courtesy of Joe Silverman)

on the underlying sets) from $\mathcal{B}(0, 1)$ to X is continuous, since the above discussion shows that the inverse image of a set of the form $U(x; \vec{v})$ is an open Berkovich disc. Conversely every Berkovich open disc is of the form $U(x; \vec{v})$, which by Proposition 1.6 implies that the inverse of ι is also continuous. $\square$

As a consequence of Proposition 1.11 and Corollary B.17, we obtain:

COROLLARY 1.12. *The space $\mathcal{B}(0, 1)$ is uniquely path-connected.*

The different branches emanating from the root in $\mathcal{B}(0, 1)$ (i.e., the tangent vectors $v \in T_{\zeta_{\text{Gauss}}}$) are naturally in one-to-one correspondence with elements of the residue field $\tilde{K}$ of K , or equivalently with the open disks of radius 1 contained in $B(0, 1)$. Each branch splits into infinitely many branches at each point of type II and each new branch behaves in the same way. This incredible collection of splitting branches forms a sort of “witch’s broom”. However, the witch’s broom has some structure: there is branching *only* at the points of type II, not those of type III; furthermore, the branches emanating from a type II point $\zeta_{a,r}$ other than the root are in one-to-one correspondence with $\mathbb{P}^1(\tilde{K})$: there is one branch going “up” to the Gauss point, and the other branches correspond in a natural way to elements of $\tilde{K}$, or equivalently open discs $B(a', r)^-$ of radius r contained in $B(0, 1)$. Some of the branches extend all the way to the bottom (terminating in points of type I), while others are “cauterized off” earlier and terminate at points of type IV, but every branch terminates either at a point of type I or type IV.

There is another metric which is only defined on $\mathcal{B}(0, 1) \setminus B(0, 1)$, i.e., on the complement of the points of Type I. It is given by the parametrization function $-\log_v(\text{diam})$ on $\mathcal{B}(0, 1) \setminus B(0, 1)$. The same verification as before shows that the pair $(\mathcal{B}(0, 1) \setminus B(0, 1), -\log_v(\text{diam}))$

is a parametrized rooted tree, and the corresponding metric ρ is called the *big metric* on the $\mathbb{R}$ -tree $\mathcal{B}(0, 1) \setminus B(0, 1)$. In particular:

LEMMA 1.13. *The metric*

$$\rho(x, y) = -\log_v(\text{diam}(x)) - \log_v(\text{diam}(y)) + 2\log_v(\text{diam}(x \vee y))$$

makes $\mathcal{B}(0, 1) \setminus B(0, 1)$ into an $\mathbb{R}$ -tree.

Since we will often make use of the tree structure of $\mathcal{B}(0, 1)$, we now make an effort to make the structures just described more concrete.

Let $x \in \mathcal{B}(0, 1)$ be a point of type I, II, or III in $\mathcal{B}(0, 1)$, corresponding to a (possibly degenerate) disc $B(a, r) \subset B(0, 1)$ of radius $r \geq 0$. For each $t \in [r, 1]$, define $x(t) = \zeta_{a,t} \in \mathcal{B}(0, 1)$. We define the *line of discs* $[x, \zeta_{\text{Gauss}}]$ to be the subset $\{x(t)\} \subset \mathcal{B}(0, 1)$ corresponding to the collection of discs $\{B(a, t) : r \leq t \leq 1\}$ in $B(0, 1)$.

LEMMA 1.14. *For $x \in \mathcal{B}(0, 1)$ of type I, II, or III, the mapping $i_{a,r} : t \mapsto x(t)$ is a continuous embedding of the real interval $[r, 1]$ into $\mathcal{B}(0, 1)$. In particular, the line of discs $[x, \zeta_{\text{Gauss}}] \subset \mathcal{B}(0, 1)$ is homeomorphic to an interval.*

PROOF. Let $\|\cdot\|_{B(a,t)}$ be the multiplicative seminorm corresponding to $x(t)$. For any $f \in K\langle T \rangle$, it is not hard to see that the mapping from $[r, 1]$ to $\mathbb{R}$ defined by $t \mapsto \|f\|_{B(a,t)}$ is continuous in the Berkovich topology; this implies the continuity of $i_{a,r}$. Since $i_{a,r}$ is clearly injective, it suffices to prove that the inverse mapping is also continuous. This in turn follows from the fact that $i_{a,r}((\alpha, \beta)) = \{x \in \mathcal{B}(0, 1) \mid \alpha < [T - a]_x < \beta\}$ for all open intervals $(\alpha, \beta) \subset [r, 1]$. $\square$

Give more details.

One can easily extend Lemma 1.14 to the situation where x is a point of Type IV. If x corresponds to the nested sequence $B(a_i, r_i)$ of closed discs, with $\lim r_i = r$, define the line of discs $[x, \zeta_{\text{Gauss}}]$ to be the set of all points $\zeta_{a',r'} \in \mathcal{B}(0, 1)$ of type II or III with $B(a_i, r_i) \subseteq B(a', r')$ for some i . Using the fact that $r_i \rightarrow r$, and that two discs in $\mathcal{B}(0, 1)$ are either disjoint or one contains the other, it follows easily that for each $r' > r$, there is a *unique* point $x(r') \in [x, \zeta_{\text{Gauss}}]$ corresponding to a ball of radius r' . We can thus define a mapping $i_x : [r, 1] \rightarrow [x, \zeta_{\text{Gauss}}]$ by sending r to x and r' to $x(r')$ for each $r < r' \leq 1$. Clearly $x(1) = \zeta_{\text{Gauss}}$. An argument similar to the proof of Lemma 1.14 shows:

LEMMA 1.15. *For $x \in \mathcal{B}(0, 1)$ of type IV, the mapping $i_x : t \mapsto x(t)$ is a continuous embedding of the real interval $[r, 1]$ into $\mathcal{B}(0, 1)$.*

This gives another way of seeing that the space $\mathcal{B}(0, 1)$ is path-connected.

There are (at least) two obvious candidates for metrics on the line of discs $[x, \zeta_{\text{Gauss}}]$. In the *small metric*, we declare that $[x, \zeta_{\text{Gauss}}]$ has length $1 - r$, and in the *big metric*, we declare that $[x, \zeta_{\text{Gauss}}]$ has length $\log_v(1/r)$. In other words, for the small metric the map $t \in [r, 1] \mapsto x(t)$ is an isometry, and for the big metric the map $s \in [0, \log_v(1/r)] \mapsto x(q_v^{-s})$ is an isometry. It turns out that even though it looks more cumbersome, the big metric is more natural and more useful than the small metric. (This will be elaborated upon in the next chapter.)

Now let $S = \{B(a_1, r_1), \dots, B(a_n, r_n)\}$ be any finite set of (closed or irrational) discs of positive radius contained in $B(0, 1)$. For simplicity, we assume that $B(0, 1) \notin S$. To each disc $B(a_i, r_i)$, there is an associated point $\zeta_{a_i, r_i} \in \mathcal{B}(0, 1)$, which is a point of type II or III.

Define the *graph of discs* $\Lambda_S \subset \mathcal{B}(0, 1)$ to be the union of the associated lines of discs $[\zeta_{a_i, r_i}, \zeta_{\text{Gauss}}]$:

$$\Lambda_S = \bigcup_{i=1}^n [\zeta_{a_i, r_i}, \zeta_{\text{Gauss}}] .$$

The subset Λ_S inherits both a small and big metric structure from the corresponding metrics on the lines of discs which comprise it. From Lemmas 1.10 and 1.13, it follows that Λ_S is an $\mathbb{R}$ -tree with respect to both of these metrics.

An alternate description of Λ_S , which may make its structure clearer, is as follows. Define the *saturation* of S to be the set $\hat{S}$ gotten by adjoining to S all discs $B(a_i, |a_i - a_j|)$ with $B(a_i, r_i), B(a_j, r_j) \in S$, and also the disc $B(0, 1)$. Note that the radius of each disc in $\hat{S} \setminus S$ belongs to the value group of K . Then Λ_S is the finite $\mathbb{R}$ -tree whose branch points correspond to the discs in $\hat{S}$, and which has an isometric path of length $|r_i - r_j|$ (in the small metric) between each pair of nodes $B(a_i, r_i), B(a_j, r_j) \in \hat{S}$ for which $B(a_i, r_i) \supseteq B(a_j, r_j)$ or $B(a_j, r_j) \supseteq B(a_i, r_i)$.

It follows from this description that every branch point of Λ_S (not including the endpoints) is a type II point of $\mathcal{B}(0, 1)$.

If S_1 and S_2 are any two finite sets of discs, then Λ_{S_1} and Λ_{S_2} are both $\mathbb{R}$ -subtrees of $\Lambda_{S_1 \cup S_2}$. Moreover, the embedding of Λ_{S_i} in $\Lambda_{S_1 \cup S_2}$ is an isometry (with respect to either metric). Thus, the collection of graphs Λ_S is a directed set. Let

$$\Lambda = \varinjlim \Lambda_S = \bigcup_S \Lambda_S$$

be the union (or direct limit) of all of the $\mathbb{R}$ -trees Λ_S ; this is itself an $\mathbb{R}$ -tree (equipped with two possible metrics). With respect to the small

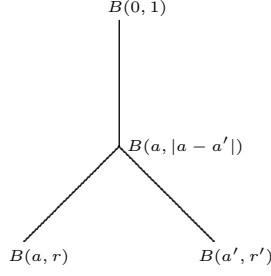


FIGURE 2. The tree Λ_S when $S = \{B(a, r), B(a', r')\}$ with $B(a, r)$ and $B(a', r')$ disjoint

metric, Λ is a bounded metric space, but with respect to the big metric, Λ is unbounded.

By construction, each disc $B(a, r) \subset B(0, 1)$ of positive radius corresponds to a unique point of Λ . In other words, as a set Λ just consists of the points of type II or III in $\mathcal{B}(0, 1)$. To incorporate the points of type I and IV, we must enlarge Λ by adding “ends”. Consider a strictly decreasing sequence of nested discs $x = \{B(a_i, r_i)\}$, and put $r = \lim r_i$. The union of the lines of discs $[\zeta_{a_i, r_i}, \zeta_{\text{Gauss}}]$ is a “half-open” line of discs, which we will write as $(\zeta_{a, r}, \zeta_{\text{Gauss}}]$. If $\cap_i B(a_i, r_i)$ is nonempty, then the intersection is either a disc $B(a, r)$ of positive radius, in which case $(\zeta_{a, r}, \zeta_{\text{Gauss}}]$ extends in a natural way to the closed path $[\zeta_{a, r}, \zeta_{\text{Gauss}}]$, or a point $a \in B(0, 1)$, in which case the half-open path $(\zeta_{a, r}, \zeta_{\text{Gauss}}]$ extends to a closed path by adding on the corresponding type I point a as an endpoint. Similarly, if $\cap_i B(a_i, r_i) = \emptyset$, we must adjoin a new endpoint of Type IV in order to close up $(\zeta_{a, r}, \zeta_{\text{Gauss}}]$. It is clear that cofinal sequences define the same half-open path, so the point closing up this path depends only on the corresponding seminorm $[\]_x$, and not on the sequence defining it.

A cleaner way to say all of this is that points of $\mathcal{B}(0, 1)$ are in one-to-one correspondence with points in the completion of Λ with respect to the small metric. Since the points of type I are infinitely far away with respect to the big metric, the completion of Λ with respect to the big metric coincides (on the level of points) with $\mathcal{B}(0, 1) \setminus B(0, 1)$.

REMARK 1.16. It is important to realize that $\mathcal{B}(0, 1)$ is *not* homeomorphic to the completion of Λ with respect to the small metric. Rather, as shown above in Proposition 1.11, $\mathcal{B}(0, 1)$ is homeomorphic to the *weak topology* on the completion of Λ .

Another way to understand the topology on $\mathcal{B}(0, 1)$ is in terms of inverse limits. Let $\mathcal{D}$ be the collection of all finite $\mathbb{R}$ -trees of the form

Λ_S as above. We have already noted that $\mathcal{D}$ is a directed set under inclusion, and we write $\Gamma \leq \Gamma'$ if $\Gamma \subseteq \Gamma'$ as subsets of $\mathcal{B}(0, 1)$. Thus whenever $\Gamma \leq \Gamma'$, there is an inclusion map

$$i_{\Gamma, \Gamma'} : \Gamma \rightarrow \Gamma' .$$

The set Λ is the direct limit of Λ_S over $\mathcal{D}$ with respect to the maps $i_{\Gamma, \Gamma'}$.

There is also a *retraction map* $r_{\Gamma', \Gamma} : \Gamma' \rightarrow \Gamma$ defined whenever $\Gamma \leq \Gamma'$. This is a general property of $\mathbb{R}$ -trees: since there is a unique path between any two points of Γ' , if $x \in \Gamma'$ we can define $r_{\Gamma', \Gamma}(x)$ to be, for any $y \in \Gamma$, the first point where the unique path in Γ' from x to y intersects Γ . This definition is independent of the choice of y , and one sees from the definition that $r_{\Gamma', \Gamma}(x) = x$ if and only if $x \in \Gamma$. (In particular, $r_{\Gamma', \Gamma}(x)$ is surjective.) Moreover, the retraction maps are compatible, in the sense that if $\Gamma \leq \Gamma' \leq \Gamma''$, then

$$(1.5) \quad r_{\Gamma'', \Gamma} = r_{\Gamma', \Gamma} \circ r_{\Gamma'', \Gamma'} .$$

By (1.5), the inverse limit

$$\hat{\Lambda} = \varprojlim \Gamma ,$$

taken over all $\Gamma \in \mathcal{D}$, is well-defined. Endowing each Γ with its metric topology (which is the same for both the big and small metrics), it follows from Tychonoff's theorem that $\hat{\Lambda}$ is a compact Hausdorff topological space. The space $\hat{\Lambda}$ is an example of what we call a *profinite $\mathbb{R}$ -tree*, i.e., an inverse limit of finite $\mathbb{R}$ -trees.

PROPOSITION 1.17. *The topological spaces $\mathcal{B}(0, 1)$ and $\hat{\Lambda}$ are homeomorphic.*

PROOF. Since $\mathcal{B}(0, 1)$ can be given the structure of an $\mathbb{R}$ -tree (for example, as the completion of Λ with respect to the small metric), for each $\Gamma \in \mathcal{D}$ there is a natural retraction map $r_{\mathcal{B}(0, 1), \Gamma} : \mathcal{B}(0, 1) \rightarrow \Gamma$. This map is continuous, since one can easily verify that if $\Gamma \in \mathcal{D}$ and $V \subset \Gamma$ is an open subset of Γ , then $r_{\mathcal{B}(0, 1), \Gamma}^{-1}(V)$ is an open affinoid in $\mathcal{B}(0, 1)$. By the universal property of inverse limits, there is a continuous map ψ from $\mathcal{B}(0, 1)$ to $\hat{\Lambda}$. This map can be described concretely as follows: if $x \in \mathcal{B}(0, 1)$ corresponds to a nested sequence $\{B(a_i, r_i)\}$ of closed discs, then for each $\Gamma \in \mathcal{D}$, we have

$$r_{\mathcal{B}(0, 1), \Gamma}(\zeta_{a_j, r_j}) = \zeta_{a_i, r_i}$$

whenever $j \geq i$, and thus the collection of points $\{\zeta_{a_i, r_i}\}$ defines an element of $\hat{\Lambda}$: this element is just $\psi(x)$.

It is easy to see that two such nested sequences define the same point of $\hat{\Lambda}$ if and only if they are equivalent; in other words, the map ψ is injective. Alternatively, ψ is injective because elements of $\mathcal{D}$ “separate points”: given two distinct points $x, y \in \mathcal{B}(0, 1)$, there is a point $\zeta_{a,r}$ of Type II such that $x \leq \zeta_{a,r}$ but $y \not\leq \zeta_{a,r}$, and if $\Gamma = [\zeta_{a,r}, \zeta_{\text{Gauss}}]$ then

$$r_{\mathcal{B}(0,1),\Gamma}(x) = \zeta_{a,r} \neq r_{\mathcal{B}(0,1),\Gamma}(y) .$$

The map ψ is also surjective, since by the definition of the indexing set $\mathcal{D}$, every point of $\hat{\Lambda}$ can be realized as the limit of a nested family of closed discs, and by Remark 1.4 every such family is equivalent to a nested *sequence* of closed discs.

Since a continuous bijection between compact Hausdorff topological spaces is automatically a homeomorphism, the desired result now follows. $\square$

In particular, we see that $\mathcal{B}(0, 1)$ is homeomorphic to a profinite $\mathbb{R}$ -tree, which we think of as rooted at the Gauss point.

5. Metrizability

If the residue field $\tilde{K}$ of K is countable, then the basis in (1.6) has a countable sub-basis, obtained by restricting to discs $\mathcal{B}(a, r)^-$ and $\mathcal{B}(a_i, r_i)$ whose centers belong to $\tilde{K} \cap B(0, 1)$ and whose radii belong to $\mathbb{Q}$.

COROLLARY 1.18. *If $\tilde{K}$ is countable, then $\mathcal{B}(0, 1)$ is a metrizable space.*

PROOF. A compact Hausdorff space is “T3”, that is, each point is closed, and for each point x and each closed set A with $x \notin A$, there are disjoint open neighborhoods U of x and V of A . Urysohn’s Metrization theorem ([41], p.125) says that any T3 space with a countable basis is metrizable. $\square$

As previously remarked, the metrics $d(x, y)$ and $\rho(x, y)$ do *not* define the Gelfand topology. For example, the set $\{x \in \mathcal{B}(0, 1) : d(x, \zeta_{\text{Gauss}}) < 1/2\}$ does not contain any point of Type I, while every neighborhood of ζ_{Gauss} in the Gelfand topology contains infinitely many such points. This same example shows that $d(x, y)$ (resp. $\rho(x, y)$) is not even continuous for the Gelfand topology. (In fact, the topology defined by $d(x, y)$ (resp. $\rho(x, y)$) is strictly finer than the Gelfand topology.)

Tracing through the proof of Urysohn’s theorem, one can construct a metric defining the Gelfand topology in the case where $\tilde{K}$ is countable

as follows. First, define a ‘separation kernel’ for discs $B(a, r)$ and points $a' \in B(0, 1)$ by

$$\Delta(B(a, r), a') = \sup_{z \in B(a, r)} (|z - a'|) = \max(r, |a - a'|) .$$

Extend it to pairs of discs $B(a, r), B(a', r')$ by

$$\Delta(B(a, r), B(a', r')) = \sup_{\substack{z \in B(a, r) \\ w \in B(a', r')}} (|z - w|) = \max(r, r', |a - a'|) ,$$

and then to arbitrary points $x, y \in \mathcal{B}(0, 1)$ by

$$\Delta(x, y) = \lim_{i \rightarrow \infty} \max(r_i, r'_i, |a_i - a'_i|) .$$

if x, y correspond to sequences of nested discs $\{B(a_i, r_i)\}, \{B(a'_i, r'_i)\}$.

Let $\{\alpha_j\}$ be an enumeration of $\tilde{K} \cap B(0, 1)$ (or more generally, take any countable dense subset of $\mathcal{B}(0, 1)$), and define a map φ from $\mathcal{B}(0, 1)$ to the infinite-dimensional unit cube $[0, 1]^{\mathbb{N}}$ by putting

$$\varphi(x) = (\Delta(x, \alpha_j))_{j \in \mathbb{N}}$$

It can be checked that φ is a topological isomorphism from $\mathcal{B}(0, 1)$ onto its image, equipped with the induced topology. Pulling back the metric on $[0, 1]^{\mathbb{N}}$, we obtain a metric defining the topology on $\mathcal{B}(0, 1)$:

$$\delta(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} |\Delta(x, \alpha_j) - \Delta(y, \alpha_j)| .$$

However, this formula seems nearly useless for understanding the Gelfand topology.

6. Borel and Radon measures

We note for future reference that every Borel subset of $\mathcal{B}(0, 1)$ is a Baire set (see §5 of Appendix A for the definition of a Baire set). Indeed, since the topology on $\mathcal{B}(0, 1)$ is generated by the sets

$$\begin{aligned} U(f, \alpha) &= \{x \in \mathcal{B}(0, 1) : [f]_x < \alpha\} \\ V(f, \alpha) &= \{x \in \mathcal{B}(0, 1) : [f]_x > \alpha\} \end{aligned}$$

for $f \in K\langle T \rangle$ and $\alpha \in \mathbb{R}$, it suffices to show that $U(f, \alpha)$ and $V(f, \alpha)$ are Baire sets, but this is clear since each map $[f]_x : \mathcal{B}(0, 1) \rightarrow \mathbb{R}$ is continuous. Therefore, by Proposition A.13, we find:

LEMMA 1.19. *Every finite Borel measure on $\mathcal{B}(0, 1)$ is Radon.*

7. Notes and further references

Much of the material in this chapter is originally due to Berkovich, and can be found in his book [9]. We have gone into more detail than Berkovich in describing the topological structure of $\mathcal{B}(0, 1)$, but of course we are working in much less generality than he does.

The notational convention $\zeta_{a,r}$ was suggested by Joe Silverman.

The notation $x \vee y$ for the least upper bound of $x, y \in \mathcal{B}(0, 1)$ has been borrowed from Favre and Rivera-Letelier [28, 29], except they use $\wedge$ instead of $\vee$. Since the Gauss point is *maximal* with respect to the partial order $\leq$ on $\mathcal{B}(0, 1)$ which we have defined, our notation $x \vee y$ is compatible with the typical usage from the theory of partially ordered sets.

We have borrowed from Chambert-Loir the term “Gauss point” for the maximal point $\zeta_{\text{Gauss}} = \zeta_{0,1}$ of $\mathcal{B}(0, 1)$.

Mattias Jonsson pointed out to us that in [6], we were using without proof the fact that every Borel measure on $\mathcal{B}(0, 1)$ is Radon. This is now proved as Lemma 1.19, and will be used later on when we invoke the Riesz Representation Theorem in order to define the Laplacian operator.

CHAPTER 2

The Berkovich projective line

In this chapter, we will give several methods for constructing the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$. Our primary definition is analogous to the “Proj” construction in algebraic geometry. Using the “Proj” definition, one sees easily that the action of a rational map φ on $\mathbb{P}^1(K)$ extends naturally to $\mathbb{P}_{\text{Berk}}^1$. Our main structural result on $\mathbb{P}_{\text{Berk}}^1$ is that it is canonically homeomorphic to an inverse limit of finite $\mathbb{R}$ -trees.

In the final sections, we define the “Berkovich hyperbolic space” $\mathbb{H}_{\text{Berk}}$ and a natural metric $\rho(x, y)$ on $\mathbb{H}_{\text{Berk}}$ which extends the “big metric” defined in the previous chapter. We prove that ρ is invariant under linear fractional transformations.

1. The Berkovich affine line $\mathbb{A}_{\text{Berk}}^1$

The Berkovich affine line $\mathbb{A}_{\text{Berk}}^1$, as a topological space, is defined as follows. As a set, the elements of $\mathbb{A}_{\text{Berk}}^1$ are multiplicative seminorms on $K[T]$ which extend the absolute value on K . The topology on $\mathbb{A}_{\text{Berk}}^1$ is the weakest one for which $x \mapsto [f]_x$ is continuous for all $f \in K[T]$. It is easy to see that a net $\langle x_\alpha \rangle$ in $\mathbb{A}_{\text{Berk}}^1$ converges to x if and only if $[f]_{x_\alpha} \rightarrow [f]_x$ for all $f \in K[T]$, so the topology on $\mathbb{A}_{\text{Berk}}^1$ is the topology of *pointwise convergence* with respect to $K[T]$.

There is a natural continuous embedding of K into $\mathbb{A}_{\text{Berk}}^1$ which associates to a point $z \in K$ the evaluation seminorm $[f]_z = |f(z)|$.

Similar to Lemma 1.1(2), we have:

LEMMA 2.1. *A multiplicative seminorm $[\]_x$ extends the absolute value on K if and only if its restriction to K is bounded.*

PROOF. Suppose the restriction of $[\]_x$ to K is bounded. We know that $[0]_x = 0$, and if $c \in K$ is nonzero, then by Lemma D.1 we have both $[c]_x \leq |c|$ and $[c^{-1}]_x \leq |c^{-1}|$. Multiplicativity gives $[c]_x \cdot [c^{-1}]_x = [c \cdot c^{-1}]_x = 1$, from which it follows that $[c]_x = |c|$.

The other direction is trivial. □

For each real number $R > 0$, we can construct the “Berkovich disc of radius R about 0” by setting

$$\mathcal{B}(0, R) = \mathcal{M}(K\langle R^{-1}T \rangle) ,$$

where

$$K\langle R^{-1}T \rangle = \left\{ \sum_{k=0}^{\infty} c_k T^k \in K[[T]] : \lim_{k \rightarrow \infty} R^k |c_k| = 0 \right\} .$$

It is easy to see that $\mathcal{B}(0, R)$ and $\mathcal{B}(0, 1)$ are homeomorphic for all R .

If $0 < r < R$, then $\mathcal{B}(0, r)$ can naturally be viewed as a subspace of $\mathcal{B}(0, R)$ via the map $\iota_{r,R}$ sending a continuous multiplicative seminorm $[\cdot]_x$ on $K\langle r^{-1}T \rangle$ to the continuous multiplicative seminorm $[\cdot]_{\iota_{r,R}(x)}$ on $K\langle R^{-1}T \rangle$ defined by

$$[f]_{\iota_{r,R}(x)} = [\pi(f)]_x ,$$

where $\pi : K\langle R^{-1}T \rangle \rightarrow K\langle r^{-1}T \rangle$ is the natural K -algebra homomorphism. This map is easily checked to be continuous, and since π is surjective, it follows that $\iota_{r,R}$ is injective. It thus makes sense to consider the direct limit, or union, of the discs $\mathcal{B}(0, R)$. This union is homeomorphic to $\mathbb{A}_{\text{Berk}}^1$:

$$(2.1) \quad \mathbb{A}_{\text{Berk}}^1 \approx \cup_{R>0} \mathcal{B}(0, R) .$$

We sketch a proof of (2.1) by specifying continuous maps in each direction which are inverse to one another. On the one hand, for each R there is an obvious map $\iota_R : \mathcal{B}(0, R) \rightarrow \mathbb{A}_{\text{Berk}}^1$ coming from the inclusion $K[T] \hookrightarrow K\langle R^{-1}T \rangle$, and if $r < R$ then $\iota_r = \iota_R \circ \iota_{r,R}$. We thus obtain a map $\iota : \varinjlim \mathcal{B}(0, R) \rightarrow \mathbb{A}_{\text{Berk}}^1$.

In the other direction, suppose $x \in \mathbb{A}_{\text{Berk}}^1$, and let $R = [T]_x$. Then one can define $\psi(x) \in \mathcal{B}(0, R)$ via

$$(2.2) \quad [f]_{\psi(x)} = \lim_{N \rightarrow \infty} \left[\sum_{i=0}^N c_i T^i \right]_x$$

for $f = \sum_{i=0}^{\infty} c_i T^i \in K\langle R^{-1}T \rangle$. To see that this is well defined, note that for $M \leq N$, a telescoping series argument using the ultrametric inequality for $[\cdot]_x$ gives

$$(2.3) \quad \left| \left[\sum_{i=0}^N c_i T^i \right]_x - \left[\sum_{i=0}^M c_i T^i \right]_x \right| \leq \max_{M \leq i \leq N} \{ |c_i| R^i \} .$$

Since $|c_i| R^i \rightarrow 0$, it follows that $\{ [\sum_{i=0}^N c_i T^i]_x \}_{N \geq 0}$ is a Cauchy sequence, and therefore that the limit in (2.2) exists. It is straightforward to check that the resulting map $\psi : \mathbb{A}_{\text{Berk}}^1 \rightarrow \varinjlim \mathcal{B}(0, R)$ is

continuous, and that ι and ψ are inverses to one another. Therefore $\mathbb{A}_{\text{Berk}}^1$ is homeomorphic to $\cup_{R>0} \mathcal{B}(0, R)$ as claimed.

It is straightforward to check that Berkovich’s classification theorem admits the following extension to $\mathbb{A}_{\text{Berk}}^1$:

THEOREM 2.2. *Every $x \in \mathbb{A}_{\text{Berk}}^1$ can be realized as*

$$(2.4) \quad [f]_x = \lim_{i \rightarrow \infty} [f]_{B(a_i, r_i)}$$

for some sequence of nested closed discs $B(a_1, r_1) \supseteq B(a_2, r_2) \supseteq \cdots$ contained in K . If this sequence has a nonempty intersection, then either

- 1) the intersection is a single point a , in which case $[f]_x = |f(a)|$, or*
- 2) the intersection is a closed disc $B(a, r)$ of radius $r > 0$, in which case $[f]_x = [f]_{B(a, r)}$.*

2. The Berkovich “Proj” construction

As a topological space, one can define the Berkovich projective line $\mathbb{P}_{\text{Berk}, K}^1$ to be the one-point compactification of the locally compact Hausdorff space $\mathbb{A}_{\text{Berk}, K}^1$ (see Appendix A). The extra point is denoted, as usual, by the symbol ∞ .

However, this definition is not always very useful. For example, it is not immediately clear from this definition how or why a rational function $\varphi \in K(T)$ induces a natural map from $\mathbb{P}_{\text{Berk}}^1$ to itself. We therefore introduce the following alternate construction of $\mathbb{P}_{\text{Berk}, K}^1$, analogous to the “Proj” construction in algebraic geometry.

We will then discuss how $\mathbb{P}_{\text{Berk}}^1$, defined via the “Proj” construction, can be thought of either as $\mathbb{A}_{\text{Berk}}^1$ together with a point at infinity, or as two copies of the Berkovich unit disc $\mathcal{B}(0, 1)$ glued together along the annulus $\mathcal{A}(1, 1) = \{x \in \mathcal{B}(0, 1) : [T]_x = 1\}$.

Let S denote the set of multiplicative seminorms on the two-variable polynomial ring $K[X, Y]$ which extend the absolute value on K , and which are not identically zero on the maximal ideal (X, Y) of $K[X, Y]$. We put an equivalence relation on S by declaring that $[\]_1 \sim [\]_2$ if and only if there exists a constant $C > 0$ such that $[G]_1 = C^d [G]_2$ for all homogeneous polynomials $G \in K[X, Y]$ of degree d .

As a set, define $\mathbb{P}_{\text{Berk}}^1$ to be the equivalence classes of elements of S .

Note that $[\]$ is automatically non-archimedean; this follows from the argument in Lemma 1.1(3). The condition that $[\]$ is not identically zero on the maximal ideal (X, Y) is therefore equivalent to requiring

that $[X]$ and $[Y]$ are not both zero. We say that a seminorm $[]$ in S is *normalized* if $\max\{[X], [Y]\} = 1$. It follows from the definition of the equivalence relation on S that there is a *unique* normalized seminorm within each equivalence class. We denote by $[]_z^*$ the normalized seminorm corresponding to a point $z \in \mathbb{P}_{\text{Berk}}^1$. Explicitly, if $[]_z$ is any representative of the equivalence class of z , then

$$[G]_z^* = [G]_z / \max\{[X]_z, [Y]_z\}^d$$

for all homogeneous polynomials $G \in K[X, Y]$ of degree d .

The topology on $\mathbb{P}_{\text{Berk}}^1$ is defined to be the weakest one such that $z \mapsto [G]_z^*$ is continuous for all $G \in K[X, Y]$.

To illustrate what's happening, we now make explicit the realization of $\mathbb{P}_{\text{Berk}}^1$ as $\mathbb{A}_{\text{Berk}}^1$ together with a point at infinity. More precisely, we will see that $\mathbb{P}_{\text{Berk}}^1$ can be obtained by gluing together two copies of $\mathbb{A}_{\text{Berk}}^1$ along the open subset $\mathbb{A}_{\text{Berk}}^1 \setminus \{0\}$ via a homeomorphism

$$\theta : \mathbb{A}_{\text{Berk}}^1 \setminus \{0\} \rightarrow \mathbb{A}_{\text{Berk}}^1 \setminus \{0\}.$$

The construction closely parallels the standard decomposition of the usual projective line $\mathbb{P}^1$ into the open charts $\mathbb{P}^1 \setminus \{\infty\}$ and $\mathbb{P}^1 \setminus \{0\}$, both of which are isomorphic to $\mathbb{A}^1$, with the transition from one chart to the other given by $t \mapsto 1/t$. Here ∞ denotes the point of $\mathbb{P}^1$ with homogeneous coordinates $(1 : 0)$. The Berkovich space version of this decomposition can be understood as follows.

Define the point ∞ in $\mathbb{P}_{\text{Berk}}^1$ to be the equivalence class of the seminorm $[]_\infty$ defined by $[G]_\infty = |G(1, 0)|$. More generally, if $P \in \mathbb{P}^1(K)$ has homogeneous coordinates $(a : b)$, the equivalence class of the evaluation seminorm $[G]_P = |G(a, b)|$ is independent of the choice of homogeneous coordinates, and therefore $[]_P$ is a well-defined point of $\mathbb{P}_{\text{Berk}}^1$. This furnishes a natural embedding of $\mathbb{P}^1(K)$ into $\mathbb{P}_{\text{Berk}}^1$.

Up to equivalence, $[]_\infty$ is the unique element of S for which $[Y] = 0$. Indeed, if $[Y] = 0$ then by multiplicativity and the ultrametric inequality, if

$$G(X, Y) = a_0 X^d + a_1 X^{d-1} Y + \cdots + a_{d-1} X Y^{d-1} + a_d Y^d$$

is homogeneous of degree d , we must have $[G(X, Y)] = |a_0| \cdot [X]^d$. Similarly, $[]_0 = []_{(0:1)}$ is the unique element of S , up to equivalence, for which $[X] = 0$.

Gluing two copies of $\mathbb{A}_{\text{Berk}}^1$. We now show that $\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$ and $\mathbb{P}_{\text{Berk}}^1 \setminus \{0\}$ are both homeomorphic to $\mathbb{A}_{\text{Berk}}^1$. By symmetry, it suffices to consider the case of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$. We define maps in each direction between $\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$ and $\mathbb{A}_{\text{Berk}}^1$ as follows.

Explain why $\mathbb{P}^1(K)$ is dense in $\mathbb{P}_{\text{Berk}}^1$.

If $G \in K[X, Y]$, define $\chi_\infty : \mathbb{A}_{\text{Berk}}^1 \rightarrow \mathbb{A}_{\text{Berk}}^1$ by the formula

$$[G]_{\chi_\infty(z)} = [G(T, 1)]_z .$$

It is easy to see that $\chi_\infty(z)$ is a continuous multiplicative seminorm on $K[T]$, and thus defines a point of $\mathbb{A}_{\text{Berk}}^1$.

If $g \in K[T]$, define $\psi_\infty : \mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\} \rightarrow \mathbb{A}_{\text{Berk}}^1$ by the formula

$$[g]_{\psi_\infty(z)} = [G]_z / [Y]_z^d ,$$

where $G \in K[X, Y]$ is homogeneous of degree d and satisfies $G(T, 1) = g(T)$. It is immediate from the definition of equivalence and the fact that $[Y]_z \neq 0$ that ψ_∞ is well-defined. It is straightforward to show that χ_∞ and ψ_∞ are inverses of one another, and that they are both continuous maps, and thus that ψ_∞ defines a homeomorphism

$$\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\} \approx \mathbb{A}_{\text{Berk}}^1 .$$

Similarly, there is a homeomorphism

$$\psi_0 : \mathbb{P}_{\text{Berk}}^1 \setminus \{0\} \rightarrow \mathbb{A}_{\text{Berk}}^1$$

defined for $h \in K[T]$ by

$$[h]_{\psi_0(z)} = [H]_z / [X]_z^d$$

for any homogeneous polynomial $H \in K[X, Y]$ satisfying $H(1, T) = h(T)$.

The continuity of χ_∞ and ψ_∞ can be verified as follows. If $\langle z_\alpha \rangle$ is a net in $\mathbb{A}_{\text{Berk}}^1$ converging to $z \in \mathbb{A}_{\text{Berk}}^1$, then for each $g \in K[T]$ we have $[g]_{z_\alpha} \rightarrow [g]_z$, so it follows that for each $G \in K[X, Y]$,

$$\frac{[G(T, 1)]_{z_\alpha}}{\max\{1, [T]_{z_\alpha}\}} \rightarrow \frac{[G(T, 1)]_z}{\max\{1, [T]_z\}} .$$

This is equivalent to the assertion that $[G]_{\chi_\infty(z_\alpha)} \rightarrow [G]_{\chi_\infty(z)}$, and thus χ_∞ is continuous.

Similarly, if $z'_\alpha \rightarrow z'$ in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$, then $[Y]_{z'_\alpha} \rightarrow [Y]_{z'} \neq 0$ and for each $H \in K[X, Y]$, $[H]_{z'_\alpha} \rightarrow [H]_{z'}$, and therefore

$$\frac{[H]_{z'_\alpha}}{[Y]_{z'_\alpha}^d} \rightarrow \frac{[H]_{z'}}{[Y]_{z'}^d} .$$

This implies that ψ_∞ is continuous.

Unravelling the definitions, we see that there is a commutative diagram

$$\begin{array}{ccc} \mathbb{P}_{\text{Berk}}^1 \setminus \{0, \infty\} & \xlongequal{\quad} & \mathbb{P}_{\text{Berk}}^1 \setminus \{0, \infty\} \\ \psi_\infty \downarrow & & \downarrow \psi_0 \\ \mathbb{A}_{\text{Berk}}^1 \setminus \{0\} & \xrightarrow{\quad \theta \quad} & \mathbb{A}_{\text{Berk}}^1 \setminus \{0\} \end{array}$$

in which the involution θ of $\mathbb{A}_{\text{Berk}}^1 \setminus \{0\}$ is defined for $g \in K[T]$ by

$$[g]_{\theta(z)} = [\tilde{g}]_z / [X]_z^d,$$

where $d = \deg(g)$ and $\tilde{g} = X^d g(1/X)$ is the *reciprocal polynomial* of g .

When restricted to $\mathbb{P}^1(K) \setminus \{0, \infty\} \cong K^*$, θ is just the map $t \mapsto 1/t$. By continuity, for any $f \in K[T]$ and $0 < a < b$,

$$\theta(\{z \in \mathbb{A}_{\text{Berk}}^1 \setminus \{0\} : a < [f]_z < b\}) = \{z \in \mathbb{A}_{\text{Berk}}^1 \setminus \{0\} : \frac{1}{b} < [f]_z < \frac{1}{a}\}.$$

In conclusion, one obtains $\mathbb{P}_{\text{Berk}}^1$ as a topological space by gluing two copies V_0, V_∞ of $\mathbb{A}_{\text{Berk}}^1$ together via the identification

$$z \in V_0 \setminus \{0\} \sim \theta(z) \in V_\infty \setminus \{0\}.$$

Gluing two copies of $\mathcal{B}(0, 1)$. One can also construct $\mathbb{P}_{\text{Berk}}^1$ as a topological space by gluing together two copies of $\mathcal{B}(0, 1)$ along the common annulus

$$\mathcal{A}(1, 1) = \{x \in \mathcal{B}(0, 1) : [T]_x = 1\}.$$

REMARK 2.3. In the terminology of Appendix D, we have

$$\mathcal{A}(1, 1) = \mathcal{M}(K\langle T, T^{-1} \rangle),$$

where

$$K\langle T, T^{-1} \rangle = \left\{ \sum_{i=-\infty}^{\infty} c_i T^i : \lim_{|i| \rightarrow \infty} |c_i| = 0 \right\}.$$

In order to understand this description of $\mathbb{P}_{\text{Berk}}^1$, first note that $\mathbb{P}_{\text{Berk}}^1 = W_0 \cup W_\infty$, where

$$\begin{aligned} W_0 &= \{z \in \mathbb{P}_{\text{Berk}}^1 : [X]_z \leq [Y]_z\}, \\ W_\infty &= \{z \in \mathbb{P}_{\text{Berk}}^1 : [Y]_z \leq [X]_z\}. \end{aligned}$$

There is a homeomorphism $\bar{\psi}_\infty : W_\infty \cong \mathcal{B}(0, 1)$ defined for $g = \sum_{i=0}^{\infty} c_i T^i \in K\langle T \rangle$ by

$$[g]_{\bar{\psi}_\infty}(z) = \lim_{N \rightarrow \infty} \left[\sum_{i=0}^N c_i X^{N-i} Y^i \right]_z / [X]_z^N,$$

which is well-defined by a telescoping series argument similar to (2.3). Similarly, one defines a homeomorphism $\bar{\psi}_0 : W_0 \cong \mathcal{B}(0, 1)$, and the intersection $W_0 \cap W_\infty = \{z \in \mathbb{P}_{\text{Berk}}^1 : [X]_z = [Y]_z\}$ is mapped by both $\bar{\psi}_\infty$ and $\bar{\psi}_0$ to $\mathcal{A}(1, 1)$. It follows that $\mathbb{P}_{\text{Berk}}^1$ can be obtained by gluing two copies W_0 and W_∞ of $\mathcal{B}(0, 1)$ together along their common

annulus $\mathcal{A}(1, 1)$ via the gluing map $\bar{\theta} : \mathcal{A}(1, 1) \rightarrow \mathcal{A}(1, 1)$ defined for $g \in K\langle T, T^{-1} \rangle$ by

$$[g(T)]_{\bar{\theta}(z)} = [g(1/T)]_z.$$

We leave detailed verifications of all of these assertions to the reader.

The topology on $\mathbb{P}_{\text{Berk}}^1$ is characterized by the fact that a subset is open if and only if its intersections with both W_0 and W_∞ are open. In particular:

Insert lemma.

LEMMA 2.4.

Decomposition into $\mathcal{B}(0, 1)$ and $\mathcal{B}(0, 1)^-$. There is one more description of $\mathbb{P}_{\text{Berk}}^1$ which will be useful to us. It follows from what we have already said that the open set

$$\mathbb{P}_{\text{Berk}}^1 \setminus \mathcal{B}(0, 1)$$

is homeomorphic to the open Berkovich unit disc

$$\mathcal{B}(0, 1)^- = \{z \in \mathcal{B}(0, 1) : [T]_z < 1\}.$$

As a set, we may therefore identify $\mathbb{P}_{\text{Berk}}^1$ with the disjoint union of a closed set X homeomorphic to $\mathcal{B}(0, 1)$ and an open subset Y homeomorphic to $\mathcal{B}(0, 1)^-$. Moreover, the closure of Y in $\mathbb{P}_{\text{Berk}}^1$ is $\bar{Y} = Y \cup \zeta_{\text{Gauss}}$. The topology on $\mathbb{P}_{\text{Berk}}^1$ is then determined by saying that a subset is open if and only if its intersections with X and $\bar{Y}$ are both open.

This provides a useful way to visualize $\mathbb{P}_{\text{Berk}}^1$: one takes the previous description of $\mathcal{B}(0, 1)$ and adds an “extra branch at infinity” corresponding to $\mathbb{P}_{\text{Berk}}^1 \setminus \mathcal{B}(0, 1)$.

The following lemma is clear from this description of $\mathbb{P}_{\text{Berk}}^1$, since we know the corresponding facts for $\mathcal{B}(0, 1)$:

LEMMA 2.5. *Both $\mathbb{P}^1(K)$ and $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$ are dense in $\mathbb{P}_{\text{Berk}}^1$.*

In addition, it follows readily from this description and the corresponding facts for $\mathcal{B}(0, 1)$ that:

LEMMA 2.6. *The space $\mathbb{P}_{\text{Berk}}^1$ is uniquely path-connected. More precisely, given any two distinct points $x, y \in \mathbb{P}_{\text{Berk}}^1$, there is a unique arc L_{xy} in $\mathbb{P}_{\text{Berk}}^1$ from x to y , and if $x \in X$ and $y \in Y$ then L_{xy} contains the Gauss point ζ_{Gauss} .*

3. The action of a rational map φ on $\mathbb{P}_{\text{Berk}}^1$

Let $\varphi \in K(T)$ be a rational function of degree $d \geq 1$. In this section, we explain how to extend the usual action of φ on $\mathbb{P}^1(K)$ to a continuous map $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$.

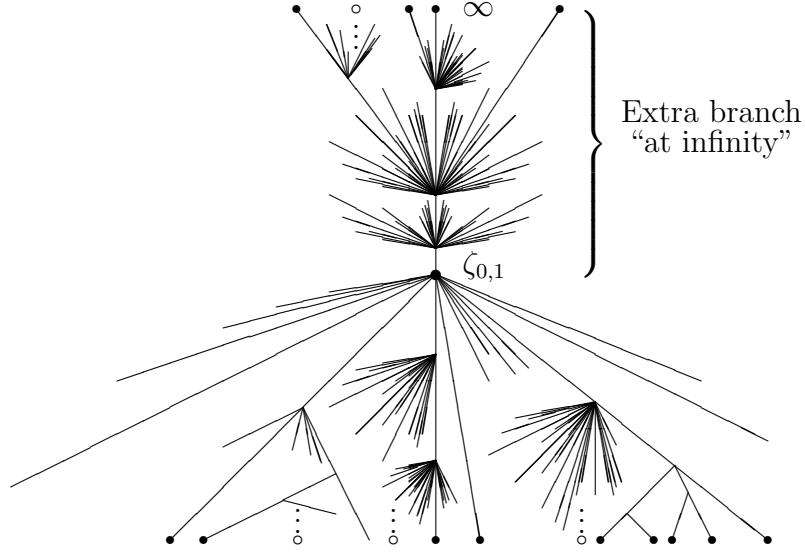


FIGURE 1. The Berkovich projective line (illustration courtesy of Joe Silverman)

Choose a homogeneous lifting $F = (F_1, F_2)$ of φ , where $F_i \in K[X, Y]$ are homogeneous of degree d and have no common zeros in K . (Recall that the field K is algebraically closed.) The condition that F_1 and F_2 have no common zeros is equivalent to requiring that the homogeneous resultant $\text{Res}(F) = \text{Res}(F_1, F_2)$ is nonzero. (See Chapter IX §4 of [45] for a definition of the resultant of two homogeneous polynomials.)

We define the action of φ on $\mathbb{P}_{\text{Berk}}^1$ as follows: Let $G \in K[X, Y]$, and define

$$(2.5) \quad [G]_{\varphi(z)} := [G(F_1(X, Y), F_2(X, Y))]_z.$$

It is readily verified that the right-hand side of (2.5) is independent of the lifting F of φ , up to equivalence of seminorms. As it is clear that the right-hand side of (2.5) gives a continuous multiplicative seminorm on $K[X, Y]$, to see that (2.5) induces a map from $\mathbb{P}_{\text{Berk}}^1$ to itself, it suffices to show that $[X]_{\varphi(z)} = [F_1(X, Y)]_z$ and $[Y]_{\varphi(z)} = [F_2(X, Y)]_z$ cannot both be zero. This follows from:

LEMMA 2.7. *There are homogenous polynomials $G_1, G_2, H_1, H_2 \in K[X, Y]$ of degree $d - 1$ such that*

$$\begin{aligned} G_1(X, Y)F_1(X, Y) + G_2(X, Y)F_2(X, Y) &= \text{Res}(F)X^{2d-1}, \\ H_1(X, Y)F_1(X, Y) + H_2(X, Y)F_2(X, Y) &= \text{Res}(F)Y^{2d-1}. \end{aligned}$$

Give reference.

By Lemma 2.7, the ultrametric inequality for $[\cdot]_z$, and the fact that $|\text{Res}(F)| \neq 0$, if $[F_1(X, Y)]_z = [F_2(X, Y)]_z = 0$ then $[X]_z = [Y]_z = 0$, a contradiction. Thus at least one of $[X]_{\varphi(z)}$ and $[Y]_{\varphi(z)}$ is nonzero, and $[\cdot]_{\varphi(z)}$ is a well-defined element of $\mathbb{P}_{\text{Berk}}^1$.

It is clear from the definitions that the action of φ on $\mathbb{P}_{\text{Berk}}^1$ which we have just defined extends the usual action of φ on $\mathbb{P}^1(K)$.

The continuity of $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ can be verified as follows: If $\langle z_\alpha \rangle$ is a net in $\mathbb{P}_{\text{Berk}}^1$ converging to z , then for all $G \in K[X, Y]$, we have $[G \circ F]_{z_\alpha}^* \rightarrow [G \circ F]_z^*$ and $[F_i]_{z_\alpha}^* \rightarrow [F_i]_z^*$, and thus

$$[G]_{\varphi(z_\alpha)}^* = [G \circ F]_{z_\alpha}^* / \max\{[F_1]_{z_\alpha}, [F_2]_{z_\alpha}\}$$

converges to

$$[G]_{\varphi(z)}^* = [G \circ F]_z^* / \max\{[F_1]_z, [F_2]_z\}$$

as required.

We will see later on (Corollaries 7.5 and 7.6) that $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is an open surjective mapping, and that every point $z \in \mathbb{P}_{\text{Berk}}^1$ has at most d preimages under φ .

If $d = 1$, then it is clear that φ has a continuous inverse, and is thus an *automorphism* of $\mathbb{P}_{\text{Berk}}^1$.

Note that if $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, then $\varphi(z) \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$ as well because the seminorm $[G]_{\varphi(z)}$ has trivial kernel (i.e., is a norm), whereas for each $a \in \mathbb{P}^1(K)$, the corresponding seminorm has nonzero kernel.

More generally, one can verify that φ takes type I points to type I points, type II points to type II points, type III points to type III points, and type IV points to type IV points. Give more details.

4. Points of $\mathcal{B}(0, 1)$ revisited

In order to understand the action of a rational map on $\mathbb{P}_{\text{Berk}}^1$ in a more concrete way, let us begin by reinterpreting the points of the Berkovich unit disc. By the results in Chapter 1, each $x \in \mathcal{B}(0, 1)$ corresponds to an equivalence class of nested discs $\{B(a_i, r_i)\}$ in $B(0, 1)$, and for $f \in K\langle T \rangle$ the corresponding seminorm $[f]_x$ is a limit of the sup norms $[f]_{B(a_i, r_i)}$. However, this association of $[f]_x$ with sup norms is misleading. A more accurate assertion is that $[f]_x$ is the *generic value* of $|f(z)|$ at x .

To understand this, suppose x is a point of type II, so that x corresponds to a disc $B(a, r)$ with r in the value group of $K^\times$. Theorem 1.2 asserts that for $f \in K\langle T \rangle$,

$$[f]_x = \max_{z \in B(a, r)} |f(z)|.$$

If the zeros of $f(z)$ in $B(a, r)$ are $a_1, \dots, a_m$, then by the Weierstrass Preparation Theorem $|f(z)|$ takes on its maximum value on $B(a, r)$ at each point of $B(a, r) \setminus \cup_{i=1}^m B(a_i, r)^-$. In other words, $[f]_x$ is the constant value which $|f(z)|$ assumes ‘almost everywhere’ on $B(a, r)$. The multiplicative seminorm $[\]_x$ extends in a unique way to the quotient field of $K\langle T \rangle$, with

$$[f/g]_x = \frac{[f]_x}{[g]_x}.$$

However, this extended seminorm is definitely *not* the supremum norm: if $(f/g)(z)$ has poles in $B(a, r)$ then $\sup_{z \in B(a, r)} |(f/g)(z)| = \infty$. Rather, if $f(z)$ has zeros $a_1, \dots, a_m$ and $g(z)$ has zeros $b_1, \dots, b_n$, then $[f/g]_x$ is the constant value which $|f(z)/g(z)|$ assumes everywhere on the *closed affinoid* $B(a, r) \setminus (\cup_{i=1}^m B(a_i, r)^-) \cup (\cup_{j=1}^n B(b_j, r)^-)$. This is best understood as the ‘generic value’ of $|(f/g)(z)|$ on $B(a, r)$.

For points x of type I, III or IV, the notion of a generic value of $|f(z)|$ at x has to be interpreted in a slightly broader way. Let $x \in \mathcal{B}(0, 1)$ be arbitrary. By continuity, for each $\varepsilon > 0$ there is a neighborhood U of x in $\mathcal{B}(0, 1)$ such that for each $t \in U$, $|[f]_t - [f]_x| < \varepsilon$. In particular, for each type I point $z \in U$,

$$||f(z)| - [f]_x| < \varepsilon.$$

By the description of the topology of $\mathcal{B}(0, 1)$ in Chapter 1, sets of the form $\mathcal{B}(a, r) \setminus \cup_{i=1}^m \mathcal{B}(a_i, r_i)^-$ are cofinal in the set of closed neighborhoods of x . Thus, $[f]_x$ is the unique number such that for each $\varepsilon > 0$, there is a closed affinoid neighborhood $B(a, r) \setminus \cup_{i=1}^m B(a_i, r_i)^-$ of x such that $|f(z)|$ is within ε of $[f]_x$ on that neighborhood. In this sense $[f]_x$ is the generic value of $|f(z)|$ at x .

In rigid analysis, each closed affinoid $V = B(a, r) \setminus (\cup_{i=1}^m B(a_i, r_i)^-)$ with $r \leq 1$ and $r, r_1, \dots, r_m$ in the value group of $K^\times$ corresponds to an *affinoid subdomain* of $B(0, 1)$. More precisely, if $b, b_1, \dots, b_m \in K$ are such that $|b| = r$ and $|b_i| = r_i$ for each i , then V is a *Laurent domain*, isomorphic to the set of maximal ideals $\text{Max}(\mathcal{A}_V)$ of the Tate algebra

$$(2.6) \quad \mathcal{A}_V = K\langle T, T_1, \dots, T_m \rangle[X]/\mathcal{I}_V.$$

Here

$$K\langle T, T_1, \dots, T_m \rangle = \left\{ \sum_{\vec{i} \geq 0} c_{\vec{i}} T^{i_0} T_1^{i_1} \dots T_m^{i_m} \in K[[\vec{T}]] : \lim_{|\vec{i}| \rightarrow \infty} |c_{\vec{i}}| = 0 \right\}$$

is the ring of power series converging on the unit polydisc $\{(z_0, z_1, \dots, z_m) \in K^{m+1} : \max(|z_i|) \leq 1\}$, and

$$\mathcal{I}_V = (bT - (X - a), (X - a_1)T_1 - b_1, \dots, (X - a_m)T_m - b_m).$$

The relations generating $\mathcal{I}_V$ mean that $\text{Max}(\mathcal{A}_V)$ is isomorphic to the set of $x \in K$ for which $(\frac{x-a}{b}, \frac{b_1}{x-a_1}, \dots, \frac{b_m}{x-a_m})$ belongs to the unit polydisc, or equivalently, that $|x-a| \leq r$ and $|x-a_i| \geq r_i$ for $i = 1, \dots, m$. This is precisely the closed affinoid V .

The localization of $K\langle T \rangle$ at $(T-a_1)/b_1, \dots, (T-a_m)/b_m$ is dense in $\mathcal{A}_V$. Hence, writing X for the element $bT+a \in K\langle T \rangle$, each bounded multiplicative seminorm $[\cdot]_x$ on $K\langle T \rangle$ for which

$$(2.7) \quad [X]_x \leq 1 \quad \text{and} \quad [b_1/(X-a_1)]_x \leq 1, \dots, [b_m/(X-a_m)]_x \leq 1$$

extends to a bounded multiplicative seminorm on $\mathcal{A}_V$. Conversely, each bounded multiplicative seminorm on $\mathcal{A}_V$ restricts to a bounded multiplicative seminorm on $K\langle T \rangle$ for which (2.7) holds. Thus, the functional analytic spectrum $\mathcal{M}(\mathcal{A}_V)$ can be identified with the subset $B(a, r) \setminus \bigcup_{i=1}^m B(a_i, r_i)^-$ of $\mathcal{B}(0, 1)$.

We have emphasized the reinterpretation of seminorms $[\cdot]_x$ as ‘generic values’ partly to explain why such an inclusion is reasonable. Since V is a more complicated domain than $\mathcal{B}(0, 1)$, one should not expect to identify seminorms $[\cdot]_x$ on $\mathcal{A}_V$ with sup norms on nested sequences of discs; however, one might hope to identify them with sup norms on nested sequences of closed affinoids contained in $B(a, r) \setminus \bigcup_{i=1}^m B(a_i, r_i)^-$. Such an identification can in fact be made, though we will not give the details. However, we do note that each closed affinoid $x = B(b, t) \setminus \bigcup_{i=1}^n B(b_i, t)^-$ in which the deleted discs have the same radius as the outer disc determines a bounded multiplicative seminorm

$$[f]_x = \sup_{z \in B(b, t) \setminus \bigcup_{i=1}^n B(b_i, t)^-} |f(z)|$$

on any ring $\mathcal{A}_V$ for which $V = B(a, r) \setminus \bigcup_{i=1}^m B(a_i, r_i)^-$ contains $B(b, t) \setminus \bigcup_{i=1}^n B(b_i, t)^-$.

Another reason for emphasizing the reinterpretation of seminorms $[\cdot]_x$ as generic values is to be able to formulate a concrete description of the action of a rational function φ on $\mathbb{P}_{\text{Berk}}^1$ in terms of Berkovich’s classification of points of $\mathbb{A}_{\text{Berk}}^1$ (Theorem 2.2). We deduce from the above discussion:

PROPOSITION 2.8. *Let $\varphi \in K(T)$ be a non-constant rational function, and suppose $x \in \mathbb{H}_{\text{Berk}}$ corresponds to a disc $B(a, r)$ in K under the correspondence given by Berkovich’s classification theorem. Then $\varphi(x)$ corresponds to a disc $B(b, R)$ if and only if there is a closed affinoid $B(b, R) \setminus \bigcup_{i=1}^m B(b_i, R)^-$ for some $b_1, \dots, b_m \in K$ which is the image under φ of a closed affinoid $B(a, r) \setminus \bigcup_{i=1}^n B(a_i, r)^-$ for some $a_1, \dots, a_n \in K$.*

Give a proof of this result.

5. Tree structure on $\mathbb{H}_{\text{Berk}}$ and $\mathbb{P}_{\text{Berk}}^1$

Recall that there is a natural embedding of $\mathbb{P}^1(K)$ in $\mathbb{P}_{\text{Berk}}^1$ with dense image.

DEFINITION 2.9. The *Berkovich hyperbolic space* $\mathbb{H}_{\text{Berk}}$ is given by

$$(2.8) \quad \mathbb{H}_{\text{Berk}} = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K) .$$

Identifying $\mathbb{P}_{\text{Berk}}^1$ with the disjoint union of $\mathcal{B}(0, 1)$ and $\mathbb{P}_{\text{Berk}}^1 \setminus \mathcal{B}(0, 1) \equiv \mathcal{B}(0, 1)^-$, it makes sense to talk about a point $x \in \mathbb{P}_{\text{Berk}}^1$ being of Type I, II, III, or IV. With this terminology, $\mathbb{H}_{\text{Berk}}$ consists of all points of $\mathbb{P}_{\text{Berk}}^1$ which are not of Type I. As with $\mathcal{B}(0, 1)$, we write $\mathbb{H}_{\text{Berk}}^{\mathbb{Q}}$ for the set of all points of Type II, and $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ for the set of all points of Type II or III.

Note that since ∞ is a Type I point of $\mathbb{P}_{\text{Berk}}^1$, we can also define $\mathbb{H}_{\text{Berk}}$ as

$$\mathbb{H}_{\text{Berk}} = \mathbb{A}_{\text{Berk}}^1 \setminus \mathbb{A}^1(K) .$$

Our next goal is to define the space of (finite metrized) subgraphs of $\mathbb{P}_{\text{Berk}}^1$, and to prove that $\mathbb{P}_{\text{Berk}}^1$ can be identified with the inverse limit of all such subgraphs. For this, recall first that

$$\mathbb{P}_{\text{Berk}}^1 = W_0 \cup W_{\infty} ,$$

where W_0, W_{∞} are each homeomorphic to the closed Berkovich unit disc $\mathcal{B}(0, 1)$, with $W_0 \cap W_{\infty}$ is homeomorphic to an annulus $\mathcal{A}(1, 1)$ in each of the two pieces. Also, recall from Lemma 2.6 that $\mathbb{P}_{\text{Berk}}^1$ is uniquely path-connected, so that we may define the *convex hull* of a finite set of points $x_1, \dots, x_n \in \mathbb{P}_{\text{Berk}}^1$ by either of the following equivalent definitions:

- (a) The smallest connected subset of $\mathbb{P}_{\text{Berk}}^1$ containing $x_1, \dots, x_n$.
- (b) The union of all arcs L_{ij} connecting x_i to x_j for $1 \leq i < j \leq n$.

By a *subgraph* of $\mathbb{P}_{\text{Berk}}^1$, we mean the convex hull of a finite set of points $\zeta_1, \dots, \zeta_n \in \mathbb{H}_{\text{Berk}}$. As we will see in §6 below, there is a canonical metric on every subgraph Γ of $\mathbb{P}_{\text{Berk}}^1$ which makes Γ into a finite $\mathbb{R}$ -tree in a natural way. However, for the moment, we can just think of Γ as a compact Hausdorff topological space.

We write $\mathcal{S}$ for the collection of all subgraphs of $\mathbb{P}_{\text{Berk}}^1$. Similarly, we write $\mathcal{S}^{\mathbb{Q}}$ (resp. $\mathcal{S}^{\mathbb{R}}$) for the subset of $\mathcal{S}$ consisting of graphs which are the convex hull of a finite set of points $\zeta_1, \dots, \zeta_n \in \mathbb{H}_{\text{Berk}}^{\mathbb{Q}}$ (resp. $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$).

In the affinoid situation, when we considered the space $\mathcal{B}(0, 1)$ we only looked at finite subgraphs of $\mathcal{B}(0, 1)$ which were rooted at the Gauss point. It is therefore convenient to define

$$\mathcal{S}_{\text{Gauss}} = \{ \Gamma \in \mathcal{S} : \zeta_{\text{Gauss}} \in \Gamma \} ,$$

and to define $\mathcal{S}_{\text{Gauss}}^{\mathbb{Q}}$ and $\mathcal{S}_{\text{Gauss}}^{\mathbb{R}}$ accordingly. There is a natural bijection between $\mathcal{S}_{\text{Gauss}} \cap W_0$ (resp. $\mathcal{S}_{\text{Gauss}} \cap W_\infty$) and the set $\mathcal{D}$ defined in §4 of Chapter I.

The set $\mathcal{S}$ is naturally a directed set under inclusion, i.e., $\Gamma \leq \Gamma'$ if and only if $\Gamma \subseteq \Gamma'$. Similarly, all of the other sets $\mathcal{S}^{\mathbb{Q}}, \mathcal{S}^{\mathbb{R}}, \mathcal{S}_{\text{Gauss}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{Q}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{R}}$ are also directed under inclusion. Thus whenever $\Gamma \leq \Gamma'$, there is an inclusion map

$$i_{\Gamma, \Gamma'} : \Gamma \rightarrow \Gamma' .$$

Just as in §4 of Chapter I, there is also a *retraction map* $r_{\Gamma, \Gamma'} : \Gamma' \rightarrow \Gamma$ defined whenever $\Gamma \leq \Gamma'$. These retraction maps are again compatible with the partial order on $\mathcal{S}$, in the sense that if $\Gamma \leq \Gamma' \leq \Gamma''$, then

$$(2.9) \quad r_{\Gamma'', \Gamma} = r_{\Gamma', \Gamma} \circ r_{\Gamma'', \Gamma'} .$$

By (2.9), the inverse limit $\varprojlim_{\Gamma \in \mathcal{S}} \Gamma$ is well-defined, and by Tychonoff's theorem, this inverse limit is a compact Hausdorff topological space.

The main structural result concerning subgraphs of $\mathbb{P}_{\text{Berk}}^1$ is the following theorem:

THEOREM 2.10. *There is a canonical homeomorphism*

$$\mathbb{P}_{\text{Berk}}^1 \cong \varprojlim_{\Gamma \in \mathcal{S}} \Gamma ,$$

i.e., $\mathbb{P}_{\text{Berk}}^1$ can be identified with the inverse limit of all of its subgraphs. Moreover, the same result holds if the directed set $\mathcal{S}$ is replaced by any of $\mathcal{S}^{\mathbb{Q}}, \mathcal{S}^{\mathbb{R}}, \mathcal{S}_{\text{Gauss}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{Q}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{R}}$.

PROOF. The last statement follows from the easily verified fact that each of the directed sets $\mathcal{S}^{\mathbb{Q}}, \mathcal{S}^{\mathbb{R}}, \mathcal{S}_{\text{Gauss}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{Q}}, \mathcal{S}_{\text{Gauss}}^{\mathbb{R}}$ is cofinal in $\mathcal{S}$. It therefore suffices to prove the theorem with $\mathcal{S}$ replaced by $\mathcal{S}_{\text{Gauss}}$. But this is a straightforward consequence of Proposition 1.17, together with the identification $\mathbb{P}_{\text{Berk}}^1 = W_0 \cup W_\infty$. $\square$

Theorem 2.10 is conceptually quite important, since it allows us to think of the very complicated, infinitely-branched tree $\mathbb{P}_{\text{Berk}}^1$ as a limit of finite trees, and this is in fact how we will think about and work with $\mathbb{P}_{\text{Berk}}^1$ throughout this book. For example, the existence of a Laplacian operator on $\mathbb{P}_{\text{Berk}}^1$ with nice properties will be deduced in a straightforward way from the existence of a natural Laplacian operator on a finite metrized graph, together with the fact that if $\Gamma_1 \leq \Gamma_2$ then the Laplacians on Γ_1 and Γ_2 are “compatible” in an appropriate sense.

Also, note that the bijection

$$\mathbb{P}_{\text{Berk}}^1 \cong \varprojlim_{\Gamma \in \mathcal{S}^{\mathbb{R}}} \Gamma$$

furnished by Theorem 2.10 can be thought of as an extension to $\mathbb{P}_{\text{Berk}}^1$ of Berkovich's classification theorem (Theorem 1.2). Moreover, the fact that this bijection is a *homeomorphism* gives us a recipe for describing the Berkovich topology in terms of the explicit description of points as (equivalence classes of) nested sequences of closed discs.

As a complement to Theorem 2.10, we can also consider the *direct limit* $\varinjlim_{\Gamma \in \mathcal{S}} \Gamma$ with respect to the inclusion maps $i_{\Gamma, \Gamma'}$. In this case, we have the following result, whose straightforward proof we omit:

THEOREM 2.11. *There are canonical bijections*

$$\begin{aligned} \mathbb{H}_{\text{Berk}} &\cong \varprojlim_{\Gamma \in \mathcal{S}} \Gamma \cong \varprojlim_{\Gamma \in \mathcal{S}_{\text{Gauss}}} \Gamma \\ \mathbb{H}_{\text{Berk}}^{\mathbb{R}} &\cong \varprojlim_{\Gamma \in \mathcal{S}^{\mathbb{R}}} \Gamma \cong \varprojlim_{\Gamma \in \mathcal{S}_{\text{Gauss}}^{\mathbb{R}}} \Gamma \\ \mathbb{H}_{\text{Berk}}^{\mathbb{R}} &\cong \varprojlim_{\Gamma \in \mathcal{S}^{\mathbb{Q}}} \Gamma \cong \varprojlim_{\Gamma \in \mathcal{S}_{\text{Gauss}}^{\mathbb{Q}}} \Gamma \end{aligned}$$

REMARK 2.12. The direct limit topology on $\mathbb{H}_{\text{Berk}}$ is not the same as the subspace topology induced by the inclusion $\mathbb{H}_{\text{Berk}} \subseteq \mathbb{P}_{\text{Berk}}^1$ (c.f. Remark 1.16). When it is necessary to distinguish between the two, we will call the topology on $\mathbb{H}_{\text{Berk}}$ induced by the Berkovich (= Gelfand) topology the *weak topology* on $\mathbb{H}_{\text{Berk}}$, and we call the direct limit topology defined using the bijection in Theorem 2.11 the *strong topology*.

As we will see in the next section, the strong topology on $\mathbb{H}_{\text{Berk}}$ coincides with the topology defined by a natural metric which we will denote by ρ .

6. The path distance metric on $\mathbb{H}_{\text{Berk}}$

Recall that $\mathbb{P}_{\text{Berk}}^1$ can be identified with a disjoint union of $\mathcal{B}(0, 1)$ and the subset $\mathbb{P}_{\text{Berk}}^1 \setminus \mathcal{B}(0, 1)$, which is homeomorphic to $\mathcal{B}(0, 1)^-$ via the map $\psi = \bar{\psi}_{\infty}$ (which on the level of Type I points takes z to $1/z$).

Using this identification, we can extend the “small” and “big” metrics from the previous chapter to $\mathbb{P}_{\text{Berk}}^1$ and $\mathbb{H}_{\text{Berk}}$, respectively. In $\mathbb{P}_{\text{Berk}}^1$ endowed with the small metric d , every point of $\mathbb{P}_{\text{Berk}}^1$ has distance at most 1 from ζ_{Gauss} . Another way to describe the small metric d is via

the parametrization map $\alpha : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R}_{\geq 0}$ defined by

$$\alpha(z) = \begin{cases} 1 - \text{diam}(z) & z \in \mathcal{B}(0, 1) \\ 1 - \text{diam}(\psi(z)) & z \notin \mathcal{B}(0, 1) \end{cases}.$$

It is easy to verify that $\mathbb{P}_{\text{Berk}}^1$, together with the parametrization α and root ζ_{Gauss} , satisfies axioms (P1)-(P5) for a parametrized rooted tree, and that the metric d we have just described is the associated distance function. In particular, $(\mathbb{P}_{\text{Berk}}^1, d)$ is itself an $\mathbb{R}$ -tree.

However, it is much more useful and important to consider the extension of the big metric to $\mathbb{H}_{\text{Berk}}$. The importance of the big metric stems from that fact that, whereas the extension of the small metric to $\mathbb{P}_{\text{Berk}}^1$ depends on our having chosen coordinates for $\mathbb{P}^1$, the extension of the small metric to $\mathbb{H}_{\text{Berk}}$ is completely canonical (see Proposition 2.13 below). In addition, we will see later that functions of the form $\log |f|$, where $f \in K(T)$ is a nonzero rational function, are *piecewise linear* with respect to the big metric on $\mathbb{H}_{\text{Berk}}$.

For these reasons and more, we will from this point consider only the big metric, and rather than using the awkward terms “small metric” and “big metric”, we will refer to the big metric $\rho(x, y)$ on $\mathbb{H}_{\text{Berk}}$ as the *path-distance metric*. We now give a concrete description of this metric which will turn out to be the most profitable way for us to think about it.

Recall from Theorem 2.11 that $\mathbb{H}_{\text{Berk}}$ can be identified with the union of all subgraphs $\Gamma \in \mathcal{S}$, and $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ can be identified with the union of all subgraphs $\Gamma \in \mathcal{S}^{\mathbb{R}}$. In the terminology of Chapter 1, this has the following concrete interpretation. For simplicity, we first describe the metric ρ on $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$, and then we will extend it to $\mathbb{H}_{\text{Berk}}$ by continuity.

Recall that the space $\mathcal{B}(0, 1) \cap \mathbb{H}_{\text{Berk}}$ was described in Chapter 1 as the union of all graphs of discs Γ_S , where $S = \{B(a_1, t_1), \dots, B(a_n, t_n)\}$ runs over all finite sets of discs contained in $B(0, 1)$. A similar construction applies for any $\mathcal{B}(0, r)$, taking

$$\Gamma_S = \bigcup_{i=1}^n [t_i, r]_{B(a_i, r_i)}$$

where $[t_i, r]_{B(a_i, r_i)} = \{B(a_i, t) : t_i \leq t \leq r\}$ is a ‘line of discs’. Under this parametrization, the edge between $B(a, r_1)$ and $B(a, r_2)$ has length $|r_1 - r_2|$. However, as we have discussed, it is better to logarithmically reparametrize each line of discs $[r_1, r_2]_{B(a, r_1)}$, putting

$$\langle \log(r_1), \log(r_2) \rangle_{B(a, r_1)} = \{B(a, q_v^t) : \log_v(r_1) \leq t \leq \log_v(r_2)\}$$

and giving it length $|\log_v(r_2) - \log_v(r_1)|$. Carrying out this reparametrization for each $\mathcal{B}(0, r)$, and viewing $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ as the union of $\mathbb{H}_{\text{Berk}}^{\mathbb{R}} \cap \mathcal{B}(0, r)$ over all $r > 0$, we obtain the path-distance metric $\rho(x, y)$ on $\mathbb{H}_{\text{Berk}}^{\mathbb{R}}$. It follows easily from the definitions that ρ extends by continuity to all of $\mathbb{H}_{\text{Berk}}$, and that ρ is indeed a *metric* on $\mathbb{H}_{\text{Berk}}$. In particular, all points of $\mathbb{H}_{\text{Berk}}$ are at finite distance from each other.

We can also extend $\rho(x, y)$ to a singular metric on all of $\mathbb{P}_{\text{Berk}}^1$ by setting $\rho(x, \infty) = \rho(\infty, x) = \infty$ for each $x \in \mathbb{A}_{\text{Berk}}^1$. We will sometimes utilize this extension, although for the most part we will think of ρ as being an honest metric defined only on $\mathbb{H}_{\text{Berk}}$.

As alluded to above, the path distance metric ρ is canonical, in the following sense.

PROPOSITION 2.13. *The path distance $\rho(x, y)$ on $\mathbb{H}_{\text{Berk}}$ is independent of the choice of homogeneous coordinates on $\mathbb{P}^1/K$, in the sense that if $h(z) \in K(z)$ is a linear fractional transformation, then $\rho(h(x), h(y)) = \rho(x, y)$ for all $x, y \in \mathbb{H}_{\text{Berk}}$.*

PROOF. Let $h(z) = (az + b)/(cz + d) \in K(z)$ be a linear fractional transformation. It induces a continuous automorphism h of $\mathbb{P}_{\text{Berk}}^1$. Since points of type II are dense in each subgraph of $\mathbb{P}_{\text{Berk}}^1$, it suffices to show that $\rho(h(x), h(y)) = \rho(x, y)$ for all $x, y \in \mathbb{H}_{\text{Berk}}$ of Type II. By the “piecewise” definition of ρ , we may assume without loss of generality that x, y correspond to discs $B(p, r_1), B(p, r_2)$ with $B(p, r_1) \subseteq B(p, r_2)$. Finally, since an arbitrary linear fractional transformation can be written as a composition of affine maps and inversions, it suffices to show that the path distance is preserved by these two types of maps.

First suppose $h(z) = az + b$ is affine. The image of $B(p, r)$ under $h(z)$ is $B(ap + b, |a|r)$. If $x, y \in \mathbb{P}_{\text{Berk}}^1$ correspond to $B(p, r_1) \subseteq B(p, r_2)$, it follows that

$$\begin{aligned} \rho(h(x), h(y)) &= |\log_v(|a|r_1) - \log_v(|a|r_2)| \\ &= |\log_v(r_1) - \log_v(r_2)| = \rho(x, y). \end{aligned}$$

Next suppose $h(z) = 1/z$, and let $x \in \mathbb{P}_{\text{Berk}}^1$ correspond to a disc $B(p, r)$.

Recall from Proposition 2.8 that if x corresponds to a disc $B(a, r)$, then $h(x)$ corresponds to a disc $B(b, R)$ if and only if there is a closed affinoid of the form $B(b, R) \setminus \cup_{i=1}^m B(b_i, R)^-$ which is the image of a closed affinoid of the form $B(a, r) \setminus \cup_{i=1}^n B(a_i, r)^-$ for an appropriate choice of $a_1, \dots, a_n$.

If $B(p, r)$ does not contain 0, then $h(B(p, r)) = B(1/p, r/|p|^2)$. If $B(p, r)$ does contain 0, put $\dot{B}(0, r) = B(0, r) \setminus B(0, r)^-$; then $h(\dot{B}(0, r)) =$

$\dot{B}(0, 1/r)$. In either case, this determines $h(x)$. Now let $x, y \in \mathbb{P}_{\text{Berk}}^1$ correspond to discs $B(p, r_1) \subseteq B(p, r_2)$.

If neither disc contains 0, then $B(1/p, r_1/|p|^2) \subset B(1/p, r_2/|p|^2)$ and so

$$\begin{aligned} \rho(h(x), h(y)) &= |\log_v(r_1/|p|^2) - \log_v(r_2/|p|^2)| \\ &= |\log_v(r_1) - \log_v(r_2)| = \rho(x, y) . \end{aligned}$$

If both $B(p, r_1)$ and $B(p, r_2)$ contain 0, then $B(0, 1/r_2) \subseteq B(0, 1/r_1)$, so

$$\begin{aligned} \rho(h(x), h(y)) &= |\log_v(1/r_1) - \log_v(1/r_2)| \\ &= |\log_v(r_1) - \log_v(r_2)| = \rho(x, y) . \end{aligned}$$

Finally, if $B(p, r_2)$ contains 0 but $B(p, r_1)$ does not, let $z \in \mathbb{P}_{\text{Berk}}^1$ correspond to $B(p, |p|)$. Then $B(p, r_1) \subseteq B(p, |p|) \subseteq B(p, r_2)$, so $\rho(x, y) = \rho(x, z) + \rho(z, y)$. Since $B(p, |p|) = B(0, |p|)$ and $B(p, r_2) = B(0, r_2)$, and since the smallest disc containing $B(1/p, r_1/|p|^2)$ and $B(0, 1/r_2)$ is $B(0, 1/|p|)$,

$$\begin{aligned} \rho(h(x), h(y)) &= |\log_v(r_1/|p|^2) - \log_v(1/|p|)| + |\log_v(1/|p|) - \log_v(1/r_2)| \\ &= |\log_v(|p|) - \log_v(r_1)| + |\log_v(r_2) - \log_v(|p|)| \\ &= \rho(x, y) . \end{aligned}$$

□

A useful point of view (and the one adopted by Rivera-Letelier in his papers on the subject) is that $\rho(x, y)$ can be interpreted in terms of moduli of annuli. Indeed, if $x, y \in \mathbb{H}_{\text{Berk}}^{\mathbb{R}}$ correspond to closed discs $B(p, r_1) \subseteq B(p, r_2)$ in K , then one naturally associates to x and y the open annulus $A_{xy}^- = B(p, r_1)^- \setminus B(p, r_2)$, and $\rho(x, y) = \log |r_2/r_1|$ is the *modulus* of this annulus. More generally, if $A^- = B_1^- \setminus B_2$ with B_1^- an open disc in $\mathbb{P}^1(K)$ and B_2 a closed disc in $\mathbb{P}^1(K)$ contained in B_1^- (i.e., if A^- is an *open annulus in $\mathbb{P}^1(K)$*), then one can define the modulus of A^- to be the modulus of $h(A^-)$ for any linear fractional transformation h which takes A^- to an open annulus of the form $B(p, r_1)^- \setminus B(p, r_2) \subset K$. Proposition 2.13 is then equivalent to the assertion that the modulus of an open annulus in $\mathbb{P}^1(K)$ is well-defined (a known fact from non-archimedean analysis which we have essentially reproved above).

7. Notes and further references

The idea for defining $\mathbb{P}_{\text{Berk}}^1$ via a “Proj” construction is due to Berkovich, and can be found in his paper [11].

The notation $\mathbb{H}_{\text{Berk}}$ for the “Berkovich hyperbolic space” $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$ is an adaptation of Rivera-Letelier’s notation in [53, 54]. Also, the idea that one can profitably think of $\mathbb{H}_{\text{Berk}}$ as an analogue of the Poincaré disc, or of hyperbolic 3-space, is something which we learned during conversations with Rivera-Letelier. One should not confuse $\mathbb{H}_{\text{Berk}}$ with the “Drinfeld upper half-plane” (see, e.g., [11] or [35]), which is a rather different object despite a few superficial similarities.

The concrete description of the action of a rational map in terms of Berkovich’s classification theorem given in Proposition 2.8 forms the foundation for Rivera-Letelier’s study of dynamics on $\mathbb{P}_{\text{Berk}}^1$.

We learned of the connection between the path-distance metric ρ and moduli of annuli from Rivera-Letelier.

CHAPTER 3

The Hsia kernel

In this chapter, we introduce the Hsia kernel, the basic kernel for potential theory on $\mathbb{A}_{\text{Berk}}^1$. It was initially defined in (Hsia, [36]) as a kernel on trees of discs. We will relate it to other objects in potential theory: the potential kernel $j_z(x, y)$ on graphs, the spherical metric $\|x, y\|$ on $\mathbb{P}^1(K)$, and the canonical distance $[x, y]_\zeta$.

1. Definition of the Hsia kernel

It is an elementary fact about ultrametric spaces that the *radius* and *diameter* of a disc coincide.

If $x \in \mathbb{A}_{\text{Berk}}^1$ corresponds to a sequence of nested discs $\{B(a_i, r_i)\}$, we call $r = \lim_{i \rightarrow \infty} r_i$ the *diameter* (or *radius*) of x , and write $r = \text{diam}(x)$. Note that the diameter depends on the choice of coordinates on $\mathbb{A}^1$.

The Hsia kernel $\delta(x, y)_\infty$ is defined for $x, y \in \mathbb{A}_{\text{Berk}}^1$ as follows. If x corresponds to a sequence of nested discs $\{B(a_i, r_i)\}$ and y to $\{B(b_i, s_i)\}$, then

$$(3.1) \quad \delta(x, y)_\infty = \lim_{i \rightarrow \infty} \max(r_i, s_i, |a_i - b_i|) .$$

Clearly the Hsia kernel is symmetric, and $\delta(x, x)_\infty = \text{diam}(x)$ for each x . If $x, y \in \mathbb{A}^1(K)$ are points of type I, then $\delta(x, y)_\infty = |x - y|$. If $x, y \in \mathbb{A}_{\text{Berk}}^1$ are points of types I, II or III, with x corresponding to $B(a, r)$ and y corresponding to $B(b, s)$, then

$$\delta(x, y)_\infty = \max(r, s, |a - b|) = \sup_{z \in B(a, r), w \in B(b, s)} |z - w| .$$

The Hsia kernel has the following geometric interpretation. Consider the paths from x to ∞ and y to ∞ , and let $x \vee y$ be the point where they first meet. Then

$$(3.2) \quad \delta(x, y)_\infty = \delta(x \vee y, x \vee y)_\infty = \text{diam}(x \vee y) .$$

To see this, note that if x lies on the path from y to ∞ then $x \vee y = x$ and $\delta(x, y)_\infty = \lim_{i \rightarrow \infty} r_i = \text{diam}(x)$. Similarly if y lies on the path from x to ∞ , then $x \vee y = y$ and $\delta(x, y)_\infty = \text{diam}(y)$. Otherwise, $x \vee y \neq x, y$, so there are disjoint discs $B(a_i, r_i)$ and $B(b_i, s_i)$ in

the sequences defining x, y . Then $|a_i - b_i| = |a_j - b_j| > \max(r_j, s_j)$ for all $j \geq i$, and the point $x \vee y$ where the paths from x and y to ∞ meet corresponds to the disc $B(a_i, |a_i - b_i|) = B(b_i, |a_i - b_i|)$, whose radius is $|a_i - b_i|$.

PROPOSITION 3.1.

A) *The Hsia kernel is nonnegative, symmetric, and continuous in each variable separately. As a function of two variables, it is upper semicontinuous. It is continuous off the diagonal, and continuous at (x_0, x_0) for each point x_0 of type I, but is discontinuous at (x_0, x_0) for each point of type II, III, or IV.*

B) *The Hsia kernel is the unique extension of $|x - y|$ to $\mathbb{A}_{\text{Berk}}^1$ such that*

$$(3.3) \quad \delta(x, y)_\infty = \limsup_{\substack{(a, b) \rightarrow (x, y) \\ a, b \in \mathbb{A}^1(K)}} |a - b|$$

for each $x, y \in \mathbb{A}_{\text{Berk}}^1$.

C) *For all $x, y, z \in \mathbb{A}_{\text{Berk}}^1$*

$$\delta(x, y)_\infty \leq \max(\delta(x, z)_\infty, \delta(y, z)_\infty)$$

with equality if $\delta(x, z)_\infty \neq \delta(y, z)_\infty$.

D) *For each $a \in \mathbb{A}_{\text{Berk}}^1$, $r \in \mathbb{R}$, the ‘open disc’ $\mathcal{B}(a, r)_\infty^- := \{x \in \mathbb{A}_{\text{Berk}}^1 : \delta(x, a)_\infty < r\}$ is open. It is empty if $r \leq \text{diam}(a)$, and coincides with an open disc $\mathcal{B}(b, r)^-$ for some $b \in \mathbb{A}^1(K)$ if $r > \text{diam}(a)$.*

For each $a \in \mathbb{A}_{\text{Berk}}^1$, $r \in \mathbb{R}$, the ‘closed disc’ $\mathcal{B}(a, r)_\infty := \{x \in \mathbb{A}_{\text{Berk}}^1 : \delta(x, a)_\infty \leq r\}$ is closed. It is empty if $r < \text{diam}(a)$, and coincides with a closed disc $\mathcal{B}(b, r)$ for some $b \in \mathbb{A}^1(K)$ if $r > \text{diam}(a)$ or if $r = \text{diam}(a)$ and a is of type II or III. If $r = \text{diam}(a)$ and a is of type I or IV, then $\mathcal{B}(a, r)_\infty = \{a\}$.

PROOF.

(A) Consider $\delta(x, y)_\infty$ as a function of two variables. We first show it is continuous off the diagonal.

Take $x_0, y_0 \in \mathbb{A}_{\text{Berk}}^1$ with $x_0 \neq y_0$. If there are disjoint open discs with $x_0 \in \mathcal{B}(a, r)^-$ and $y_0 \in \mathcal{B}(b, s)^-$, then $|a - b| > \max(r, s)$, so for each $x \in \mathcal{B}(a, r)^-, y \in \mathcal{B}(b, s)^-$,

$$\delta(x, y)_\infty = |a - b| = \delta(x_0, y_0)_\infty,$$

and $\delta(x, y)_\infty$ is continuous at (x_0, y_0) .

Otherwise, either each open disc $\mathcal{B}(a, r)^-$ containing x_0 contains y_0 , or each open disc $\mathcal{B}(b, s)^-$ containing y_0 contains x_0 . Suppose the former holds. Then x_0 must be a point of type II or III corresponding to a disc $B(a, r)$ with $y_0 \in \mathcal{B}(a, r)$. After relabeling the center, we can

assume that $y_0 \in \mathcal{B}(a, r)^-$, and that $y_0 \in B(a, s)^-$ for some $s < r$. Take $0 < \varepsilon < r - s$. Then $U := \mathcal{B}(a, r + \varepsilon)^- \setminus \mathcal{B}(a, r - \varepsilon)^-$ and $V := \mathcal{B}(a, s)^-$ are disjoint open neighborhoods of x_0, y_0 respectively. Given $x \in U$, put $r_1 = \inf\{t : x \in \mathcal{B}(a, t)\}$. Then $r - \varepsilon < r_1 < r + \varepsilon$, and the geometric description of the Hsia kernel (3.2) implies that $\delta(x, y)_\infty = r_1$ for each $y \in V$, so

$$|\delta(x, y)_\infty - \delta(x_0, y_0)_\infty| = |r_1 - r| < \varepsilon.$$

Thus, again $\delta(x, y)_\infty$ is continuous at (x_0, y_0) .

Now consider points on the diagonal. Suppose $x_0 \in \mathbb{A}_{\text{Berk}}^1$ has radius r . For each $\varepsilon > 0$, there is an open disc $\mathcal{B}(a, r + \varepsilon)^-$ containing x_0 , and (3.2) shows that $\delta(x, y)_\infty < r + \varepsilon$ for each $x, y \in \mathcal{B}(a, r + \varepsilon)^-$. If $r = 0$, this implies $\delta(x, y)_\infty$ is continuous at (x_0, x_0) .

If $r > 0$, then $\delta(x, y)_\infty$ is not continuous at (x_0, x_0) , since every neighborhood of (x_0, x_0) contains points (a, a) with a of type I, and for such points $\delta(a, a)_\infty = 0$, while $\delta(x_0, x_0)_\infty = r$. On the other hand, the discussion in the previous paragraph shows that

$$(3.4) \quad \limsup_{(x, y) \rightarrow (x_0, x_0)} \delta(x, y)_\infty = r = \delta(x_0, x_0)_\infty.$$

Thus $\delta(x, y)_\infty$ is upper semicontinuous as a function of two variables.

Now fix x , and consider $\delta(x, y)_\infty$ as a function of y . By what has been shown above, $\delta(x, y)_\infty$ is continuous for $y \neq x$. For continuity at x , put $r = \text{diam}(x)$ and consider a neighborhood $\mathcal{B}(a, r + \varepsilon)^-$ of x as discussed above. For each $y \in \mathcal{B}(a, r + \varepsilon)^-$, the paths from x and y to ∞ meet at a point $w = x \vee y \in \mathcal{B}(a, r + \varepsilon)^-$, so

$$r = \text{diam}(x) = \delta(x, x)_\infty \leq \delta(x, y)_\infty = \text{diam}(w) \leq r + \varepsilon.$$

Thus $|\delta(x, y)_\infty - \delta(x, x)_\infty| < \varepsilon$.

(B) Note that each neighborhood $\mathcal{B}(a, r + \varepsilon)^-$ of x_0 contains type I points $a \neq b$ with $r < |a - b| < r + \varepsilon$, so (3.4) remains true even if the lim sup is restricted to $x, y \in \mathbb{A}^1(K)$. The characterization of $\delta(x, y)_\infty$ follows from this, together with continuity off the diagonal.

(C) For discs $B(a, r), B(b, s)$ it is easy to see that

$$\max(r, s, |a - b|) = \sup_{\substack{p \in B(a, r) \\ q \in B(b, s)}} (|p - q|).$$

The ultrametric inequality for $\delta(x, y)$ follows from this, (3.1), and the classical ultrametric inequality.

(D) Given $a \in \mathbb{A}_{\text{Berk}}^1$ and $r \in \mathbb{R}$, consider the ‘open disc’ $\mathcal{B}(a, r)_\infty^- := \{x \in \mathbb{A}_{\text{Berk}}^1 : \delta(x, a)_\infty < r\}$. If $r \leq \text{diam}(a)$, it is clearly empty. If $r > \text{diam}(a)$ it contains a , and is open by the upper semicontinuity

of $\delta(x, a)_\infty$. Each nonempty open set contains points of type I. If $b \in \mathcal{B}(a, r)^- \cap \mathbb{A}^1(K)$, the ultrametric inequality for $\delta(x, y)_\infty$ shows that $\mathcal{B}(a, r)_\infty^- = \mathcal{B}(b, r)^-$.

Likewise, consider the ‘closed disc’ $\mathcal{B}(a, r)_\infty := \{x \in \mathbb{A}_{\text{Berk}}^1 : \delta(x, a)_\infty \leq r\}$. If $r < \text{diam}(a)$, it is empty. If $r = \text{diam}(a)$ and a is of type I or IV it is easy to see that $\mathcal{B}(a, r)_\infty = \{a\}$. If $r > \text{diam}(a)$ then it contains $\mathcal{B}(a, r)_\infty^-$ and hence contains points b of type I; this also holds if $r = \text{diam}(a)$ and a is of type II or III. In either case the ultrametric inequality for $\delta(x, y)_\infty$ shows that $\mathcal{B}(a, r)_\infty = \mathcal{B}(b, r)$, and in particular that $\mathcal{B}(a, r)_\infty$ is closed. $\square$

The function-theoretic meaning of the Hsia Kernel is as follows:

COROLLARY 3.2. *For each $a \in K$, the function $T - a \in K(T)$ satisfies*

$$[T - a]_x = \delta(x, a)_\infty .$$

PROOF. By Proposition 3.1 (B), if $x \in \mathbb{A}_{\text{Berk}}^1$ corresponds to the nested sequence of discs $\{B(a_i, r_i)\}$ then

$$[T - a]_x = \lim_{i \rightarrow \infty} \|T - a\|_{B(a_i, r_i)} = \limsup_{\substack{b \rightarrow x \\ b \in K}} |b - a| = \delta(x, a)_\infty .$$

$\square$

The Hsia kernel $\delta(x, y)_\infty$ has a pole at ∞ . We now set out to generalize it to a kernel $\delta(x, y)_\zeta$ with a pole at an arbitrary point $\zeta \in \mathbb{P}_{\text{Berk}}^1$. To do so we will use some facts from the theory of metrized graphs (see Appendix C).

2. Extension of $j_z(x, y)$ to $\mathbb{P}_{\text{Berk}}^1$

Recall from §5 and §6 of Chapter 2 that a *subgraph* Γ of $\mathbb{P}_{\text{Berk}}^1$ is a finite $\mathbb{R}$ -tree contained in $\mathbb{H}_{\text{Berk}}$ which is equipped with the (logarithmic) path distance metric $\rho(x, y)$. Since there is a unique path between any two points of $\mathbb{P}_{\text{Berk}}^1$, there is a natural retraction map $r_\Gamma : \mathbb{P}_{\text{Berk}}^1 \rightarrow \Gamma$ (c.f. Theorem 2.10).

The relation between the Hsia kernel and the potential kernel $j_z(x, y)$ on a metrized graph (see Appendix C) is as follows. Fix a coordinate system on $\mathbb{P}^1$, so that $\mathbb{P}_{\text{Berk}}^1 = \mathbb{A}_{\text{Berk}}^1 \cup \{\infty\}$. Recall that ζ_{Gauss} denotes the ‘Gauss point’, the point in $\mathbb{P}_{\text{Berk}}^1$ corresponding to the sup norm over the closed unit disc $B(0, 1)$.

PROPOSITION 3.3. *Let $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, and let Γ be any subgraph of $\mathbb{P}_{\text{Berk}}^1$ containing z . Put $\tilde{\infty} = r_\Gamma(\infty)$. Then for all $x, y \in \Gamma$,*

$$(3.5) \quad -\log_v(\delta(x, y)_\infty) = j_z(x, y) - j_z(x, \tilde{\infty}) - j_z(y, \tilde{\infty}) - \log_v(\text{diam}(z)) .$$

In particular, if $z = \zeta_{\text{Gauss}} \in \mathbb{P}_{\text{Berk}}^1$ is the Gauss point, then $\text{diam}(\zeta_{\text{Gauss}}) = 1$, so

$$(3.6) \quad -\log_v(\delta(x, y)_\infty) = j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \tilde{\infty}) - j_{\zeta_{\text{Gauss}}}(y, \tilde{\infty}) .$$

PROOF. Fix y , and set $s = \text{diam}(y)$. The intersection of the path from y to ∞ with Γ is a subgraph $\Gamma_y = [y, \tilde{\infty}]$ of Γ isometric to the closed segment $[s, S]$.

Consider the function $f_y(x) = -\log_v(\delta(x, y)_\infty)$ on Γ . The geometric description of $\delta(x, y)_\infty$ (3.2) shows that for each $x \in \Gamma$

$$\delta(x, y)_\infty = \delta(r_{\Gamma_y}(x), y)_\infty .$$

Thus, $f_y(x)$ is constant on branches off Γ_y . For $x \in \Gamma_y$,

$$-\log_v(\delta(x, y)_\infty) = -\log_v(t) , \quad \text{where } t = \text{diam}(x) .$$

The arclength parameter along Γ_y is $\log_v(t)$ for $s \leq t \leq S$, so the restriction of $f_y(x)$ to Γ_y has constant slope -1 . Hence $f_y(x) \in \text{CPA}(\Gamma)$, and

$$\Delta_x(f_y(x)) = \delta_y(x) - \delta_{\tilde{\infty}}(x) .$$

By (C.1),

$$\Delta_x(j_z(x, y) - j_z(x, \tilde{\infty})) = \delta_y(x) - \delta_{\tilde{\infty}}(x) .$$

It follows from Proposition C.2 (B) that there is a constant $C_z(y)$ such that

$$(3.7) \quad -\log_v(\delta(x, y)_\infty) = f_y(x) = j_z(x, y) - j_z(x, \tilde{\infty}) - C_z(y) .$$

Fixing x , and letting y vary, we see from (3.7) that the function $h(y) = C_z(y)$ belongs to $\text{CPA}(\Gamma)$. Let Δ_y be the Laplacian with respect to the variable y . Applying Δ_y to both sides of (3.7) gives

$$\delta_x(y) - \delta_{\tilde{\infty}}(y) = (\delta_x(y) - \delta_z(y)) - 0 - \Delta_y(h(y)) ,$$

so $\Delta_y(h(y)) = -(\delta_{\tilde{\infty}}(y) - \delta_z(y))$. By Proposition C.2 (B), $h(y) = -j_{\zeta_{\text{Gauss}}}(y, \tilde{\infty}) + C_z$ for some constant C_z . Hence

$$-\log_v(\delta(x, y)_\infty) = f_y(x) = j_z(x, y) - j_z(x, \tilde{\infty}) - j_z(y, \tilde{\infty}) + C_z$$

Taking $x = y = z$, we see that $C_z = -\log_v(\delta(z, z)_\infty) = -\log_v(\text{diam}(z))$. $\square$

Fix $z \in \mathbb{H}_{\text{Berk}}$. We will now extend $j_z(x, y)$ to a function from $\mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1$ to $\mathbb{R} \cup \{+\infty\}$ which is singular only along the diagonal $\text{Diag}(K)$ in $\mathbb{P}^1(K) \times \mathbb{P}^1(K)$.

Fix $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, and let Γ vary over subgraphs of $\mathbb{P}_{\text{Berk}}^1$ containing z ; temporarily write $j_z(x, y)_\Gamma$ for the potential kernel on Γ . Suppose $\Gamma_1 \subset \Gamma_2$. Since Γ_1 and Γ_2 are both trees, their potential kernels are compatible: by the description of the potential kernel on a tree, given $x, y \in \Gamma_1$,

$$j_z(x, y)_{\Gamma_1} = \rho(z, w) = j_z(x, y)_{\Gamma_2}$$

where $w = w_z(x, y)$ is the point where the paths from x and y to z meet. Thus the functions $j_z(x, y)_\Gamma$ cohere to give a well-defined function $j_z(x, y)$ on $\mathbb{H}_{\text{Berk}}$.

We can extend $j_z(x, y)$ to all $x, y \in \mathbb{P}_{\text{Berk}}^1$ by setting

$$(3.8) \quad j_z(x, y) = \begin{cases} j_z(r_\Gamma(x), r_\Gamma(y))_\Gamma & \text{if } (x, y) \notin \text{Diag}(K), \text{ and} \\ +\infty & \text{if } (x, y) \in \text{Diag}(K) , \end{cases}$$

where Γ is any subgraph containing z and the point $w_z(x, y)$ where the (unique) paths from x to z and y to z first meet.

Explicitly, for $x, y \in \mathbb{P}^1(K)$, we have

$$j_z(x, y) = \begin{cases} \rho(z, w_z(x, y)) & \text{if } x \neq y, \text{ and} \\ \infty & \text{if } x = y . \end{cases}$$

By Proposition 3.3 and the continuity of $\delta(x, y)_\infty$ off the diagonal, for all $x, y \in \mathbb{A}_{\text{Berk}}^1$ we have

$$(3.9) \quad -\log_v(\delta(x, y)_\infty) = j_z(x, y) - j_z(x, \infty) - j_z(y, \infty) - \text{diam}(z) .$$

In particular, if $z = \zeta_{\text{Gauss}}$ is the Gauss point, then

$$(3.10) \quad -\log_v(\delta(x, y)_\infty) = j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \infty) - j_{\zeta_{\text{Gauss}}}(y, \infty) .$$

Similarly, by Proposition C.3(C), for each $z, \zeta \in \mathbb{H}_{\text{Berk}}$ we have

$$(3.11) \quad j_\zeta(x, y) = j_z(x, y) - j_z(x, \zeta) - j_z(\zeta, y) + j_z(\zeta, \zeta) .$$

COROLLARY 3.4. *Let $0 \neq \varphi \in K(T)$ be a rational function with divisor $\text{div}(\varphi) = \sum_{i=1}^m n_i(a_i)$, and fix $z \in \mathbb{H}_{\text{Berk}}$. Then:*

A) *For all $x \in \mathbb{P}_{\text{Berk}}^1$, we have*

$$(3.12) \quad -\log_v([\varphi]_x) = -\log_v([\varphi]_z) + \sum_{i=1}^m n_i j_z(x, a_i) .$$

B) Let Γ be a subgraph of $\mathbb{P}_{\text{Berk}}^1$ containing z , and put $\tilde{a}_i = r_\Gamma(a_i)$. Then for all $x \in \Gamma$,

$$(3.13) \quad -\log_v([\varphi]_x) = -\log_v([\varphi]_z) + \sum_{i=1}^m n_i j_z(x, \tilde{a}_i) .$$

PROOF. (A) There is a $B \in K$ such that $\varphi(T) = B \cdot \prod_{a_i \neq \infty} (T - a_i)^{n_i}$. By Corollary 3.2, for each $x \in \Gamma$

$$-\log_v([\varphi]_x) = -\log_v(|B|) + \sum_{a_i \neq \infty} -\log_v(\delta(x, a_i)_\infty) .$$

Inserting formula (3.9) and using $\sum_{i=1}^m n_i = 0$ gives

$$-\log_v([\varphi]_x) = -\sum_{i=1}^m n_i \log_v(j_z(x, a_i)) + C$$

for some constant C . Taking $x = z$ gives $C = -\log_v([\varphi]_z)$.

Part (B) follows from (3.12), using (3.8). $\square$

For future reference, we note one situation where (3.8) holds automatically:

PROPOSITION 3.5. (Retraction Formula) *Let Γ be a subgraph of $\mathbb{P}_{\text{Berk}}^1$, and suppose $x, z \in \Gamma$. Then for any $y \in \mathbb{P}_{\text{Berk}}^1$*

$$j_z(x, y) = j_z(x, r_\Gamma(y)) .$$

PROOF. Since x and z belong to Γ , so does the path from x to z . Hence the point w where the paths from x and y to z meet, which lies on the path from x to z , belongs to Γ . $\square$

3. The spherical meet

Recall that $\mathcal{O}$ denotes the ring of integers of K . The spherical metric $\|x, y\|$ on $\mathbb{P}^1(K)$ is the unique $GL_2(\mathcal{O})$ -invariant metric on $\mathbb{P}^1(K)$ such that $\|x, y\| = |x - y|$ for $x, y \in B(0, 1)$. In homogeneous coordinates, if $x = (x_0 : x_1)$ and $y = (y_0 : y_1)$, then

$$(3.14) \quad \|x, y\| = \frac{|x_0 y_1 - x_1 y_0|}{\max(|x_0|, |x_1|) \max(|y_0|, |y_1|)} .$$

In affine coordinates, for $x, y \in \mathbb{A}^1(K)$ this becomes

$$(3.15) \quad \|x, y\| = \frac{|x - y|}{\max(1, |x|) \max(1, |y|)} .$$

Thus if we identify $\mathbb{P}^1(K)$ with $\mathbb{A}^1(K) \cup \{\infty\}$ and put $1/\infty = 0$, then

$$(3.16) \quad \|x, y\| = \begin{cases} |x - y| & \text{if } x, y \in B(0, 1), \\ |1/x - 1/y| & \text{if } x, y \in \mathbb{P}^1(K) \setminus B(0, 1), \\ 1 & \text{if exactly one of } x, y \in B(0, 1). \end{cases}$$

It is well known, and easy to check using (3.16), that the spherical metric satisfies the strict ultrametric inequality.

We will now extend $\|x, y\|$ to a function from $\mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1$ to $\mathbb{R}$.

For $x, y \in \mathbb{P}^1(K)$, we claim that

$$-\log_v \|x, y\| = j_{\zeta_{\text{Gauss}}}(x, y),$$

i.e., that

$$\|x, y\| = q_v^{-j_{\zeta_{\text{Gauss}}}(x, y)}.$$

To see this, consider the various cases in (3.16). If $x, y \in B(0, 1)$, then the point w where the paths from x and y to ζ_{Gauss} first meet corresponds to the ball $B(x, r) = B(y, r)$ with $r = |x - y|$. The path distance $\rho(\zeta_{\text{Gauss}}, w)$ is $\log_v(1/r)$, so $j_{\zeta_{\text{Gauss}}}(x, y) = -\log_v(r)$ and $q_v^{-j_{\zeta_{\text{Gauss}}}(x, y)} = |x - y| = \|x, y\|$. If $x, y \in \mathbb{P}^1(K) \setminus B(0, 1)$, a similar argument applies, using the local parameter $1/T$ at ∞ . Finally, if one of x, y belongs to $B(0, 1)$ and the other to $\mathbb{P}^1(K) \setminus B(0, 1)$, then the paths from x and y to ζ_{Gauss} meet at $w = \zeta_{\text{Gauss}}$, so $j_{\zeta_{\text{Gauss}}}(x, y) = 0$ and $q_v^{-j_{\zeta_{\text{Gauss}}}(x, y)} = 1 = \|x, y\|$.

This motivates the following extension of $\|x, y\|$ to $\mathbb{P}_{\text{Berk}}^1$: for all $x, y \in \mathbb{P}_{\text{Berk}}^1$, define the *spherical meet* by the formula

$$(3.17) \quad \|x, y\| = q_v^{-j_{\zeta_{\text{Gauss}}}(x, y)}.$$

The spherical meet has the following geometric interpretation, which motivates its name. Consider the paths from x and y to ζ_{Gauss} , and let w be the first point where they meet. Then by (3.17), it follows that

$$\|x, y\| = \|w, w\| = q_v^{-\rho(\zeta_{\text{Gauss}}, w)},$$

and for $x \in \mathcal{B}(0, 1)$, we have

$$\|x, x\| = \text{diam}(x).$$

REMARK 3.6. Note that although $\|x, y\|$ is a metric on $\mathbb{P}^1(K)$, the spherical meet $\|x, y\|$ is *not* a metric on $\mathbb{P}_{\text{Berk}}^1$ because $\|x, x\| > 0$ if x is not of type I. Nonetheless, for each $a \in \mathbb{P}^1(K)$, the balls

$$\begin{aligned} \mathcal{B}(a, r)_0^- &= \{x \in \mathbb{P}_{\text{Berk}}^1 : \|x, a\| < r\}, \\ \mathcal{B}(a, r)_0 &= \{x \in \mathbb{P}_{\text{Berk}}^1 : \|x, a\| \leq r\} \end{aligned}$$

are indeed open (resp. closed) in the Berkovich topology. If $r > 1$ then $\mathcal{B}(a, r)_0^- = \mathbb{P}^1(K)$. If $r \leq 1$ and $a \in B(0, 1)$ then $\mathcal{B}(a, r)_0^- = \mathcal{B}(a, r)^-$, while if $a \notin B(0, 1)$ then $\mathcal{B}(a, r)_0^- = \{x \in \mathbb{P}_{\text{Berk}}^1 : [1/T - 1/a]_x < r\}$. Similar formulas hold for the closed balls.

PROPOSITION 3.7.

A) *The spherical meet $\|x, y\|$ is nonnegative, symmetric, continuous in each variable separately, and bounded above by 1. As a function of two variables, it is upper semicontinuous. It is continuous off the diagonal, and continuous at (x_0, x_0) for each point x_0 of type I, but is discontinuous at (x_0, x_0) for each point of type II, III, or IV.*

B) *For all $x, y \in \mathbb{P}_{\text{Berk}}^1$*

$$(3.18) \quad \|x, y\| = \limsup_{\substack{(a,b) \rightarrow (x,y) \\ a,b \in \mathbb{P}^1(K)}} \|a, b\| .$$

C) *For all $x, y, z \in \mathbb{P}_{\text{Berk}}^1$*

$$\|x, y\| \leq \max(\|x, z\|, \|y, z\|) ,$$

with equality if $\|x, z\| \neq \|y, z\|$.

D) *For each $a \in \mathbb{P}_{\text{Berk}}^1$ and $r \in \mathbb{R}$, the ‘open ball’ $\mathcal{B}(a, r)_0^- := \{x \in \mathbb{P}_{\text{Berk}}^1 : \|x, a\| < r\}$ is open. It is empty if $r \leq \text{diam}(a)$, and coincides with an open ball $\mathcal{B}(b, r)_0^-$ for some $b \in \mathbb{P}^1(K)$ if $r > \text{diam}(a)$.*

Likewise, the ‘closed ball’ $\mathcal{B}(a, r)_0 := \{x \in \mathbb{P}_{\text{Berk}}^1 : \|x, a\| \leq r\}$ is closed. It is empty if $r < \|a, a\|$, and coincides with $\mathcal{B}(b, r)_0$ for some $b \in \mathbb{P}^1(K)$ if $r > \text{diam}(a)$ or if $r = \text{diam}(a)$ and a is of type II or III. If $r = \text{diam}(a)$ and a is of type I or IV, then $\mathcal{B}(a, r)_0 = \{a\}$.

PROOF. Similar to Proposition 3.1. □

The Hsia kernel and the spherical meet can be obtained from each other using the following formulae.

PROPOSITION 3.8.

$$A) \text{ For } x, y \in \mathbb{A}_{\text{Berk}}^1, \quad \delta(x, y)_\infty = \frac{\|x, y\|}{\|x, \infty\| \|y, \infty\|} .$$

B) *For $x, y \in \mathbb{P}_{\text{Berk}}^1$,*

$$\|x, y\| = \begin{cases} \frac{\delta(x, y)_\infty}{\max(1, \text{diam}(x)) \max(1, \text{diam}(y))} & \text{if } x, y \neq \infty, \\ 1 / \max(1, \text{diam}(x)) & \text{if } y = \infty, \\ 1 / \max(1, \text{diam}(y)) & \text{if } x = \infty . \end{cases}$$

PROOF. Part (A) follows from (3.10). Part (B) follows from (3.15) using Proposition 3.1 (B) and Proposition 3.7 (B). □

4. The generalized Hsia kernel

Proposition 3.8 motivates a definition of the Hsia kernel relative to an arbitrary point $\zeta \in \mathbb{P}_{\text{Berk}}^1$.

For $x, y \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, we define the generalized Hsia kernel by

$$(3.19) \quad \delta(x, y)_\zeta = \frac{\|x, y\|}{\|x, \zeta\| \|y, \zeta\|}.$$

If $\zeta \notin \mathbb{P}^1(K)$, the formula (3.19) makes sense for all $x, y \in \mathbb{P}_{\text{Berk}}^1$, since $\|x, \zeta\|, \|y, \zeta\| \geq \text{diam}(\zeta) > 0$, giving $\delta(x, \zeta)_\zeta = \delta(\zeta, y)_\zeta = 1/\text{diam}(\zeta)$. If $\zeta \in \mathbb{P}^1(K)$, we put $\delta(x, \zeta)_\zeta = \delta(\zeta, y)_\zeta = \infty$. In this way, we can regard $\delta(x, y)_\zeta$ as a function from $\mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R} \cup \{+\infty\}$.

Actually, it is more natural to regard the Hsia kernel as only defined up to scaling. By the definition of the spherical meet, we have

$$\delta(x, y)_\zeta = q_v^{-j_{\zeta_{\text{Gauss}}}(x, y) + j_{\zeta_{\text{Gauss}}}(x, \zeta) + j_{\zeta_{\text{Gauss}}}(y, \zeta)}.$$

However, for any $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, formula (3.11) shows that there is a constant C_ζ such that

$$(3.20) \quad C_\zeta \cdot \delta(x, y)_\zeta = q_v^{-j_z(x, y) + j_z(x, \zeta) + j_z(y, \zeta)}.$$

For each $C_\zeta > 0$, we will also call $C_\zeta \cdot \delta(x, y)_\zeta$ a Hsia kernel relative to the point ζ .

When $\zeta = \infty$, Proposition 3.8 shows that definition (3.19) is consistent with our earlier definition of $\delta(x, y)_\infty$. When $\zeta = \zeta_{\text{Gauss}}$ is the Gauss point, we have $\|x, \zeta_{\text{Gauss}}\| = \|\zeta_{\text{Gauss}}, y\| = 1$, so that

$$\delta(x, y)_{\zeta_{\text{Gauss}}} = \|x, y\|$$

is just the spherical meet of x and y .

For an arbitrary point $\zeta \in \mathbb{H}_{\text{Berk}}$, Proposition C.3(C) shows that

$$(3.21) \quad \delta(x, y)_\zeta = C_\zeta \cdot q_v^{-j_\zeta(x, y)}, \quad \text{where } C_\zeta = q_v^{j_{\zeta_{\text{Gauss}}}(\zeta, \zeta)}.$$

Thus for $\zeta \in \mathbb{H}_{\text{Berk}}$, $\delta(x, y)_\zeta$ can be thought of as a generalized spherical meet.

REMARK 3.9. For $\zeta \in \mathbb{P}^1(K)$, the reader familiar with ([62]) will recognize (3.19) as the ‘canonical distance’ $[x, y]_\zeta$ for $x, y \in \mathbb{P}^1(K)$. Thus, the generalized Hsia kernel is a natural extension of the canonical distance to the Berkovich projective line.

For each $x \in \mathbb{P}_{\text{Berk}}^1$, put $\text{diam}_\zeta(x) = \delta(x, x)_\zeta$. Then the generalized Hsia kernel has the following geometric interpretation which extends

(3.2). Given $x, y \in \mathbb{P}_{\text{Berk}}^1$, let $w = w_\zeta(x, y)$ be the point where the paths from x and y to ζ first meet. Then

$$(3.22) \quad \delta(x, y)_\zeta = \text{diam}_\zeta(w) .$$

For $\zeta \in \mathbb{H}_{\text{Berk}}$, it follows from (3.21) and the definitions that

$$-\log_v \delta(x, y)_\zeta = \rho(w, \zeta) - \log_v \text{diam}(\zeta) .$$

The following result generalizes Proposition 3.1:

PROPOSITION 3.10.

A) For each $\zeta \in \mathbb{P}_{\text{Berk}}^1$, the generalized Hsia kernel $\delta(x, y)_\zeta$ is non-negative, symmetric and continuous in each variable separately. If $\zeta \in \mathbb{H}_{\text{Berk}}$, then $\delta(x, y)_\zeta$ is bounded. For $\zeta \in \mathbb{P}^1(K)$ it is unbounded, and extends the canonical distance $[x, y]_\zeta$ from [62].

As a function of two variables, $\delta(x, y)_\zeta$ is upper semicontinuous. It is continuous off the diagonal, and is continuous at (x_0, x_0) for each point x_0 of type I, but is discontinuous at (x_0, x_0) for each point of type II, III, or IV.

B) For each $x, y \in \mathbb{P}_{\text{Berk}}^1$

$$(3.23) \quad \delta(x, y)_\zeta = \limsup_{\substack{(a, b) \rightarrow (x, y) \\ a, b \in \mathbb{P}^1(K)}} \delta(a, b)_\zeta .$$

C) For all $x, y, z \in \mathbb{P}_{\text{Berk}}^1$,

$$\delta(x, y)_\zeta \leq \max(\delta(x, z)_\zeta, \delta(y, z)_\zeta) ,$$

with equality if $\delta(x, z)_\zeta \neq \delta(y, z)_\zeta$.

D) For each $a \in \mathbb{P}_{\text{Berk}}^1$ and $r \in \mathbb{R}$, the ‘open ball’ $\mathcal{B}(a, r)_\zeta^- := \{x \in \mathbb{P}_{\text{Berk}}^1 : \delta(x, a)_\zeta < r\}$ is in fact open. It is empty if $r \leq \text{diam}_\zeta(a)$, and coincides with an open ball $\mathcal{B}(b, r)_\zeta^-$ for some $b \in \mathbb{P}^1(K)$ if $r > \text{diam}_\zeta(a)$.

Likewise, the ‘closed ball’ $\mathcal{B}(a, r)_\zeta := \{x \in \mathbb{P}_{\text{Berk}}^1 : \delta(x, a)_\zeta \leq r\}$ is closed. It is empty if $r < \text{diam}_\zeta(a)$, and coincides with $\mathcal{B}(b, r)_\zeta$ for some $b \in \mathbb{P}^1(K)$ if $r > \text{diam}_\zeta(a)$ or if $r = \text{diam}_\zeta(a)$ and a is of type II or III. If $r = \text{diam}_\zeta(a)$ and a is of type I or IV, then $\mathcal{B}(a, r)_\zeta = \{a\}$.

PROOF. Parts (A), (B), and (D) follow by arguments similar to those in the proof of Proposition 3.1. Part (C) follows from (3.22), or can be shown using Proposition 3.7(C) using a case-by-case analysis similar to the proof of ([62], Theorem 2.5.1, p.125). $\square$

REMARK 3.11. If $r \leq \text{diam}_\zeta(a)$, then the balls $\mathcal{B}(a, r)_\zeta$ and $\mathcal{B}(a, r)_\zeta^-$ have been described in Proposition 3.10(D). If $r > \text{diam}_\zeta(\zeta)$ then $\mathcal{B}(a, r)_\zeta = \mathcal{B}(a, r)_\zeta^- = \mathbb{P}_{\text{Berk}}^1$. Suppose $\text{diam}_\zeta(a) \leq r \leq \text{diam}_\zeta(\zeta)$. Then

the balls have the following geometric interpretation. Consider the function $\text{diam}_\zeta(x)$ for x in the path $[a, \zeta]$ from a to ζ . By Proposition 3.10(A), it is continuous on that path, since $\text{diam}_\zeta(x) = \delta(x, x)_\zeta = \delta(x, a)_\zeta$. And by (3.22), it is monotone increasing. Hence there is a unique $x \in [a, \zeta]$ with $\text{diam}_\zeta(x) = r$. The closed ball $\mathcal{B}(a, r)_\zeta$ is the set of all $z \in \mathbb{P}_{\text{Berk}}^1$ such that the path $[z, \zeta]$ from z to ζ contains x , and the open ball $\mathcal{B}(a, r)_\zeta^-$ is the connected component of $\mathcal{B}(a, r)_\zeta \setminus \{x\}$ containing a .

Finally, we note that the generalized Hsia kernel can be used to decompose absolute values of rational functions on $\mathbb{P}_{\text{Berk}}^1$, just as the canonical distance does on $\mathbb{P}^1(K)$:

COROLLARY 3.12. *Let $0 \neq \varphi \in K(T)$ have divisor $\text{div}(\varphi) = \sum_{i=1}^m n_i(a_i)$. For each $\zeta \in \mathbb{P}_{\text{Berk}}^1$ disjoint from the support of $\text{div}(\varphi)$, there is a constant $C = C(\varphi, \zeta)$ such that for all $x \in \mathbb{P}_{\text{Berk}}^1$,*

$$(3.24) \quad [\varphi]_x = C \cdot \prod_{i=1}^m \delta(x, a_i)_\zeta^{n_i}.$$

PROOF. Similar to the proof of Corollary 3.4. □

REMARK 3.13. After defining the Laplacian operator Δ on $\mathbb{P}_{\text{Berk}}^1$, we will see that if we fix $y, \zeta \in \mathbb{P}_{\text{Berk}}^1$ and think of $\delta(x, y)_\zeta$ as a function of x , then

$$\Delta(-\log_v(\delta(x, y)_\zeta)) = \delta_\zeta(x) - \delta_y(x).$$

Using this, one can give an alternate proof of Corollary 3.12 by observing that the logarithms of both sides of (3.24) have the same Laplacian.

5. Notes and further references

CHAPTER 4

Capacity theory

Fix $\zeta \in \mathbb{P}_{\text{Berk}}^1$, and let $E \subset \mathbb{P}_{\text{Berk}}^1$ be a set not containing ζ . In this chapter, we will develop the theory of the *logarithmic capacity* of E with respect to ζ (or more correctly, with respect to a choice of the Hsia kernel $\delta(x, y)_\zeta$). The exposition below is adapted from ([62], Chapter 4.1).

1. Logarithmic capacities

Recall that a probability measure is a non-negative Borel measure of total mass 1. Given a probability measure ν with support contained in E , define the energy integral

$$I_\zeta(\nu) = \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) .$$

Here, the integral is a Lebesgue integral. The kernel $-\log_v \delta(x, y)_\zeta$ is lower semicontinuous, and hence Borel measurable, since $\delta(x, y)_\zeta$ is upper semicontinuous.

Let ν vary over probability measures with support contained in E , and define the *Robin constant*

$$V_\zeta(E) = \inf_{\nu} I_\zeta(\nu) .$$

Define the *logarithmic capacity*

$$\gamma_\zeta(E) = q_v^{-V_\zeta(E)} .$$

By its definition, the logarithmic capacity is monotonic in E : if $E_1 \subset E_2$, then $\gamma_\zeta(E_1) \leq \gamma_\zeta(E_2)$. It also follows from the definition that for each E

$$(4.1) \quad \gamma_\zeta(E) = \sup_{\substack{E' \subset E \\ E' \text{ compact}}} \gamma_\zeta(E')$$

Indeed, the support of each probability measure ν is compact, so tautologically for each ν supported on E there is a compact set $E' \subset E$ with $I_\zeta(\nu) \leq \gamma_\zeta(E') \leq \gamma_\zeta(E)$.

An important distinction is between sets of capacity 0 and sets of positive capacity. If E contains a point a of types II, III, or IV, then

$\gamma_\zeta(E) > 0$, since the point mass $\delta_a(x)$ satisfies $I_\zeta(\delta_a) = -\log(\text{diam}_\zeta(a)) < \infty$. Thus, every set of capacity 0 is contained in $\mathbb{P}^1(K)$. The converse is not true; there are many sets in $\mathbb{P}^1(K)$ with positive capacity.

The property that a set has capacity 0 or positive capacity is independent of the point ζ .

PROPOSITION 4.1. *Suppose $E \subset \mathbb{P}_{\text{Berk}}^1$. Then $\gamma_\zeta(E) = 0$ for some $\zeta \notin E$ if and only if $\gamma_\xi(E) = 0$ for each $\xi \in \mathbb{P}_{\text{Berk}}^1 \setminus E$.*

PROOF. By (4.1) it suffices to consider the case where E is compact. Take $\xi \notin E$. Since $\|x, \xi\|$ is continuous on E , there is a constant $C_\xi > 0$ such that

$$1/C_\xi \leq \|x, \xi\| \leq C_\xi$$

for all $x \in E$. Since

$$\delta(x, y)_\xi = \frac{\|x, y\|}{\|x, \xi\| \|y, \xi\|}, \quad \delta(x, y)_\zeta = \frac{\|x, y\|}{\|x, \zeta\| \|y, \zeta\|}.$$

it follows that

$$\frac{1}{(C_\zeta C_\xi)^2} \delta(x, y)_\zeta \leq \delta(x, y)_\xi \leq (C_\zeta C_\xi)^2 \delta(x, y)_\zeta.$$

Hence for each probability measure ν on E , either $I_\zeta(\nu) = I_\xi(\nu) = \infty$, or $I_\zeta(\nu)$ and $I_\xi(\nu)$ are both finite. $\square$

Here are some examples of capacities.

EXAMPLE 4.2. If $E \subset \mathbb{P}^1(K)$ is a countable set, and $\zeta \notin E$, then $\gamma_\zeta(E) = 0$. To see this, note first that if ν is a probability measure supported on E , then necessarily ν has point masses (if $\nu(\{x\}) = 0$ for each $x \in E$, then by countable additivity $\nu(E) = 0$, which contradicts $\nu(E) = 1$). If $p \in E$ is a point with $\nu(\{p\}) > 0$, then

$$\begin{aligned} I_\zeta(\nu) &= \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) \\ &\geq -\log_v \delta(p, p)_\zeta \cdot \nu(\{p\})^2 = \infty \end{aligned}$$

so $V_\zeta(E) = \infty$.

EXAMPLE 4.3. If $E = \{a\}$ is a point not of type I, and $\zeta \neq a$, then

$$\gamma_\zeta(E) = \text{diam}_\zeta(a).$$

The only probability measure supported on E is the point mass $\nu = \delta_a(x)$, for which

$$\begin{aligned} I_\zeta(\nu) &= \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) \\ &= -\log_v \delta(a, a)_\zeta = -\log_v \text{diam}_\zeta(a). \end{aligned}$$

Hence $V_\zeta(E) = -\log_v \text{diam}_\zeta(a)$ and $\gamma_\zeta(a) = \text{diam}_\zeta(a)$.

EXAMPLE 4.4. If $\zeta = \infty$ and $E = \mathbb{Z}_p \subset \mathbb{A}^1(\mathbb{C}_p)$, then $\gamma_\infty(E) = p^{-1/(p-1)}$. More generally, if $E = \mathcal{O}_v$ is the ring of integers of a finite extension K_v/Q_p with ramification index e and residue degree f , then $\gamma_\infty(E) = (p^f)^{-1/e(p^f-1)}$.

For $x, y \in \mathbb{A}^1(\mathbb{C}_p)$, $\delta(x, y)_\infty = |x - y|$, so the capacity is given by the same computation as in the classical case; see ([62], Example 5.2.13, p. 347).

2. The equilibrium distribution

If E is compact, and if $\gamma_\zeta(E) > 0$, there is a probability measure $\mu = \mu_\zeta$ on E for which $I_\zeta(\mu) = V_\zeta(E)$. In Proposition 5.40, we will show it is unique; it will be called the *equilibrium distribution*, or the *equilibrium measure*, of E with respect to ζ . To prove its existence, we need some preliminary lemmas.

LEMMA 4.5. *If U is a connected open subset of $\mathbb{P}_{\text{Berk}}^1$ and $\bar{U}$ is its closure, then every finite Borel measure on $\mathbb{P}_{\text{Berk}}^1$ is Radon.*

PROOF. This follows easily from Lemma 1.19 and the definition of the topology on $\mathbb{P}_{\text{Berk}}^1$. $\square$

LEMMA 4.6. *Let E be a compact Hausdorff space, and suppose $\langle \nu_\alpha \rangle$ is a net of probability measures on E which converge weakly to a measure μ . Then $\langle \nu_\alpha \times \nu_\alpha \rangle$ converges weakly to $\mu \times \mu$ on $E \times E$.*

PROOF. We fill in the details of the argument given in ([38], p.283). Since E is a compact Hausdorff space, the Stone-Weierstrass theorem asserts that linear combinations of functions of the form $f(x)g(y)$, with $f, g \in \mathcal{C}(E)$, are dense in $\mathcal{C}(E \times E)$ under the sup norm $\|\cdot\|_{E \times E}$. For such products,

$$\begin{aligned} \lim_{n \rightarrow \infty} \iint_{E \times E} f(x)g(y) d\nu_\alpha(x) d\nu_\alpha(y) &= \int_E f(x) d\mu(x) \cdot \int_E g(y) d\mu(y) \\ &= \iint_{E \times E} f(x)g(y) d\mu(x) d\mu(y) \end{aligned}$$

so the same holds for their linear combinations.

Now let $F(x, y) \in \mathcal{C}(E \times E)$ be arbitrary. Given $\varepsilon > 0$, take $f(x, y) = \sum_i c_i f_i(x) g_i(y)$ so that $h(x, y) = F(x, y) - f(x, y)$ satisfies $\|h\|_{E \times E} < \varepsilon$. For any probability measure ν on E , clearly

$$\left| \iint_{E \times E} h(x, y) d\nu(x) d\nu(y) \right| \leq \|h\|_{E \times E}.$$

Let α_0 be large enough that

$$\left| \iint_{E \times E} f(x, y) d\nu_\alpha(x) d\nu_\alpha(y) - \iint_{E \times E} f(x, y) d\mu(x) d\mu(y) \right| < \varepsilon$$

for $\alpha \geq \alpha_0$. By a three-epsilons argument, for such α we have

$$\left| \iint_{E \times E} F(x, y) d\nu_\alpha(x) d\nu_\alpha(y) - \iint_{E \times E} F(x, y) d\mu(x) d\mu(y) \right| \leq 3\varepsilon.$$

Since ε was arbitrary, it follows that

$$\lim_{\alpha} \iint_{E \times E} F(x, y) d\nu_\alpha(x) d\nu_\alpha(y) = \iint_{E \times E} F(x, y) d\mu(x) d\mu(y).$$

It follows that $\langle \nu_\alpha \times \nu_\alpha \rangle$ converges weakly to $\mu \times \mu$. $\square$

We can now show the existence of an equilibrium measure.

PROPOSITION 4.7. *Let $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ be a compact set with positive capacity. Then there is a probability measure μ supported on E such that $I_\zeta(\mu) = V_\zeta(E)$.*

PROOF. Take a sequence of probability measures ν_n on E for which $\lim_{n \rightarrow \infty} I_\zeta(\nu_n) = V_\zeta(E)$. After passing to a subsequence, if necessary, we can assume that $\{\nu_n\}$ converges weakly to a measure μ . Clearly μ is a probability measure supported on E .

Since E is a compact subset of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, it follows easily that $-\log_v \delta(x, y)_\zeta$ is bounded below on $E \times E$ by some constant $-M$. By Proposition A.3 applied to $-\log_v \delta(x, y)_\zeta$ on $E \times E$, we see that

$$\iint -\log_v \delta(x, y)_\zeta d\mu(x) d\mu(y) = \sup_{\substack{g \in \mathcal{C}(E \times E) \\ -M \leq g \leq -\log_v \delta(x, y)_\zeta}} \iint_{E \times E} g(x, y) d\mu(x) d\mu(y)$$

Furthermore, for each $g \in \mathcal{C}(E \times E)$, it follows from Lemma 4.6 that

$$\iint_{E \times E} g(x, y) d\mu(x) d\mu(y) = \lim_{n \rightarrow \infty} \iint_{E \times E} g(x, y) d\nu_n(x) d\nu_n(y).$$

If, moreover, we have $g(x, y) \leq -\log_v \delta(x, y)_\zeta$, then for each n

$$\begin{aligned} \iint_{E \times E} g(x, y) d\nu_n(x) d\nu_n(y) &\leq \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\nu_n(x) d\nu_n(y) \\ &= I_\zeta(\nu_n), \end{aligned}$$

and thus for each such g , we have

$$\iint_{E \times E} g(x, y) d\mu(x) d\mu(y) \leq \lim_{n \rightarrow \infty} I_\zeta(\nu_n) = V_\zeta(E).$$

Therefore

$$\begin{aligned}
I_\zeta(\mu) &= \iint_{E \times E} -\log_v \delta(x, y)_\zeta d\mu(x) d\mu(y) \\
&= \sup_{\substack{g \in \mathcal{C}(E \times E) \\ -M \leq g \leq -\log_v \delta(x, y)_\zeta}} \iint_{E \times E} g(x, y) d\mu(x) d\mu(y) \\
&\leq V_\zeta(E) .
\end{aligned}$$

The opposite inequality is trivial, so $I_\zeta(\mu) = V_\zeta(E)$. $\square$

REMARK 4.8. In the classical proof over $\mathbb{C}$, one considers the truncated logarithm $-\log^{(t)}(|x - y|) = \min(t, -\log |x - y|)$ which is continuous, and which approaches $-\log(|x - y|)$ pointwise as $t \rightarrow \infty$. Here $-\log_v^{(t)}(\delta(x, y)_\zeta)$ is lower semicontinuous, but not continuous, so it was necessary to introduce the approximating functions $g(x, y)$.

Let U_ζ be the connected component of $\mathbb{P}_{\text{Berk}}^1 \setminus E$ containing ζ . Write $\partial E_\zeta = \partial U_\zeta$ for the boundary of U_ζ , the part of ∂E in common with $\bar{U}$.

PROPOSITION 4.9. *Let $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ be compact with positive capacity. Then the equilibrium distribution μ_ζ is supported on ∂E_ζ .*

PROOF. Suppose μ_ζ is not supported on ∂E_ζ , and fix $x_0 \in \text{supp}(\mu_\zeta) \setminus \partial E_\zeta$. Let $r_\zeta : E \rightarrow \partial E_\zeta$ be the retraction map which takes each $x \in E$ to the last point in E on the path from x to ζ . Then r_ζ is continuous. Put $\mu_0 = (r_\zeta)_*(\mu_\zeta)$. We claim that $I_\zeta(\mu_0) < I_\zeta(\mu_\zeta)$.

For each $x, y \in E$, if $\bar{x} = r_\zeta(x)$, $\bar{y} = r_\zeta(y)$, then $\delta(x, y)_\zeta \leq \delta(\bar{x}, \bar{y})_\zeta$. This follows from the geometric interpretation of $\delta(x, y)_\zeta$: if w is the point where the paths from x to ζ and y to ζ meet, and $\bar{w}$ is the point where the paths from $\bar{x}$ to ζ and $\bar{y}$ to ζ meet, then $\bar{w}$ lies on the path from w to ζ .

Now consider the point x_0 , and put $x_1 = r_\zeta(x_0)$. Then $\text{diam}_\zeta(x_0) < \text{diam}_\zeta(x_1)$. Fix r with $\text{diam}_\zeta(x_0) < r < \text{diam}_\zeta(x_1)$, and put $U = \mathcal{B}(x_0, r)_\zeta^-$. For each $x \in U$, $r_\zeta(x) = x_1$. If $x, y \in U$,

$$\delta(x, y)_\zeta < r < \text{diam}_\zeta(x_1) = \delta(x_1, x_1)_\zeta$$

Since $x_0 \in \text{supp}(\mu)$, necessarily $\mu_\zeta(U) > 0$. Hence

$$\begin{aligned}
I_\zeta(\mu_\zeta) &= \int -\log_v \delta(x, y)_\zeta d\mu_\zeta(x) d\mu_\zeta(y) \\
&> \int -\log_v \delta(r_\zeta(x), r_\zeta(y))_\zeta d\mu_\zeta(x) d\mu_\zeta(y) \\
&= \int -\log_v \delta(z, w)_\zeta d\mu_0(z) d\mu_0(w) = I_\zeta(\mu_0) .
\end{aligned}$$

This contradicts the minimality of $I_\zeta(\mu_\zeta)$, so μ_ζ is supported on ∂E_ζ . $\square$

The existence of the equilibrium measure has the following important consequence.

COROLLARY 4.10. *Let $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ be compact. Then for any $\varepsilon > 0$, there is a compact neighborhood W containing E such that $\gamma_\zeta(W) \leq \gamma_\zeta(E) + \varepsilon$.*

In fact, there is such a neighborhood which is a finite union of discs $\bigcup_{k=1}^m \mathcal{B}(a_k, r_k)_\zeta$, where each a_k is of type II or type III, and where $r_k = \text{diam}_\zeta(a_k)$.

PROOF. Let $\{W_\alpha\}$ be any family of (not necessarily connected) compact neighborhoods containing E whose intersection is precisely E . (Such a family necessarily exists: for example, for each point $z \notin E$, there is a covering of E by open affinoids whose closure does not contain z , and by compactness this open cover has a finite subcover. For each such z , take W_z to be the corresponding finite union of closed affinoids.) We may consider this family as a net, since the sets W_α are partially ordered by containment. Without loss of generality, we may assume that each W_α is supported on a compact set $E' \supseteq E$ not containing ζ . As each W_α has non-empty intersection with $\mathbb{H}_{\text{Berk}}$, we have $\gamma_\zeta(W_\alpha) > 0$ for all α . Therefore each W_α has an equilibrium measure μ_α for which $I_\zeta(\mu_\alpha) = V_\zeta(\mu_\alpha) < \infty$. (The measure μ_α is in fact unique, but we don't need this.)

Passing to a subnet, we may assume without loss of generality that the net $\langle \mu_\alpha \rangle$ converges weakly to some probability measure μ^* . As $\bigcap W_\alpha = E$, it follows easily that μ^* is supported on E . We claim that

$$(4.2) \quad \lim_{\alpha} V_\zeta(W_\alpha) \geq I_\zeta(\mu^*) .$$

Given this claim, we conclude as follows. Since for all α we have

$$V_\zeta(W_\alpha) \leq V_\zeta(E) \leq I_\zeta(\mu^*) ,$$

we must have

$$\lim_{\alpha} V_\zeta(W_\alpha) = I_\zeta(\mu^*) = V_\zeta(E) .$$

This implies that

$$\lim_{\alpha} \gamma_\zeta(W_\alpha) = \gamma_\zeta(E) ,$$

so that there exists a $W = W_\alpha$ with $\gamma_\zeta(W) < \gamma_\zeta(E) + \varepsilon$ as required.

It remains to justify (4.2). We do this in several steps. First of all, since E' is a compact subset of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, we know that $-\log_v \delta(x, y)_\zeta$

is bounded below on $E' \times E'$ by some constant $-M$. By Proposition A.3, for any probability measure μ we have

$$I_\zeta(\mu) = \sup_{\substack{g \in \mathcal{C}(E' \times E') \\ -M \leq g \leq -\log_v \delta(x, y)_\zeta}} \iint_{E' \times E'} g(x, y) d\mu(x) d\mu(y) .$$

By Lemma 4.6, for each $g \in \mathcal{C}(E' \times E')$ we have

$$\iint_{E' \times E'} g(x, y) d\mu^*(x) d\mu^*(y) = \lim_\alpha \iint_{E' \times E'} g(x, y) d\mu_\alpha(x) d\mu_\alpha(y) .$$

When in addition $g(x, y) \leq -\log_v \delta(x, y)_\zeta$ on $E' \times E'$, for each α we have

$$\iint_{E' \times E'} g(x, y) d\mu_\alpha(x) d\mu_\alpha(y) \leq I_\zeta(\mu_\alpha) .$$

For each such g , we therefore have

$$\begin{aligned} \lim_\alpha V_\zeta(W_\alpha) &= \lim_\alpha I_\zeta(\mu_\alpha) \\ &\geq \lim_\alpha \iint_{E' \times E'} g(x, y) d\mu_\alpha(x) d\mu_\alpha(y) \\ &= \iint_{E \times E} g(x, y) d\mu^*(x) d\mu^*(y) . \end{aligned}$$

Taking the supremum over all such g , we obtain

$$\lim_\alpha V_\zeta(W_\alpha) \geq \sup_g \iint_{E \times E} g(x, y) d\mu^*(x) d\mu^*(y) = I_\zeta(\mu^*)$$

as claimed.

To show that the neighborhood can be taken in the special form described, fix any closed neighborhood W of E with $\gamma_\zeta(W) \leq \gamma_\zeta(E) + \varepsilon$. Let $r_\zeta : E \rightarrow \partial W_\zeta$ be the retraction map which takes each x in E to the last point on the path from x to ζ which belongs to W .

Since E is contained in the interior of W , none of the points $a \in r_\zeta(E)$ belongs to E , and each lies on the interior of the path from some point x in E to ζ . Such a point a is necessarily of type II or III. If $a = r_\zeta(x)$ and $r = \text{diam}_\zeta(a)$, then $\mathcal{B}(a, r)_\zeta = \mathcal{B}(x, r)_\zeta$, and $x \in \mathcal{B}(x, r)_\zeta^-$. Since E is compact, a finite number of the discs $\mathcal{B}(x, r)_\zeta^-$ cover E . It follows that $r_\zeta(E)$ is a finite set of points $\{a_1, \dots, a_m\}$, each of which is of type II or III. For each a_k , put $r_k = \text{diam}_\zeta(a_k)$, and put $\tilde{W} = \cup_{i=1}^m \mathcal{B}(a_k, r_k)_\zeta$. Clearly $\tilde{W}$ is compact, and has E in its interior. Since $\tilde{W}$ contains points of type II or III, $\gamma_\zeta(\tilde{W}) > 0$.

We claim that $\gamma_\zeta(\tilde{W}) \leq \gamma_\zeta(E) + \varepsilon$. To see this, let $\tilde{\mu}$ be any equilibrium distribution for $\tilde{W}$, and put $\tilde{\mu} = r_\zeta(\tilde{\mu}_0)$. By Proposition

4.9, $\tilde{\mu}_0$ is another equilibrium distribution for $\tilde{W}$. It is supported on $\{a_1, \dots, a_m\} \subset W$, so $V_\zeta(\tilde{W}) = I_\zeta(\tilde{\mu}) \geq V_\zeta(W)$. This is equivalent to $\gamma_\zeta(\tilde{W}) \leq \gamma_\zeta(W)$. Since $\gamma_\zeta(W) \leq \gamma_\zeta(E) + \varepsilon$, our claim follows. $\square$

Assuming the uniqueness of the equilibrium measure of a compact set E of positive capacity (which will be proved in Proposition 5.40), the proof of Corollary 4.10 yields:

COROLLARY 4.11. *Let $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ be a compact set with positive capacity, and let $\langle W_\alpha \rangle$ be a net of compact neighborhoods of E (partially ordered by inclusion). Let μ_α be the equilibrium measure on W_α , and let μ_E be the equilibrium measure on E . Then $\langle \mu_\alpha \rangle$ converges weakly to μ_E .*

PROOF. The proof of Corollary 4.10 shows that if μ^* is any weak limit of a convergent subnet of $\langle \mu_\alpha \rangle$, then μ^* is supported on E and

$$I_\zeta(\mu^*) = V_\zeta(E) .$$

By uniqueness, we must have $\mu^* = \mu_E$, i.e., every convergent subnet of $\langle \mu_\alpha \rangle$ converges weakly to μ_E . This implies that $\langle \mu_\alpha \rangle$ itself converges weakly to μ_E . $\square$

In what follows we will speak of ‘the’ equilibrium measure μ_ζ of a compact set E of positive capacity, anticipating the uniqueness of the equilibrium measure. However, the arguments below would apply to any equilibrium measure.

3. Potential functions

For each probability measure ν supported on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, define the potential function

$$u_\nu(z, \zeta) = \int -\log_v \delta(z, w)_\zeta d\nu(w) .$$

PROPOSITION 4.12. *Let ν be a probability measure on $\mathbb{P}_{\text{Berk}}^1$, and take $\zeta \notin \text{supp}(\nu)$. Then $u_\nu(z, \zeta)$ is lower semi-continuous, and is continuous at each $z \notin \text{supp}(\nu)$ (including ζ , if continuity is understood relative the extended reals, $\mathbb{R} \cup \{-\infty\}$). Moreover, for each $p \in \mathbb{P}^1(K)$, if x approaches p along a path $[y, p]$, then*

$$\lim_{\substack{x \rightarrow p \\ x \in [y, p]}} u_\nu(x, \zeta) = u_\nu(p, \zeta) .$$

PROOF. We will first show $u_\nu(z, \zeta)$ is lower semi-continuous. Lower semi-continuity is automatic at $z = \zeta$, since for all z, w

$$-\log_v \delta(z, w)_\zeta \geq -\log_v \delta(\zeta, w)_\zeta .$$

Let E' be any compact neighborhood of $\text{supp}(\nu)$ which does not contain ζ . By Proposition A.3, we have

$$u_\nu(z, \zeta) = \sup \left\{ \int_{E'} g(z, w) d\nu(w) : g \in \mathcal{C}(E' \times E'), 0 \leq g \leq -\log_v \delta(z, w)_\zeta \right\} .$$

Since E' is compact, each function $\int g(z, w) d\nu(w)$ is continuous on E' . Thus $u_\nu(z, \zeta)$ is lower semi-continuous on E' by Lemma A.2. For any $z \neq \zeta$, we can choose E' so as to contain z in its interior. Thus $u_\nu(z, \zeta)$ is lower semi-continuous everywhere.

Put $E = \text{supp}(\nu)$, and consider the decomposition

$$\delta(z, w)_\zeta = \frac{\|z, w\|}{\|z, \zeta\| \|w, \zeta\|} .$$

Inserting this in the definition of $u_\nu(z, \zeta)$ we see that

$$\begin{aligned} u_\nu(z, \zeta) &= \int_E -\log_v \|z, w\| d\nu(w) + \int_E \log_v \|z, \zeta\| d\nu(w) \\ &\quad + \int_E \log_v \|w, \zeta\| d\nu(w) . \end{aligned}$$

The second integral is $\log_v \|z, \zeta\|$ since the integrand does not involve w . The third integral is a finite constant C_ζ , because $\zeta \notin E$, so $\log_v \|w, \zeta\|$ is a bounded, continuous function of $w \in E$. Thus

$$u_\nu(z, \zeta) = \int_E -\log_v \|z, w\| d\nu(w) + \log_v \|z, \zeta\| + C_\zeta .$$

For each $z \notin E$, the first integral is continuous at z since $\|x, y\|$ is continuous off the diagonal and z has a closed neighborhood disjoint from E . The function $\log_v \|z, \zeta\|$ is continuous as a function to the extended reals, since $\|x, y\|$ is continuous as a function of each variable separately. Hence $u_\nu(z, \zeta)$ is continuous off $\text{supp}(\nu)$.

Finally, suppose $p \in \mathbb{P}^1(K)$. Let $y \in \mathbb{P}_{\text{Berk}}^1$ be arbitrary, and let x approach p along the path $[y, p]$. Then $\|x, w\|$ decreases monotonically to $\|p, w\|$, and thus $-\log_v \|x, w\|$ increases monotonically to $-\log_v \|p, w\|$. By the Monotone Convergence Theorem, $\int_E -\log_v \|x, w\| d\nu(w)$ converges to $\int_E -\log_v \|p, w\| d\nu(w)$. It follows that

$$\lim_{\substack{x \rightarrow p \\ x \in [y, p)}} u_\nu(x, \zeta) = u_\nu(p, \zeta)$$

as desired. □

In the classical theory, the two main facts about potential functions are Maria's theorem and Frostman's theorem. We will now establish their analogues for the Berkovich line.

THEOREM 4.13. (Maria) *Let ν be a probability measure supported on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. If there is a constant $M < \infty$ such that $u_\nu(z, \zeta) \leq M$ on $\text{supp}(\nu)$, then $u_\nu(z, \zeta) \leq M$ for all $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.*

PROOF. Put $E = \text{supp}(\nu)$ and fix $z \in \mathbb{P}_{\text{Berk}}^1 \setminus (E \cup \{\zeta\})$. Since $\delta(z, w)_\zeta$ is continuous off the diagonal, there is a point $\bar{z} \in E$ such that $\delta(z, \bar{z})_\zeta \leq \delta(z, w)_\zeta$ for all $w \in E$. By the ultrametric inequality, for each $w \in E$

$$\delta(\bar{z}, w)_\zeta \leq \max(\delta(z, \bar{z})_\zeta, \delta(z, w)_\zeta) = \delta(z, w)_\zeta .$$

Hence

$$\begin{aligned} u_\nu(z, \zeta) &= \int -\log_v \delta(z, w)_\zeta d\nu(w) \\ &\leq \int -\log_v \delta(\bar{z}, w)_\zeta d\nu(w) \\ &= u_\nu(\bar{z}, \zeta) \leq M . \end{aligned}$$

□

The following lemma asserts that sets of capacity 0 are ‘small’ in a measure-theoretic sense.

LEMMA 4.14. *If $f \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ is a set of capacity 0, then $\nu(f) = 0$ for any probability measure ν supported on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ with $I_\zeta(\nu) < \infty$.*

PROOF. Recall that $\text{supp}(\nu)$ is compact. After scaling $\delta(x, y)_\zeta$ if necessary, we can assume that $\delta(x, y)_\zeta \leq 1$ on $\text{supp}(\nu)$. If $\nu(f) > 0$, then $\nu(e) > 0$ for some compact subset e of f , and $\eta := (1/\nu(e)) \cdot \nu$ is probability measure η on e . It follows that

$$\begin{aligned} I_\zeta(\eta) &= \iint_{e \times e} -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) \\ &\leq \frac{1}{\nu(e)^2} \iint -\log_v \delta(x, y)_\zeta d\nu(x) d\nu(y) < \infty . \end{aligned}$$

□

COROLLARY 4.15. *Let $\{f_n\}_{n \geq 1}$ be a countable collection of Borel sets in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ such that each f_n has capacity 0. Put $f = \cup_{n=1}^\infty f_n$. Then f has capacity 0.*

PROOF. If $\gamma_\zeta(f) > 0$, there is a probability measure ν supported on f such that $I_\zeta(\nu) < \infty$. Since each f_n is Borel measurable, so is f , and

$$\sum_{n=1}^{\infty} \nu(f_n) \geq \nu(f) = 1.$$

Hence $\nu(f_n) > 0$ for some n , contradicting Lemma 4.14. $\square$

Recall that an F_σ set is a countable union of compact sets.

THEOREM 4.16. (Frostman) *Let $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ be a compact set of positive capacity. Then the equilibrium potential $u_E(z, \zeta)$ satisfies*

A) $u_E(z, \zeta) \leq V_\zeta(E)$ for all $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

B) $u_E(z, \zeta) = V_\zeta(E)$ for all $z \in E$, except possibly on an F_σ set $f \subset E$ of capacity 0.

C) $u_E(z, \zeta)$ is continuous at each point z_0 where $u_E(z_0, \zeta) = V_\zeta(E)$.

PROOF. First, using a quadraticity argument, we will show that $u_E(z, \zeta) \geq V_\zeta(E)$ on E except on the (possibly empty) set f . Then, we will show that $u_E(z, \zeta) \leq V_\zeta(E)$ on the support $\text{supp}(\mu)$ of the equilibrium measure $\mu = \mu_\zeta$. Since $u_E(z, \zeta) = u_\mu(z, \zeta)$, it follows from Maria's theorem that $u_E(z, \zeta) \leq V_\zeta(E)$ for all z .

Put

$$\begin{aligned} f &= \{z \in E : u_E(z, \zeta) < V_\zeta(E)\}, \\ f_n &= \{z \in E : u_E(z, \zeta) \leq V_\zeta(E) - 1/n\}, \quad \text{for } n = 1, 2, 3, \dots \end{aligned}$$

Since $u_E(z, \zeta)$ is lower semicontinuous, each f_n is closed, hence compact, so f is an F_σ set. By Corollary 4.15, $\gamma_\zeta(f) = 0$ if and only if $\gamma_\zeta(f_n) = 0$ for each n .

Suppose $\gamma_\zeta(f_n) > 0$ for some n ; then there is a probability measure σ supported on f_n such that $I_\zeta(\sigma) < \infty$. On the other hand, since

$$V_\zeta(E) = \iint -\log_v \delta(z, w)_\zeta d\mu(w) d\mu(z) = \int u_E(z, \zeta) d\mu(z),$$

there is a point $q \in \text{supp}(\mu)$ with $u_E(q, \zeta) \geq V_\zeta(E)$. Since $u_E(z, \zeta)$ is lower semicontinuous, there is a neighborhood U of q on which $u_E(z, \zeta) > V_\zeta(E) - 1/(2n)$. After shrinking U if necessary, we can assume that its closure $\bar{U}$ is disjoint from f_n . Put $e_n = E \cap U$. By the definition of $\text{supp}(\mu)$, it follows that $M := \mu(U) = \mu(e_n) > 0$. Define a measure σ_1 of total mass 0 on E by

$$\sigma_1 = \begin{cases} M \cdot \sigma & \text{on } f_n, \\ -\mu & \text{on } e_n, \\ 0 & \text{elsewhere.} \end{cases}$$

We claim that $I_\zeta(\sigma_1)$ is finite. Indeed

$$\begin{aligned} I_\zeta(\sigma_1) &= M^2 \cdot \iint_{f_n \times f_n} -\log_v \delta(z, w)_\zeta d\sigma(z) d\sigma(w) \\ &\quad - 2M \cdot \iint_{f_n \times e_n} -\log_v \delta(z, w)_\zeta d\sigma(z) d\mu(w) \\ &\quad + \iint_{e_n \times e_n} -\log_v \delta(z, w)_\zeta d\mu(z) d\mu(w) . \end{aligned}$$

The first integral is finite by hypothesis. The second is finite because e_n and $\bar{U}$ are disjoint, so $-\log_v \delta(z, w)_\zeta$ is bounded on $e_n \times f_n$. The third is finite because $I_\zeta(\mu)$ is finite.

For each $0 \leq t \leq 1$, the measure $\mu_t := \mu + t\sigma_1$ is a probability measure on E . By an expansion like the one above,

$$\begin{aligned} I_\zeta(\mu_t) - I_\zeta(\mu) &= 2t \cdot \int_E u_E(z, \zeta) d\sigma_1(z) + t^2 \cdot I_\zeta(\sigma_1) \\ &\leq 2t \cdot ((V_\zeta(E) - 1/n) - (V_\zeta(E) - 1/(2n)) \cdot M) + t^2 \cdot I_\zeta(\sigma_1) \\ &= (-M/n) \cdot t + I_\zeta(\sigma_1) \cdot t^2 . \end{aligned}$$

For sufficiently small $t > 0$, the right side is negative. This contradicts the fact that μ minimizes $I_\zeta(\nu)$ for all probability measures ν supported on E . It follows that $\gamma_\zeta(f_n) = 0$, and hence that $\gamma_\zeta(f) = 0$.

The second part requires showing that $u_E(z, \zeta) \leq V_\zeta(E)$ for all $z \in \text{supp}(\mu)$. If $u_E(q, \zeta) > V_\zeta(E)$ for some $q \in \text{supp}(\mu)$, take $\varepsilon > 0$ such that $u_E(q, \zeta) > V_\zeta(E) + \varepsilon$. The lower semicontinuity of $u_E(z, \zeta)$ shows there is a neighborhood U of q on which $u_E(z, \zeta) > V_\zeta(E) + \varepsilon$. Put $e = U \cap E$. Since $q \in \text{supp}(\mu)$, $T := \mu(e) = \mu(U) > 0$. By Lemma 4.14, $\mu(f) = 0$. Therefore, since $u_E(z, \zeta) \geq V_\zeta(E)$ on $E \setminus f$,

$$\begin{aligned} V_\zeta(E) &= \int_e u_E(z, \zeta) d\mu(z) + \int_{E \setminus e} u_E(z, \zeta) d\mu(z) \\ &\geq T \cdot (V_\zeta(E) + \varepsilon) + (1 - T) \cdot V_\zeta(E) = V_\zeta(E) + T\varepsilon \end{aligned}$$

which is obviously false.

Thus $u_E(z, \zeta) \leq V_\zeta(E)$ on $\text{supp}(\mu)$, and Maria's theorem implies that $u_E(z, \zeta) \leq V_\zeta(E)$ for all z .

The final assertion, that $u_E(z, \zeta)$ is continuous at each point z_0 where $u_E(z_0, \zeta) = V_\zeta(E)$, is now trivial. By lower semicontinuity,

$$\liminf_{z \rightarrow z_0} u_E(z, \zeta) \geq u_E(z_0, \zeta) = V_\zeta(E) .$$

On the other hand, since $u_E(z, \zeta) \leq V_\zeta(E)$ for all z ,

$$\limsup_{z \rightarrow z_0} u_E(z, \zeta) \leq V_\zeta(E) = u_E(z_0, \zeta) .$$

□

COROLLARY 4.17. *For any compact set $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ of positive capacity, and any probability measure ν supported on E ,*

$$\inf_{z \in E} u_\nu(z, \zeta) \leq V_\zeta(E) \leq \sup_{z \in E} u_\nu(z, \zeta) .$$

PROOF. This follows immediately from the identity

$$\begin{aligned} \int_E u_\nu(z, \zeta) d\mu(z) &= \iint_{E \times E} -\log_v \delta(z, w)_\zeta d\nu(w) d\mu(z) \\ &= \int_E u_E(w, \zeta) d\nu(w) = V_\zeta(E) . \end{aligned}$$

Here the second equality follows from Tonelli's theorem, and the third from Frostman's theorem and the fact that if f is the exceptional set on which $u_E(z, \zeta) < V_\zeta(E)$, then $\mu(f) = 0$ by Lemma (4.14). □

We will now show that adjoining or removing a set of capacity 0 from a given set F does not change its capacity. This is a consequence of a quantitative bound which we prove first.

PROPOSITION 4.18. *Let $\{F_m\}_{m \geq 1}$ be a countable collection of sets contained in $\mathbb{P}_{\text{Berk}}^1 \setminus \zeta$, and put $F = \bigcup_{m=1}^\infty F_m$. Suppose there is an $R < \infty$ such that $\delta(x, y)_\zeta \leq R$ for all $x, y \in F$ (or equivalently, that there is a disc $\mathcal{B}(a, R)_\zeta$ containing F). Then*

$$(4.3) \quad \frac{1}{V_\zeta(F) + \log_v(R) + 1} \leq \sum_{m=1}^\infty \frac{1}{V_\zeta(F_m) + \log_v(R) + 1} .$$

PROOF. First suppose $R < 1$. In this case we will show that

$$(4.4) \quad \frac{1}{V_\zeta(F)} \leq \sum_{m=1}^\infty \frac{1}{V_\zeta(F_m)} .$$

Note that our hypothesis implies that $V_\zeta(F) \geq -\log(R) > 0$.

If $\gamma_\zeta(F) = 0$, then $\gamma_\zeta(F_m) = 0$ for each m , so $V_\zeta(F) = V_\zeta(F_m) = \infty$ and (4.4) is trivial. Hence we can assume that $\gamma_\zeta(F) > 0$. Let $E \subset F$ be a compact set with $\gamma_\zeta(E) > 0$, and let μ be its equilibrium distribution. For each m , put $E_m = E \cap F_m$. Then

$$(4.5) \quad \sum_{m=1}^\infty \mu(E) \geq \mu(E) = 1 .$$

We claim that for each m , $V_\zeta(E)/V_\zeta(E_m) \geq \mu(E_m)$. If $\mu(E_m) = 0$ there is nothing to prove, so suppose $\mu(E_m) > 0$. There are compact

sets $e_{m,1} \subset e_{m,2} \subset \cdots \subset E_m$ such that $\lim_{i \rightarrow \infty} \mu(e_{m,i}) = \mu(E_m)$ and $\lim_{i \rightarrow \infty} V_\zeta(e_{m,i}) = V_\zeta(E_m)$. Without loss we can assume $\mu(e_{m,i}) > 0$ for each i . Put $\nu_{m,i} = (1/\mu(e_{m,i})) \cdot \mu|_{e_{m,i}}$. Applying Corollary 4.17, we see that

$$(4.6) \quad \sup_{z \in e_{m,i}} u_{\nu_{m,i}}(z, \zeta) \geq V_\zeta(e_{m,i}) .$$

But also, since $-\log_v \delta(z, w)_\zeta \geq -\log_v(R) > 0$ on E , for each $z \in e_{m,i}$

$$(4.7) \quad \begin{aligned} \frac{1}{\mu(e_{m,i})} u_E(z, \zeta) &= \frac{1}{\mu(e_{m,i})} \int_E -\log_v \delta(z, w)_\zeta d\mu(w) \\ &\geq \int_{e_{m,i}} -\log_v \delta(z, w)_\zeta d\nu_{m,i}(w) = u_{\nu_{m,i}}(z, \zeta) \end{aligned}$$

Because $\mu(e_{m,i}) > 0$, Theorem 4.16 shows there exist points $z \in e_{m,i}$ where $u_E(z, \zeta) = V_\zeta(E)$. Combining (4.6) and (4.7) gives

$$V_\zeta(E)/\mu(e_{m,i}) \geq V_\zeta(e_{m,i}) .$$

Transposing terms and letting $i \rightarrow \infty$ shows $V_\zeta(E)/V_\zeta(E_m) \geq \mu(E_m)$. By (4.5)

$$\frac{1}{V_\zeta(E)} \leq \sum_{m=1}^{\infty} \frac{1}{V_\zeta(E_m)} .$$

But E can be chosen so that $V_\zeta(E)$ is arbitrarily close to $V_\zeta(F)$, and also such that as many of the $V_\zeta(E_m)$ as we wish are arbitrarily close to $V_\zeta(F_m)$. Taking a limit over such E , we obtain (4.4).

The general case follows by scaling the Hsia kernel. Replace $\delta(x, y)_\zeta$ by $\delta'(x, y)_\zeta = 1/(q_v R) \cdot \delta(x, y)_\zeta$, so $\delta'(x, y)_\zeta \leq 1/q_v$ for all $x, y \in F$. For each $E \subset F$, this changes $V_\zeta(E)$ to $V'_\zeta(E) = V_\zeta(E) + \log_v(R) + 1$. Applying (4.4) to $V'_\zeta(F)$ and the $V'_\zeta(F_m)$ gives (4.3). $\square$

COROLLARY 4.19. *Let $e \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ have capacity 0. Then for any $F \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$,*

$$\gamma_\zeta(F \cup e) = \gamma_\zeta(F \setminus e) = \gamma_\zeta(F) .$$

PROOF. By Proposition 4.18, if F and e are contained in a ball $\mathcal{B}(a, R)_\zeta$ with $R < \infty$, then

$$\begin{aligned} \frac{1}{V_\zeta(F \cup e) + \log(R) + 1} &\leq \frac{1}{V_\zeta(F) + \log(R) + 1} + \frac{1}{V_\zeta(e) + R + 1} \\ &= \frac{1}{V_\zeta(F) + \log(R) + 1} , \end{aligned}$$

giving $V_\zeta(F) \leq V_\zeta(F \cup e)$. The opposite inequality is trivial, so $V_\zeta(F \cup e) = V_\zeta(F)$. The general case follows from this, by replacing F and e

by $F \cap \mathcal{B}(a, R)_\zeta$ and $e \cap \mathcal{B}(a, R)_\zeta$, fixing a center a and letting $R \rightarrow \infty$. Since any compact subset of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ is contained in $\mathcal{B}(a, r)_\zeta$ for some R ,

$$\begin{aligned} \gamma_\zeta(F \cup e) &= \lim_{R \rightarrow \infty} \gamma_\zeta((F \cup e) \cap \mathcal{B}(a, R)_\zeta) \\ &= \lim_{R \rightarrow \infty} \gamma_\zeta(F \cap \mathcal{B}(a, R)_\zeta) = \gamma_\zeta(F) . \end{aligned}$$

For the equality $\gamma_\zeta(F \setminus e) = \gamma_\zeta(F)$, apply what has been just shown to e and $F \setminus e$, noting that $(F \setminus e) \cup e = F \cup e$:

$$\gamma_\zeta(F \setminus e) = \gamma_\zeta(F \cup e) = \gamma_\zeta(F) .$$

□

As in the classical theory, there are two other important capacity functions: the *transfinite diameter* and the *Chebyshev constant*. The existence of these quantities, and the fact that for compact sets they are equal to the logarithmic capacity, will be established below.

4. The transfinite diameter $d_\infty(E)_\zeta$

For each $n = 2, 3, 4, \dots$ put

$$d_n(E)_\zeta = \sup_{x_1, \dots, x_n \in E} \left(\prod_{i \neq j} \delta(x_i, x_j)_\zeta \right)^{1/(n(n-1))} .$$

Note that unlike the classical case, $d_n(E)_\zeta$ can be nonzero even if some of the x_i coincide; this will happen, for example, if $E = \{a\}$ for a point of type II, III, or IV. We claim that the sequence $\{d_n(E)_\zeta\}$ is monotone decreasing. Fix n . If $d_{n+1}(E)_\zeta = 0$ then certainly $d_n(E)_\zeta \geq d_{n+1}(E)_\zeta$. Otherwise, take ε with $0 < \varepsilon < d_{n+1}(E)_\zeta$. There are points $z_1, \dots, z_{n+1} \in E$ such that

$$\prod_{\substack{i,j=1 \\ i \neq j}}^{n+1} \delta(z_i, z_j)_\zeta \geq (d_{n+1}(E)_\zeta - \varepsilon)^{(n+1)n} .$$

By the definition of $d_n(E)_\zeta$, for each $k = 1, \dots, n+1$

$$(d_n(E)_\zeta)^{n(n-1)} \geq \prod_{\substack{i \neq j \\ i,j \neq k}} \delta(z_i, z_j)_\zeta .$$

Taking the product over all k , we see that

$$(d_n(E)_\zeta)^{(n+1)n(n-1)} \geq \left(\prod_{\substack{i,j=1 \\ i \neq j}}^{n+1} \delta(z_i, z_j)_\zeta \right)^{n-1} = (d_{n+1}(E)_\zeta - \varepsilon)^{(n+1)n(n-1)} .$$

Since $\varepsilon > 0$ is arbitrary, $d_n(E)_\zeta \geq d_{n+1}(E)_\zeta$. Put

$$d_\infty(E)_\zeta = \lim_{n \rightarrow \infty} d_n(E)_\zeta .$$

5. The Chebyshev Constant $CH(E)_\zeta$

We will define three variants of the Chebyshev constant. For each positive integer n and $a_1, \dots, a_n \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ (which need not be distinct), define the ‘pseudo-polynomial’

$$P_n(x; a_1, \dots, a_n) = \prod_{i=1}^n \delta(x, a_i)_\zeta .$$

Put $\|P_n(x; a_1, \dots, a_n)\|_E = \sup_{x \in E} P_n(x; a_1, \dots, a_n)$ and let

$$\begin{aligned} \text{CH}_n^*(E)_\zeta &= \inf_{a_1, \dots, a_n \in E} (\|P_n(x; a_1, \dots, a_n)\|_E)^{1/n} , \\ \text{CH}_n^a(E)_\zeta &= \inf_{a_1, \dots, a_n \in \mathbb{P}^1(K) \setminus \{\zeta\}} (\|P_n(x; a_1, \dots, a_n)\|_E)^{1/n} , \\ \text{CH}_n(E)_\zeta &= \inf_{a_1, \dots, a_n \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}} (\|P_n(x; a_1, \dots, a_n)\|_E)^{1/n} . \end{aligned}$$

We will show the three numbers

$$\begin{aligned} \text{CH}^*(E)_\zeta &= \lim_{n \rightarrow \infty} \text{CH}_n^*(E)_\zeta , \\ \text{CH}^a(E)_\zeta &= \lim_{n \rightarrow \infty} \text{CH}_n^a(E)_\zeta , \\ \text{CH}(E)_\zeta &= \lim_{n \rightarrow \infty} \text{CH}_n(E)_\zeta \end{aligned}$$

exist, and that $\text{CH}^*(E)_\zeta \geq \text{CH}^a(E)_\zeta = \text{CH}(E)_\zeta$. They will be called the restricted Chebyshev constant, the algebraic Chebyshev constant, and the unrestricted Chebyshev constant respectively.

The proofs that $\text{CH}^*(E)_\zeta$, $\text{CH}^a(E)_\zeta$ and $\text{CH}(E)_\zeta$ exist are similar; we give the argument only for $\text{CH}(E)_\zeta$. Put $\alpha = \inf_n \text{CH}_n(E)$. If $\alpha = \infty$, then $\text{CH}(E)_\zeta = \infty$ and there is nothing to prove. Otherwise, fix $\varepsilon > 0$. Then there are an N and $a_1, \dots, a_N \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ such that

$$\sup_{x \in E} P_N(x; a_1, \dots, a_N) \leq (\alpha + \varepsilon)^N .$$

Let $\{b_j\} = \{a_1, \dots, a_N, a_1, \dots, a_N, \dots\}$ be the sequence which cyclicly repeats $\{a_1, \dots, a_N\}$. Put $M_0 = 1$ and for each $r = 1, \dots, N-1$, put $M_r = \|P_r(x; a_1, \dots, a_r)\|_E$. Since M_N is finite, each M_r is finite as well. Put $M = \max_{0 \leq r < N} M_r / (\alpha + \varepsilon)^r$. For each n we can write $n = qN + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < N$. Then

$$\begin{aligned} \text{CH}_n(E)_\zeta &\leq (\|P_n(x; b_1, \dots, b_n)\|_E)^{1/N} \\ &\leq (M_N^q \cdot M_r)^{1/n} \leq M^{1/n} \cdot (\alpha + \varepsilon) . \end{aligned}$$

It follows that $\limsup_{n \rightarrow \infty} CH_n(E)_\zeta \leq \alpha + \varepsilon$. Since $\varepsilon > 0$ is arbitrary,

$$\limsup_{n \rightarrow \infty} CH_n(E)_\zeta \leq \alpha = \liminf_{n \rightarrow \infty} CH_n(E)_\zeta$$

and $CH(E)_\zeta = \lim_{n \rightarrow \infty} CH_n(E)_\zeta$ exists.

Clearly $CH(E)_\zeta^* \geq CH(E)_\zeta$ and $CH(E)_\zeta^a \geq CH(E)_\zeta$. We will now show that $CH^a(E)_\zeta = CH(E)_\zeta$. If $CH(E)_\zeta = \infty$ there is nothing to prove. Otherwise put $\alpha = CH(E)_\zeta$ and fix $\varepsilon > 0$. Then there are an N and $a_1, \dots, a_N \in \mathbb{P}_{\text{Berk}}^1$ such that $\|P_N(x; a_1, \dots, a_N)\|_E \leq (\alpha + \varepsilon)^N$. Take $\eta > 0$. We claim that for each i , there is an a'_i of type I such that $\delta(x, a'_i)_\zeta \leq (1 + \eta)^2 \delta(x, a_i)_\zeta$ for all $x \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. If a_i of type I, put $a'_i = a_i$. If a_i is not of type I, let $r_i = \text{diam}_\zeta(a_i)$; then there is an a'_i of type I in the ball $\mathcal{B}(a_i, r_i(1 + \eta))_\zeta$. If $x \notin \mathcal{B}(a_i, r_i(1 + \eta))_\zeta$ then $\delta(x, a'_i)_\zeta = \delta(x, a_i)_\zeta$ by the ultrametric inequality. If $x \in \mathcal{B}(a_i, r_i(1 + \eta))_\zeta$ then again by the ultrametric inequality

$$\delta(x, a'_i)_\zeta \leq (1 + \eta)r_i \leq (1 + \eta)\delta(x, a_i)_\zeta ,$$

and in either case the claim is true. It follows that

$$\begin{aligned} \|P_N(x; a'_1, \dots, a'_N)\|_E &\leq \|P_N(x; a_1, \dots, a_N)\|_E \cdot (1 + \eta)^N \\ &\leq (\alpha + \varepsilon)^N \cdot (1 + \eta)^N . \end{aligned}$$

Since η is arbitrary, $CH_N^a(E)_\zeta \leq \alpha + \varepsilon$, so $CH^a(E)_\zeta = CH(E)_\zeta$.

THEOREM 4.20. *For any compact set $E \subset \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$*

$$\gamma_\zeta(E) = d_\infty(E)_\zeta = CH(E)_\zeta = CH^*(E)_\zeta = CH^a(E)_\zeta .$$

PROOF. We have seen that $CH_\zeta^a(E) = CH(E)_\zeta \leq CH^*(E)_\zeta$. We now show that $\gamma_\zeta(E) \leq CH^a(E)_\zeta$, $CH^*(E)_\zeta \leq d_\infty(E)$, and $d_\infty(E)_\zeta \leq \gamma_\zeta(E)$.

I. $\gamma_\zeta(E) \leq CH^a(E)_\zeta$.

If $\gamma_\zeta(E) = 0$ there is nothing to prove. Suppose $\gamma_\zeta(E) > 0$. We will show that $\gamma_\zeta(E) \leq CH_n^a(E)_\zeta$ for each n . Fix n and take $\varepsilon > 0$; let $a_1, \dots, a_n \in \mathbb{P}^1(K) \setminus \{\zeta\}$ be points such that

$$\sup_{z \in E} \prod_{i=1}^n \delta(z, a_i)_\zeta \leq (CH_n^a(E)_\zeta + \varepsilon)^n .$$

Any finite set of Type I points has capacity 0. By Corollary 4.19, replacing E by $E \cup \{a_1, \dots, a_n\}$ does not change its capacity. Let ν be the probability measure on E with a point mass $1/n$ at each a_i . Then

for each $z \in E$,

$$-\log_v(\text{CH}_n^a(E)_\zeta + \varepsilon) \leq (1/n) \sum_{i=1}^n -\log_v \delta(z, a_i)_\zeta = u_\nu(z, \zeta) .$$

By Corollary 4.17 $\inf_{z \in E} u_\nu(z, \zeta) \leq V_\zeta(E)$. Thus $-\log_v(\text{CH}_n^a(E)_\zeta + \varepsilon) \leq V_\zeta(E)$, which gives $\gamma_\zeta(E) \leq \text{CH}_n^a(E)_\zeta + \varepsilon$.

II. $\text{CH}^*(E)_\zeta \leq d_\infty(E)_\zeta$.

We will show that $d_{n+1}(E)_\zeta \geq \text{CH}_n^*(E)_\zeta$ for each n . Since E is compact, the sup defining $d_{n+1}(E)_\zeta$ is achieved: there are points $x_1, \dots, x_{n+1} \in E$ such that

$$d_{n+1}(E)^{(n+1)n} = \prod_{i \neq j} \delta(x_i, x_j)_\zeta .$$

If $x_1, \dots, x_n$ are fixed, then by the definition of $d_{n+1}(E)_\zeta$,

$$H(z; x_1, \dots, x_n) := \prod_{1 \leq i < j \leq n} \delta(x_i, x_j)_\zeta \cdot \prod_{i=1}^n \delta(z, x_i)_\zeta$$

achieves its maximum for $z \in E$ at $z = x_{n+1}$. But by the definition of $\text{CH}_n(E)_\zeta$,

$$\max_{z \in E} \prod_{i=1}^n \delta(z, x_i)_\zeta \geq (\text{CH}_n^*(E)_\zeta)^n .$$

Thus,

$$d_{n+1}(E)_\zeta^{(n+1)n/2} \geq \prod_{1 \leq i < j \leq n} \delta(x_i, x_j)_\zeta \cdot (\text{CH}_n^*(E)_\zeta)^n .$$

A similar inequality holds if $\{x_1, \dots, x_n\}$ is replaced by $\{x_1, \dots, x_{n+1}\} \setminus \{x_k\}$ for each $k = 1, \dots, n+1$. Multiplying these inequalities together, we find

$$\begin{aligned} (d_{n+1}(E)_\zeta^{(n+1)n/2})^{n+1} &\geq (\text{CH}_n^*(E)_\zeta)^{(n+1)n} \cdot \prod_{k=1}^{n+1} \left(\prod_{\substack{1 \leq i < j \leq n+1 \\ i, j \neq k}} \delta(x_i, x_j)_\zeta \right) \\ &= (\text{CH}_n^*(E)_\zeta)^{(n+1)n} \cdot \left(\prod_{1 \leq i < j \leq n+1} \delta(x_i, x_j)_\zeta \right)^{n-1} \\ &= (\text{CH}_n^*(E)_\zeta)^{(n+1)n} \cdot (d_{n+1}(E)_\zeta^{(n+1)n/2})^{n-1} . \end{aligned}$$

Cancelling terms and taking roots gives $d_{n+1}(E)_\zeta \geq \text{CH}_n^*(E)_\zeta$.

III. $d_\infty(E)_\zeta \leq \gamma_\zeta(E)$.

We will show that $d_\infty(E)_\zeta \leq \gamma_\zeta(E) + \varepsilon$ for each $\varepsilon > 0$.

Fix ε . By Corollary 4.10 there is a closed neighborhood W of E of the form $W = \cup_{k=1}^m \mathcal{B}(a_k, r_k)_\zeta$, where each a_k is of type II or type III and $r_k = \text{diam}_\zeta(a_k)$, such that $\gamma_\zeta(W) \leq \gamma_\zeta(E) + \varepsilon$. Let $r_\zeta : E \rightarrow \partial W_\zeta$ be the retraction map which takes each x in E to the last point on the path from x to ζ which belongs to W ; thus $r_\zeta(E) \subset \{a_1, \dots, a_m\}$. Put $r = \min(r_k) > 0$.

Note that if $x, y \in E$ and $r_\zeta(x) = a_k$, $r_\zeta(y) = a_\ell$, then $\delta(x, y)_\zeta \leq \delta(a_k, a_\ell)_\zeta$. Indeed, if $k = \ell$ then $x, y \in B(a_k, r_k)_\zeta$ so $\delta(x, y)_\zeta \leq d_k = \delta(a_k, a_\ell)_\zeta$. If $k \neq \ell$ then $\delta(x, y)_\zeta = \delta(a_k, a_\ell)_\zeta$ since the paths from x and y to ζ encounter a_k, a_ℓ before they meet, so $\delta(x, y)_\zeta = \delta(a_k, a_\ell)_\zeta$.

Fix n . Since E is compact, there are points $x_1, \dots, x_n \in E$ such that

$$(d_n(E)_\zeta)^{n(n-1)} = \prod_{\substack{i,j=1 \\ i \neq j}}^n \delta(x_i, x_j)_\zeta$$

For each a_k , let m_k be the number of points x_i for which $r_\zeta(x_i) = a_k$, and let ν be the probability measure on W given by $\nu = \sum_{k=1}^m (m_k/n) \delta_{a_k}(z)$. Then $I_\zeta(\nu) \geq V_\zeta(W)$. It follows that

$$\begin{aligned} -(1 - \frac{1}{n}) \log_v d_n(E)_\zeta &= \frac{1}{n^2} \sum_{i \neq j} -\log_v \delta(x_i, x_j)_\zeta \\ &\geq \frac{1}{n^2} \sum_{i \neq j} -\log_v \delta(r_\zeta(x_i), r_\zeta(x_j))_\zeta \\ &= I_\zeta(\nu) + \frac{1}{n^2} \sum_{k=1}^m m_k \log_v(r_k) \\ &\geq V_\zeta(W) + (1/n) \log_v(r) . \end{aligned}$$

Letting $n \rightarrow \infty$, we see that $-\log_v d_\infty(E)_\zeta \geq V_\zeta(W)$. Equivalently, $d_\infty(E)_\zeta \leq \gamma_\zeta(W) \leq \gamma_\zeta(E) + \varepsilon$ as claimed. $\square$

As an application, we prove the following result comparing our capacities with those defined in [62]. Suppose $\zeta \in \mathbb{P}^1(K)$. For $F \subset \mathbb{P}^1(K) \setminus \{\zeta\}$, we let $\bar{\gamma}_\zeta^R(F)$ be the *outer capacity* of F , in the sense of [62], §4.3. Since the generalized Hsia kernel $\delta(x, y)_\zeta$ restricts to the canonical distance $[x, y]_\zeta$ on $\mathbb{P}^1(K) \times \mathbb{P}^1(K)$, it follows from the definitions in [62] that

$$\bar{\gamma}_\zeta^R(F) = \lim_{n \rightarrow \infty} \inf_{a_1, \dots, a_n \in \mathbb{P}^1(K) \setminus \{\zeta\}} (\|P_n(x; a_1, \dots, a_n)\|_F)^{1/n} ,$$

where as before

$$P_n(x; a_1, \dots, a_n) = \prod_{i=1}^n \delta(x, a_i)_\zeta .$$

COROLLARY 4.21. *If E is a compact subset of $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ such that $E \cap \mathbb{P}^1(K)$ is dense in E , then*

$$\gamma_\zeta(E) = \bar{\gamma}_\zeta^R(E \cap \mathbb{P}^1(K)) .$$

PROOF. Since $\gamma_\zeta(E) = \lim_{n \rightarrow \infty} \text{CH}_n^a(E)_\zeta$ by Theorem 4.20, it suffices to note that since $P_n(x; a_1, \dots, a_n)$ is continuous in x on $\mathbb{P}^1 \setminus \{\zeta\}$ and $E' = E \cap \mathbb{P}^1(K)$ is dense in E , we have

$$(\|P_n(x; a_1, \dots, a_n)\|_E)^{1/n} = (\|P_n(x; a_1, \dots, a_n)\|_{E'})^{1/n}$$

for all n and all $a_1, \dots, a_n \in \mathbb{P}^1(K) \setminus \{\zeta\}$. $\square$

REMARK 4.22. In particular, the hypothesis of Corollary 4.21 is satisfied when E is a *closed affinoid* in $\mathbb{P}_{\text{Berk}}^1$, so that $E \cap \mathbb{P}^1(K)$ is an “ RL_ζ -domain” in the sense of [62].

6. A Generalization of the Fekete-Szegő Theorem

In this section, we give, as an application of capacity theory on $\mathbb{P}_{\text{Berk}}^1$, a generalization of Rumely’s adelic version of the Fekete and Fekete-Szegő theorems (Theorems 6.3.1 and 6.3.2 of [62]). This section will not be used in the sequel. We assume some familiarity with basic concepts from algebraic number theory.

Let k be a global field (i.e., a finite extension of either $\mathbb{Q}$ or $\mathbb{F}_q(T)$ for some finite field $\mathbb{F}_q$). For each place v of k , let k_v be the completion of k at v , let $\bar{k}_v$ be an algebraic closure of k_v , and let $\mathbb{C}_v$ denote the completion of $\bar{k}_v$. It is well known that $\mathbb{C}_v$ is algebraically closed. We let $\mathbb{P}_{\text{Berk},v}^1$ denote the Berkovich projective line over $\mathbb{C}_v$ (which we take to mean $\mathbb{P}^1(\mathbb{C})$ when v is archimedean).

Fix a point $\zeta \in \mathbb{P}^1(k)$, and define an “adelic set” $\mathbb{E}$ (relative to ζ) to be a collection E_v of nonempty compact subsets of $\mathbb{P}_{\text{Berk},v}^1 \setminus \{\zeta\}$, one for each place v of k , such that for all v outside a finite set S containing all archimedean places, E_v is just the closed disc $\mathbb{P}_{\text{Berk},v}^1 \setminus \mathcal{B}(\zeta, 1)_\zeta^-$ consisting of all points whose reduction modulo v is disjoint from the reduction of ζ . An “adelic neighborhood” of $\mathbb{E}$ is a set $\mathbb{U}$ of the form

$$\prod_{v \in S} U_v \times \prod_{v \notin S} E_v$$

for some finite set of places S , where U_v is an open neighborhood of E_v in $\mathbb{P}_{\text{Berk},v}^1$ for each $v \in S$. The *logarithmic capacity* (relative to ζ) of $\mathbb{E}$,

denoted $\gamma_\zeta(\mathbb{E})$, is just

$$\gamma_\zeta(\mathbb{E}) = \prod_v \gamma_\zeta(E_v) ,$$

which is well-defined since $\gamma_\zeta(\mathbb{P}_{\text{Berk},v}^1 \setminus \mathcal{B}(\zeta, 1)_\zeta^-) = 1$. We say that $\mathbb{E}$ is *k-symmetric* if $E_v \cap \mathbb{P}^1(\bar{k}_v)$ is stable under all continuous automorphisms of $\bar{k}_v/k_v$ for each place v .

Our generalization of the Fekete and Fekete-Szegö theorems is as follows:

THEOREM 4.23. *Let $\mathbb{E}$ be a k -symmetric adelic set (relative to ζ).*

- (a) *If $\gamma_\zeta(\mathbb{E}) < 1$, then there is an adelic neighborhood $\mathbb{U}$ of $\mathbb{E}$ for which the set of points of $\mathbb{P}^1(\bar{k})$, all of whose conjugates are contained in $\mathbb{U}$, is finite.*
- (b) *If $\gamma_\zeta(\mathbb{E}) > 1$, then for every adelic neighborhood $\mathbb{U}$ of $\mathbb{E}$, the set of points of $\mathbb{P}^1(\bar{k})$, all of whose conjugates are contained in $\mathbb{U}$, is infinite.*

This theorem is proved in §6.3 of [62] with $\gamma(\mathbb{E}) = \prod_v \gamma_\zeta(E_v)$ replaced by $\prod_v \bar{\gamma}_\zeta^R(E_v \cap \mathbb{P}^1(\bar{k}_v))$, and assuming that $E_v \cap \mathbb{P}^1(\bar{k}_v)$ is *algebraically capacitable* for all v (we refer to [62] for the definition of algebraic capacitability). It is enough for our purposes to know that compact subsets of $\mathbb{P}^1(\mathbb{C}) \setminus \{\zeta\}$ and closed affinoids in $\mathbb{P}^1(\bar{k}_v) \setminus \{\zeta\}$ (which are called “RL $_\zeta$ -domains” in [62]) are algebraically capacitable. Our proof of Theorem 4.23 will use Corollary 4.10 and the following lemma to reduce to the already-proved result from [62].

LEMMA 4.24. *Suppose E is a compact subset of $\mathbb{P}_{\text{Berk}}^1$ and U is an open neighborhood of E . Then there exists a closed affinoid F with $E \subseteq F \subseteq U$. Furthermore, if $K = \mathbb{C}_v$ and $E \cap \mathbb{P}^1(\bar{k}_v)$ is k_v -symmetric, then F can be taken to be k_v -symmetric as well.*

PROOF. Let C be an open cover of F by open affinoids whose closures are contained in U . By compactness, C has a finite subcover; let F be the union of the closures of these finitely many open affinoids. Then F is a closed affinoid containing E and contained in U , as desired. For the last statement, note that each element V of C has finitely many k_v -conjugates, and by continuity one can assume for each such V that all conjugates of $\bar{V}$ lie in U . $\square$

PROOF OF THEOREM 4.23. By the remark following Corollary 4.21, together with the version of Theorem 4.23 proved in [62], the result is true when each non-archimedean E_v is a closed affinoid. Corollary 4.10 and Lemma 4.24 allow us to reduce to this case.

Indeed, for part (a), Corollary 4.10 allows us to replace each E_v for $v \in S$ by a slightly larger closed affinoid F_v for which $\gamma_\zeta(F_v)$ is arbitrarily close to $\gamma_\zeta(E_v)$. The corresponding $\gamma_\zeta(\mathbb{F})$ can thus be taken to be strictly less than 1, and every adelic neighborhood of $\mathbb{F}$ is also an adelic neighborhood of $\mathbb{E}$. Since it suffices to prove (a) with k replaced by a finite extension k' , and since F_v has only finitely many k_v -conjugates for each $v \in S$, one reduces easily to the case where $\mathbb{F}$ is k -symmetric. Part (a) follows in a straightforward way from these observations.

For part (b), given an adelic neighborhood $\mathbb{U}$ of $\mathbb{E}$, Lemma 4.24 allows us to replace each E_v for $v \in S$ by a slightly larger closed affinoid F_v which is still contained in $\mathbb{U}$, in such a way that the corresponding adelic set $\mathbb{F}$ is still k -symmetric. Part (b) follows easily from this. $\square$

REMARK 4.25. Compact subsets of $\mathbb{P}^1(\bar{k}_v)$ are algebraically capacitable. Note that a closed affinoid is not compact as a subset of $\mathbb{P}^1(\bar{k}_v)$, but both closed affinoids and compact subsets of $\mathbb{P}_{\text{Berk}}^1$ are intersections of compact subsets of $\mathbb{P}_{\text{Berk}}^1$ with $\mathbb{P}^1(\bar{k}_v)$. So Theorem 4.23 generalizes both the RL-domain and compact cases of the Fekete and Fekete-Szegő theorems from [62].

7. Notes and further references

In [62], capacity theory is developed relative to an arbitrary finite set $\mathcal{X}$ (instead of just relative to a point ζ). With some extra work, we could have developed capacity theory on $\mathbb{P}_{\text{Berk}}^1$ in the same generality. However, we chose to make the theory slightly less general for the sake of simplicity.

The Fekete and Fekete-Szegő Theorems proved in [62] hold with $\mathbb{P}^1$ replaced by an arbitrary curve C/k , and with ζ replaced by any k -symmetric finite set $\mathcal{X} \subset C(\bar{k})$. There is undoubtedly an analogue of Theorem 4.23 which holds in this generality.

CHAPTER 5

Harmonic functions

In this chapter, we develop a theory of harmonic functions on subdomains of the Berkovich projective line. (By a *domain* in Berkovich space we mean a nonempty, connected, open subset of $\mathbb{P}_{\text{Berk}}^1$.) We define a Laplacian operator and prove analogues of the Maximum Principle, Poisson's formula, Harnack's theorem, the Riemann Extension theorem, and the theory of Green's functions. We characterize harmonic functions as limits of logarithms of norms of rational functions, and show their stability under uniform limits and pullbacks by meromorphic functions. As a byproduct, we obtain the uniqueness of the equilibrium measure, asserted in Chapter 4.

Our idea for constructing the Laplacian on a domain in Berkovich space is to use the Laplacian on metrized graphs, and to take a limit over all metrized graphs contained in the domain. For a function f satisfying suitable hypotheses, the Laplacian of the restriction of f to each subgraph of the domain is a measure, so it defines a linear functional on the space of continuous functions on the graph. The Laplacians on the graphs satisfy a compatibility condition called *coherence*, which will be explained below. Passing to the limit, we obtain a canonical linear functional on the space of continuous functions on the domain. By the Riesz representation theorem, this corresponds to a measure; we define this measure to be the Laplacian of f .

1. Continuous functions

This section should be moved to Chapter 2.

Given a domain $U \subset \mathbb{P}_{\text{Berk}}^1$, let $\overline{U}$ be its closure. Our goal in this section is to understand the space $\mathcal{C}(\overline{U})$ of continuous functions $f : \overline{U} \rightarrow \mathbb{R}$.

When $\overline{U}$ is a Berkovich disc, there is a direct description of $\mathcal{C}(\overline{U})$ using the definition of the Berkovich topology.

PROPOSITION 5.1. *Let $\mathcal{B}(0, R)$ be a closed Berkovich disc. Then linear combinations of the functions $x \rightarrow [F]_x$, for polynomials $F(T) \in K[T]$, are dense in $\mathcal{C}(\mathcal{B}(0, R))$.*

PROOF. By the definition of the Berkovich topology, sets of the form $U_f(a, b) = \{x \in \mathcal{B}(0, R) : [f]_x \in (a, b)\}$ for $f \in K\langle R^{-1}T \rangle$ give a basis for the open sets of $\mathcal{B}(0, R)$. Since $\mathcal{B}(0, R)$ is compact, the Stone-Weierstrass theorem tells us that the set of polynomials in the functions $[f]_x$ is dense in $\mathcal{C}(\mathcal{B}(0, R))$. The multiplicativity of the seminorms $[\cdot]_x$ shows that each monomial $[f_1]_x^{k_1} \cdots [f_n]_x^{k_n}$ reduces to a single term $[f_1^{k_1} \cdots f_n^{k_n}]_x$. Finally, the Weierstrass Preparation theorem shows that for each $f \in K\langle R^{-1}T \rangle$ there is a polynomial $F \in K[T]$ with $[f]_x = [F]_x$ for all $x \in \mathcal{B}(0, r)$. $\square$

We now seek a description of $\mathcal{C}(\overline{U})$ which works in general.

As in Chapter 3, by a *subgraph* Γ of $\mathbb{P}_{\text{Berk}}^1$ we mean a connected, finitely branched metrized subgraph, whose metric is given by the logarithmic path distance $\rho(x, y)$ induced from the big model, and having finite total length. Such a graph Γ is necessarily a tree.

Move this proof
to Chapter 2.

LEMMA 5.2. *Let Γ be a subgraph of $\mathbb{P}_{\text{Berk}}^1$. Then*

A) *Γ is a closed subset of $\mathbb{P}_{\text{Berk}}^1$.*

B) *The metric topology on Γ coincides with the relative topology induced from $\mathbb{P}_{\text{Berk}}^1$.*

PROOF. Part (A) follows from part (B): since Γ is compact in the metric topology, it is compact in the relative topology, hence closed.

We will now prove part (B). Fix a system of coordinates on $\mathbb{P}_{\text{Berk}}^1$, and regard Γ as a subset of $\mathbb{A}_{\text{Berk}}^1$. Thus, we can view Γ as a tree of discs, partially ordered by the function $x \rightarrow \text{rad}(x)$:

$$\Gamma = \bigcup_{i=1}^m [r_i, R]_{x_i}$$

where the x_i are the points of minimal radius in Γ , and $r_i = \text{rad}(x_i)$ for each i .

We will first show that each $x \in \Gamma$ has a basis of neighborhoods in the metric topology which are open in the relative topology. There are several cases to consider.

(1) $x = x_i$ is an endpoint of Γ , and is a point of minimal radius in its branch. In this case, a basis of the open neighborhoods of x given by the half-open segments $[r_i, r_i + \varepsilon)_{x_i}$ for sufficiently small $\varepsilon > 0$. Such a segment is the intersection of the disc $\mathcal{B}(x_i, r_i + \varepsilon)^- = \{z \in \mathbb{A}_{\text{Berk}}^1 : \delta(z, x_i)_\infty < r_i + \varepsilon\}$ with Γ .

(2) x is an interior point of an edge of Γ . Suppose $x \in [r_i, R]_{x_i}$, and put $r = \text{rad}(x)$. Then a basis of the open neighborhoods of x is given by the open segments $(r - \varepsilon, r + \varepsilon)_{x_i}$ for sufficiently small $\varepsilon > 0$. Such a segment is the intersection of the open annulus $\mathcal{B}(x_i, r + \varepsilon)^- \setminus \mathcal{B}(x_i, r - \varepsilon)$ with Γ .

(3) x is a branch point of Γ , but is not the point of maximal radius. In this case, after relabeling the branches if necessary, we can assume that $[r_1, R]_{x_1}, \dots, [r_k, R]_{x_k}$ come together at x . Put $r = \text{rad}(x)$. Then a basis of the open neighborhoods of x is given by star-shaped sets of the form $\cup_{i=1}^k (r - \varepsilon, r + \varepsilon)_{x_i}$ for sufficiently small $\varepsilon > 0$. Such a set is the intersection of the simple domain $\mathcal{B}(x_1, r + \varepsilon)^- \setminus (\cup_{i=1}^k \mathcal{B}(x_i, r - \varepsilon))$ with Γ .

(4) x is the point of maximal radius R in Γ . Some of the branches of Γ may come together at points below x ; after relabeling the branches if necessary, we can assume that $[r_1, R]_{x_1}, \dots, [r_k, R]_{x_k}$ come together at x . Then a basis of the open neighborhoods of x is given by star-shaped sets of the form $\cup_{i=1}^\ell (R - \varepsilon, R]_{x_i}$ for sufficiently small $\varepsilon > 0$. Such a set is the intersection of $\mathbb{P}_{\text{Berk}}^1 \setminus (\cup_{i=1}^\ell \mathcal{B}(x_i, R - \varepsilon))$ with Γ .

Next we will show that each subset of Γ which is open in the relative topology is also open in the metric topology. It suffices to consider the intersection of Γ with a basic open set of the form $\mathcal{B}(a, s)^- \setminus (\cup_{j=1}^k \mathcal{B}(a_j, s_j))$. Using the notation introduced above, for each $j = 1, \dots, k$ let T_j be the set of x_i for which $[r_i, R]_{x_i} \cap \mathcal{B}(a_j, r_j)$ is nonempty, and let T be the set of x_i , not belonging to any T_j , for which $[r_i, R]_{x_i} \cap \mathcal{B}(a, r)^-$ is nonempty. Then the intersection of $\mathcal{B}(a, r)^- \setminus (\cup_{j=1}^k \mathcal{B}(a_j, r_j))$ with Γ is the union of the intervals of discs

$$\bigcup_{j=1}^k \left(\bigcup_{i \in T_j} (s_i, s)_{x_i} \right) \cup \left(\bigcup_{i \in T} [r_i, s)_{x_i} \right)$$

which is open in Γ . □

Now let $E \subset \mathbb{P}_{\text{Berk}}^1$ be an arbitrary nonempty connected closed set, equipped with the relative topology. Recall that a set is connected under the topology of $\mathbb{P}_{\text{Berk}}^1$ if and only if it is path-connected, and there is unique path between any two points of $\mathbb{P}_{\text{Berk}}^1$.

There is a retraction map $r_E : \mathbb{P}_{\text{Berk}}^1 \rightarrow E$ defined as follows. Fix a point $p_0 \in E$. Given $x \in \mathbb{P}_{\text{Berk}}^1$, let $r_E(x)$ be the first point p in E on the path from x to p_0 . Clearly $r_E(x) = x$ if $x \in E$. To see that $r_E(x)$ is independent of the choice of p_0 when $x \notin E$, suppose $p'_0 \in E$ is another point, and p' is the first point in E on the path from x to p'_0 . If $p' \neq p$, then since E is connected there is a path from p' to p contained in E .

There is another path from p' to p gotten by concatenating the path from p' to x with the path from x to p , and eliminating backtracking. This second path lies outside of E , apart from its endpoints. This contradicts the fact that there is a unique path between any two points of $\mathbb{P}_{\text{Berk}}^1$, so it must be that $p' = p$.

LEMMA 5.3. *For each nonempty closed connected subset $E \subset \mathbb{P}_{\text{Berk}}^1$, the retraction map $r_E : \mathbb{P}_{\text{Berk}}^1 \rightarrow E$ is continuous.*

PROOF. The connected open sets form a basis for the topology of $\mathbb{P}_{\text{Berk}}^1$. If $U \subset \mathbb{P}_{\text{Berk}}^1$ is a connected open set, then $U \cap E$ is also connected, since if $x, y \in U \cap E$ the path P from x to y is contained in U since U is connected, and it is contained in E since E is connected. Hence it is contained in $U \cap E$. It follows that the relative topology on E has a basis consisting of connected open sets.

Let $V \subset E$ be a connected open set, and let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set with $V = U \cap E$. If U_0 is the connected component of U containing V , then U_0 is also open; so we can assume without loss of generality that U is connected. Let $\tilde{U}$ be the union of all the connected open subsets $U \subset \mathbb{P}_{\text{Berk}}^1$ with $U \cap E = V$. Then $\tilde{U}$ is itself connected and open, and $\tilde{U} \cap E = V$, so it is the maximal set with these properties.

We claim that $r_E^{-1}(V) = \tilde{U}$. First, we will show that $r_E^{-1}(V) \subset \tilde{U}$. Suppose $x \in r_E^{-1}(V)$ and put $p = r_E(x) \in V$. Consider the path P from x to p . If $x \notin \tilde{U}$, let $\bar{x}$ be the first point in the closure of $\tilde{U}$ in P . Then $\bar{x} \notin \tilde{U}$, since $\tilde{U}$ is open; also, $\bar{x} \notin E$, since p is the only point of P in E and $p \in \tilde{U}$. Since E is compact and $\mathbb{P}_{\text{Berk}}^1$ is Hausdorff, there is a neighborhood W of $\bar{x}$ which is disjoint from E . Without loss of generality, we can assume W is connected. Since $\bar{x}$ is in the closure of $\tilde{U}$, $W \cap \tilde{U}$ is nonempty. It follows that $W \cup \tilde{U}$ is connected and open, and $(W \cup \tilde{U}) \cap E = V$. By the maximality of $\tilde{U}$ we must have $W \subset \tilde{U}$. This contradicts the fact that $\bar{x} \notin \tilde{U}$. Hence $x \in \tilde{U}$.

Next, we will show that $\tilde{U} \subset r_E^{-1}(V)$. Fix a point $p_0 \in V$, and let $x \in \tilde{U}$ be arbitrary. Since $\tilde{U}$ is connected, the path from x to p_0 must be entirely contained in $\tilde{U}$. Hence, the point $p = r_E(x)$, which is the first point in E along that path, belongs to $\tilde{U}$. But then $p \in \tilde{U} \cap E = V$, so $x \in r_E^{-1}(V)$. $\square$

If $E_1 \subset E_2 \subset \mathbb{P}_{\text{Berk}}^1$ are two nonempty connected closed subsets, the retraction map $r_{E_1} : \mathbb{P}_{\text{Berk}}^1 \rightarrow E_1$ induces a retraction map $r_{E_2, E_1} : E_2 \rightarrow E_1$. Clearly

$$r_{E_1}(x) = r_{E_2, E_1}(r_{E_2}(x))$$

for all x . If E_1 and E_2 both have the relative topology, it follows from Lemma 5.3 that r_{E_2, E_1} is continuous.

PROPOSITION 5.4. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain.*

A) As Γ ranges over all subgraphs of U , and as f ranges over $\mathcal{C}(\Gamma)$, the functions of the form $f \circ r_{\overline{U}, \Gamma}(x)$ are dense in $\mathcal{C}(\overline{U})$.

B) As Γ ranges over all subgraphs of U , and as f ranges over $\text{CPA}(\Gamma)$, the functions of the form $f \circ r_{\overline{U}, \Gamma}(x)$ are dense in $\mathcal{C}(\overline{U})$.

PROOF. For part (A), we will apply the Stone-Weierstrass theorem. Functions of the form $f \circ r_{\overline{U}, \Gamma}(x)$ separate points in $\overline{U}$: given distinct points $x, y \in \overline{U}$, take Γ to be a closed segment $[p, q]$ in the path from x to y , and let $f \in \mathcal{C}([p, q])$ be any function with $f(p) \neq f(q)$. Then $f \circ r_{\overline{U}, \Gamma}(x) \neq f \circ r_{\overline{U}, \Gamma}(y)$. Likewise, each constant function on $\overline{U}$ has the form $f \circ r_{\overline{U}, \Gamma}(x)$, where $\Gamma \subset U$ is arbitrary and f is the corresponding constant function on Γ .

Take $F \in \mathcal{C}(\overline{U})$. By the Stone-Weierstrass theorem (see [41], p. 244), since $\overline{U}$ is compact the algebra of functions generated by the $f \circ r_{\Gamma}(x)$ is dense in $\mathcal{C}(\overline{U})$. Hence, for any $\varepsilon > 0$, there are a finite number of subgraphs $\Gamma_{ij} \subset U$, $i = 1, \dots, m$, $j = 1, \dots, n_i$, and functions $f_{ij} \in \mathcal{C}(\Gamma_{ij})$, for which

$$|F(x) - (\sum_{i=1}^m \prod_{j=1}^{n_i} f_{ij} \circ r_{\overline{U}, \Gamma_{ij}}(x))| < \varepsilon$$

for all $x \in \overline{U}$. Let Γ be the smallest connected subgraph containing all the Γ_{ij} : then $\Gamma \subset U$. For each i, j , put $g_{ij} = f_{ij} \circ r_{\Gamma, \Gamma_{ij}}(x)$; then $g_{ij} \in \mathcal{C}(\Gamma)$ and $g_{ij} \circ r_{\overline{U}, \Gamma}(x) = f_{ij} \circ r_{\overline{U}, \Gamma_{ij}}(x)$ for all $x \in \overline{U}$. Put

$$g = \sum_{i=1}^m (\prod_{j=1}^{n_i} g_{ij}) \in \mathcal{C}(\Gamma).$$

Then $|F(x) - g \circ r_{\overline{U}, \Gamma}(x)| < \varepsilon$ for all $x \in \overline{U}$.

Part (B) follows from part (A), since $\text{CPA}(\Gamma)$ is dense in $\mathcal{C}(\Gamma)$ under the sup norm, for each Γ . $\square$

2. Coherent systems of measures

For each subgraph $\Gamma \subset U$, let μ_{Γ} be a finite signed Borel measure on Γ .

Give a name, for example $\mathcal{S}_U$, to the collection of subgraphs of U , and use this notation throughout.

DEFINITION 5.5. A system of measures $\{\mu_\Gamma\}$ on the subgraphs of U is called coherent if

A) For each pair of subgraphs with $\Gamma_1 \subset \Gamma_2$, $(r_{\Gamma_2, \Gamma_1})_*(\mu_{\Gamma_2}) = \mu_{\Gamma_1}$, that is, for each Borel subset $T \subset \Gamma_1$,

$$\mu_{\Gamma_2}(r_{\Gamma_2, \Gamma_1}^{-1}(T)) = \mu_{\Gamma_1}(T) .$$

B) There is a constant B such that $|\mu_\Gamma|(\Gamma) \leq B$ for each Γ .

The collection of subgraphs $\Gamma \subset U$ forms a directed set under containment: for any two graphs Γ_1, Γ_2 , there is a unique minimal subgraph Γ_3 containing Γ_1 and Γ_2 .

If μ is a finite signed Borel measure on $\overline{U}$, and $\mu_\Gamma = (r_{\overline{U}, \Gamma})_*(\mu)$ for each $\Gamma \subset U$, then $\{\mu_\Gamma\}$ is clearly a coherent system of measures on the subgraphs of U . The following proposition shows that every coherent system arises in this way: there is a 1-1 correspondence between measures μ on $\overline{U}$ and coherent systems of measures on subgraphs of U .

PROPOSITION 5.6. *If $\{\mu_\Gamma\}$ is a coherent system of measures in U , the map*

$$\Lambda(F) = \lim_{\Gamma} \int_{\Gamma} F(x) d\mu_\Gamma(x)$$

defines a bounded linear functional on $\mathcal{C}(\overline{U})$, so there is a unique finite signed Borel measure μ on $\overline{U}$ such that

$$\Lambda(F) = \int_{\overline{U}} F(x) d\mu(x)$$

for each $F \in \mathcal{C}(\overline{U})$. This measure is characterized by the property that $(r_{\overline{U}, \Gamma})_(\mu) = \mu_\Gamma$ for each subgraph $\Gamma \subset U$.*

PROOF. Fix $\Gamma_0 \subset U$ and $f_0 \in \mathcal{C}(\Gamma_0)$. Put $F_0 = f_0 \circ r_{\overline{U}, \Gamma_0}(x)$.

Since $\{\mu_\Gamma\}$ is a coherent system of measures, for each subgraph $\Gamma \subset U$ containing Γ_0

$$\int_{\Gamma} f_0(r_{\Gamma, \Gamma_0}(x)) d\mu_\Gamma(x) = \int_{\Gamma_0} f_0(x) d\mu_{\Gamma_0}(x) .$$

Then, since $F_0|_\Gamma = f_0 \circ r_{\Gamma, \Gamma_0}$ for each Γ containing Γ_0 ,

$$\Lambda(F_0) = \lim_{\Gamma} \int_{\Gamma} F_0(x) d\mu_\Gamma(x) = \int_{\Gamma_0} f_0(x) d\mu_{\Gamma_0}(x)$$

exists.

By Proposition 5.4, functions of the form $F = f \circ r_{\overline{U}, \Gamma}$ for $f \in \mathcal{C}(\Gamma)$ and $\Gamma \subset U$ are dense in $\mathcal{C}(\overline{U})$ under the sup norm, so Λ is defined on a dense subset of $\mathcal{C}(\overline{U})$.

On the other hand, since $|\mu_\Gamma|$ has total mass at most B for each Γ , for each $G \in \mathcal{C}(\overline{U})$

$$\limsup_{\Gamma} \left| \int_{\Gamma} G(x) d\mu_{\Gamma}(x) \right| \leq B \cdot \|G\|_{\overline{U}}$$

It follows that Λ extends to a bounded linear functional on $\mathcal{C}(\overline{U})$. The Riesz representation theorem (Theorem A.8) tells us there is a unique finite Borel measure μ on $\overline{U}$

$$\Lambda(F) = \int_{\overline{U}} F(x) d\mu(x)$$

for each $F \in \mathcal{C}(\overline{U})$. (Recall from Lemma 4.5 that every finite Borel measure on $\overline{U}$ is Radon.)

For the final assertion, it suffices to show that $(r_{\overline{U},\Gamma})_*(\mu) = \mu_\Gamma$ for each Γ , since this means that μ determines the measures μ_Γ , which in turn determine the functional $\Lambda(F)$. This follows from the uniqueness portion of the Riesz Representation Theorem, since for any $f \in \mathcal{C}(\Gamma)$, we have

$$\int_{\Gamma} f d(r_{\overline{U},\Gamma})_*\mu = \int_{\overline{U}} (f \circ r_{\overline{U},\Gamma}) d\mu = \int_{\Gamma} f d(\mu_\Gamma) .$$

□

3. The Laplacian on a subdomain of $\mathbb{P}_{\text{Berk}}^1$

We now set about defining a measure-valued Laplacian operator on a suitable class of functions $f : U \rightarrow \mathbb{R} \cup \{\pm\infty\}$, where U is a domain in $\mathbb{P}_{\text{Berk}}^1$. If $f|_\Gamma \in \text{BDV}(\Gamma)$, we write $\Delta_\Gamma(f)$ for $\Delta(f|_\Gamma)$.

DEFINITION 5.7. Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain. We will say that $f : U \rightarrow \mathbb{R} \cup \{\pm\infty\}$ is of *bounded differential variation* on U , and write $f \in \text{BDV}(U)$, if

- A) $f|_\Gamma \in \text{BDV}(\Gamma)$ for each subgraph $\Gamma \subset U$, and
- B) there is a constant $B = B(f)$ such that for all Γ ,

$$|\Delta_\Gamma(f)|(\Gamma) \leq B .$$

PROPOSITION 5.8. *If $f \in \text{BDV}(U)$, the system of measures $\{\Delta_\Gamma(f)\}_{\Gamma \subset U}$ is coherent.*

PROOF. The boundedness condition in the definition of coherence is built into the definition of $\text{BDV}(U)$; it suffices to check the compatibility under pushforwards.

Let subgraphs $\Gamma_1 \subset \Gamma_2 \subset U$ be given. Since Γ_2 can be obtained by sequentially attaching a finite number of edges to Γ_1 , it suffices to

consider the case where $\Gamma_2 = \Gamma_1 \cup T$, and T is a segment attached to Γ_1 at a point p . By the definition of the Laplacian on a graph, for any Borel set e contained in $T \setminus \{p\}$ we have $\Delta_{\Gamma_2}(f)(e) = \Delta_T(f)(e)$. For the point p , the set of direction vectors at p in Γ_2 is the union of the corresponding sets in Γ_1 and T . It follows that $\Delta_{\Gamma_2}(f)(\{p\}) = \Delta_T(f)(\{p\}) + \Delta_{\Gamma_1}(f)(\{p\})$. Since $(r_{\Gamma_2, \Gamma_1})^{-1}(\{p\}) = T$ and $\Delta_T(f)(T) = 0$,

$$\Delta_{\Gamma_2}(f)(r_{\Gamma_2, \Gamma_1})^{-1}(\{p\}) = \Delta_T(f)(T) + \Delta_{\Gamma_1}(f)(\{p\}) = \Delta_{\Gamma_1}(f)(\{p\}) .$$

Trivially $\Delta_{\Gamma_2}(f)(e) = \Delta_{\Gamma_1}(f)(e)$ for any Borel set e contained in $\Gamma_1 \setminus \{p\}$. Hence $(r_{\Gamma_2, \Gamma_1})_*(\Delta_{\Gamma_2}(f)) = \Delta_{\Gamma_1}(f)$. $\square$

DEFINITION 5.9. If $f \in \text{BDV}(U)$, the Laplacian

$$\Delta(f) = \Delta_{\overline{U}}(f)$$

is the measure on $\overline{U}$ associated to the coherent system $\{\Delta_{\Gamma}(f)\}_{\Gamma \subset U}$ by Proposition 5.6, characterized by the property that for each subgraph $\Gamma \subset U$

$$(r_{\overline{U}, \Gamma})_*(\Delta_{\overline{U}}(f)) = \Delta_{\Gamma}(f) .$$

PROPOSITION 5.10. For each $f \in \text{BDV}(U)$, $\Delta_{\overline{U}}(f)$ has total mass 0.

PROOF. For any subgraph $\Gamma \subset U$,

$$\Delta_{\overline{U}}(f)(\overline{U}) = (r_{\overline{U}, \Gamma})_*(\Delta_{\overline{U}}(f))(\Gamma) = \Delta_{\Gamma}(f)(\Gamma) = 0 .$$

$\square$

Here are some examples of functions in $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and their Laplacians.

EXAMPLE 5.11. If $f(x) = C$ is constant then $f \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and $\Delta(f) \equiv 0$.

Indeed $\Delta_{\Gamma}(f) = 0$ for each subgraph γ , so the associated functional $\Lambda(F)$ is the 0 functional.

Conversely, if U is a domain, $f \in \text{BDV}(U)$ and $\Delta_{\overline{U}}(f) \equiv 0$, then f is constant on $U \cap \mathbb{H}_{\text{Berk}}$. Indeed, for each subgraph $\Gamma \subset \overline{U}$ we have $\Delta_{\Gamma}(f) = (r_{\overline{U}, \Gamma})_*(\Delta_{\overline{U}}(f)) \equiv 0$, so that f is constant on Γ . Since there exists a subgraph $\Gamma \subset \overline{U}$ containing two given points $x, y \in U \cap \mathbb{H}_{\text{Berk}}$, f is constant on $U \cap \mathbb{H}_{\text{Berk}}$.

EXAMPLE 5.12. If $f(x) = -\log_v(\delta(x, y)_{\zeta})$, then $f \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and

$$\Delta(f) = \delta_y(x) - \delta_{\zeta}(x) .$$

To see this, let $\zeta_{\text{Gauss}} \in \mathbb{P}_{\text{Berk}}^1$ be the Gauss point, the point corresponding to $B(0, 1)$. Given a subgraph Γ containing ζ_{Gauss} , put $\tilde{y} = r_\Gamma(y)$ and $\tilde{\zeta} = r_\Gamma(\zeta)$. For $x \in \Gamma$, it follows from the definition of the Hsia kernel and the restriction formula for $j_{\zeta_{\text{Gauss}}}(x, y)$ in Chapter 3 that

$$\begin{aligned} -\log_v(\delta(x, y)_\zeta) &= j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \zeta) - j_{\zeta_{\text{Gauss}}}(y, \zeta) \\ (5.1) \quad &= j_{\zeta_{\text{Gauss}}}(x, \tilde{y}) - j_{\zeta_{\text{Gauss}}}(x, \tilde{\zeta}) - j_{\zeta_{\text{Gauss}}}(y, \zeta) \end{aligned}$$

Hence $\Delta_\Gamma(f) = \delta_{\tilde{y}}(x) - \delta_{\tilde{\zeta}}(x) = (r_\Gamma)_*(\delta_y(x) - \delta_\zeta(x))$. The only measure on $\mathbb{P}_{\text{Berk}}^1$ with this property for all Γ is $\mu = \delta_y(x) - \delta_\zeta(x)$.

Give more precise reference.

EXAMPLE 5.13. Take $0 \neq g \in K(T)$; suppose $\text{div}(g) = \sum_{i=1}^m n_i(a_i)$. Then $f(x) = -\log_v([g]_x) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and

$$\Delta(-\log_v([g]_x)) = \sum_{i=1}^m n_i \delta_{a_i}(x) .$$

This follows from Example 5.12. Let $\zeta \in \mathbb{P}_{\text{Berk}}^1$ be arbitrary. By the decomposition formula for the Hsia kernel, there is a constant C_ζ such that

$$[g]_x = C_\zeta \cdot \prod_{a_i \neq \zeta} \delta(x, a_i)_\zeta .$$

Taking logarithms, applying Example 5.12, and using $\sum_{i=1}^m n_i = 0$ gives the result.

EXAMPLE 5.14. If ν is any probability measure on $\mathbb{P}_{\text{Berk}}^1$ and $\zeta \in \mathbb{H}_{\text{Berk}}$ or $\zeta \notin \text{supp}(\nu)$, consider the potential function

$$u_\nu(x, \zeta) = \int -\log_v(\delta(x, y)_\zeta) d\nu(y) .$$

Then $u_\nu(x, \zeta) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and

$$(5.2) \quad \Delta(u_\nu(x, \zeta)) = \nu - \delta_\zeta(x) .$$

To see this, let ζ_{Gauss} be the Gauss point, and let Γ be any subgraph containing ζ_{Gauss} . For $x \in \Gamma$, using the definition of $\delta(x, y)_\zeta$, the restriction formula $j_{\zeta_{\text{Gauss}}}(x, y) = j_{\zeta_{\text{Gauss}}}(x, r_\Gamma(y))$, and the fact that ν has total mass 1, we have

$$\begin{aligned} u_\nu(x, \zeta) &= \int j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \zeta) - j_{\zeta_{\text{Gauss}}}(y, \zeta) d\nu(y) \\ &= \int j_{\zeta_{\text{Gauss}}}(x, r_\Gamma(y)) d\nu(y) - \int j_{\zeta_{\text{Gauss}}}(x, r_\Gamma(\zeta)) d\nu(y) - C_\zeta \\ &= \int_\Gamma j_{\zeta_{\text{Gauss}}}(x, t) d((r_\Gamma)_*(\nu))(t) - j_{\zeta_{\text{Gauss}}}(x, r_\Gamma(\zeta)) - C_\zeta \end{aligned}$$

where C_ζ is a finite constant. (It is finite because we are assuming that either $\zeta \in \mathbb{H}_{\text{Berk}}$ or $\zeta \notin \text{supp}(\nu)$, which implies that $j_{\zeta_{\text{Gauss}}}(y, \zeta)$ is continuous and real-valued as a function of y on the compact set $\text{supp}(\nu)$.) By Proposition C.10, $u_\nu(x, \zeta)|_\Gamma \in \text{BDV}(\Gamma)$ and

$$\Delta_\Gamma(u_\nu(z, \zeta)) = (r_\Gamma)_*(\nu) - \delta_{r_\Gamma(\zeta)} = (r_\Gamma)_*(\nu - \delta_\zeta) .$$

It follows that $\Delta(u_\nu(x, \zeta)) = \nu - \delta_\zeta$.

EXAMPLE 5.15. More generally, if ν is any finite Borel measure on $\mathbb{P}_{\text{Berk}}^1$ and either $\zeta \in \mathbb{H}_{\text{Berk}}$ or $\zeta \notin \text{supp}(\nu)$, consider the potential function

$$u_\nu(x, \zeta) = \int -\log_v(\delta(x, y)_\zeta) d\nu(y) .$$

Then $u_\nu(x, \zeta) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and

$$\Delta(u_\nu(x, \zeta)) = \nu - \nu(\bar{V})\delta_\zeta .$$

This follows from the same calculation as above, using Proposition C.10.

REMARK 5.16. The definition of $f \in \text{BDV}(U)$ only concerns the restrictions $f|_\Gamma$ to subgraphs $\Gamma \subset U$. Such subgraphs can only contain points of type II, III, or IV (in fact, as the subgraphs vary they exhaust $U \setminus \mathbb{P}^1(K)$), so requiring that $f \in \text{BDV}(U)$ imposes no conditions at all on its behavior on $\mathbb{P}^1(K)$. That behavior must be deduced from auxiliary hypotheses, such as continuity or upper-semicontinuity.

The Laplacian is compatible with restriction to subdomains.

PROPOSITION 5.17. *Suppose $U_1 \subset U_2 \subset \mathbb{P}_{\text{Berk}}^1$ are domains, and $f \in \text{BDV}(U_2)$. Then $f \in \text{BDV}(U_1)$ and*

$$\Delta_{\bar{U}_1}(f) = (r_{\bar{U}_2, \bar{U}_1})_*(\Delta_{\bar{U}_2}(f)) .$$

PROOF. If $f \in \text{BDV}(U_2)$, then trivially $f \in \text{BDV}(U_1)$. By the compatibility of the retraction maps, for each $\Gamma \subset U_1$

$$(r_{\bar{U}_1, \Gamma})_*((r_{\bar{U}_2, \bar{U}_1})_*(\Delta_{\bar{U}_2}(f))) = (r_{\bar{U}_2, \Gamma})_*(\Delta_{\bar{U}_2}(f)) = \Delta_\Gamma(f) .$$

By the characterization of $\Delta_{\bar{U}_1}(f)$, it follows that $\Delta_{\bar{U}_1}(f) = (r_{\bar{U}_2, \bar{U}_1})_*(\Delta_{\bar{U}_2}(f))$. $\square$

4. Harmonic Functions

If U is a domain and $f \in \text{BDV}(U)$, the measure $\Delta_{\bar{U}}(f)$ can be decomposed into two parts: a part supported on ∂U , and a part supported on U . As will be seen, the part supported on ∂U is analogous to the classical normal derivative on the boundary, while the part supported on U is analogous to the classical Laplacian.

DEFINITION 5.18. If U is a domain, $f : U \rightarrow \mathbb{R}$ is strongly harmonic on U if it is continuous on U , belongs to $\text{BDV}(U)$, and satisfies $\Delta_{\overline{U}}(f)|_U \equiv 0$.

If V is an arbitrary open set, $f : V \rightarrow \mathbb{R}$ is harmonic in V if for each $x \in V$ there is a connected open neighborhood U_x of x on which f is strongly harmonic.

If f and g are harmonic (or strongly harmonic) in U , then so are $-f$ and $a \cdot f + b \cdot g$, for any $a, b \in \mathbb{R}$. In view of the computations of Laplacians given above, we have the following examples of strongly harmonic functions:

EXAMPLE 5.19. Each constant function $f(x) = C$ on a domain U is strongly harmonic.

EXAMPLE 5.20. For fixed y, ζ , $-\log_v(\delta(x, y)_\zeta)$ is strongly harmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{y, \zeta\}$.

EXAMPLE 5.21. If $0 \neq g \in K(T)$ has divisor $\text{div}(g) = \sum_{i=1}^m n_i(a_i)$, then $f(x) = -\log_v([g]_x)$ is strongly harmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{a_1, \dots, a_m\}$.

EXAMPLE 5.22. Given a probability measure ν and a point $\zeta \notin \text{supp}(\nu)$, the potential function $u_\nu(z, \zeta)$ is strongly harmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus (\text{supp}(\nu) \cup \{\zeta\})$.

LEMMA 5.23.

A) If $U_1 \subset U_2$ are domains, and f is strongly harmonic in U_2 , then f is strongly harmonic in U_1 .

B) If f is harmonic in an open set V , and U is a subdomain of V with $\overline{U} \subset V$, then f is strongly harmonic in U .

C) If f is harmonic on V and $E \subset V$ is compact and connected, there is a subdomain $U \subset V$ containing E such that f is strongly harmonic on U . In particular, if V is a domain, there is an exhaustion of V by subdomains U on which f is strongly harmonic.

PROOF. For (A) note that $f \in \text{BDV}(U_1)$ and $(r_{\overline{U}_2, \overline{U}_1})_*(\Delta_{\overline{U}_2}(f)) = \Delta_{\overline{U}_1}(f)$ by Proposition 5.17. Since $\Delta_{\overline{U}_2}(f)$ is supported on $\partial \overline{U}_2$ and $r_{\overline{U}_2, \overline{U}_1}$ maps $\partial \overline{U}_2$ to $\partial \overline{U}_1$, it follows that $(r_{\overline{U}_2, \overline{U}_1})_*(\Delta_{\overline{U}_2}(f))$ is supported on $\partial \overline{U}_1$.

For (B), note that for each $x \in \overline{U}$ there is a connected neighborhood U_x such that f is strongly harmonic in U_x . Since $\overline{U}$ is compact, a finite number of the U_x cover $\overline{U}$, say $U_{x_1}, \dots, U_{x_m}$. For each U_{x_i} , there is a constant B_i such that for every subgraph $\Gamma_i \subset U_{x_i}$,

$$|\Delta_{\Gamma_i}(f)|(\Gamma_i) \leq B_i.$$

If $\Gamma \subset U$ is a subgraph, then there are subgraphs $\Gamma_1, \dots, \Gamma_m$ with $\Gamma_i \subset U_{x_i}$ such that $\Gamma \subset \bigcup_{i=1}^m \Gamma_i$. (These subgraphs are not in general disjoint). We have $f|_\Gamma \in \text{BDV}(\Gamma)$ since $f|_{\Gamma_i} \in \text{BDV}(\Gamma_i)$ for each i , and

$$|\Delta_\Gamma(f)|(\Gamma) \leq \sum_{i=1}^m |\Delta_{\Gamma_i}(f)|(\Gamma_i) \leq \sum_{i=1}^m B_i .$$

Thus $f|_U \in \text{BDV}(U)$.

For each x_i , $U \cap U_{x_i}$ is a subdomain of U_{x_i} . (It is connected because $U \cap U_{x_i} \neq \emptyset$ and U, U_{x_i} are both path-connected.) By part (A),

$$\Delta_{\overline{U} \cap \overline{U}_{x_i}}(f)|_{U \cap U_{x_i}} \equiv 0 .$$

However, the $U \cap U_{x_i}$ cover U , and

$$(r_{\overline{U}, \overline{U} \cap \overline{U}_{x_i}})_*(\Delta_{\overline{U}}(f)) = \Delta_{\overline{U} \cap \overline{U}_{x_i}}(f) .$$

It follows that $\Delta_{\overline{U}}(f)|_{U \cap U_{x_i}} \equiv 0$. This holds for each i , so $\Delta_{\overline{U}}(f)|_U \equiv 0$.

For (C), note that for each $x \in E$ there is a connected neighborhood $U_x \subset V$ on which f is strongly harmonic. Since E is compact, a finite number of the U_x cover Γ . Let U be the union of those neighborhoods. Then U is connected, and by the same argument as above, f is strongly harmonic on U . $\square$

Note that if U is open but is not connected, and f is harmonic in U , we have not defined $\Delta_{\overline{U}}(f)$ even if f is strongly harmonic in each component of U . Different components may share boundary points, creating unwanted interaction between the Laplacians on the components. For example, let $\zeta_{\text{Gauss}} \in \mathbb{P}_{\text{Berk}}^1$ be the Gauss point, and consider the open set $U = \mathbb{P}_{\text{Berk}}^1 \setminus \{0, \zeta_{\text{Gauss}}\}$. It has one distinguished component $U_0 = \mathcal{B}(0, 1)^- \setminus \{0\}$ and infinitely many other components $U_i = \mathcal{B}(a_i, 1)_0^-$. They all have ζ_{Gauss} as a boundary point. The function $f(z)$ which is $-\log_v(\delta(x, 0)_\infty)$ in U_0 and is 0 elsewhere has Laplacian $\Delta_{\overline{U}_0}(f) = \delta_0(x) - \delta_{\zeta_{\text{Gauss}}}(x)$, while $\Delta_{\overline{U}_i}(f) = 0$ for $i \neq 0$.

A key observation is that the behavior of a harmonic function on a domain U is controlled by its behavior on a special subset.

DEFINITION 5.24. If U is a domain, the *main dendrite* $D = D(U) \subset U$ is set of all $x \in U$ belonging to paths between boundary points $y, z \in \partial U$.

If the main dendrite is nonempty, it is a tree which is finitely branched at each point. To see this, we require the following lemmas.

LEMMA 5.25. *Let $W \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, let $x \in W$, and let $y \in \mathbb{P}_{\text{Berk}}^1 \setminus W$. Then the path Γ from x to y contains some boundary point of W .*

PROOF. Let $W' = \mathbb{P}_{\text{Berk}}^1 \setminus \overline{W}$. If $\Gamma \cap \partial W = \emptyset$, then $\Gamma = (\Gamma \cap W) \cup (\Gamma \cap W')$ is a union of two relatively open subsets, contradicting the fact that Γ is connected. Thus $\Gamma \cap \partial W \neq \emptyset$. $\square$

PROPOSITION 5.26. *The main dendrite D of U is a locally finite tree.*

PROOF. Fix $z \in D$, and choose a covering of ∂U by simple domains not containing z . Since ∂U is compact, a finite number of simple domains $W_1, \dots, W_n$ will suffice. Let $W = W_1 \cup \dots \cup W_n$, and let B be the finite set consisting of all boundary points of $W_1, \dots, W_n$. It follows easily from Lemma 5.25 that $D \cap (\mathbb{P}_{\text{Berk}}^1 \setminus W)$ coincides with the union P of all paths between z and points in B . This implies that D is finitely branched at z , as desired. $\square$

Note that the main dendrite can be empty: that will be the case iff U has one or no boundary points, namely if U is an open disc, if $U \cong \mathbb{A}_{\text{Berk}}^1$, or if $U = \mathbb{P}_{\text{Berk}}^1$.

LEMMA 5.27. *Let $f(x)$ be harmonic in a domain U . If the main dendrite is empty, then $f(x)$ is constant; otherwise, $f(x)$ is constant on branches off the main dendrite.*

PROOF. First suppose the main dendrite is empty, so ∂U is empty or consists of a single point. Then $U = \mathbb{P}_{\text{Berk}}^1$, $U \cong \mathbb{A}_{\text{Berk}}^1$, or U is an open disc. Let $x, y \in U$ be arbitrary points not of type I, and let Γ be the path connecting them. By the description of U , there is a disc V whose closure is contained in U , with $\Gamma \subset V$.

By Lemma 5.23, f is strongly harmonic in V . We claim that $\Delta_{\overline{V}}(f) \equiv 0$. By definition, $\Delta_{\overline{V}}(f)|_V \equiv 0$. Also, $\Delta_{\overline{V}}(f)(\overline{V}) = 0$ by Proposition 5.10, so $\Delta_{\overline{V}}(f)(\partial V) = \Delta_{\overline{V}}(f)(\overline{V}) - \Delta_{\overline{V}}(f)(V) = 0$. Since ∂V consists of one point, our claim is established.

It follows that $\Delta_{\Gamma}(f) = (r_{\overline{V}, \Gamma})_*(\Delta_{\overline{V}}(f)) = 0$. By Proposition C.13(A), $f|_{\Gamma}$ is constant; thus $f(x) = f(y)$. Since $\mathbb{H}_{\text{Berk}}$ is dense in $\mathbb{P}_{\text{Berk}}^1$ and f is continuous, f must be constant. Give reference.

Next suppose the main dendrite is nonempty. Let $x \in U \setminus D$ be arbitrary. If x is not of type I, fix a point $y_0 \in D$, and let w be the first point in D along the path from x to y_0 ; we claim that $f(x) = f(w)$. Let Γ be the path from x to w ; then there is a subdomain V of U containing Γ , on which f is strongly harmonic. Since $\Delta_{\overline{V}}(f)$ is supported on ∂V and the retraction of ∂V to Γ is the point w , it follows

that $\Delta_\Gamma(f) = (r_{\overline{V}, \Gamma})_*(\Delta_{\overline{V}}(f))$ is supported on w . However, $\Delta_\Gamma(f)(\Gamma) = 0$, so $\Delta_\Gamma(\{w\}) = 0$. Thus $\Delta_\Gamma(f) \equiv 0$, and by Proposition C.13(A), $f(x) = f(w)$. If x is of type I, then it is a limit of non-type I points along the path from x to w , so by continuity $f(x) = f(w)$. $\square$

Here is an example of a domain U and a function $f : U \rightarrow \mathbb{R}$ which is harmonic on U but not strongly harmonic.

Fix coordinates so that $\mathbb{P}_{\text{Berk}}^1 = \mathbb{A}_{\text{Berk}}^1 \cup \{\infty\}$, and take $U = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{Z}_p$. Then the main dendrite D is a tree whose root node is the Gauss point ζ_{Gauss} , and whose other nodes are at the points corresponding to balls $B(a, p^{-n})$ for $a, n \in \mathbb{Z}$, with $n \geq 1$ and $0 \leq a \leq p^n - 1$. It has p branches extending down from each node, and each edge has length 1.

To describe f , it suffices to give its values on the main dendrite. It will have constant slope on each edge. Let $f(\zeta_{\text{Gauss}}) = 0$; we will recursively give its values on the other nodes. Suppose z_a is a node on which $f(z_a)$ has already been defined. If z_a is the root node, put $N_a = 0$; otherwise, let N_a be the slope of f on the edge entering z_a from above. Of the p edges extending down from z_a , choose two distinguished ones, and let $f(z)$ have slope $N_a + 1$ on one and slope -1 on the other, until the next node. On the $p - 2$ other edges, let $f(z)$ have the constant value $f(z_a)$ until the next node.

By construction, the sum of the slopes of f on the edges leading away from each node is 0, so f is harmonic in U . However, there are edges of D on which f has arbitrarily large slope; if Γ is an edge of D on which f has slope m_Γ , then $|\Delta_\Gamma(f)|(\Gamma) = 2|m_\Gamma|$. Thus, $f \notin \text{BDV}(U)$, and f is not strongly harmonic on U .

5. The Maximum Principle

We will now show that harmonic functions on Berkovich space share many properties of classical harmonic functions.

PROPOSITION 5.28. (Maximum Principle)

If f is a nonconstant harmonic function on a domain $U \subset \mathbb{P}_{\text{Berk}}^1$, then f does not achieve a maximum or a minimum value on U .

In particular, if f is harmonic on U and $\limsup_{x \rightarrow \partial U} f(x) \leq M$, then $f(x) \leq M$ for all $x \in U$; if $\liminf_{x \rightarrow \partial U} f(x) \geq m$, then $f(x) \geq m$ for all $x \in U$.

PROOF. It suffices to deal with the case of a maximum.

Suppose f takes on its maximum at a point $x \in U$. Let $V \subset U$ be a subdomain containing x on which f is strongly harmonic. We will show that f is constant on V . It follows that $\{z \in U : f(z) = f(x)\}$ is

both open and closed, and hence equals U since U is connected, so f is constant on U .

By Lemma 5.27, we can assume the main dendrite D for V is nonempty. Let T be the branch off the main dendrite containing x , and let w be the point where T attaches to D . By Lemma 5.27, $f(w) = f(x)$.

Let $\Gamma \subset D$ be an arbitrary subgraph with w in its interior. By the definition of the main dendrite, $r_{\overline{V}, \Gamma}(\partial V)$ consists of the endpoints of Γ . Since $\Delta_{\overline{V}}(f)$ is supported on ∂V , $\Delta_{\Gamma}(f) = (r_{\overline{V}, \Gamma})_*(\Delta_{\overline{V}}(f))$ is supported on the endpoints of Γ . By Proposition C.10, $f|_{\Gamma} \in \text{CPA}(\Gamma)$. Let Γ_w be the connected component of $\{z \in \Gamma : f(z) = f(w)\}$ containing w . If $\Gamma_w \neq \Gamma$, let p be a boundary point of Γ_w . Then p cannot be an endpoint of Γ , because Γ_w is closed. Since $f(p)$ is maximal, we must have $d_{\vec{v}}f(p) \leq 0$ for all $\vec{v}$ at p . Since $f|_{\Gamma} \in \text{CPA}(\Gamma)$, and $f|_{\Gamma}$ is nonconstant near p , there must be some $\vec{v}$ with $d_{\vec{v}}(f)(p) < 0$. It follows that

$$\Delta_{\Gamma}(f)(p) = - \sum_{\vec{v} \text{ at } p} d_{\vec{v}}f(p) > 0 .$$

This contradicts the fact that $\Delta_{\Gamma}(f)$ is supported on the endpoints of Γ , and we conclude that $\Gamma_w = \Gamma$.

Since Γ can be taken arbitrarily large, f must be constant on the main dendrite. By Lemma 5.27, it is constant everywhere on V .

The last part of the proposition is deduced as follows. Since the function $f^* : \overline{U} \rightarrow \mathbb{R}$ defined by

$$f^*(x) = \begin{cases} f(x) & x \in U \\ \limsup_{y \in U, y \rightarrow x} f(y) & x \in \partial U \end{cases}$$

is upper semicontinuous, it attains its maximum value on $\overline{U}$. Since f is harmonic, the maximum value cannot occur on U , which implies that if $\limsup_{x \rightarrow \partial U} f(x) \leq M$, then $f^*(x) \leq M$ for all $x \in \overline{U}$, and thus $f(x) \leq M$ for all $x \in U$. A similar argument using lower semicontinuity implies that if $\liminf_{x \rightarrow \partial U} f(x) \geq m$, then $f(x) \geq m$ for all $x \in U$. $\square$

Fix the $\lim \sup$ subscripts.

There is an important strengthening of the Maximum Principle which allows us to ignore sets of capacity 0 in ∂U .

PROPOSITION 5.29. (Strong Maximum Principle)

Let f be harmonic on a domain $U \subset \mathbb{P}_{\text{Berk}}^1$. Assume either that f is nonconstant, or that ∂U has positive capacity.

If f is bounded above on U , and there are a number M and a set $e \subset \partial U$ of capacity 0 such that $\limsup_{x \rightarrow z} f(x) \leq M$ for all $z \in \partial U \setminus e$, then $f(x) \leq M$ for all $x \in U$.

Similarly, if f is bounded below on U , and there are a number m and a set $e \subset \partial U$ of capacity 0 such that $\liminf_{x \rightarrow z} f(x) \geq m$ for all $z \in \partial U \setminus e$, then $f(x) \geq m$ for all $x \in U$.

For the proof, we will need the existence of an ‘Evans function’.

LEMMA 5.30. *Let $e \subset \mathbb{P}_{\text{Berk}}^1$ be a compact set of capacity 0, and let $\zeta \notin e$. Then there is a probability measure ν supported on e for which the potential function $u_\nu(z, \zeta)$ (which is defined on all of $\mathbb{P}_{\text{Berk}}^1$ as a function to $\mathbb{R} \cup \{\pm\infty\}$) satisfies*

$$\lim_{z \rightarrow x} u_\nu(z, \zeta) = \infty$$

for all $x \in e$. A function with this property is called an Evans function.

PROOF. By Theorem 4.20, the restricted Chebyshev constant $\text{CH}^*(E)_\zeta$ is 0. This means that for each $\epsilon > 0$, there are points $a_1, \dots, a_N \in e$ for which the pseudo-polynomial

$$P_N(z; a_1, \dots, a_N) = \prod_{i=1}^N \delta(z, a_i)_\zeta$$

satisfies $(\|P_N(z; a_1, \dots, a_N)\|_e)^{1/N} < \epsilon$.

Let $q = q_v$ be the base of the logarithm $\log_v(t)$, and take $\epsilon = 1/q^{2^k}$ for $k = 1, 2, 3, \dots$. For each k we obtain points $a_{k,1}, \dots, a_{k,N_k}$ such that the corresponding pseudopolynomial $P_{N_k}(z)$ satisfies $(\|P_{N_k}\|_e)^{1/N_k} < 1/q^{2^k}$. Put

$$\nu = \sum_{k=1}^{\infty} \frac{1}{2^k} \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \delta_{a_{k,i}}(x) \right).$$

Then ν is a probability measure supported on e , and

$$u_\nu(z, \zeta) = \sum_{k=1}^{\infty} \frac{1}{2^k} \left(-\frac{1}{N_k} \log_v(P_{N_k}(z)) \right).$$

Since $P_{N_k}(z)$ is continuous, there is a neighborhood U_k where $P_{N_k}(z)^{1/N_k} < 1/q^{2^k}$. Without loss of generality, we can assume $U_1 \supset U_2 \supset U_3 \dots$. On U_m , for each $k = 1, \dots, m$ we have

$$-\frac{1}{2^k} \cdot \frac{1}{N_k} \log_v(P_{N_k}(z)) \geq -\frac{1}{2^k} \log_v(1/q^{2^k}) = 1.$$

Hence $u_\nu(z, \zeta) \geq m$ on U_m . Letting $m \rightarrow \infty$, we see that $\lim_{z \rightarrow x} u_\nu(z, \zeta) = \infty$ for each $x \in e$. $\square$

We can now prove Proposition 5.29.

PROOF. It suffices to deal with the case of a maximum. Suppose $f(x_0) > M$ for some $x_0 \in U$. If f is nonconstant, there is a point $\zeta \in U$ with $f(\zeta) \neq f(x_0)$, and after interchanging x_0 and ζ if necessary we can assume $f(x_0) > f(\zeta)$. If ∂U has positive capacity, then for each $z \in \partial U \setminus e$ we have $\limsup_{x \rightarrow z} f(x) \leq M$, so there is a point $\zeta \in U$ with $f(\zeta) < f(x_0)$.

Fix $M_1 > M$ with $f(x_0) > M_1 > f(\zeta)$, and let $W = \{x \in U : f(x) > M_1\}$. Let V be the connected component of W containing x_0 . Since f is continuous, V is open, and is itself a domain in $\mathbb{P}_{\text{Berk}}^1$. Consider its closure $\bar{V}$. If $\bar{V} \subset U$, then each $y \in \partial V$ would have a neighborhood containing points of V and points of $U \setminus W$. By the continuity of f , this implies $f(y) = M_1$. However, $f(x_0) > M_1$. This violates the Maximum Principle for V .

Hence $e_1 = \bar{V} \cap \partial U$ is nonempty; clearly it is compact. By the definition of W , e_1 is contained in e , the exceptional set where $\limsup_{x \rightarrow y} f(x) > M$. Since e has capacity 0, e_1 has capacity 0 as well.

Let $h(x)$ be an Evans function for e_1 with respect to ζ . Then $h(x)$ is harmonic in V , and $\lim_{x \rightarrow y} h(y) = \infty$ for each $y \in e_1$. Since $\zeta \notin \bar{V}$, $h(x)$ is bounded below on $\bar{V}$. Let $-B$ be a lower bound for $h(x)$ on V .

For each $\eta > 0$, put

$$f_\eta(x) = f(x) - \eta \cdot h(x)$$

on V . Then $f_\eta(x)$ is harmonic in V . Since $f(x)$ is bounded above, for each $y \in e_1$

$$\limsup_{x \rightarrow y} f_\eta(y) = -\infty.$$

On the other hand, for each $y \in \partial V \cap U$, f is continuous at y and $f(y) = M_1$, as before. Hence

$$\limsup_{x \rightarrow y} f_\eta(x) \leq M_1 + \eta \cdot B.$$

Since each $y \in \partial V$ either belongs to e_1 or to $\partial V \cap U$, the Maximum Principle shows that $f_\eta(x) \leq M_1 + \eta \cdot B$ on V , hence that

$$f(x) \leq M_1 + \eta \cdot (B + h(x)).$$

Fixing x and letting $\eta \rightarrow 0$, we see that $f(x) \leq M_1$ for each $x \in V$. However, this contradicts $f(x_0) > M_1$.

Thus, $f(x) \leq M$ for all $x \in U$, as was to be shown. $\square$

PROPOSITION 5.31. (Riemann Extension Theorem) *Let U be a domain, and let $e \subset U$ be a compact set of capacity 0. Suppose f is harmonic and bounded in $U \setminus e$. Then f extends to a harmonic function on U .*

PROOF. Note that $U \setminus e$ is indeed a domain: it is open since e is closed, and it is connected since a set of capacity 0 is necessarily contained in $\mathbb{P}^1(K)$, and removing Type I points cannot disconnect any connected set.

Include a proof of this fact somewhere.

We claim that f is locally constant in a neighborhood of each point $a \in e$. To see this, given $a \in e$, let $\mathcal{B}$ be an open ball containing a , and whose closure is contained in U (such a ball exists, because $a \in \mathbb{P}^1(K)$), and consider the restriction of f to the domain $V = \mathcal{B} \setminus e$. Since the intersection of ∂V with the complement of e is equal to $\partial \mathcal{B} = \{z\} \subset \mathbb{H}_{\text{Berk}}$, and since a harmonic function is continuous at each point of $\mathbb{H}_{\text{Berk}}$, it follows that

$$\lim_{\substack{x \rightarrow z \\ x \in V}} f(x) = f(z).$$

Applying Proposition 5.29 to f on V , we see that for each $x \in V$, both $f(x) \leq f(z)$ and $f(x) \geq f(z)$. Thus $f(x) = f(z)$ for all $z \in V$.

If we extend f by putting $f(x) = f(z)$ for all $x \in \mathcal{B}$, it is easy to verify that the extended function is harmonic on U , as desired. $\square$

COROLLARY 5.32. *Let $\{a_1, \dots, a_m\}$ be a finite set of points in $\mathbb{P}^1(K)$, or in $\mathbb{P}^1(K) \cap \mathcal{B}(a, r)_{\zeta}^-$ for some disc. Then the only bounded harmonic functions in $\mathbb{P}_{\text{Berk}}^1 \setminus \{a_1, \dots, a_m\}$, or in $\mathcal{B}(a, r)_{\zeta}^- \setminus \{a_1, \dots, a_m\}$, are the constant functions.*

PROOF. We have seen in Lemma 5.27 that if U is $\mathbb{P}_{\text{Berk}}^1$ or an open disc, the only harmonic functions on U are the constant functions. Since a finite set of type I points has capacity 0, Proposition 5.31 shows that a bounded harmonic function on $U \setminus \{a_1, \dots, a_m\}$ extends to a function harmonic on U . $\square$

6. The Poisson Formula

In the classical theory of harmonic functions in the complex plane, if f is harmonic on an open disc D and has a continuous extension to its closure, the Poisson Formula expresses the values of f on D in terms of its values of f on the boundary of D .

In Berkovich space, the basic open neighborhoods are the simple domains. A simple domain has only a finite number of boundary points, and its main dendrite is the interior of a graph Γ . We will now show that every harmonic function f on a simple domain has a continuous extension to its closure, and give a formula for $f(x)$ in terms of its values on the boundary. In one sense, the Berkovich Poisson formula is simpler than the classical one: it is a finite sum, obtained using linear

algebra. However, in another sense, it is more complicated, because it depends on the structure of the graph Γ .

Consider a simple domain $U = \mathcal{B}(a_1, r_1)_\zeta^- \setminus \cup_{i=2}^m \mathcal{B}(a_i, r_i)_\zeta$. Assume $r_1 < \text{diam}_\zeta(\zeta)$, and each $r_i > 0$, so that each disc in the representation of U corresponds to a boundary point $x_i \in \partial U$, namely the point corresponding to the classical disc $B(a_i, r_i)_\zeta$.

If $m = 1$, U is simply the open disc $\mathcal{B}(a_1, r_1)_\zeta^-$, and ∂U is the point x_1 corresponding to $B(a_1, r_1)_\zeta$. By Lemma 5.27 the only harmonic functions on $\mathcal{B}(a_1, r_1)_\zeta^-$ are constant functions, and such functions clearly extend continuously to $\mathcal{B}(a_1, r_1)_\zeta$. Thus if $f(x_1) = A_1$, the Poisson formula is trivially

$$(5.3) \quad f(x) = A_1 \quad \text{for all } x \in \mathcal{B}(a_1, r_1)_\zeta^-.$$

Now assume $m \geq 2$. Let Γ be the tree spanned by $\partial U = \{x_1, \dots, x_m\}$, so $\Gamma^0 := \Gamma \setminus \{x_1, \dots, x_m\}$ is the main dendrite of U . If $f(x)$ is harmonic in U , $\Delta_\Gamma(f)(p) = -\sum_{\bar{v}} d_{\bar{v}} f(p) = 0$ for each $p \in \Gamma^0$. In particular, the restriction of $f(x)$ to each edge of Γ^0 is affine. It follows that $f(x)$ has a continuous extension to $\bar{U}$.

Fix $z \in \mathbb{P}_{\text{Berk}}^1$. We will give a formula for $f(x)$ on U in terms of the values $A_i = f(x_i)$, using the Hsia kernel $\delta(x, y)_z$. By Lemma 5.27, $f(x)$ is determined by its restriction to Γ . Fix $z_1 \in \Gamma$ and let $\bar{z} = r_\Gamma(z)$ be the retraction of z to Γ . As shown in Section 2 of Chapter 3, there are constants C, C_z such that for all $x, y \in \Gamma$,

Give more precise reference.

$$(5.4) \quad \begin{aligned} -\log_v(\delta(x, y)_z) &= j_{z_1}(x, y) - j_{z_1}(x, z) - j_{z_1}(y, z) - C \\ &= (j_{z_1}(x, y) - j_{z_1}(x, \bar{z}) - j_{z_1}(y, \bar{z}) + j_{z_1}(\bar{z}, \bar{z})) \\ &\quad - (C + j_{z_1}(\bar{z}, \bar{z})) \\ &= j_{\bar{z}}(x, y) - C_z \end{aligned}$$

where $C_z = C + j_{z_1}(\bar{z}, \bar{z})$.

Consider functions on Γ of the form

$$f(x) = f_{\bar{c}}(x) = c_0 + \sum_{i=1}^m c_i j_{\bar{z}}(x, x_i)$$

where $\sum_{i=1}^m c_i = 0$. If $f(x) \equiv 0$, then

$$0 = \Delta_\Gamma(f) = \sum_{i=1}^m c_i \delta_{x_i}(x),$$

so $c_1 = \dots = c_m = 0$, and then $0 \equiv f(x) = c_0$ so $c_0 = 0$ as well. Let $L = \{(c_0, c_1, \dots, c_m) \in \mathbb{R}^{m+1} : \sum_{i=1}^m c_i = 0\}$. It follows that the map

$F : L \rightarrow \mathbb{R}^m$ given by

$$F(\vec{c}) = (f_{\vec{c}}(x_1), \dots, f_{\vec{c}}(x_m))$$

is injective. By a dimension count, it is surjective. Let $\tilde{M}$ be the $(m+1) \times (m+1)$ matrix

$$(5.5) \quad \tilde{M} = \begin{pmatrix} 0 & 1 & \cdots & 1 \\ 1 & j_{\bar{z}}(x_1, x_1) & \cdots & j_{\bar{z}}(x_1, x_m) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & j_{\bar{z}}(x_m, x_1) & \cdots & j_{\bar{z}}(x_m, x_m) \end{pmatrix}$$

Then $\tilde{M}$ is nonsingular, and for each $(A_1, \dots, A_m) \in \mathbb{R}^m$, the numbers $c_0, c_1, \dots, c_m$ for which $f_{\vec{c}}(x_i) = A_i$ are uniquely determined by the system of equations

$$(5.6) \quad \begin{pmatrix} 0 \\ A_1 \\ \vdots \\ A_m \end{pmatrix} = \tilde{M} \cdot \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_m \end{pmatrix}.$$

In the matrix $\tilde{M}$, the $j_{\bar{z}}(x_i, x_j)$ can be replaced by $-\log_v(\delta(x_i, x_j)_z)$ since $-\log_v(\delta(x_i, x_j)_z) = j_{\bar{z}}(x_i, x_j) - C_z$ for all i, j and the first row of $\tilde{M}$ asserts that $\sum_{i=1}^m c_i = 0$. Thus, put

$$(5.7) \quad M(z) = \begin{pmatrix} 0 & 1 & \cdots & 1 \\ 1 & -\log_v(\delta(x_1, x_1)_z) & \cdots & -\log_v(\delta(x_1, x_m)_z) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & -\log_v(\delta(x_m, x_1)_z) & \cdots & -\log_v(\delta(x_m, x_m)_z) \end{pmatrix},$$

and for each $i = 0, 1, \dots, m$ let $M_i(z, \vec{A})$ be the matrix gotten by replacing the i^{th} column of $M(z)$ by ${}^t(0, A_1, \dots, A_m)$. The matrix $M(z)$ first appeared in (Cantor, [21]); we will call it the Cantor matrix.

PROPOSITION 5.33. (Poisson Formula I) *Let $U = \mathcal{B}(a_1, r_1)_\zeta^- \setminus \cup_{i=2}^m \mathcal{B}(a_i, r_i)_\zeta$ be a simple domain with boundary points $x_1, \dots, x_m$. Each harmonic function $f(x)$ on U has continuous extension to $\bar{U}$, and there is a unique such function with prescribed boundary values $A_1, \dots, A_m$. It is given as follows.*

Fix $z \in \mathbb{P}_{\text{Berk}}^1$, and put $c_i = \det(M_i(z, \vec{A})) / \det(M(z))$ for $i = 0, 1, \dots, m$. Then

$$(5.8) \quad f(x) = c_0 + \sum_{i=1}^m c_i \cdot (-\log_v(\delta(x, x_i)_z)).$$

(This should be understood as a limit, if z is of type I and $x = z$.)
Moreover

$$\Delta_{\overline{U}}(f) = \sum_{i=1}^m c_i \delta_{x_i}(x) .$$

PROOF. By the discussion above, (5.8) holds for all $x \in \Gamma$. Now let $x \in U$ be arbitrary. If x does not belong to the same branch off Γ as z , it is easy to see from (5.4) that $-\log_v(\delta(x, x_i)_z) = j_{\overline{z}}(r_{\Gamma}(x), x_i) - C_z$, so (5.8) holds for such x . If x belongs to the same branch off Γ as z , then $j_{\overline{z}}(x, x_i)$ is independent of x_i . It follows that $f(x) = c_0$ on that branch, except at possibly when $x = z$. If z is not of type I, then $f(z) = c_0$ by the continuity of $\delta(x, x_i)_z$ at $x = z$. If z is of type I, then (5.8) is undefined at $x = z$, but $\lim_{t \rightarrow z} f(t) = c_0$.

The assertion regarding $\Delta_{\overline{U}}(f)$ follows immediately from (5.8). $\square$

Note that formula (5.8) holds even when $m = 1$, as is easily seen by inspection.

There is another formula for f , which has the merit of being independent of auxiliary variables. Since f is unique, the expression (5.8) is valid for each x and z . Taking $x = z$ shows $f(z) = c_0 = \det(M_0(z, \vec{A})) / \det(M(z))$. Note that by (5.4), (5.5) and (5.7), the determinants are well-defined and finite, and $\det(M(z)) \neq 0$, even when z is of type I.

For each $i = 1, \dots, m$, let $\hat{e}_i \in \mathbb{R}^{m+1}$ be the vector which is 1 in the $(i+1)^{st}$ place and 0 elsewhere, and put

$$h_i(z) = \det(M_0(z, \hat{e}_i)) / \det(M(z)) .$$

Then $h_i(z)$ is harmonic in U and continuous on $\overline{U}$. It takes the value 1 at x_i , and is 0 at each x_j with $j \neq i$. It is the analogue of the classical *harmonic measure* for the boundary component x_i of U . By the maximum principle, $0 \leq h_i(z) \leq 1$ for each $z \in \overline{U}$, and

$$(5.9) \quad \sum_{i=1}^m h_i(z) \equiv 1 .$$

In sum, we have

PROPOSITION 5.34. (Poisson Formula II) *Let $U = \mathcal{B}(a_1, r_1)_{\zeta}^- \setminus \cup_{i=2}^m \mathcal{B}(a_i, r_i)_{\zeta}$ be a simple domain with boundary points $x_1, \dots, x_m$. Let $A_1, \dots, A_m \in \mathbb{R}$ be given. Then the unique function $f(z)$ which is harmonic on U , continuous on $\overline{U}$, and satisfies $f(x_i) = A_i$ for each*

$i = 1, \dots, m$ is given by

$$(5.10) \quad f(z) = \det(M_0(z, \vec{A})) / \det(M(z))$$

$$(5.11) \quad = \sum_{i=1}^m A_i \cdot h_i(z) ,$$

for all $z \in \bar{U}$, where $h_i(z) = \det(M_0(z, \hat{e}_i)) / \det(M(z))$ is the harmonic measure for $x_i \in \partial U$.

7. Uniform Convergence

Poisson's formula implies that the limit of a sequence of harmonic functions is harmonic, under a much weaker condition than is required classically.

PROPOSITION 5.35. *Let V be an open subset of $\mathbb{P}_{\text{Berk}}^1$. Suppose $f_1, f_2, \dots$ are harmonic in V and converge pointwise to a function $f : V \rightarrow \mathbb{R}$. Then $f(z)$ is harmonic in V , and the $f_i(z)$ converge uniformly to $f(z)$ on compact subsets of V .*

PROOF. Given $x \in V$, take a simple domain U_x containing x , with closure $\bar{U}_x \subset V$. If $\partial U_x = \{x_1, \dots, x_m\}$ then by Proposition 5.34

$$f_k(z) = \sum_{i=1}^m f_k(x_i) h_i(z)$$

for each $z \in \bar{U}_x$. It follows that the $f_k(z)$ converge uniformly to $f(z)$ on $\bar{U}_x$, and $f(z) = \sum_{i=1}^m f(x_i) h_i(z)$ is strongly harmonic in U_x .

Thus f is harmonic in V . Any compact $E \subset V$ is covered by finitely many sets U_x , so the $f_i(z)$ converge uniformly to $f(z)$ on E . $\square$

Using this, we can characterize harmonic functions as local uniform limits of logarithms of norms of rational functions.

COROLLARY 5.36. *If $U \subset \mathbb{P}_{\text{Berk}}^1$ is a domain and f is harmonic in U , there are rational functions $g_1(T), g_2(T), \dots \in K(T)$ and numbers $R_1, R_2, \dots \in \mathbb{Q}$ such that*

$$f(x) = \lim_{k \rightarrow \infty} R_k \cdot \log_v([g_k]_x)$$

uniformly on compact subsets of U .

PROOF. The assertion is trivial if the main denrite of U is empty, since the only harmonic functions on U are constants.

Hence we can assume the main dendrite is nonempty. Choose an exhaustion of U by simple domains $\{U_k\}$ with closures $\bar{U}_k \subset U$. (For example,

Need to use nets here rather than sequences.

such an exhaustion can be gotten by exhausting the main dendrite D by subgraphs Γ_k , and putting $U_k = r_{\Gamma_k}^{-1}(\Gamma_{k,0})$.

Fix k , and write $\partial U_k = \{x_{k,1}, \dots, x_{k,m_k}\}$. Taking $z = \infty$ in Proposition 5.33, there are numbers $c_{k,i} \in \mathbb{R}$, with $\sum_{i=1}^{m_k} c_{k,i} = 0$, such that on U_k

$$f(x) = c_{k,0} - \sum_{i=1}^{m_k} c_{k,i} \log_v(\delta(x, x_{k,i})_\infty) .$$

For each $i = 1, \dots, m_k$ fix a type I point $a_{k,i}$ whose retraction to $\bar{U}_k$ is $x_{k,i}$. Then $\delta(x, x_{k,i})_\infty = \delta(x, a_{k,i})_\infty = [T - a_{k,i}]_x$ for each $x \in \bar{U}_k$.

Choose rational numbers $d_{k,i}$, with $\sum_{i=1}^{m_k} d_{k,i} = 0$, which are close enough to the $c_{k,i}$ that

$$f_k(x) := d_{k,0} - \sum_{i=1}^{m_k} d_{k,i} \log_v(\delta(x, a_{k,i})_\infty)$$

satisfies $|f_k(x) - f(x)| < 1/k$ on $\bar{U}_k$. Let N_k be a common denominator for the $d_{k,i}$, and put $n_{k,i} = N_k \cdot d_{k,i}$. Let $b_k \in K$ be a constant with $|b_k| = q_v^{-n_{k,0}}$, and put

$$(5.12) \quad g_k(T) = b_k \cdot \prod_{i=1}^{m_k} (T - a_{k,i})^{n_{k,i}} .$$

Then $f_k(x) = -\frac{1}{N_k} \log_v([g_k]_x)$ on U_k . The result follows. $\square$

8. Pullbacks

Harmonicities are preserved under pullbacks by meromorphic functions. To show this, we first need a lemma asserting that logarithms of norms of meromorphic functions are harmonic.

LEMMA 5.37. *Let U be a domain, and suppose $g(x)$ is meromorphic in U . Then $f(x) := \log_v([g]_x)$ is harmonic in $U \setminus \text{supp}(\text{div}(g))$.*

PROOF. Choose a covering of U by simple domains U_k with $\bar{U}_k \subset U$.

Let $\mathcal{A}_k$ be a Tate algebra for which $\bar{U}_k = \mathcal{M}(\mathcal{A}_k)$. For each k , there are coprime polynomials $P_k(z)$ and $Q_k(z)$, and a unit power series $u_k(z) \in \mathcal{A}_k$, such that $g(z) = (P_k(z)/Q_k(z)) \cdot u_k(z)$ on $\bar{U}_k \cap K$. A unit power series satisfies $[u_k]_x = 1$ for all x , so

$$\log_v([g]_x) = \log_v([P_k]_x) - \log_v([Q_k]_x)$$

is strongly harmonic on $U_k \setminus (\text{div}(P_k/Q_k))$. $\square$

COROLLARY 5.38. *Let U and V be domains in $\mathbb{P}_{\text{Berk}}^1$. Suppose $F(z)$ is meromorphic in U , with $F(U) \subset V$. If f is harmonic in V , then $f \circ F$ is harmonic in U .*

PROOF. Let x be the variable on U , and y the variable on V . By Lemma 5.37 there are rational functions g_k and numbers R_k such that

$$\lim_{k \rightarrow \infty} R_k \cdot \log_v([g_k]_y) = f(y)$$

uniformly on compact subsets of V . For each $y \in V$, let V_y be a neighborhood of y with $\overline{V}_y \subset V$; since f is harmonic there is a constant C_y so that g_k has no zeros or poles in $\overline{V}_y$ if $k \geq C_y$.

Suppose $x \in U$ satisfies $F(x) = y$. Choose a simple neighborhood U_x of x contained in $F^{-1}(V_y)$. The functions $g_k \circ F$ are meromorphic in U_x , and if $k \geq C_y$ they have no zeros or poles in U_x . The functions $R_k \cdot \log_v([g_k \circ F]_z)$ converge to $f \circ F(z)$ for each $z \in U_x$.

By Corollary 5.35, $f \circ F$ is harmonic on U_x . By Proposition 5.33, $f \circ F$ is strongly harmonic on U_x . Since $x \in U$ is arbitrary, $f \circ F$ is harmonic on U . $\square$

9. Harnack's Principle

Harnack's principle holds as well:

PROPOSITION 5.39. (Harnack's Principle) *Let U be a domain, and suppose $f_1, f_2, \dots$ are harmonic in U with $0 \leq f_1 \leq f_2 \leq \dots$. Then either*

- A) $\lim_{i \rightarrow \infty} f_i(z) = \infty$ for each $z \in U$, or
- B) $f(z) = \lim_{i \rightarrow \infty} f_i(z)$ is finite for all z , the $f_i(z)$ converge uniformly to $f(z)$ on compact subsets of U , and $f(z)$ is harmonic in U .

PROOF. If the main dendrite D for U is empty, then by Lemma 5.27 each $f_i(z)$ is constant, and our assertions are trivial. Hence we can assume D is nonempty.

Suppose there is some $x \in U$ for which $\lim_{i \rightarrow \infty} f_i(x)$ is finite. Since each f_i is constant on branches off the main dendrite, there is a point x_0 on D where $\lim_{i \rightarrow \infty} f_i(x_0)$ is finite.

Let $\rho(x, y)$ be the logarithmic path distance on $\mathbb{P}_{\text{Berk}}^1$. Since the main dendrite is everywhere finitely branched, for each $p \in D$ there is an $\varepsilon > 0$ such that the closed neighborhood of p in D defined by $\Gamma(p, \varepsilon) = \{x \in D : \rho(x, p) \leq \varepsilon\}$ is a star, e.g. is a union of n closed segments of length ε emanating from p , for some n . Write $q_i = p + \varepsilon \vec{v}_i$, $i = 1, \dots, n$, for the endpoints of $\Gamma(p, \varepsilon)$.

Suppose $h(z)$ is harmonic and non-negative in U . The restriction of $h(z)$ to the subgraph $\Gamma(p, \varepsilon)$ is linear on each of the segments $[p, q_i]$,

and $\Delta_\Gamma(h)(p) = 0$. Thus for each i , and $z = p + t\vec{v}_i \in [p, q_i]$

$$0 = \Delta_\Gamma(h)(p) = \frac{h(z) - h(p)}{t} + \sum_{j \neq i} \frac{h(q_j) - h(p)}{\varepsilon}.$$

Since $h(q_j) \geq 0$ for each j , it follows that $h(x) \leq n \cdot h(p)$. This holds for each $z \in \Gamma(p, \varepsilon)$. Let $\Gamma \subset D$ be an arbitrary subgraph containing x_0 . Proceeding stepwise from x_0 , it follows that there is a constant $C = C_\Gamma$ such that for each $x \in \Gamma$,

$$0 \leq h(x) \leq C_\Gamma \cdot h(x_0).$$

Applying this to the functions $f_i(x)$, since $f(x_0) = \lim_{i \rightarrow \infty} f_i(x_0)$ is finite, the $f_i(x)$ are uniformly bounded on each subgraph $\Gamma \subset D$ containing x_0 . Hence the $f_i(x)$ converge uniformly to $f(x)$ on Γ . If E is any compact subset of U , then the image of E under the retraction to the main dendrite is contained in some subgraph Γ . Hence the $f_i(x)$ converge locally uniformly to $f(x)$ on U , and $f(x)$ is harmonic by Proposition 5.35. $\square$

10. Green's functions

We will now prove the uniqueness of the equilibrium distribution, as promised in Chapter 4, and use this to build a theory of Green's functions.

PROPOSITION 5.40. *Let $E \subset \mathbb{P}_{\text{Berk}}^1$ be a compact set with positive capacity, and let $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus E$. Then the equilibrium distribution μ_ζ of E with respect to ζ is unique.*

PROOF. Suppose μ_1 and μ_2 are two equilibrium distributions for E with respect to ζ , so that

$$I_\zeta(\mu_1) = I_\zeta(\mu_2) = V_\zeta(E).$$

Let $u_1(x) = u_{\mu_1}(x, \zeta)$ and $u_2(x) = u_{\mu_2}(x, \zeta)$ be the corresponding potential functions, and let U be the connected component of $\mathbb{P}_{\text{Berk}}^1 \setminus E$ containing ζ .

Both $u_1(x)$ and $u_2(x)$ satisfy Frostman's Theorem (Theorem 4.16): each is bounded above by $V_\zeta(E)$ for all z , each is equal to $V_\zeta(E)$ for all $z \in E$ except possibly on an F_σ set f_i of capacity 0, and each is continuous on E except on f_i . Each is bounded in a neighborhood of E .

Consider $u(z) = u_1(z) - u_2(z)$. By Example 5.14, $u(z)$ is in $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(u) = (\mu_1 - \delta_\zeta(x)) - (\mu_2 - \delta_\zeta(x)) = \mu_1 - \mu_2.$$

By Proposition 4.9, μ_1 and μ_2 are both supported on $\partial E_\zeta = \partial \bar{U} \subset \bar{U}$. Since the retraction map $r_{\mathbb{P}_{\text{Berk}}^1, \bar{U}}$ fixes $\bar{U}$, it follows that

$$(5.13) \quad \Delta_{\bar{U}}(u) = (r_{\mathbb{P}_{\text{Berk}}^1, \bar{U}})_*(\Delta_{\mathbb{P}_{\text{Berk}}^1}(u)) = \mu_1 - \mu_2.$$

Thus $\Delta_{\bar{U}}(u)$ is supported on $\partial \bar{U}$.

Suppose $\zeta \notin \mathbb{P}^1(K)$. By Proposition 4.9, each $u_i(z)$ is defined and bounded on all of $\mathbb{P}_{\text{Berk}}^1$, and is continuous on U . Hence $u(z)$ has these properties as well. By (5.13) $u(z)$ is strongly harmonic on U . Put $f = f_1 \cup f_2$, which has capacity 0 by Corollary 4.19. By the discussion above, for each $x \in \partial U \setminus f$, $u(z)$ is continuous at x and $u(x) = 0$. In particular

$$(5.14) \quad \lim_{\substack{z \rightarrow x \\ z \in U}} u(z) = 0.$$

By the Strong Maximum Principle (Proposition 5.29), $u(z) \equiv 0$ on U . Hence $\Delta_{\bar{U}}(u) = 0$, so $\mu_1 = \mu_2$.

If $\zeta \in \mathbb{P}^1(K)$, there is a slight complication because $u_1(\zeta) - u_2(\zeta) = \infty - \infty$ is undefined. However, we claim that $u(z)$ is constant, and in fact is identically 0, in a deleted neighborhood of ζ . Assuming this, since Proposition 4.9 shows that each $u_i(z)$ is bounded in a neighborhood of E and is continuous in $U \setminus \{\zeta\}$, it follows that $u(z)$ is harmonic and bounded in $U \setminus \{\zeta\}$. The boundary limits (5.14) continue to hold for all $x \in \partial U \setminus f$, and in addition

$$(5.15) \quad \lim_{\substack{z \rightarrow \zeta \\ z \in U}} u(z) = 0.$$

Applying the Strong Maximum Principle to $u(z)$ on the domain $U \setminus \{\zeta\}$, relative to the exceptional set $f \cup \{\zeta\}$ contained in its boundary $\partial U \cup \{\zeta\}$, we conclude $u(z) \equiv 0$ in $U \setminus \{\zeta\}$. Since the closure of $U \setminus \{\zeta\}$ is $\bar{U}$, it follows as before that $\Delta_{\bar{U}}(u) = 0$, and $\mu_1 = \mu_2$.

To establish our claim, fix $a \in \mathbb{P}^1(K)$ with $a \neq \zeta$ and note that the balls $\mathcal{B}(a, R)_\zeta^-$ exhaust $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. Since E is compact there is an R with $E \subset \mathcal{B}(a, R)_\zeta$. For each $z \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathcal{B}(a, R)_\zeta$ and each $w \in \mathcal{B}(a, R)_\zeta$, the ultrametric inequality for $\delta(x, y)_\zeta$ shows that $\delta(z, w)_\zeta = \delta(z, a)_\zeta$. Since μ_1 and μ_2 are both supported on $E \subset \mathcal{B}(a, R)_\zeta$, for $z \notin \mathcal{B}(a, R)_\zeta$, $z \neq \zeta$, we have

$$u_i(z) = \int -\log_v(\delta(z, w)_\zeta) d\mu_i(w) = -\log_v(\delta(z, a)_\zeta).$$

Hence $u(z) \equiv 0$ in $\mathbb{P}_{\text{Berk}}^1 \setminus (\mathcal{B}(a, R)_\zeta \cup \{\zeta\})$. □

Since the equilibrium distribution μ_ζ is unique, the potential function $u_E(z, \zeta) = u_{\mu_\zeta}(z, \zeta)$ is well-defined.

DEFINITION 5.41. If $E \subset \mathbb{P}_{\text{Berk}}^1$ is a compact set of positive capacity, then for each $\zeta \notin E$, the Green's function of E is

$$G(z, \zeta; E) = V_\zeta(E) - u_E(z, \zeta) \quad \text{for all } z \in \mathbb{P}_{\text{Berk}}^1.$$

If $U \subset \mathbb{P}_{\text{Berk}}^1$ is a domain for which ∂U has positive capacity, the Green's function of U is

$$G_U(z, \zeta) = G(z, \zeta; \mathbb{P}_{\text{Berk}}^1 \setminus U) \quad \text{for } z, \zeta \in U.$$

PROPOSITION 5.42. *Let $E \subset \mathbb{P}_{\text{Berk}}^1$ be a compact set of positive capacity. Then $G(z, \zeta; E)$ has the following properties:*

A) *For each fixed $\zeta \notin E$, as a function of z ,*

(1) $G(z, \zeta; E) \geq 0$ *for all $z \in \mathbb{P}_{\text{Berk}}^1$.*

(2) $G(z, \zeta; E) > 0$ *for $z \in U_\zeta$, where U_ζ is the connected component of $\mathbb{P}_{\text{Berk}}^1 \setminus E$ containing ζ .*

(3) $G(z, \zeta; E)$ *is finite on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, and harmonic in $U_\zeta \setminus \{\zeta\}$. For each $a \neq \zeta$, $G(z, \zeta; E) + \log_v(\delta(z, a)_\zeta)$ extends to a function harmonic on a neighborhood of ζ .*

(4) $G(z, \zeta; E) = 0$ *on $\mathbb{P}_{\text{Berk}}^1 \setminus U_\zeta$, except on a (possibly empty) F_σ set $e \subset \partial U_\zeta$ of capacity 0 which depends only on U_ζ .*

(5) $G(z, \zeta; E)$ *is continuous on $\mathbb{P}_{\text{Berk}}^1 \setminus e$, and is upper semicontinuous everywhere.*

(6) $G(z, \zeta; E)$ *is bounded if $\zeta \notin \mathbb{P}^1(K)$.*

It is unbounded, with $G(\zeta, \zeta; E) = +\infty$, if $\zeta \in \mathbb{P}^1(K)$.

(7) $G(z, \zeta; E) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, *with $\Delta_{\mathbb{P}_{\text{Berk}}^1}(G(z, \zeta; E)) = \delta_\zeta(z) - \mu_\zeta(z)$.*

B) *As a function of two variables, $G(z_1, z_2; E) = G(z_2, z_1; E)$ for all $z_1, z_2 \notin E$.*

PROOF. For (A), part (1) follows from Frostman's theorem (Theorem 4.16). Parts (4) and (5) (except for the fact that e is independent of ζ , which we will prove later) follow from Proposition 4.12 and Frostman's theorem. Part (6) follows from the definition of the potential function $u_{\mu_\zeta}(z, \zeta)$ and properties of the Hsia kernel. These same facts show $G(\zeta, \zeta; E) > 0$. Part (7) is a reformulation of Example 5.14. For the first part of (3), we know that $G(z, \zeta; E)$ is finite and harmonic in $U_\zeta \setminus \{\zeta\}$ by Example 5.22. For the second part of (3), fix $a \neq \zeta$ and consider the function $f(z) = G(z, \zeta; E) + \log_v(\delta(z, a)_\zeta)$. By Examples 5.12 and 5.14, it belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and satisfies

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(f) = \delta_a(z) - \mu_\zeta.$$

Let V be a connected neighborhood of ζ with $\overline{V} \cap (E \cup \{a\}) = \emptyset$. If $\zeta \notin \mathbb{P}^1(K)$ then $f(z)$ is defined everywhere, and is strongly harmonic

in V . If $\zeta \in \mathbb{P}^1(K)$ then $f(z)$ is harmonic and bounded in $V \setminus \{\zeta\}$, so it extends to a function strongly harmonic in V by the Riemann Extension Theorem (Proposition 5.31). For (2), if $G(x, \zeta; E) = 0$ for some $x \in U_\zeta \setminus \{\zeta\}$, let W be the component of $U_\zeta \setminus \{\zeta\}$ containing x . Then $G(z, \zeta; E)$ would achieve its minimum value 0 at an interior point of W , and so would be identically 0 on W by the Maximum Principle (Proposition 5.28). Thus

$$\lim_{\substack{z \rightarrow \zeta \\ z \in W}} G(z, \zeta; E) = 0$$

which contradicts the fact that $G(\zeta, \zeta; E) > 0$ and $G(z, \zeta; E)$ is continuous at ζ .

To see that the set e in part (4) is independent of ζ , temporarily write e_ζ for it, and take $\xi \neq \zeta$ in U_ζ . Let V be a neighborhood of ζ such that $\bar{V} \subset U_\zeta$ and $\xi \notin \bar{V}$. Since $\partial V \subset U_\zeta \setminus \{\zeta, \xi\}$ there is a constant $C > 0$ such that $G(z, \xi; E) > C \cdot G(z, \zeta; E)$ on ∂V . Let W be a connected component of $U_\zeta \setminus \bar{V}$. Then $\partial W \subset \partial U_\zeta \cup \partial V$. Consider $f(z) = G(z, \xi; E) - C \cdot G(z, \zeta; E)$, which is harmonic on W . For each $x \in \partial W \cap \partial V$

$$\lim_{\substack{z \rightarrow x \\ z \in W}} f(z) > 0,$$

while for each $x \in \partial W \cap (\partial U_\zeta \setminus e_\zeta)$

$$\liminf_{\substack{z \rightarrow x \\ z \in W}} f(z) \geq 0.$$

By the Strong Maximum Principle (Proposition 5.29), $f(z) \geq 0$ for all $z \in W$. Hence, if $x \in \partial W \cap e_\zeta$, then

$$\liminf_{\substack{z \rightarrow x \\ z \in W}} G(z, \xi; E) \geq C \cdot \liminf_{\substack{z \rightarrow x \\ z \in W}} G(z, \zeta; E) > 0.$$

Since each $x \in e_\zeta$ belongs to ∂W for some W , it follows that $e_\zeta \subset e_\xi$. By symmetry, $e_\zeta = e_\xi$.

For part (B), we can assume without loss of generality that z_1, z_2 belong to the same connected component U of $\mathbb{P}_{\text{Berk}}^1 \setminus E$, otherwise by (4) we have $G(z_1, z_2; E) = G(z_2, z_1; E) = 0$.

First, suppose both z_1 and z_2 are in $\mathbb{H}_{\text{Berk}}$. Let $\mathcal{S}_{\bar{U}}$ be a directed set indexing the subgraphs of $\bar{U}$, and let Γ_α be the subgraph of $\bar{U}$ corresponding to $\alpha \in \mathcal{S}_{\bar{U}}$. For $\alpha \in \mathcal{S}_{\bar{U}}$, let $r_\alpha(x) := r_{\bar{U}, \Gamma_\alpha}(x) : \bar{U} \rightarrow \Gamma_\alpha$ be the retraction map. Then $\lim_\alpha r_\alpha(z) = z$ for all $z \in \bar{U}$.

Put $g_1(x) = G(x, z_1; E)$, $g_2(x) = G(x, z_2; E)$, and write $\mu_1 = \mu_{z_1}$, $\mu_2 = \mu_{z_2}$. Then $g_1, g_2 \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ by (7), and by (5) we have $g_1, g_2 \in \mathcal{C}(\Gamma_\alpha)$ for each $\alpha \in \mathcal{S}_{\bar{U}}$, since $z_1, z_2 \notin \mathbb{P}^1(K)$.

For each subgraph Γ_α of $\overline{U}$, we have $g_1|_{\Gamma_\alpha}, g_2|_{\Gamma_\alpha} \in \text{CPA}(\Gamma_\alpha)$ since each $g_i(x)$ is harmonic in $U \setminus \{z_i\}$. By Proposition C.2(D),

$$(5.16) \quad \int_{\Gamma_\alpha} g_1 \Delta_{\Gamma_\alpha}(g_2) = \int_{\Gamma_\alpha} g_2 \Delta_{\Gamma_\alpha}(g_1) .$$

Assume α is chosen so that $z_1, z_2 \in \Gamma_\alpha$. Then for $\{i, j\} = \{1, 2\}$, we have

$$\Delta_{\Gamma_\alpha}(g_i) = r_\alpha(z_i) - (r_\alpha)_*(\mu_i)$$

and

$$\int_{\Gamma_\alpha} g_i d(r_\alpha)_*(\mu_j) = \int_{\overline{U}} g_i \circ r_\alpha d\mu_j .$$

Since μ_j is supported on ∂U , it follows that

$$(5.17) \quad g_1(z_2) - \int_{\partial U} g_1(r_\alpha(x)) d\mu_2(x) = g_2(z_1) - \int_{\partial U} g_2(r_\alpha(x)) d\mu_1(x) .$$

We claim that for each $\varepsilon > 0$, there exists $\alpha \in \mathcal{S}_{\overline{U}}$ such that

$$\left| \int_{\partial U} g_i(r_\alpha(x)) d\mu_j(x) \right| < \varepsilon/2 .$$

Assuming the claim, it follows from (5.17) that $|g_1(z_2) - g_2(z_1)| < \varepsilon$ for every $\varepsilon > 0$, and thus that $G(z_1, z_2; E) = G(z_2, z_1; E)$ as desired.

To prove the claim, let e be the exceptional set of capacity 0 in ∂U given in (4). By Lemma 4.14, $\mu_j(e) = 0$. Recall that $g_i(x) = G(x, z_i; E)$ is bounded on $E \supseteq \partial U$, say by $M > 0$. By the regularity of μ_j , there exists an open set $\tilde{e}$ containing e such that $\mu_j(\tilde{e}) < \varepsilon/4M$. Since $\lim_\alpha r_\alpha(x) = x$ and $g_i \circ r_\alpha$ is continuous for all α , for each $x \in \partial U \setminus e$ we have

$$\lim_\alpha g_i(r_\alpha(x)) = g_i(x) = 0 .$$

Since $\partial U \setminus \tilde{e}$ is compact, there exists $\alpha_0 \in \mathcal{S}_{\overline{U}}$ such that if $\alpha \geq \alpha_0$, then $|g_i(r_\alpha(x))| < \varepsilon/4$ for all $x \in \partial U \setminus \tilde{e}$. For $\alpha \geq \alpha_0$, we therefore have

$$\begin{aligned} \left| \int_{\partial U} g_i(r_\alpha(x)) d\mu_j(x) \right| &\leq \sup_{x \in \partial U \setminus \tilde{e}} |g_i(r_\alpha(x))| + M \cdot \mu_j(\tilde{e}) \\ &< \varepsilon/4 + M \cdot \varepsilon/4M = \varepsilon/2 \end{aligned}$$

as claimed.

Now suppose $z_1 \in U$ is of type I, but $z_2 \in U$ is not of type I. Let $t \rightarrow z_1$ along a path in $\mathbb{H}_{\text{Berk}}$. Since $G(x, z_2; E)$ is continuous for $x \in U$,

$$G(z_1, z_2; E) = \lim_{t \rightarrow z_1} G(t, z_2; E) = \lim_{t \rightarrow z_1} G(z_2, t; E) .$$

We claim that $\lim_{t \rightarrow z_1} G(z_2, t; E) = G(z_2, z_1; E)$. To see this, fix a point z in the main dendrite D for U . There is a constant C_t such that

$$-\log_v(\delta(x, y)_t) = j_z(x, y) - j_z(x, t) - j_z(y, t) + C_t$$

for all $x, y \in \mathbb{P}_{\text{Berk}}^1$. Let w be the point where the path from z_1 to z first meets D . If t lies on the path from z_1 to w , then for all $x, y \in E$, $j_z(x, t) = j_z(x, w)$ and $j_z(y, t) = j_z(y, w)$. In other words, for $x, y \in E$, $\log_v(\delta(x, y)_t)$ depends on t only through the constant C_t . Hence the same probability measure μ minimizes the energy integral

$$V_t(E) = \iint_{E \times E} -\log_v(\delta(x, y)_t) d\mu(x) d\mu(y)$$

for all t on the path from w to z_1 , that is, $\mu_t = \mu_{z_1} = \mu$.

Similarly, $-\log_v(\delta(z_2, y)_t)$ depends on t only through C_t , if $y \in E$ and t is sufficiently near z_1 . Hence

$$G(z_2, t; E) = V_t(E) - \int_E -\log_v(\delta(z_2, w)_t) d\mu(w)$$

is independent of t , for t sufficiently near z_1 . Thus $\lim_{t \rightarrow z_1} G(z_2, t; E) = G(z_2, z_1; E)$.

Finally, let $z_1, z_2 \in U$ be arbitrary. Let t approach z_1 through points of type I. Using continuity and an argument like the one above we find

$$\begin{aligned} G(z_1, z_2; E) &= \lim_{t \rightarrow z_1} G(t, z_2; E) \\ &= \lim_{t \rightarrow z_1} G(z_2, t; E) = G(z_2, z_1; E). \end{aligned}$$

□

We will now show that the equilibrium measure μ_ζ has an interpretation as the reproducing kernel for harmonic functions:

PROPOSITION 5.43. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain such that ∂U has positive capacity. Suppose f is harmonic in U and extends to a continuous function on $\bar{U}$. For each $\zeta \in U$, let μ_ζ be the equilibrium measure on ∂U relative to ζ . Then*

$$f(\zeta) = \int_{\partial U} f(z) d\mu_\zeta(z).$$

PROOF. First suppose $\zeta \in U$ is not of type I. Choose an exhaustion on U by a net of simple subdomains $V_\alpha \subset U$. Without loss of generality, we can assume $\zeta \in V_\alpha$ for all α . Put $g_\alpha(x) = G(x, \zeta; V_\alpha)$. Since ∂V_α consists of a finite number of points, none of which is of type I, $g_\alpha(x)$ is continuous on $\bar{V}_\alpha$ and $g_\alpha(x) = 0$ for each $x \in \partial V_\alpha$. Note that $g_\alpha(x) \in$

$\text{BDV}(V_\alpha)$, $g_\alpha \in \mathcal{C}(\overline{V}_\alpha)$ for each α , and f is continuous on $\overline{V}_\alpha$ and strongly harmonic in V_α by Lemma 5.23; in particular $f \in \text{BDV}(V_\alpha)$.

Fix α , and let $\Gamma \subset V_\alpha$ be a subgraph. Then $f|_\Gamma, g|_\Gamma \in \text{CPA}(\Gamma)$. By Proposition C.2(D),

$$(5.18) \quad \int_\Gamma f \Delta_\Gamma(g_\alpha) = \int_\Gamma g_\alpha \Delta_\Gamma(f) .$$

Taking a limit over subgraphs Γ , we find that

$$(5.19) \quad \int_{\overline{V}_\alpha} f \Delta_{\overline{V}_\alpha}(g_\alpha) = \int_{\overline{V}_\alpha} g_\alpha \Delta_{\overline{V}_\alpha}(f) .$$

Here $\Delta_{\overline{V}_\alpha}(g_\alpha) = \delta_\zeta(x) - \mu_{\zeta,\alpha}$ where $\mu_{\zeta,\alpha}$ is the equilibrium measure of $E_\alpha = \mathbb{P}_{\text{Berk}}^1 \setminus V_\alpha$ with respect to ζ . It is supported on ∂V_α . Since f is strongly harmonic on V_α , $\Delta_{\overline{V}_\alpha}(f)$ is supported on ∂V_α , where $g_\alpha(x) = 0$. Hence the integral on the right side of (5.19) is 0. It follows that

$$(5.20) \quad f(\zeta) = \int f(x) d\mu_{\zeta,\alpha}(x) .$$

By Corollary 4.11 (applied to $W_\alpha = \mathbb{P}_{\text{Berk}}^1 \setminus V_\alpha$), the net $\langle \mu_{\zeta,\alpha} \rangle$ converges weakly to μ_ζ , the equilibrium measure of $E = \mathbb{P}_{\text{Berk}}^1 \setminus U$. Since $f(x)$ is continuous on $\overline{U}$,

$$f(\zeta) = \lim_\alpha \int f(x) d\mu_{\zeta,\alpha}(x) = \int f(x) d\mu_\zeta(x) ,$$

yielding the result in this case.

If ζ is of type I, let t approach ζ along a path in $\mathbb{H}_{\text{Berk}}$. As in the proof of Proposition 5.42(B), $\mu_\zeta = \mu_t$ for t is sufficiently near ζ . Since f is continuous, it follows from the previous case that $f(\zeta) = \int_{\partial U} f(x) d\mu_\zeta(x)$. $\square$

In the classical theory over $\mathbb{C}$, the reproducing kernel is the inward normal derivative of $G_U(z, \zeta; E)$ on ∂E_ζ ,

$$\frac{1}{2\pi} \frac{\partial}{\partial n} G_U(z, \zeta) .$$

On Berkovich space, if we take $E = \mathbb{P}_{\text{Berk}}^1 \setminus U$, then part (A7) of Proposition 5.42 suggests that $\mu_\zeta = \Delta(G(z, \zeta; E))|_{\partial U}$ should be identified with the normal derivative of $G(z, \zeta; E)$ on $\partial E_\zeta = \partial U$.

11. Notes and further references

CHAPTER 6

Subharmonic functions

In this section we develop a theory of subharmonic functions on the Berkovich projective line.

We first recall some terminology. A *domain* in $\mathbb{P}_{\text{Berk}}^1$ is a connected open subset of $\mathbb{P}_{\text{Berk}}^1$. A domain $V \subset \mathbb{P}_{\text{Berk}}^1$ is called *simple* if V is an open disc or punctured disc, and is not $\mathbb{P}_{\text{Berk}}^1$ itself. Thus V is a simple domain if and only if V is connected and ∂V is a nonempty finite set $\{x_1, \dots, x_m\}$, where each $x_i \in \mathbb{H}_{\text{Berk}}$.

Make sure this is all defined earlier.

DEFINITION 6.1. If $U \subset \mathbb{P}_{\text{Berk}}^1$ is open, then $V \subset U$ is a *simple subdomain* of U if V is a simple domain and $\bar{V} \subset U$.

1. Subharmonic and strongly subharmonic functions

We make the following definitions, by analogy with (5.18) and the complex case.

DEFINITION 6.2. Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain.

A function $f : U \rightarrow [-\infty, \infty)$ is *strongly subharmonic* on U if:

(A) f is upper semicontinuous on U , and for each $x \in U \cap \mathbb{P}^1(K)$,

$$f(x) = \limsup_{\substack{z \rightarrow x \\ z \in U \cap \mathbb{H}_{\text{Berk}}}} f(z) ;$$

(B) $f \in \text{BDV}(U)$ and $\Delta_{\bar{U}}(f)|_U \leq 0$.

If V is an arbitrary open set, $f : V \rightarrow [-\infty, \infty)$ is *subharmonic* on V if for each $x \in V$ there is a connected open neighborhood U_x of x on which f is strongly subharmonic.

It is clear from the definition that subharmonicity is a local property.

REMARK 6.3. The condition in (A) controlling f on $U \cap \mathbb{P}^1(K)$ is necessary, as shown by the following example. Take $U = \mathcal{B}(0, 1)^-$, and put $f(z) = 0$ on $U \setminus \mathbb{Z}_p$, $f(z) = 1$ on $\mathbb{Z}_p$. Then $f(z)$ is upper semicontinuous on U , since for each $x \in U \setminus \mathbb{Z}_p$ there is a neighborhood V of x with $V \cap \mathbb{Z}_p = \emptyset$. It also belongs to $\text{BDV}(U)$, with $\Delta_{\bar{U}}(f) \equiv 0$. However, $f(z)$ should not be considered as subharmonic on U , since, for

example, it does not satisfy the maximum principle (Proposition 6.10 below).

A function $f : U \rightarrow \mathbb{R} \cup \{\infty\}$ will be called *strongly superharmonic* on U if $-f$ is strongly subharmonic. It will be called *superharmonic* if $-f$ is subharmonic.

Here are some examples of subharmonic and superharmonic functions.

EXAMPLE 6.4. A function $f : U \rightarrow \mathbb{R}$ is harmonic in an open set $U \subset \mathbb{P}_{\text{Berk}}^1$ if and only if it is both subharmonic and superharmonic.

The only functions which are subharmonic on all of $\mathbb{P}_{\text{Berk}}^1$ are the constant functions. Indeed, a function $f(x)$ subharmonic on all of $\mathbb{P}_{\text{Berk}}^1$ is strongly subharmonic by Lemma 6.9, hence $\Delta_{\mathbb{P}_{\text{Berk}}^1}(f) \leq 0$. Since the total mass of the Laplacian is 0, this means $\Delta_{\mathbb{P}_{\text{Berk}}^1}(f) = 0$. Thus, f is harmonic on $\mathbb{P}_{\text{Berk}}^1$ and so is constant.

EXAMPLE 6.5. For fixed $a, \zeta \in \mathbb{P}_{\text{Berk}}^1$ with $a \neq \zeta$, $f(x) = \log_v(\delta(x, a)_\zeta)$ is strongly subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, and strongly superharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{a\}$. Indeed, $\delta(x, a)_\zeta$ is continuous by Proposition 3.10, and

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(\log_v(\delta(x, a)_\zeta)) = \delta_\zeta(x) - \delta_a(x)$$

by Example 5.12.

Correspondingly, $-\log_v(\delta(x, a)_\zeta)$ is strongly superharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, and is strongly subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{a\}$.

EXAMPLE 6.6. If $f \in K(T)$ is a nonzero rational function with divisor $\text{div}(f) = \sum_{i=1}^m n_i \delta_{a_i}(x)$, let $\text{supp}^-(\text{div}(f))$, $\text{supp}^+(\text{div}(f))$ be its polar locus and zero locus, respectively. Then $\log_v([f]_x)$ is strongly subharmonic on the complement of $\text{supp}^-(\text{div}(f))$ and strongly superharmonic on the complement of $\text{supp}^+(\text{div}(f))$. Likewise $-\log_v([f]_x)$ is strongly superharmonic on the complement of $\text{supp}^-(\text{div}(f))$ and strongly subharmonic on the complement of $\text{supp}^+(\text{div}(f))$.

These assertions follow from the continuity of $[f]_x$ and from Example 5.13.

EXAMPLE 6.7. If ν is a probability measure on $\mathbb{P}_{\text{Berk}}^1$ and $\zeta \notin \text{supp}(\nu)$, then the potential function $u_\nu(x, \zeta)$ is strongly superharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ and is strongly subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \text{supp}(\nu)$. These assertions follow from Proposition 4.12 and Example 5.14.

EXAMPLE 6.8. If $E \subset \mathbb{P}_{\text{Berk}}^1$ is a compact set of positive capacity and $\zeta \notin E$, then the Green's function $G(z, \zeta; E)$ is strongly subharmonic on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. Indeed, if μ_ζ is the equilibrium distribution of E with respect to ζ , then $G(z, \zeta; E) = V_\zeta(E) - u_{\mu_\zeta}(z, \zeta)$.

The proof of the following lemma is quite similar to that of Lemma 5.23, so we omit the details.

LEMMA 6.9.

A) If $U_1 \subset U_2$ are domains, and f is strongly subharmonic on U_2 , then f is strongly subharmonic on U_1 .

B) If f is subharmonic on an open set V , and U is a subdomain of V with $\bar{U} \subset V$, then f is strongly subharmonic on U .

C) If f is subharmonic on V and $E \subset V$ is compact and connected, there is a subdomain $U \subset V$ containing E such that f is strongly subharmonic on U . In particular, if V is a domain, there is an exhaustion of V by subdomains U on which f is strongly subharmonic.

The only thing which should be mentioned is the compatibility of the Jordan decomposition with restriction, which tells us that if U is a domain and U_i is an open covering of U , then a measure is ≤ 0 on U iff it is ≤ 0 on U_i for all i .

Subharmonic functions obey the following maximum principle:

PROPOSITION 6.10. (Maximum Principle) Let $U \subseteq \mathbb{P}_{\text{Berk}}^1$ be a domain, and suppose that f is subharmonic on U . Then:

(A) If f attains a global maximum on U , then f is constant.

(B) If M is a bound such that for each $q \in \partial U$,

$$\limsup_{\substack{z \rightarrow q \\ z \in U}} f(z) \leq M,$$

then $f(z) \leq M$ for all $z \in U$.

The proof is based on part (C) of Proposition C.13.

PROOF OF PROPOSITION 6.10. (A) Suppose first that f attains its maximum value on U at a point $\zeta \in U \cap \mathbb{H}_{\text{Berk}}$. Let W be a simple subdomain of U containing ζ . Let Γ be a subgraph of W containing ζ such that $\zeta \notin r_{\bar{W}, \Gamma}(\partial W)$. Since f is strongly subharmonic on W , it follows that $\Delta_{\bar{W}}(f)^+$ is supported on ∂W . By construction, we have $\Delta_{\Gamma}(f) = (r_{\bar{W}, \Gamma})_*(\Delta_{\bar{W}}(f))$, so that $\Delta_{\Gamma}(f)^+$ is supported on $r_{\bar{W}, \Gamma}(\partial W)$. But the maximum value of f on Γ is achieved at $\zeta \notin r_{\bar{W}, \Gamma}(\partial W)$, contradicting Proposition C.13(C).

Now suppose f attains its maximum value on U at a point $z \in U \cap \mathbb{P}^1(K)$. Let W be a simple subdomain of U containing z , which we may assume to be a disc. Let q be the unique boundary point of W . If $f(q) = f(z)$, then we're done by the previous case. So we may assume

that $f(q) < f(z)$. Since f is strongly subharmonic on W , we have

$$f(z) = \limsup_{\substack{p \rightarrow z \\ p \in W \cap \mathbb{H}_{\text{Berk}}} } f(p)$$

and thus there exists $p \in W \cap \mathbb{H}_{\text{Berk}}$ such that $f(q) < f(p)$.

Let Γ be the path from q to p ; it is a subgraph of U contained in $\overline{W}$, and $r_{\overline{U}, \Gamma}(\partial U) = \{q\}$. As above, we see that $\Delta_{\Gamma}(f)^+$ is supported on $\{q\}$, and thus the maximum value of f on Γ occurs at q by Proposition C.13(C). This contradicts the fact that $f(q) < f(p)$.

(B) Define $f^* : \overline{U} \rightarrow [-\infty, \infty)$ by

$$f^*(q) = \limsup_{\substack{z \rightarrow q \\ z \in U}} f(z).$$

Then it is easy to verify that f^* is upper semicontinuous on $\overline{U}$, and $f^* = f$ on U . The result therefore follows from (A), together with the fact that an upper semicontinuous function on a compact set E achieves its maximum value on E . $\square$

COROLLARY 6.11. *If f is subharmonic on U and $V \subset U$ is a domain with $\overline{V} \subset U$, then the maximum value of f on $\overline{V}$ is attained on ∂V .*

PROOF. The maximum value M of f on $\overline{V}$ is attained, since $\overline{V}$ is compact and g is upper semicontinuous on $\overline{V}$. By the Maximum Principle, either f is constant on V or $f(x) < M$ for all $x \in V$. If $f(x) \equiv M$ on V , then by semicontinuity $f(q) \geq M$ and therefore $f(q) = M$ for all $q \in \overline{V}$. So in both cases we may assume that $f(x) < M$ for all $x \in V$, which implies that $f(q) = M$ for some $q \in \overline{V}$. $\square$

The following observation about the behavior of subharmonic functions is quite useful.

PROPOSITION 6.12. *Let f be subharmonic on a domain U . Then $f(x)$ is non-increasing on paths leading away from the main dendrite of U . If U is a disc, then $f(x)$ is non-increasing on paths leading away from the boundary $\partial U = \{q\}$.*

PROOF. Let $z \in U \cap \mathbb{H}_{\text{Berk}}$ be a point which is not of type IV. If D is nonempty, let $q = r_{U,D}(z)$ be the point where the path from z to D first meets D . (If $z \in D$, then $q = z$.) Let $p \in U \cap \mathbb{H}_{\text{Berk}}$ be any point not lying on D such that z lies on the path Γ from q to p . Let $\vec{v}$ be the tangent vector at z pointing towards p . We need to show that $d_{\vec{v}}f(z) \leq 0$. By construction of Γ , we have $r_{\overline{U}, \Gamma}(\partial U) = \{q\}$. Since f is subharmonic, the support of $\Delta_{\overline{U}}(f)^+$ is contained in ∂U . By the coherence property of Laplacians, we conclude that $\Delta_{\Gamma}(f)^+$ is

supported on $\{q\}$, so that $\Delta_\Gamma(f) \leq 0$ on $\Gamma \setminus \{q\}$. On the other hand, the Mass Formula (C.12) implies that

$$\Delta_\Gamma(f)((z, p]) = d_{\vec{v}}f(z),$$

so that $d_{\vec{v}}f(z) \leq 0$ as required.

If U is a disc with boundary $\{q\}$, then the same argument applies with $\Gamma = [p, q]$ for any point $p \in U$ and any $z \in \Gamma$. $\square$

REMARK 6.13. According to Proposition 5.26, the main dendrite of a domain is finitely branched. Proposition 6.12 therefore implies that at any given point, there are only finitely many tangent directions in which a subharmonic function can be increasing.

The following consequence of Proposition 6.12 will be used in §8.

PROPOSITION 6.14. *Suppose f is subharmonic on a domain $U \subseteq \mathbb{P}_{\text{Berk}}^1$, and let $z_0 \in U \cap \mathbb{H}_{\text{Berk}}$. Then there exists a point $q \in \partial U$ and a path Λ from z_0 to q such that f is non-decreasing along Λ .*

PROOF. If U is a disk, this follows immediately from Proposition 6.12. We may assume, therefore, that the main dendrite D of U is non-empty. By Proposition 6.12, we may assume without loss of generality that $z_0 \in D$.

Let S be the partially ordered (by containment) set consisting of all paths $\Lambda = [z_0, p]$ from z_0 to a point $p \in \overline{D} = D \cup \partial U$ for which f is non-decreasing along the interior Λ^0 of Λ . (We allow the trivial path $\Lambda = \{z_0\}$ with $p = z_0$.) Since $\overline{D}$ is uniquely path-connected, we have $\Lambda \subseteq \overline{D}$. Every totally ordered subset of S has an upper bound in S , since the closure of the union of such paths is again a path in S . Also, $S \neq \emptyset$, since the trivial path $\{z_0\}$ is in S . By Zorn's Lemma, there exists a maximal path $\Lambda_q = [z_0, q] \in S$.

If $q \in \partial U$, then we're done. So assume for the sake of contradiction that $q \notin \partial U$. Then $q \in \mathbb{H}_{\text{Berk}}$, since $\overline{D} \cap \mathbb{P}^1(K) \subseteq \partial U$. Thus Λ_q is a finite subgraph of D . Let Γ be a subgraph of D containing Λ_q for which $q \notin r_{\overline{U}, \Gamma}(\partial U)$ and $q \in \Gamma^0$. (Such a subgraph Γ exists because D is *finitely branched* at q , and because $q \in D$ implies that q is not of type IV.) As $q \in \Gamma^0$, we have $|T_q(\Gamma)| \geq 2$, i.e., Γ has at least 2 distinct tangent directions at q . By the maximality of Λ_q , we have $\Delta_\Gamma(f)(q) > 0$, and thus $q \in \text{supp}(\Delta_\Gamma(f)^+)$. As $q \notin r_{\overline{U}, \Gamma}(\partial U)$, we obtain a contradiction from the fact that $\text{supp}(\Delta_\Gamma(f)^+) \subseteq r_{\overline{U}, \Gamma}(\partial U)$ by the subharmonicity of f on U . $\square$

2. Domination by harmonic functions

In this section, we will show that subharmonic functions can be characterized in terms of domination by harmonic functions.

By Proposition 5.33, if V is a simple domain, each harmonic function $h(x)$ on V extends to a continuous function on $\bar{V}$, and the Poisson formula expresses $h(x)$ on V in terms of its values on ∂V . We will sometimes use this implicitly, and if $h(x)$ is harmonic on a simple domain, we will speak of its values on ∂V .

The following definition is motivated by analogy with the classical theory over $\mathbb{C}$, and explains the name “subharmonic”:

DEFINITION 6.15. Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set.

A function $f : U \rightarrow [-\infty, \infty)$ is domination subharmonic on U if

(A') $f(x)$ is upper semicontinuous, and $f(x) \not\equiv -\infty$ on any connected component of U ;

(B') for each simple subdomain $V \subset U$ and each function $h(x)$ harmonic on V , if $h(x) \geq f(x)$ on ∂V , then $h(x) \geq f(x)$ on V .

The following result shows that the notions of “subharmonic” and “domination subharmonic” are one and the same.

THEOREM 6.16. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set. Then a function $f : U \rightarrow [-\infty, \infty)$ is subharmonic on U if and only if it is domination subharmonic on U .*

Before giving the proof of Theorem 6.16, we will need several lemmas. The hardest part of the proof will be showing that if f is domination subharmonic on U , then every $x \in U$ has a neighborhood U_x for which $f \in \text{BDV}(U_x)$.

If V is a simple domain, then either V is a disc with a single boundary point x , or a punctured disc $\mathcal{B}(a_1, r_1)_\zeta^- \setminus \cup_{i=2}^m \mathcal{B}(a_i, r_i)_\zeta$ with boundary points $x_1, \dots, x_m$. In that case the main dendrite of V is the graph Γ whose endpoints are $x_1, \dots, x_m$. (By an endpoint of Γ , we mean a point with a single edge emanating from it.) We will call $\partial\Gamma = \{x_1, \dots, x_m\}$ the *boundary* of Γ , and $\Gamma^0 = \Gamma \setminus \partial\Gamma$ the *interior* of Γ . Note that V can be recovered from Γ , indeed if $r_\Gamma : \mathbb{P}_{\text{Berk}}^1 \rightarrow \Gamma$ is the retraction map, then

$$V = r_\Gamma^{-1}(\Gamma^0) .$$

Conversely, if $\Gamma \subset \mathbb{P}_{\text{Berk}}^1$ is a subgraph, we will call $V = r_\Gamma^{-1}(\Gamma^0)$ the *domain associated to Γ* . Clearly $\partial V = \partial\Gamma$, and Γ^0 is the main dendrite of V .

LEMMA 6.17. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain.*

Fix the primes here and in Appendix C.

Introduce better notation for this.

If $\Gamma \subset U$ is a subgraph, then the associated domain $V = r_\Gamma^{-1}(\Gamma^0)$ is a simple subdomain of U if and only if $r_\Gamma(\partial U) \subset \partial \Gamma$.

A disc V is a simple subdomain of U and only if its boundary point x belongs to U and $\partial U \cap V = \emptyset$.

PROOF. If $r_\Gamma(\partial U) \subset \partial \Gamma$, then $V = r_\Gamma^{-1}(\Gamma^0)$ is a connected open set with $\partial U \cap V = \emptyset$. Since U is connected and $\Gamma^0 \subset U$, necessarily $V \subset U$. Since $\Gamma \subset U$ and $\partial V = \partial \Gamma$, it follows that $\bar{V} \subset U$. Conversely, if V is a simple subdomain of U and Γ is the main dendrite of V , then $r_\Gamma^{-1}(\Gamma^0) \subset U$. Since $\partial U \cap U = \emptyset$, it follows that $\partial U \subset r_\Gamma^{-1}(\partial \Gamma)$. Thus $r_\Gamma(\partial U) \subset \partial \Gamma$.

If a disc $V = \mathcal{B}(a, r)_\zeta^-$ is a simple subdomain of V then by definition $\bar{V} \subset U$ so $\{x\} = \partial V \subset U$, and $\partial U \cap \bar{V} = \emptyset$ so $\partial U \cap V = \emptyset$. Conversely, if V is a disc with $\{x\} = \partial V \subset U$, then $V \cap U$ is nonempty. Since V is one of the connected components of $r_\Gamma^{-1}(x) \setminus \{x\}$, if $\partial U \cap V = \emptyset$, necessarily $V \subset U$. It follows that $\bar{V} \subset U$. $\square$

If U is a domain in $\mathbb{P}_{\text{Berk}}^1$ and $\Gamma \subset U$ is a subgraph of U , we will say that Γ is a *simple subgraph* of U if the domain $V = r_\Gamma^{-1}(\Gamma^0)$ is a simple subdomain of U . The first part of Lemma 6.17 can thus be rephrased by saying that Γ is a simple subgraph of U if and only if $r_\Gamma(\partial U) \subset \partial \Gamma$.

LEMMA 6.18. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be open.*

If V is a simple subdomain of U , then there is a simple subdomain W of U with $\bar{V} \subset W \subset U$. Moreover, W can be chosen so that for each $x_i \in \partial V$ which is not of type IV, there is a point $w_j \in \partial W$ with $r_{\bar{V}}(w_j) = x_i$.

Do we ever use the last sentence of Lemma 6.18? Make clear somewhere whether ∂W for a simple domain W is allowed to have Type IV boundary points. Perhaps we want to exclude this in the definition?

PROOF. Write $\partial V = \{x_1, \dots, x_m\}$. For each x_i , choose a simple subdomain V_i of U which contains x_i . Such a V_i exists, since ∂U is closed and disjoint from x_i , and punctured discs are cofinal in the open neighborhoods of x_i . Put $W = V \cup_{i=1}^m V_i$. Then W is connected and open, and $\partial W \subset \cup_{i=1}^m \partial V_i \subset U$ is finite, so W is a simple subdomain of U . Clearly $\bar{V} \subset W$.

Write $\partial W = \{w_1, \dots, w_M\}$. If there is some x_i which is not the retraction to $\bar{V}$ of any w_j , and x_i is not of type IV, then the component of $\mathbb{P}_{\text{Berk}}^1 \setminus V$ containing x_i is a nonempty closed disc contained in W . By removing a proper closed subdisc of that disc from W , we would obtain

a simple subdomain $\tilde{W}$ of U with $\overline{V} \subset \tilde{W}$, and having a boundary point whose retraction was x_i . Doing this sequentially for each x_i gives the result. $\square$

LEMMA 6.19. *Let f be domination subharmonic on a domain $U \subset \mathbb{P}_{\text{Berk}}^1$. Then for all $x \in U \cap \mathbb{H}_{\text{Berk}}$, we have $f(x) \neq -\infty$.*

PROOF. Suppose $f(x_1) = -\infty$ for some $x_1 \in U \cap \mathbb{H}_{\text{Berk}}$. We will show that $f(x) \equiv -\infty$ on U , which contradicts the definition of domination subharmonicity.

Fix $x \in U$ with $x \neq x_1$, and take a simple subdomain V_1 of U which contains x and x_1 . Let V be the connected component of $V_1 \setminus \{x_1\}$ which contains x . Then V is a simple subdomain of U with x_1 as a boundary point, which contains x in its interior. (Note that this uses $x_1 \notin \mathbb{P}^1(K)$).

If x_1 is the only boundary point of V , then V is a disc. In that case, by Lemma 5.27, each harmonic function on V is constant and is determined by its value on $\partial V = \{x_1\}$. Since $f(z)$ is domination subharmonic and $f(x_1) = -\infty$, we have $f(z) \leq C$ on V for each $C \in \mathbb{R}$. Thus $f(x) = -\infty$.

If x_1 is not the only boundary point of V , let Γ be the main dendrite of V , and let $\partial\Gamma = \{x_1, \dots, x_m\}$. Fix numbers $A_2, \dots, A_m$ with $A_i \geq f(x_i)$ for each i . Given $A_1 \in \mathbb{R}$, Poisson's formula (Proposition 5.34) constructs a harmonic function

$$h_{A_1}(z) = \sum_{i=1}^m A_i h_i(z)$$

on V , where $h_i(z)$ is the harmonic measure with $h_i(x_i) = 1$, $h_i(x_j) = 0$ for each $j \neq i$. Here $0 < h_i(z) < 1$ on V , otherwise $h_i(z)$ would achieve its maximum or minimum value on the interior of V , contradicting the Maximum Principle (Proposition 5.28). Fixing z and letting $A_1 \rightarrow -\infty$, we see that $h_{A_1}(z) \rightarrow -\infty$. Since f is domination subharmonic on V , again $f(x) = -\infty$.

Since $x \in U$ is arbitrary, we have shown that $f(x) \equiv -\infty$ on U . $\square$

The next two results are pretty hard for ‘‘lemmas’’ Maybe they should be rechristened as propositions.

LEMMA 6.20. *Let f be domination subharmonic on a domain $U \subset \mathbb{P}_{\text{Berk}}^1$. If Γ is a simple subgraph of U , then for each $p \in \Gamma$ and each direction $\vec{v} \in T_p(\Gamma)$, the directional derivative $d_{\vec{v}}f(p)$ exists and is finite. In particular, the restriction of f to Γ is continuous.*

PROOF. By Lemma 6.17, $r_\Gamma(\partial U) \subset \partial\Gamma$. In the argument below we will want to consider subgraphs $\Gamma' \subset \Gamma$ for which the associated domain $V' = r_{\Gamma'}^{-1}((\Gamma')^0)$ is also a simple subdomain of U .

There are two types of Γ' for which this is assured. One is if $\Gamma' = [a, b]$ is a segment contained in an edge of Γ (that is, no branch point of Γ is contained in $(\Gamma')^0$). The other is if $\Gamma' = \overline{\Gamma_{x_0}(\varepsilon)} = \{x \in \Gamma : \rho(x, x_0) \leq \varepsilon\}$ is a closed neighborhood of a point $x_0 \in \Gamma$, for some $\varepsilon > 0$. If ε is small enough, then Γ' is a star, a union of closed segments $[x_0, x_i]$ for $i = 1, \dots, m$. In either case, $r_{\Gamma, \Gamma'}(\partial\Gamma) \subset \partial\Gamma'$, so $r_{\Gamma'}(\partial U) \subset \partial\Gamma'$.

By Lemma 6.19, $f(x) \in \mathbb{R}$ for all $x \in \Gamma$.

By abuse of notation, write f for $f|_\Gamma$. We will first show that for each $p \in \Gamma$, and each $\vec{v}$ at p , the derivative $d_{\vec{v}}f(p)$ exists in $\mathbb{R} \cup \{-\infty\}$. Fix p and $\vec{v}$. For each sufficiently small $T > 0$, the segment $\Gamma' = [p, p + T\vec{v}]$ corresponds to a simple subdomain of U . The harmonic function on U whose values agree with those of $f(x)$ at p and $p + T\vec{v}$ is linear on Γ' and is given by

$$(6.1) \quad h(p + t\vec{v}) = \left(1 - \frac{t}{T}\right) \cdot f(p) + \frac{t}{T} \cdot f(p + T\vec{v}) .$$

for $0 \leq t \leq T$. Since f is domination subharmonic, $f(p + t\vec{v}) \leq h(p + t\vec{v})$ for all t . Using (6.1), this gives

$$(6.2) \quad \frac{f(p + t\vec{v}) - f(p)}{t} \leq \frac{f(p + T\vec{v}) - f(p)}{T} .$$

Hence

$$(6.3) \quad d_{\vec{v}}f(p) = \lim_{t \rightarrow 0^+} \frac{f(p + t\vec{v}) - f(p)}{t}$$

exists in $\mathbb{R} \cup \{-\infty\}$, since the limit on the right side is non-increasing.

Next, we will show that if $p \notin \partial\Gamma$, then $d_{\vec{v}}f(p) \neq -\infty$. Since $p \notin \partial\Gamma$, for sufficiently small $T > 0$, the star $\Gamma' = \Gamma_p(T)$ is a subgraph of Γ with $p \in (\Gamma')^0$. Let $x_1, \dots, x_m$ be its endpoints, where $m \geq 2$, and let $\vec{v}_1, \dots, \vec{v}_m$ be the direction vectors at p , so $\Gamma' = \cup_{i=1}^m [p, p + T\vec{v}_i]$. Without loss, suppose $\vec{v} = \vec{v}_1$. We claim that for each $0 < t < T$,

$$(6.4) \quad \frac{f(p + t\vec{v}) - f(p)}{t} \geq -\sum_{i=2}^m \frac{f(x_i) - f(p)}{T} .$$

If this fails for some t , then

$$(6.5) \quad f(p) > \frac{Tf(p + t\vec{v}) + t \cdot \sum_{i=2}^m f(x_i)}{T + (m-1) \cdot t} .$$

Consider the subgraph $\Gamma'' = [p, p + t\vec{v}] \cup (\cup_{i=2}^m [p, p + T\vec{v}_i])$, which corresponds to a simple subdomain $V'' \subset U$ with boundary points

$x_1'' = p + t\vec{v}, x_2, \dots, x_m$. The harmonic function $h(z)$ on V'' with

$$h(x_1'') = f(x_1'') , \quad h(x_i) = f(x_i) \quad \text{for } i \geq 2 ,$$

satisfies

$$0 = \Delta_\Gamma h(p) = \frac{f(x_1'') - h(p)}{t} + \sum_{i=2}^m \frac{f(x_i) - h(p)}{T} ,$$

which leads to

$$h(p) = \frac{Tf(p + t\vec{v}) + t \cdot \sum_{i=2}^m f(x_i)}{T + (m-1) \cdot t} < f(p) .$$

This contradicts the domination subharmonicity of $f(x)$. By (6.4), the right side of

$$d_{\vec{v}}f(p) = \lim_{t \rightarrow 0^+} \frac{f(p + t\vec{v}) - f(p)}{t}$$

is bounded from below, so $d_{\vec{v}}f(p) > -\infty$.

It remains to consider points $p \in \partial\Gamma$.

First suppose $p \in \partial\Gamma \setminus r_\Gamma(\partial U)$. There is a single edge e of Γ emanating from p ; let x approach p along this edge. For each x , let V_x be the connected component of $\mathbb{P}_{\text{Berk}}^1 \setminus \{x\}$ containing p . Then V_x is a disc with $\partial V_x = \{x\}$. Since $x \notin r_\Gamma(\partial U)$, it follows that $\bar{V}_x \subset U$, and so V_x is a simple subdomain of U . Let h be the harmonic function on V_x with $h(z) = f(x)$ for all $z \in V_x$. Since f is domination subharmonic, $f(z) \leq f(x)$ for all $z \in V_x$. Thus, $f(x)$ is decreasing as $x \rightarrow p$. If $\vec{v}$ is the unique direction vector at p , it follows that

$$(6.6) \quad d_{\vec{v}}f(p) = \lim_{t \rightarrow 0^+} \frac{f(p + t\vec{v}) - f(p)}{t} \geq 0 .$$

Finally, suppose $p \in r_\Gamma(\partial U) \subset \partial\Gamma$. Let $\vec{v}$ be the unique direction vector at p , and write $\partial\Gamma = \{x_1, \dots, x_m\}$. Without loss, suppose $p = x_1$. Let $W \subset U$ be a punctured disc containing p but not $x_2, \dots, x_m$, such that $\bar{W}$ is contained in U . Recalling that $V = r_\Gamma^{-1}(\Gamma^0)$, put $\tilde{V} = W \cup V$. Then $\tilde{V}$ is a connected open set with a finite number of boundary points whose closure is contained in U , so it is a simple subdomain of U . Let $\tilde{\Gamma}$ be the subgraph spanned by $\partial\tilde{V}$. Then $\Gamma \subset \tilde{\Gamma}$, since $\{x_2, \dots, x_m\}$ remain boundary points of $\tilde{V}$, and if $\tilde{x}$ is a boundary points of $\tilde{V}$ which is not contained in V , then the path from $\tilde{x}$ to each x_i , $i = 2, \dots, m$, passes through p .

From this we see that p is an interior point of $\tilde{\Gamma}$. By what has been shown above, $d_{\vec{v}}f(p)$ is finite, when f is regarded as a function on $\tilde{\Gamma}$. However, $d_{\vec{v}}f(p)$ depends only on the restriction of f to Γ . Thus it is finite. $\square$

LEMMA 6.21. *Let f be domination subharmonic on a domain $U \subset \mathbb{P}_{\text{Berk}}^1$. If $\Gamma \subset U$ is a subgraph for which $V = r_\Gamma^{-1}(\Gamma^0)$ is a simple subdomain of U , then $f|_\Gamma \in \text{BDV}(\Gamma)$ and $\text{supp}(\Delta_\Gamma(f)^+) \subset r_\Gamma(\partial U) \subset \partial \Gamma$.*

PROOF. We have seen in Lemma 6.19 that $f(x) \in \mathbb{R}$ for all $x \in \Gamma$, and in Lemma 6.20 that $d_{\vec{v}}f(p)$ exists and is finite, for all $p \in \Gamma$ and all directions $\vec{v}$ at p . Thus, $\Delta_\Gamma(f)$ exists as a finitely additive set function on the Boolean algebra $\mathcal{A}(\Gamma)$ generated by the open, closed and half-open segments in Γ .

We first claim that $\Delta_\Gamma(f)(p) \leq 0$ for each $p \in \Gamma^0$. Suppose not, and fix $p \in \Gamma^0$ with $\Delta_\Gamma(f)(p) > \eta > 0$. Since $p \in \Gamma^0$, there are at least two edges emanating from p . Let $\varepsilon > 0$ be small enough that the closed neighborhood $\overline{\Gamma_p(\varepsilon)}$ is a star. If $\vec{v}_1, \dots, \vec{v}_m$ are the direction vectors at p , then

$$(6.7) \quad \sum_{i=1}^m d_{\vec{v}_i}f(p) = -\Delta_\Gamma(f)(p) < -\eta.$$

Since $d_{\vec{v}_i}f(p) = \lim_{t \rightarrow 0^+} (f(p + t\vec{v}_i) - f(p))/t \in \mathbb{R}$, for each i there is a number $0 < t_i < \varepsilon$ such that for $q_i = p + t_i\vec{v}_i$

$$\left| \frac{f(q_i) - f(p)}{t_i} - d_{\vec{v}_i}f(p) \right| < \frac{\eta}{2m}.$$

Hence $\sum_{i=1}^m (f(q_i) - f(p))/t_i < 0$, which gives

$$(6.8) \quad f(p) > \frac{\sum_i f(q_i)/t_i}{\sum_i 1/t_i}.$$

Put $\Gamma' = \cup_{i=1}^m [p, q_i]$. Then $V' = r_{\Gamma'}^{-1}((\Gamma')^0)$ is a simple subdomain of U . Let $h(z)$ be the harmonic function on V' with $h(q_i) = f(q_i)$. Since $\Delta_\Gamma(h)(p) = 0$, we have

$$(6.9) \quad h(p) = \frac{\sum_i f(q_i)/t_i}{\sum_i 1/t_i}.$$

Combining (6.8) and (6.9) contradicts the domination subharmonicity of f .

Thus for each $p \in \Gamma^0$,

$$(6.10) \quad \Delta_\Gamma(f)(p) = -\sum_i d_{\vec{v}_i}f(p) \leq 0.$$

If p is an endpoint of Γ which does not belong to $r_\Gamma(\partial U)$, and if $\vec{v}$ is the direction vector at p , it has already been shown in (6.6) that

$d_{\vec{v}}f(p) \geq 0$. Hence for such points as well,

$$(6.11) \quad \Delta_{\Gamma}(f)(p) = -d_{\vec{v}}f(p) \leq 0.$$

Now consider an open segment (x, y) contained in an edge of Γ . Put $\Gamma'' = [x, y]$. The associated subgraph V'' is a simple subdomain of U with boundary points x, y . Let $h(z)$ be the harmonic function on V'' with $h(x) = f(x)$, $h(y) = f(y)$. Let $T = \rho(x, y)$ be the length of Γ'' and put $\alpha = (f(y) - f(x))/T$. The restriction of $h(z)$ to Γ'' is linear, and $f(z) \leq h(z)$ for all $z \in \Gamma$. Let $\vec{v}_+$ be the direction vector at x pointing to y , and let $\vec{v}_-$ be the direction vector at y pointing to x . Then

$$\begin{aligned} d_{\vec{v}_+}f(x) &= \lim_{t \rightarrow 0+} \frac{f(x + t\vec{v}_+) - f(x)}{t} \leq \lim_{t \rightarrow 0+} \frac{h(x + t\vec{v}_+) - h(x)}{t} = \alpha, \\ d_{\vec{v}_-}f(y) &= \lim_{t \rightarrow 0+} \frac{f(y + t\vec{v}_-) - f(y)}{t} \leq \lim_{t \rightarrow 0+} \frac{h(y + t\vec{v}_-) - h(y)}{t} = -\alpha. \end{aligned}$$

so the segment (x, y) has measure

$$(6.12) \quad \Delta_{\Gamma}(f)((x, y)) = d_{\vec{v}_+}f(x) + d_{\vec{v}_-}f(y) \leq 0.$$

We can now show that $f \in \text{BDV}(\Gamma)$, i.e. that $\Delta_{\Gamma}(f)$ extends to a bounded Borel measure on Γ . By definition (see (C.13) in Appendix C), we must show that there is a number B such that for any countable collection $\{T_i\}$ of pairwise disjoint sets in $\mathcal{A}(\Gamma)$,

$$(6.13) \quad \sum_{i=1}^{\infty} |\Delta_{\Gamma}(f)(T_i)| \leq B.$$

Since each T_i can be decomposed as a finite disjoint union of points and open intervals, it suffices to prove (6.13) under the assumption that each T_i is a point or an open interval. Since Γ has only finite many edges, endpoints, and branch points, it also suffices to prove (6.13) assuming that all the T_i are contained in the interior of an edge $e = [a, b]$.

In this case the fact that $\Delta_{\Gamma}(f)$ is finitely additive, with $\Delta_{\Gamma}(f)(p) \leq 0$ and $\Delta_{\Gamma}(f)((x, y)) \leq 0$ for each point p and open interval (x, y) contained in (a, b) , means that for any finite sum

$$\sum_{i=1}^n |\Delta_{\Gamma}(f)(T_i)| = - \sum_{i=1}^n \Delta_{\Gamma}(f)(T_i) \leq |\Delta_{\Gamma}(f)((a, b))|.$$

Letting $n \rightarrow \infty$ gives (6.13). The argument also shows that $\Delta_{\Gamma}(f)$ is ≤ 0 on $\Gamma \setminus r_{\Gamma}(\partial U)$. Hence $\text{supp}(\Delta_{\Gamma}(f)^+) \subset r_{\Gamma}(\partial U)$. $\square$

We can now prove Theorem 6.16.

PROOF OF THEOREM 6.16. First suppose that f is subharmonic on U ; we want to show that it is domination subharmonic. Let W be a connected component of U , and let V be a simple subdomain of W . Since $f \in \text{BDV}(V)$ by Lemma 6.9(B), it follows that $f(x)$ is finite on each subgraph $\Gamma \subset V$, hence $f(x) \not\equiv -\infty$ on W . It remains to show that if V is a simple subdomain of W , and if $h(z)$ is harmonic on V and satisfies $f(x) \leq h(x)$ on ∂V , then $f(z) \leq h(z)$ on V .

Let $g(x) = f(x) - h(x)$ for $x \in \bar{V}$. Then g is strongly subharmonic on V , so by the corollary to the Maximum Principle (Corollary 6.11), the maximum value of g on $\bar{V}$ is attained on ∂V . But $g(x) \leq 0$ on ∂V , and therefore $g(x) \leq 0$ on V , or equivalently $f(x) \leq h(x)$ for all $x \in V$ as desired.

Thus, if f is subharmonic on U , then it is domination subharmonic on U .

Note that the proof works for any domain $V \subset U$ with $\bar{V} \subset U$ and any harmonic function h on V which extends continuously to $\bar{V}$.

Now suppose f is domination subharmonic on U . We will show it is strongly subharmonic on each subdomain $V \subset U$ with $\bar{V} \subset U$.

By Lemma 6.9, it suffices to prove this for simple subdomains, since an arbitrary subdomain V with $\bar{V} \subset U$ is contained in a simple subdomain V' . (Cover $\bar{V}$ with a finite number of simple domains whose closures are contained in U , and let V' be their union. V' is connected since $\bar{V}$ is connected and each simple domain is connected; $\bar{V}' \subset U$; and $\partial V'$ is finite, since the boundary of each punctured disc is finite.)

Make this into a separate lemma?

So, let V be a simple subdomain contained in a component W of U . We will first show that $f \in \text{BDV}(V)$. Let $\Gamma \subset V$ be an arbitrary subgraph. We must show that $f \in \text{BDV}(\Gamma)$, and that there is a bound B independent of Γ such that $|\Delta_\Gamma(f)|(\Gamma) \leq B$. We will do this by enlarging Γ to a graph $\tilde{\Gamma}$ which contains all the boundary points of V , and applying Lemma 6.21.

Let $\tilde{\Gamma}$ be the smallest subgraph of W containing both Γ and ∂V . Since $\Gamma \subset V$,

$$r_{\tilde{\Gamma}}(\partial W) \subseteq \partial V \subseteq \partial \tilde{\Gamma},$$

and thus $\tilde{\Gamma}$ is a simple subgraph of W by Lemma 6.17.

Write $\partial V = \{x_1, \dots, x_m\}$. Since each x_i is an endpoint of $\tilde{\Gamma}$, there is a single direction vector $\vec{v}_i$ at x_i , and

$$\Delta_{\tilde{\Gamma}}(f)(x_i) = -d_{\vec{v}_i} f(x_i).$$

The x_i are the only points where $\Delta_{\tilde{\Gamma}}(f)$ can have positive mass. Since $\Delta_{\tilde{\Gamma}}(f)$ has total mass 0, we see that

$$(6.14) \quad |\Delta_{\tilde{\Gamma}}(f)|(\tilde{\Gamma}) = 2 \cdot \sum_{i=1}^m |d_{\tilde{v}_i} f(x_i)|$$

The right side of (6.14) is a bound B independent of the graph $\tilde{\Gamma}$.

Taking the retraction to Γ , it follows that

$$\Delta_{\Gamma}(f) = (r_{\tilde{\Gamma}, \Gamma})_*(\Delta_{\tilde{\Gamma}}(f))$$

has total mass at most B . Since Γ is arbitrary, $f \in \text{BDV}(V)$. Furthermore $\Delta_{\Gamma}(f)^+$ is supported on $r_{\bar{V}, \Gamma}(\partial V)$. Since

$$\Delta_{\Gamma}(f) = (r_{\bar{V}, \Gamma})(\Delta_{\bar{V}}(f))$$

for each Γ , taking the limit over subgraphs Γ shows $\Delta_{\bar{V}}(f)^+$ is supported on ∂V , that is, $\Delta_{\bar{V}}(f)|_V \leq 0$.

Explain in more detail the process of passing to the limit over all Γ .

It remains to show that

$$(6.15) \quad f(p) = \limsup_{\substack{z \rightarrow p \\ z \in V \cap \mathbb{H}_{\text{Berk}}}} f(z)$$

for each $p \in V \cap \mathbb{P}^1(K)$. By assumption f is upper semicontinuous, so for each $p \in V$

$$(6.16) \quad f(p) \geq \limsup_{z \rightarrow p} f(z) .$$

Fix $p \in V$; first suppose $p \in \mathbb{P}^1(K)$. Fix $y \in V$, and let x approach p along the path $[y, p]$. If x is close enough to p , then the connected component V_x of $\mathbb{P}_{\text{Berk}}^1 \setminus \{x\}$ containing p is an open disc whose closure is $V_x \cup \{x\} \subset W$. Since f is subharmonic, $f(z) \leq f(x)$ for all $z \in V_x$. These discs V_x are a fundamental system of neighborhoods of p , so

$$f(p) \leq \lim_{\substack{x \rightarrow p \\ x \in [y, p]}} f(x) .$$

Combined with (6.16), this establishes (6.15), and indeed shows that

$$(6.17) \quad f(p) = \lim_{\substack{x \rightarrow p \\ x \in [y, p]}} f(x) .$$

□

3. Stability properties

We will now show that subharmonic functions on $\mathbb{P}_{\text{Berk}}^1$ are stable under the same operations as classical subharmonic functions (see [43], p.49).

If $U \subset \mathbb{P}_{\text{Berk}}^1$ is open, we will write $\text{SH}(U)$ for the set of subharmonic functions on U . Recall from §2 that if $f : U \rightarrow [\infty, \infty)$ is locally bounded above, the *upper semicontinuous regularization* f^* of f is defined as

$$f^*(x) = \limsup_{y \rightarrow x} f(y).$$

PROPOSITION 6.22. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set. Then:*

A) $\text{SH}(U)$ is a convex cone: if $0 \leq \alpha, \beta \in \mathbb{R}$, and if $f, g \in \text{SH}(U)$, then $\alpha f + \beta g \in \text{SH}(U)$.

B) If U is connected, and if $\mathcal{F}$ is a family of functions in $\text{SH}(U)$, put $f(x) = \inf_{g \in \mathcal{F}} g(x)$. Then either $f \in \text{SH}(U)$, or $f(x) \equiv -\infty$ on U .

C) If $\langle f_\alpha \rangle_{\alpha \in A}$ is a net of functions in $\text{SH}(U)$ which converge uniformly to a function $f : U \rightarrow \mathbb{R}$ on compact subsets of U , then $f \in \text{SH}(U)$.

D) If $f, g \in \text{SH}(U)$, then $\max(f, g) \in \text{SH}(U)$.

E) If $\{f_\alpha\}_{\alpha \in A}$ is a family of functions in $\text{SH}(U)$ which is locally bounded from above, and if $f(x) = \sup_\alpha f_\alpha(x)$, then $f^*(x) \in \text{SH}(U)$. Furthermore $f^*(x) = f(x)$ for all $x \in U \cap \mathbb{H}_{\text{Berk}}$.

F) If U is connected and $\{f_n\}_{n \geq 0}$ is a sequence of functions in $\text{SH}(U)$ which is locally bounded from above, put $f(x) = \limsup_{n \rightarrow \infty} f_n(x)$. Then either $f(x) \equiv -\infty$ on U , or $f^*(x) \in \text{SH}(U)$. Furthermore $f^*(x) = f(x)$ for all $x \in U \cap \mathbb{H}_{\text{Berk}}$.

Replace sequences by families in (F).

PROOF. Except for the last assertion in (E), the proofs of these are the essentially same as their classical counterparts, and rely on general properties of semicontinuity and domination by harmonic functions.

(A) If $f, g \in \text{SH}(U)$ then f and g are upper semicontinuous, and $\alpha f + \beta g$ is upper semicontinuous since $\alpha, \beta \geq 0$. Neither f nor g is $-\infty$ on $U \cap \mathbb{H}_{\text{Berk}}$, so $\alpha f + \beta g \neq -\infty$ on $U \cap \mathbb{H}_{\text{Berk}}$, and certainly $\alpha f + \beta g \not\equiv -\infty$ on any component of U . If V is a simple subdomain of U , let $\partial V = \{x_1, \dots, x_m\}$. Suppose h is harmonic on V with $h(x_i) \geq \alpha f(x_i) + \beta g(x_i)$ for each i . Let h_1 be the harmonic function on V with $h_1(x_i) = f(x_i)$ on ∂V , and let h_2 be the harmonic function on V with $h_2(x_i) = g(x_i)$ on ∂V . Then $h_1(z) \geq f(z)$ on V , and $h_2(z) \geq g(z)$ on V . Put $H = h - \alpha h_1 - \beta h_2$. Then $H(x_i) \geq 0$ on ∂V , so by the

Maximum Principle for harmonic functions, $H(z) \geq 0$ on V . Hence $\alpha f(z) + \beta g(z) \leq \alpha h_1(z) + \beta h_2(z) \leq h(z)$ on V .

(B) The fact that f is upper semicontinuous follows from Lemma A.2. By Lemma 6.19, either $f(x) \neq -\infty$ on $U \cap \mathbb{H}_{\text{Berk}}$, or $f(x) \equiv -\infty$ on U .

Suppose $f \not\equiv -\infty$. If V is a simple subdomain of U and h is a harmonic function on V with $f(x_i) \leq h(x_i)$ for each x_i in the finite set ∂V , then for each $\varepsilon > 0$ there is a $g \in \mathcal{F}$ such that $g(x_i) \leq h(x_i) + \varepsilon$ and all $x_i \in \partial V$. It follows that $f(z) \leq g(z) \leq h(z) + \varepsilon$ on V . Since $\varepsilon > 0$ is arbitrary, $f(z) \leq h(z)$ for all $z \in V$.

(C) Suppose $\langle f_\alpha \rangle_{\alpha \in A}$ is a net of subharmonic functions converging uniformly to a function $f : U \rightarrow \mathbb{R}$ on compact subsets of U . Since each f_α is upper semicontinuous, f is upper semicontinuous. By hypothesis $f \not\equiv -\infty$ on any component of U .

Let V be a simple subdomain of U , and let h be a harmonic function on V with $f(x_i) \leq h(x_i)$ for all $x_i \in \partial V$. Take $\varepsilon > 0$. Since ∂V is finite, there is a β_1 so that $f_\alpha(x_i) \leq h(x_i) + \varepsilon$ on ∂V , for all $\alpha \geq \beta_1$. Thus $f_\alpha(z) \leq h(z) + \varepsilon$ on V for each $\alpha \geq \beta_1$. Since the f_α converge uniformly to f on compact subsets, there is a β_2 such that $|f(z) - f_\alpha(z)| < \varepsilon$ on $\bar{V}$ for all $\alpha \geq \beta_2$. Taking $\alpha \in A$ such that $\alpha \geq \beta_1$ and $\alpha \geq \beta_2$, we see that $f(z) \leq h(z) + 2\varepsilon$ on V . Since $\varepsilon > 0$ is arbitrary, $f(z) \leq h(z)$ on V .

(D) Suppose $f, g \in \text{SH}(U)$, and put $F(z) = \max(f(z), g(z))$. Since f and g are upper semicontinuous, so is F . Since neither f nor g is $\equiv -\infty$ on any component of U , the same is true for F . If V is a simple subdomain of U and h is a harmonic function on U with $h(x_i) \geq F(x_i)$ on ∂V , then $h(x_i) \geq f(x_i)$ and $h(x_i) \geq g(x_i)$ on ∂V , so $h(z) \geq f(z)$ and $h(z) \geq g(z)$ on V , which means that $h(z) \geq \max(f(z), g(z)) = F(z)$ on V .

(E) Let $\{f_\alpha\}_{\alpha \in A}$ be a family of subharmonic functions on U , and put $f(z) = \sup_\alpha (f_\alpha(z))$ on U . Then $f^*(z)$ is upper semicontinuous on U . Since no f_α is $\equiv -\infty$ on any component of U , the same is true for f , and also for $f^* \geq f$. If V is a simple subdomain of U and h is harmonic on U with $h(x_i) \geq f^*(x_i)$ on ∂V , then $h(x_i) \geq f_\alpha(x_i)$ on ∂V for each $\alpha \in A$. It follows that $h(z) \geq f_\alpha(z)$ on V for each α , so $h(z) \geq f(z)$ on V . However, $h(z)$ is continuous on V , hence certainly upper semicontinuous, so $h(z) \geq f^*(z)$ on V by the properties of the upper semicontinuous regularization.

We will now show that for each $x \in U \cap \mathbb{H}_{\text{Berk}}$,

$$(6.18) \quad f(x) \geq \limsup_{z \rightarrow x} f(z) .$$

We first construct a simple subdomain V of U such that x lies on the main dendrite of V .

If x is on the main dendrite of U , let V be any simple subdomain of U containing x . Then x is on the main dendrite of V , and in particular is an interior point of the graph Γ with $V = r_\Gamma^{-1}(\Gamma^0)$. If x is not on the main dendrite of U , let V_0 be an open subdisc of U containing x . Let x_0 be the boundary point of V_0 , and let $x_1 \in V_0 \cap \mathbb{H}_{\text{Berk}}$ be a point such that x is in the interior of the path $\Gamma = [x_0, x_1]$. Put $V = r_\Gamma^{-1}(\Gamma^0)$; then x lies on the main dendrite of V .

Next, we claim that $f|_{\Gamma^0}$ is continuous. Indeed, for each subgraph $\Gamma' \subset \Gamma^0$ of the type considered in the proof of Lemma 6.20, let V' be the corresponding simple subdomain of V . Let $h(x)$ be the harmonic function on V' whose value at each $p \in \partial V' = \partial \Gamma'$ is $f(p)$. For each $\alpha \in A$ and each $p \in \partial \Gamma'$, $f_\alpha(p) \leq f(p)$. Hence $f_\alpha(z) \leq h(z)$ for all $z \in V'$, and in turn $f(z) = \sup_\alpha f_\alpha(z) \leq h(z)$. By the same argument as in proof of Lemma 6.20, the directional derivative $d_{\vec{v}}f(p)$ exists for each $p \in \Gamma^0$ and each $\vec{v}$ at p . Thus f is continuous on Γ^0 .

Now fix $\varepsilon > 0$ and let $\Gamma_p(\delta) = \{q \in \Gamma : \rho(q, x) < \delta\}$ be a neighborhood of x in Γ on which $\overline{f(q)} < f(x) + \varepsilon$. Let V'' be the simple subdomain of V associated to $\Gamma_p(\delta)$. For each $\alpha \in A$ and each $q \in \Gamma_p(\delta)$, $f_\alpha(q) \leq f(x) + \varepsilon$. By Proposition 6.12, f_α is non-increasing on paths off the main dendrite $\Gamma_p(\delta)$ of V'' , so $f_\alpha(z) \leq f(x) + \varepsilon$ on V'' . Hence $f(z) = \sup_\alpha f_\alpha(z) \leq f(x) + \varepsilon$ for each $z \in V''$. Since ε is arbitrary, this gives (6.18), and shows that $f^*(x) = f(x)$.

(F) For each n , put $F_n(z) = \sup_{m \geq n} f_m(x)$ and let $F_n^*(z)$ be the upper semicontinuous regularization of F_n . By the final assertion in part (E), $F_n^*(x) = F_n(x)$ for each $x \in U \cap \mathbb{H}_{\text{Berk}}$. Then, $F_1^*(z) \geq F_2^*(z) \geq \dots$ is a decreasing sequence of subharmonic functions, and $f(x) = \lim_{n \rightarrow \infty} F_n^*(x)$, so by part (B) either $f(x) \equiv -\infty$, or $f(x) \in \text{SH}(U)$. $\square$

Subharmonic functions are also stable under integration over suitably bounded families on a parameter space, at least when the integral over the family is assumed to be upper semicontinuous.

PROPOSITION 6.23. *Let μ be a positive measure on a measure space T , and let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain. Suppose that $F : U \times T \rightarrow [-\infty, \infty)$ is a measurable function such that*

A) For each $t \in T$, the function $F_t(z) = F(z, t) : U \rightarrow [-\infty, \infty)$ is subharmonic in U ; and

B) the function

$$f(z) := \int_T F(z, t) d\mu(t)$$

is upper semicontinuous.

Then f is either subharmonic in U , or is $\equiv -\infty$ on U .

PROOF. (Compare with [43], Theorem 2.6.5, p.51.) Suppose $f(z) \not\equiv -\infty$, and let $V \subset U$ be a simple subdomain with boundary $\partial V = \{x_1, \dots, x_m\}$. Let h be harmonic on V with $h(x_i) \geq f(x_i)$ on ∂V . Let $h_1(z), \dots, h_m(z)$ be the unique harmonic functions on V with $h_i(x_j) = \delta_{ij}$. Recall that, by the maximum modulus principle for harmonic functions, for each $z \in V$ we have $0 \leq h_i(z) \leq 1$ and $\sum_{i=1}^m h_i(z) = 1$. Since each $F_t(x)$ is subharmonic, for each $z \in V$

$$F_t(z) \leq \sum_{i=1}^m h_i(z) F_t(x_i) .$$

Integrating over μ gives

$$f(z) \leq \sum_{i=1}^m f(x_i) h_i(z) \leq \sum_{i=1}^m h(x_i) h_i(z) = h(z) .$$

Thus f is subharmonic. $\square$

REMARK 6.24. In the classical setting over $\mathbb{C}$, one can show using Fatou's lemma that if μ is σ -finite and there exists a majorizing function $g : T \rightarrow (-\infty, \infty]$ in $L^1(\mu)$ such that $F_t(z) \leq g(t)$ for all $t \in T$ and all $z \in X$, then f is automatically upper semicontinuous if each $F_t(z)$ is upper semicontinuous. However, Fatou's lemma does not hold in general for nets, so we are forced to add as a hypothesis that f is upper semicontinuous.

Here are more ways of getting new subharmonic functions from old ones:

LEMMA 6.25. Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be open. If f is subharmonic on U , and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex and non-decreasing, then $\varphi \circ f$ is subharmonic on U . (Here, $\varphi(-\infty)$ is to be interpreted as $\lim_{t \rightarrow -\infty} \varphi(t)$.)

PROOF. (See [43], Theorem 2.6.6, p.51.) Note that $\varphi(t)$ can be written as

$$\varphi(t) = \sup(\{a \cdot t + b : a \geq 0, b \in \mathbb{R}, \text{ and } a \cdot t + b \leq \varphi(t) \text{ for all } t \in \mathbb{R}\}) .$$

Let A be the corresponding set of pairs $(a, b) \in \mathbb{R}^2$; then

$$\varphi \circ f(z) = \sup_{(a,b) \in A} a \cdot f(z) + b .$$

For each $(a, b) \in A$, we have $a \cdot f(z) + b \in \text{SH}(U)$. By Proposition 6.22(E), if $F(z) = \varphi \circ f(z)$, then $F^*(z)$ is subharmonic.

A convex function on $\mathbb{R}$ is automatically continuous, since its one-sided derivatives exist at each point. Since φ is continuous and non-decreasing, and $f(z)$ is upper semicontinuous, it follows that $F(z) = \varphi(f(z))$ is upper semicontinuous. Hence $F^*(z) = F(z)$. $\square$

COROLLARY 6.26. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be open.*

A) If f is subharmonic on U and $q \geq 1$, then the function $F(z) = q^{f(z)}$ is subharmonic on U .

B) If f is subharmonic and non-negative on U , then for any $\alpha \geq 1$, the function $F(z) = f(z)^\alpha$ is subharmonic on U .

PROOF. (See [43], Corollary 2.6.8, p.52). Note that $t \rightarrow q^t$ and $t \rightarrow t^\alpha$ are convex and non-decreasing, and apply Lemma 6.25. $\square$

We can now give additional examples of subharmonic functions.

EXAMPLE 6.27. Fix $\zeta \in \mathbb{P}_{\text{Berk}}^1$. For each $\alpha > 0$, and each $a \neq \zeta$, the function $f(x) = \delta(x, a)_\zeta^\alpha$ is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$. In particular this applies to $\delta(x, a)_\zeta$. More generally, for each $\alpha_1, \dots, \alpha_n \geq 0$, and each $a_1, \dots, a_n \in \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$, the generalized pseudo-polynomial

$$P(x, \vec{\alpha}, \vec{a}) = \prod_{i=1}^m \delta(x, a_i)_\zeta^{\alpha_i}$$

is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

This follows from Corollary 6.26(A) and Proposition 6.22(A), taking $q = q_v$, since $\alpha_i \cdot \log_v(\delta(x, a_i)_\zeta)$ is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

In particular, consider $f(x) = \delta(x, 0)_\infty$ on $\mathbb{A}_{\text{Berk}}^1$. It is constant on branches off the path $[0, \infty]$. Give $(0, \infty)$ the arclength parametrization, so that x_t is the point corresponding to the disc $B(0, q_v^t)$ for $-\infty < t < \infty$. Then $f(x_t) = q_v^t$. For each disc $V_T := \mathcal{B}(0, q_v^T)^-$, the Laplacian $\Delta_{V_T}(f)$ is supported on $[0, x_T]$. Relative to the arclength parametrization,

$$\begin{aligned} \Delta_{V_T}(f) &= f'(x_T) \cdot \delta_T(t) - f''(x_t) dt \\ &= q_v^T \log(q_v) \cdot \delta_T(t) - q_v^t (\log(q_v))^2 dt. \end{aligned}$$

The total variation of these measures grows to ∞ as $T \rightarrow \infty$. Thus $f(x)$ is subharmonic, but not strongly subharmonic, on $\mathbb{A}_{\text{Berk}}^1$.

EXAMPLE 6.28. Consider the function $\psi(t) = \arcsin(q_v^t)$. It is bounded, increasing and convex on $[-\infty, 0)$, with a vertical tangent at $t = 0$.

Put $f(x) = \psi(\log_v(\delta(x, 0)_\infty))$ on $U := \mathcal{B}(0, 1)^-$. By Lemma 6.25, $f(x)$ is bounded and subharmonic on U . However, it is not strongly subharmonic, and it cannot be extended to a subharmonic function on any larger domain.

EXAMPLE 6.29. For a nonzero rational function $f \in K(T)$ with divisor $\text{div}(f)$, the function $F(x) = [f]_x$ is subharmonic on $\mathbb{P}_{\text{Berk}}^1 \setminus \text{supp}(\text{div}(f)^-)$.

This follows from Corollary 6.26(A), taking $q = q_v$, since $\log_v([f]_x)$ is subharmonic on the complement of $\text{supp}(\text{div}(f)^-)$.

EXAMPLE 6.30. Let $E \subset \mathbb{P}_{\text{Berk}}^1$ be a compact set of positive capacity, and take $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus E$. Then for each $\alpha \geq 1$, the function $G(x, \zeta; E)^\alpha$ is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

This follows from Corollary 6.26(B), taking $\varphi(t) = t^\alpha$, since $G(x, \zeta; E)$ is subharmonic and non-negative on $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$.

EXAMPLE 6.31. Let $f_1, \dots, f_m \in K(T)$ be nonzero rational functions with poles supported on $\{\zeta_1, \dots, \zeta_d\}$, and let $N_1, \dots, N_m$ be positive integers. Then

$$g(x) = \max\left(\frac{1}{N_1} \log_v([f_1]_x), \dots, \frac{1}{N_m} \log_v([f_m]_x)\right)$$

is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta_1, \dots, \zeta_d\}$.

This follows from Example 6.29 above, and Proposition 6.22(D).

4. The Comparison Theorem

If $f(z)$ and $g(z)$ are subharmonic functions on an open set U , one can also ask for conditions which assure $f(z) \leq g(z)$ on U . The desired result is called the Comparison Theorem.

In order to formulate it, we need to generalize the Laplacian. By Lemma 6.9, if f is subharmonic on U , then it is strongly subharmonic on each simple subdomain $V \subset U$. Hence $\Delta_{\overline{V}}(f)$ is defined, and $\Delta_{\overline{V}}(f)|_V \leq 0$. It makes sense to write $\Delta_V(f)$ for $\Delta_{\overline{V}}(f)|_V$. However, we have not defined $\Delta_U(f)$ in general.

Let W be a component of U . The simple subdomains of W exhaust W , and form a directed set under containment. If $V_1 \subset V_2$, then

$$\Delta_{\overline{V_1}}(f)|_{V_1} = (r_{\overline{V_2}, \overline{V_1}})_*(\Delta_{\overline{V_2}}(f))|_{V_1} = \Delta_{\overline{V_2}}(f)|_{V_1}.$$

Thus, the measures $\Delta_{\overline{V}}(f)|_V$ cohere to give a well-defined Borel measure on W . We will call this measure $\Delta_W(f)$. By its construction, $\Delta_W(f)$ is supported on W , and $\Delta_W(f) \leq 0$. It may or may not have finite total mass; its mass is finite if and only if $f \in \text{BDV}(W)$. We will write $\Delta_U(f)$ for the measure on U whose restriction to each component is $\Delta_W(f)$.

Be consistent
with $\Delta_U(f)$
versus $\Delta_{\overline{U}}(f)$.

PROPOSITION 6.32. (Comparison Theorem). *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set with nonempty boundary. Suppose f and g are subharmonic on U , and*

$$A) \text{ for each } q \in \partial U, \limsup_{\substack{z \rightarrow q \\ z \in U}} (f(z) - g(z)) \leq 0 ;$$

$$B) \Delta_U(f) \leq \Delta_U(g) \text{ on } U.$$

Then $f(z) \leq g(z)$ on U .

PROOF. Consider the function $h(z) = f(z) - g(z)$ on U .

If we knew that $h(z)$ were subharmonic on U , the result would follow from the maximum principle. Indeed, $\limsup_{z \rightarrow q} h(z) \leq 0$ for each $q \in \partial U$, and for any simple subdomain V of U , we have $f, g \in \text{BDV}(V)$. Thus $\Delta_{\bar{V}}(h) = \Delta_{\bar{V}}(f) - \Delta_{\bar{V}}(g)$, and $\Delta_{\bar{V}}(h)|_V \leq 0$. Unfortunately, the difference of two upper semicontinuous functions need not be upper semicontinuous, so we must go back to first principles.

Fix a simple subdomain V of U , and fix $p \in V$. As noted above, $h \in \text{BDV}(V)$ and $\Delta_{\bar{V}}(h)|_V \leq 0$. We claim that

$$(6.19) \quad h(p) \leq \max_{q_i \in \partial V} h(q_i) .$$

To see this, let Γ be any subgraph of $\bar{V}$ with $\partial V \subseteq \partial \Gamma$. Then $h|_{\Gamma} \in \text{BDV}(\Gamma)$ and $\Delta_{\Gamma}(h) = (r_{\bar{V}, \Gamma})_*(\Delta_{\bar{V}}(h))$ is supported on $\partial V \subset \partial \Gamma$. By Proposition C.13(C), $h|_{\Gamma}$ achieves its maximum value on $r_{\bar{V}, \Gamma}(\partial V) = \partial V$, so that $h(x) \leq \max_{q_i \in \partial V} h(q_i)$ for all $x \in \Gamma$.

If $p \in \mathbb{H}_{\text{Berk}}$, we may choose Γ so that $p \in \Gamma$, and thus (6.19) is satisfied in this case.

If $p \in \mathbb{P}^1(K)$, let $y = r_{\bar{V}, \Gamma}(p)$, and consider the path $[y, p] \subset \bar{V}$. We've just shown that $h(x) \leq \max_{q_i \in \partial V} h(q_i)$ for all $x \in [y, p]$. But in (6.17), we saw that

$$f(p) = \lim_{\substack{x \rightarrow p \\ x \in [y, p]}} f(x) , \quad g(p) = \lim_{\substack{x \rightarrow p \\ x \in [y, p]}} g(x) .$$

Thus $h(p) = \lim_{\substack{x \rightarrow p \\ x \in [y, p]}} h(x) \leq \max_{q_i \in \partial V} h(q_i)$, which verifies (6.19) in all cases.

Now let $p \in U$, and let V_{α} be the directed system of all simple subdomains of U containing p . For each α , choose a point $x_{\alpha} \in \partial V_{\alpha}$ such that $h(p) \leq h(x_{\alpha})$. This defines a net $\langle x_{\alpha} \rangle$ in V . Since $\bar{U}$ is compact, there is a subnet $\langle y_{\beta} \rangle$ of $\langle x_{\alpha} \rangle$ converging to a point $q \in \bar{U}$. As $\cup_{\alpha} V_{\alpha} = U$, it follows that $q \in \partial U$. But then

$$h(p) \leq \lim_{\beta} h(y_{\beta}) \leq \limsup_{z \rightarrow q} h(z) \leq 0$$

as desired.

□

As a special case, the Comparison Theorem gives:

COROLLARY 6.33. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be an open set with nonempty boundary. Suppose f and g are subharmonic functions on U such that*

$$A) \text{ for each } q \in \partial U, \quad -\infty < \limsup_{\substack{z \rightarrow q \\ z \in U}} f(z) \leq \liminf_{\substack{z \rightarrow q \\ z \in U}} g(z) < \infty,$$

and

$$B) \Delta_U(f) \leq \Delta_U(g) \text{ on } U.$$

Then $f(z) \leq g(z)$ on U .

The proof of the Comparison Theorem yields the following useful criterion for equality of subharmonic functions.

COROLLARY 6.34. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, and let V be a simple subdomain of U . Suppose f and g are subharmonic on U with $\Delta_V(f) = \Delta_V(g)$ and $f(x) = g(x)$ on ∂V . Then $f(z) \equiv g(z)$ on V .*

PROOF. By symmetry, it suffices to show that $f(z) \leq g(z)$. Put $h(z) = f(z) - g(z)$. The proof of Proposition 6.32 shows that for all $z \in V$, we have

$$h(z) \leq \max_{q_i \in \partial V} h(q_i) = 0.$$

Thus $f(z) \leq g(z)$ as desired. □

5. The Riesz Decomposition Theorem

Let U be an open set in $\mathbb{P}_{\text{Berk}}^1$, and let $V \subset U$ be a simple subdomain. Suppose f is subharmonic in U . By Lemma 6.9, $f \in \text{BDV}(V)$. Put $\nu = -\Delta_V(f)$, i.e., $\nu = -\Delta_{\bar{V}}(f)|_V$. Then $\nu \geq 0$. Fix $\zeta \notin \bar{V}$, and consider the potential function

$$u_\nu(z, \zeta) = \int -\log_v(\delta(x, y)_\zeta) d\nu(y).$$

PROPOSITION 6.35. (Riesz Decomposition Theorem)

Let V be a simple subdomain of the open set $U \subset \mathbb{P}_{\text{Berk}}^1$. Fix $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus \bar{V}$. Suppose f is subharmonic on U , and let ν be the positive measure $\nu = -\Delta_V(f)$. Then there is a continuous function $h(z)$ on $\bar{V}$ whose restriction to V is harmonic such that

$$f(z) = h(z) - u_\nu(z, \zeta) \quad \text{for all } z \in V.$$

PROOF. Define $F(z) = -u_\nu(z, \zeta)$. By Example 5.15, $F(z) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(F) = \nu(\mathbb{P}_{\text{Berk}}^1)\delta_\zeta(x) - \nu.$$

Choose a subgraph $\Gamma \subset \bar{V}$ such that $\partial V \subseteq \partial\Gamma$. By the retraction property of the Laplacian, $\Delta_\Gamma(f) = (r_{\bar{V}, \Gamma})_*(\Delta_{\bar{V}}(f))$, and similarly for $\Delta_\Gamma(F)$. It follows that $\Delta_\Gamma(F) - \Delta_\Gamma(f)$ is a discrete measure supported on $\bar{V}$.

Define $h(x) = f(x) - F(x)$ on Γ , and extend $h(z)$ to $\bar{V}$ by setting $h(z) = h \circ r_{\bar{V}, \Gamma}(z)$ for all $z \in \bar{V}$. Then $h(z)$ is harmonic on V , since it is continuous on $\bar{V}$ and $\Delta_{\bar{V}}(h) = \Delta_\Gamma(F) - \Delta_\Gamma(f)$ is supported on $\bar{V}$. Now consider the functions $f(z)$ and $h(z) + F(z)$. By construction, both functions are subharmonic on V and satisfy $\Delta_V(f) = \Delta_V(h + F) = -\nu$. In addition, they agree on $\partial V \subset \Gamma$. By Corollary 6.34, $f(z) \equiv h(z) + F(z)$. $\square$

For future applications, it is useful to know that if f is continuous and subharmonic on U , and if $\nu = -\Delta_V(f)$, then the potential function $u_\nu(z, \zeta)$ is continuous everywhere.

PROPOSITION 6.36. *Suppose f is continuous and subharmonic on an open set $U \subset \mathbb{P}_{\text{Berk}}^1$. Let V be a simple subdomain of U , and fix $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus \bar{V}$. Put $\nu = -\Delta_{\bar{V}}(f)|_V = -\Delta_V(f)$. Then $u_\nu(z, \zeta)$ is continuous on all of $\mathbb{P}_{\text{Berk}}^1$.*

PROOF. By Proposition 6.35, there is a continuous function $h(z)$ on $\bar{V}$, harmonic on V , such that $u_\nu(z, \zeta) = h(z) - f(z)$ on V . In particular, $u_\nu(z, \zeta)$ is continuous on V . Since $\text{supp}(\nu) \subset \bar{V}$, $u_\nu(z, \zeta)$ is also continuous on $\mathbb{P}_{\text{Berk}}^1 \setminus \bar{V}$ by Proposition 4.12 (which includes continuity at ζ). It remains to show that $u_\nu(z, \zeta)$ is continuous on the finite set ∂V .

Fix $x_i \in \partial V$. Since f and h are continuous on $\bar{V}$,

$$\lim_{\substack{z \rightarrow x_i \\ z \in V}} u_\nu(z, \zeta) = u_\nu(x_i, \zeta).$$

Now consider the behavior of $u_\nu(z, \zeta)$ on each connected component of $\mathbb{P}_{\text{Berk}}^1 \setminus \{x_i\}$, as $z \rightarrow x_i$. Each such component is an open disc with x_i as its boundary point. One component contains V ; we have already dealt with it. Suppose W is a component which does not contain V . If W contains ζ , choose a point p on the interior of the path from x_i to ζ ; otherwise, let $p \in U \setminus \mathbb{P}^1(K)$ be arbitrary. Put $\Gamma = [x_i, p]$. Since $u_\nu(z, \zeta) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, its restriction to Γ belongs to $\text{BDV}(\Gamma)$, hence

is continuous on Γ . Thus

$$(6.20) \quad \lim_{\substack{z \rightarrow x_i \\ z \in \Gamma}} u_\nu(z, \zeta) = u_\nu(x_i, \zeta) .$$

If W is a component which does not contain ζ , then $u_\mu(z, \zeta)$ is harmonic on the open disc W , hence constant. By (6.20), $u_\nu(z, \zeta) \equiv u_\nu(x_i, \zeta)$ on W for such a component.

If W is the component containing ζ , put $W_0 = r_\Gamma^{-1}(\Gamma^0)$; then W_0 is a simple subdomain of $\mathbb{P}_{\text{Berk}}^1$ contained in $W \setminus \{\zeta\}$, and $u_\nu(z, \zeta)$ is harmonic on W_0 . The main dendrite of W_0 is Γ^0 . Since harmonic functions are constant on branches off the main dendrite,

$$(6.21) \quad \lim_{\substack{z \rightarrow x_i \\ z \in W}} u_\nu(z, \zeta) = \lim_{\substack{z \rightarrow x_i \\ z \in \Gamma}} u_\nu(z, \zeta) = u_\nu(x_i, \zeta) .$$

Combining all cases, we see that $u_\nu(z, \zeta)$ is continuous at x_i . $\square$

The proof of the Riesz Decomposition Theorem also gives interesting information about the structure of functions in $\text{BDV}(U)$. Definition 5.7 asserts that $f \in \text{BDV}(U)$ iff $f|_\Gamma \in \text{BDV}(\Gamma)$ for every subgraph $\Gamma \subset U$, and there is a uniform bound B for the measures $|\Delta_\Gamma(f)|(\Gamma)$ for all $\Gamma \subset U$. Nothing is said about continuity on U , or about the behavior of f on $\mathbb{P}^1(K)$.

PROPOSITION 6.37. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, and let $f \in \text{BDV}(U)$. For any simple subdomain $V \subset U$, there are subharmonic functions g_1, g_2 on V such that $f(z) = g_1(z) - g_2(z)$ for all $z \in V \cap \mathbb{H}_{\text{Berk}}$.*

PROOF. Since $f \in \text{BDV}(U)$, we also have $f \in \text{BDV}(V)$. Put $\nu = \Delta_{\overline{V}}(f)$, and let ν^+, ν^- be the positive and negative measures in the Jordan decomposition of ν , so $\nu = \nu^+ - \nu^-$. As ν has total mass zero, we have $\nu^+(\overline{V}) = \nu^-(\overline{V}) = M$ for some $M \geq 0$. Fix $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus \overline{V}$, and put

$$g_-(z) = -u_{\nu^-}(z, \zeta) , \quad g_+(z) = -u_{\nu^+}(z, \zeta) .$$

Then g_- and g_+ are subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta\}$ and belong to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, with $\Delta_{\mathbb{P}_{\text{Berk}}^1}(g_-) = M\delta_\zeta(x) - \nu^-$ and $\Delta_{\mathbb{P}_{\text{Berk}}^1}(g_+) = M\delta_\zeta(x) - \nu^+$. Thus

$$\Delta_{\overline{V}}(g_- - g_+) = \nu^+ - \nu^- = \nu .$$

Set $F(z) = f(z) - (g_- - g_+)$. Then $F \in \text{BDV}(V)$ and $\Delta_{\overline{V}}(F) \equiv 0$. It follows from Example 5.11 that $F \equiv C$ is constant on $\overline{V} \cap \mathbb{H}_{\text{Berk}}$. Set $g_1 = g_- + C$ and $g_2 = g_+$. $\square$

6. Convergence of Laplacians

Let U be a domain in $\mathbb{P}_{\text{Berk}}^1$, and let $\langle f_\alpha \rangle_\alpha$ be a net of subharmonic functions on U which converges to a subharmonic function F . One can ask under what conditions the net of measures $\langle \Delta_U(f_\alpha) \rangle$ converges to $\Delta_U(F)$.

PROPOSITION 6.38. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, and let $\langle f_\alpha \rangle_{\alpha \in A}$ be a sequence of subharmonic functions on U . Suppose one of the following conditions holds:*

- A) *The $f_\alpha(z)$ converge uniformly to $F(z)$ on each subgraph $\Gamma \subset U$;*
- B) *The $f_\alpha(z)$ decrease monotonically to $F(z)$ on each subgraph $\Gamma \subset U$, and $F(z) \not\equiv -\infty$ on U ;*
- C) *The $f_\alpha(z)$ increase monotonically on each subgraph $\Gamma \subset U$, and the family $\{f_\alpha\}$ is locally bounded above. Put $f(z) = \lim_\alpha f_\alpha(z)$ and let $F(z) = f^*(z)$ be the upper semicontinuous regularization of $f(z)$.*

Then $F(z)$ is subharmonic on U , and the measures $\Delta_{\bar{V}}(f_\alpha)$ converge weakly to $\Delta_{\bar{V}}(F)$ for each simple subdomain $V \subset U$.

PROOF. In all three cases, Proposition 6.22 shows that $F(z)$ is subharmonic. Only the convergence of the measures needs to be established.

Proof of (A). Let $V \subset U$ be a simple subdomain.

We first claim that the measures $|\Delta_{\bar{V}}(f_n)|$ have uniformly bounded total mass. This depends on the fact that V can be enlarged within U .

Let W be a simple subdomain of U containing $\bar{V}$, and let $\Gamma \subset \tilde{\Gamma} \subset W$ be subgraphs with $\Gamma \subset \tilde{\Gamma}^0$, $\partial W \subseteq \partial \tilde{\Gamma}$ and $\partial V \subseteq \partial \Gamma$. Write $\partial V = \{x_1, \dots, x_m\}$. When x_i is regarded as a point of Γ , it has a unique tangent vector $\vec{v}_i$ leading into Γ . As in (6.14) in the proof of Lemma 6.9, for each $\alpha \in A$ we have

$$|\Delta_\Gamma(f_\alpha)|(\bar{V}) = |\Delta_\Gamma(f_\alpha)|(\Gamma) = 2 \cdot \sum_{i=1}^m |d_{\vec{v}_i} f_\alpha(x_i)|.$$

Thus, to bound the $|\Delta_{\bar{V}}(f_\alpha)|(\bar{V})$, it suffices to show that the $|d_{\vec{v}_i} f_\alpha(x_i)|$ are uniformly bounded, for all i and α .

Fix $x_i \in \partial V$. When x_i is regarded as a point of $\tilde{\Gamma}$, it will have several tangent vectors $\vec{v}_{i,j}$, $j = 1, \dots, n_i$, with each $n_i \geq 2$. Without loss, we can assume the $\vec{v}_{i,j}$ are indexed so that $\vec{v}_i = \vec{v}_{i,1}$ for each i . Fix $T > 0$ small enough that for each i, j , the point $q_{i,j} = x_i + t\vec{v}_{i,j}$ lies on the edge emanating from x_i in the direction $\vec{v}_{i,j}$.

Consider the limit function F , and fix $\varepsilon > 0$. Since the f_α converge uniformly to F on $\tilde{\Gamma}$, there is an α_0 such that $|f_\alpha(q_{i,j}) - F(q_{i,j})| < \varepsilon$ and

Justify this in Appendix A. $|f_\alpha(x_i) - F(x_i)| < \varepsilon$ for all $\alpha \geq \alpha_0$ and all i, j . It follows that for such i, j and α , we have

$$(6.22) \quad \frac{f_\alpha(q_{ij}) - f_\alpha(x_i)}{T} \leq \frac{(F(q_{ij}) + \varepsilon) - (F(x_i) - \varepsilon)}{T}.$$

Since each f_α is convex on edges of $\tilde{\Gamma}$, for each i, j

$$(6.23) \quad d_{\vec{v}_{ij}} f_\alpha(x_i) \leq \frac{f_\alpha(q_{ij}) - f_\alpha(x_i)}{T}.$$

Thus (6.22) provides an upper bound B_{ij} for each $d_{\vec{v}_{ij}} f_\alpha(x_i)$ which is independent of α .

Since x_i is an interior point of $\tilde{\Gamma}$ and $n_i \geq 2$, Lemma 6.21 gives $0 \geq \Delta_{\tilde{\Gamma}}(f_\alpha)(x_i) = -\sum_{j=1}^{n_i} d_{\vec{v}_{ij}} f_\alpha(x_i)$, so

$$(6.24) \quad d_{\vec{v}_{i1}} f_\alpha(x_i) \geq -\sum_{j=2}^{n_i} d_{\vec{v}_{ij}} f_\alpha(x_i) \geq -\sum_{j=2}^{n_i} B_{ij}.$$

Since $d_{\vec{v}_i} f_\alpha(x_i) = d_{\vec{v}_{i1}} f_\alpha(x_i)$, we have shown that $|d_{\vec{v}_i} f_\alpha(x_i)|$ is uniformly bounded for all i and all $\alpha \geq \alpha_0$. In particular, there exists $B > 0$ such that $|\Delta_{\bar{V}}(f_\alpha)|(\bar{V}) \leq B$ for all $\alpha \geq \alpha_0$.

We will now show the net of measures $\Delta_{\bar{V}}(f_\alpha)$ converges weakly to a measure μ on $\bar{V}$. Recall that by Proposition 5.4(B), as Γ ranges over all subgraphs of V and G ranges over all functions in $\text{CPA}(\Gamma)$, the functions $g(x) = G \circ r_\Gamma(x)$ are dense in $\mathcal{C}(\bar{V})$. For each subgraph $\Gamma' \supset \Gamma$, each $g(x)$ and each f_α , Lemma C.13(D) gives

$$\int_{\Gamma'} g \Delta_{\Gamma'}(f_\alpha) = \int_{\Gamma'} f_\alpha \Delta_{\Gamma'}(g) = \int_{\Gamma} f_\alpha \Delta_{\Gamma}(G),$$

where the last equality holds because $\Delta_{\Gamma'}(g)|_\Gamma = \Delta_\Gamma(G)$. Passing to the limit over all subgraphs $\Gamma \subset V$, we find that

$$(6.25) \quad \int_{\bar{V}} g \Delta_{\bar{V}}(f_\alpha) = \int_{\bar{V}} f_\alpha \Delta_{\bar{V}}(g) = \int_{\Gamma} f_\alpha \Delta_{\Gamma}(G).$$

Since the f_α converge uniformly to F on Γ , we can define a linear functional

$$(6.26) \quad \Lambda_F(g) = \lim_{\alpha} \int_{\bar{V}} f_\alpha \Delta_{\bar{V}}(g) = \int_{\bar{V}} F \Delta_{\bar{V}}(g) = \int_{\Gamma} F \Delta_{\Gamma}(G)$$

on the dense space of functions $g(x) = G \circ r_\Gamma(x)$ in $\mathcal{C}(\bar{V})$.

On the other hand, by (6.25), we have

$$(6.27) \quad \Lambda_F(g) = \lim_{\alpha} \int_{\bar{V}} g \Delta_{\bar{V}}(f_\alpha).$$

Since the measures $|\Delta_{\bar{V}}(f_\alpha)|(\bar{V})$ are uniformly bounded, Λ_F extends to a bounded linear functional on $\mathcal{C}(\bar{V})$. By the Riesz Representation theorem, there is a unique measure μ such that

$$\Lambda_F(g) = \int_{\bar{V}} g(x) d\mu(x) .$$

for all $g \in \mathcal{C}(\bar{V})$. This measure is the weak limit of the $\Delta_{\bar{V}}(f_\alpha)$.

To complete the proof, we must show that $\mu = \Delta_{\bar{V}}(F)$. Since $\Delta_{\bar{V}}(F)$ is a bounded measure on $\bar{V}$, it suffices to check that

$$\Lambda_F(g) = \int_{\bar{V}} g \Delta_{\bar{V}}(F)(x)$$

on the dense space of functions considered above. This follows immediately from (6.25), with f_α replaced by F . This proves (A).

For (B), fix a subgraph $\Gamma \subset U$. By part (A), it suffices to show that f_α converges uniformly to F on Γ . This follows from Dini's lemma (Lemma A.7), together with the observation that by Lemma 6.20, the restriction to Γ of a subharmonic function on U is continuous.

For (C), a similar argument shows that since $F(z) = f^*(z)$ is subharmonic on U , for each subgraph $\Gamma \subset U$ the restriction of $F(z)$ to Γ is continuous. By Proposition 6.22(E), $F(z)$ coincides with $f(z) = \lim_\alpha f_\alpha(z)$ on $U \cap \mathbb{H}_{\text{Berk}}$. Hence $f|_\Gamma = F|_\Gamma$ is continuous, and the f_α converge uniformly to F on Γ by Dini's lemma. $\square$

The following variant of Proposition 6.38 will be used in §7.

PROPOSITION 6.39. *Let U be a domain in $\mathbb{P}_{\text{Berk}}^1$, and let $h_\alpha, h \in \text{BDV}(U)$ for α in some directed set I . Suppose that $h_\alpha|_\Gamma \rightarrow h|_\Gamma$ for every subgraph $\Gamma \subset U$, and that the measures $\Delta_{\bar{U}}h_\alpha$ have uniformly bounded total mass. Then $\Delta_{\bar{U}}h_\alpha$ converges weakly to $\Delta_{\bar{U}}h$ on $\bar{U}$.*

PROOF. Write $r_\Gamma = r_{\bar{U}, \Gamma}$, $\mu_\alpha = \Delta_{\bar{U}}h_\alpha$, and $\mu = \Delta_{\bar{U}}h$. Let M be a bound such that $|\mu_\alpha| \leq M$ for all α .

By Proposition 5.4(B), as Γ ranges over all subgraphs of U and G ranges over all functions in $\text{CPA}(\Gamma)$, the “smooth” functions of the form $g(x) = G \circ r_\Gamma(x)$ are dense in $\mathcal{C}(\bar{U})$.

Let $f \in \mathcal{C}(\overline{U})$ be arbitrary. Writing $\|\cdot\|$ for the sup-norm on $\mathcal{C}(\overline{U})$, for any smooth $g \in \mathcal{C}(\overline{U})$ we have

$$\begin{aligned} \left| \int f \mu_\alpha - \int f \mu \right| &\leq \left| \int f \mu_\alpha - \int g \mu_\alpha \right| \\ &\quad + \left| \int g \mu_\alpha - \int g \mu \right| + \left| \int g \mu - \int f \mu \right| \\ &\leq (M + |\mu|(\overline{U})) \|f - g\| + \left| \int g \mu_\alpha - \int g \mu \right|. \end{aligned}$$

Choosing g so that $\|f - g\|$ is sufficiently small, we see that it suffices to prove that

$$\int g \mu_\alpha \rightarrow \int g \mu$$

when $g = G \circ r_\Gamma$ is a smooth function.

Since

$$\int_{\overline{U}} (G \circ r_\Gamma) \mu_\alpha = \int_\Gamma G (r_\Gamma)_* \mu_\alpha = \int_\Gamma G \Delta_\Gamma h_\alpha,$$

and similarly

$$\int_{\overline{U}} (G \circ r_\Gamma) \mu = \int_\Gamma G \Delta_\Gamma h,$$

we have reduced to proving that

$$\int_\Gamma F(\Delta_\Gamma h_\alpha) \rightarrow \int_\Gamma F(\Delta_\Gamma h).$$

But

$$\int_\Gamma F(\Delta_\Gamma h_\alpha) = \int_\Gamma h_\alpha \Delta_\Gamma F,$$

and similarly

$$\int_\Gamma F(\Delta_\Gamma h) = \int_\Gamma h \Delta_\Gamma F.$$

The proposition now follows from the hypothesis that $h_\alpha \rightarrow h$ uniformly on Γ . $\square$

We augment Proposition 6.38 with two simple results which deal with other cases in Proposition 6.22. The first is immediate by the linearity of the Laplacian.

PROPOSITION 6.40. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be open. Suppose $f, g \in \text{SH}(U)$, and $0 \leq \alpha, \beta \in \mathbb{R}$. Then $\Delta_U(\alpha f + \beta g) = \alpha \cdot \Delta_U(f) + \beta \cdot \Delta_U(g)$.*

Rewrite the proof of the following Proposition.

PROPOSITION 6.41. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be open. Suppose $f(z)$ and $g(z)$ are continuous subharmonic functions on U , with $f(z) > g(z)$ outside a compact subset of U . Put $h(z) = \max(f(z), g(z))$ and let $Z = \{z \in U : f(z) \leq g(z)\}$. Then*

- A) $\Delta_U(h)|_{U \setminus Z} \equiv \Delta_U(f)|_{U \setminus Z}$, and
- B) $\Delta_U(h)(Z) = \Delta_U(f)(Z)$.

PROOF. We can assume without loss that U is a domain, since the result holds if and only if it holds for each component of U .

The proof of (A) is simple. For each $x \in U \setminus \bar{V}$, there is a simple subdomain $W \subset U$ such that $x \in W$ and $\bar{W} \cap \bar{V} = \varnothing$. We have $h(z) = f(z)$ on W , so $\Delta_{\bar{W}}(h) = \Delta_{\bar{W}}(f)$, and

$$\Delta_U(f)|_W = \Delta_{\bar{W}}(f)|_W = \Delta_{\bar{W}}(h)|_W = \Delta_U(h)|_W.$$

Since $U \setminus U$ can be covered by such W , the result follows.

For part (B), note that Z is compact; fix a simple subdomain $V \subset U$ with $Z \subset V$. By Lemma 6.9, the restrictions of f , g and h to V belong to $\text{BDV}(V)$. Note that ∂V is compact. Fix $\zeta \in Z$, and cover ∂V with a finite number of balls $\mathcal{B}(x_i, r_i)_{\zeta}^-$, where the closure of each $\mathcal{B}(x_i, r_i)_{\zeta}^-$ is disjoint from Z . Without loss, we can assume the $\mathcal{B}(x_i, r_i)_{\zeta}^-$ are pairwise disjoint. Let p_i be the unique boundary point of $\mathcal{B}(x_i, r_i)_{\zeta}^-$; it belongs to $V \setminus Z$. Put $V_i = U \cap \mathcal{B}(x_i, r_i)_{\zeta}^-$; then $h(z) = f(z)$ on V_i , and $\partial V_i \subset \partial V \cup \{p_i\}$.

By the retraction formula for Laplacians,

$$(r_{\bar{V}, \bar{V}_i})_*(\Delta_{\bar{V}}(f)) = \Delta_{\bar{V}_i}(f) = \Delta_{\bar{V}_i}(h) = (r_{\bar{V}, \bar{V}_i})_*(\Delta_{\bar{V}}(h)).$$

Since $r_{\bar{V}, \bar{V}_i}(x)$ is the identity map on $\bar{V}_i \setminus \{p_i\}$ and $r_{\bar{V}, \bar{V}_i}(\bar{V} \setminus \bar{V}_i) = \{p_i\}$,

$$\Delta_{\bar{V}}(f)|_{\bar{V}_i \setminus \{p_i\}} = \Delta_{\bar{V}}(h)|_{\bar{V}_i \setminus \{p_i\}}.$$

Combined with the result from (A), this gives $\Delta_{\bar{V}}(f)|_{\bar{V} \setminus Z} = \Delta_{\bar{V}}(h)|_{\bar{V} \setminus Z}$. Since both $\Delta_{\bar{V}}(f)$ and $\Delta_{\bar{V}}(h)$ have total mass 0,

$$\Delta_{\bar{V}}(f)(Z) = -\Delta_{\bar{V}}(f)(\bar{V} \setminus Z) = -\Delta_{\bar{V}}(h)(\bar{V} \setminus Z) = \Delta_{\bar{V}}(h)(Z).$$

□

7. Smoothing

In the classical theory over the complex numbers, each subharmonic function f is locally a decreasing limit of $\mathcal{C}^\infty$ subharmonic functions. These functions are obtained by convolving f with a certain smoothing kernel.

Here there seems to be no smoothing kernel, but nonetheless each subharmonic function can be approximated by continuous subharmonic

functions of a special form. The “nicest” functions we have encountered on $\mathbb{P}_{\text{Berk}}^1$ are those of the form $f(z) = F \circ r_\Gamma(z)$, where Γ is a subgraph and $F \in \text{CPA}(\Gamma)$. We deem them to be the smooth functions on $\mathbb{P}_{\text{Berk}}^1$. More generally, we make the following definition:

DEFINITION 6.42. Let $U \subseteq \mathbb{P}_{\text{Berk}}^1$ be a domain, and let $f : U \rightarrow \mathbb{R}$ be a real-valued function. We say that f is *smooth* if there exists a subgraph $\Gamma \subset U$ and a function $F \in \text{CPA}(\Gamma)$ such that $f = F \circ r_{\overline{U}, \Gamma}$.

LEMMA 6.43. *If $f \in \text{BDV}(U)$ is continuous on U , then f is smooth if and only if $\Delta_{\overline{U}}(f)$ is a discrete measure supported on $U \cap \mathbb{H}_{\text{Berk}}$.*

PROOF. If $f = F \circ r_{\overline{U}, \Gamma}$ is smooth, then $\Delta_{\overline{U}}(f) = \Delta_\Gamma(F)$ is a discrete measure supported on $\Gamma \subset U \cap \mathbb{H}_{\text{Berk}}$ since $F \in \text{CPA}(\Gamma)$. On the other hand, suppose $\Delta_{\overline{U}}(f)$ is a discrete measure supported on $U \cap \mathbb{H}_{\text{Berk}}$, and choose a subgraph $\Gamma \subset U$ containing $\text{supp}(\Delta_{\overline{U}}(f))$. Then $\Delta_\Gamma(f) = (r_{\overline{U}, \Gamma})_* \Delta_{\overline{U}}(f) = \Delta_{\overline{U}}(f)$ is a discrete measure supported on Γ . If $F := f|_\Gamma$, then $F \in \text{CPA}(\Gamma)$. Since $\Delta_{\overline{U}}(F \circ r_{\overline{U}, \Gamma}) = \Delta_{\overline{U}}(f)$ and $f = F \circ r_{\overline{U}, \Gamma}$ on Γ , it follows that $f = F \circ r_{\overline{U}, \Gamma}$ on $U \cap \mathbb{H}_{\text{Berk}}$. Since f is continuous on U , $f = F \circ r_{\overline{U}, \Gamma}$ on all of U and therefore f is smooth. $\square$

PROPOSITION 6.44. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, and let f be subharmonic in U . Then for every simple subdomain V of U , there is a net $\langle f_\alpha \rangle$ of smooth subharmonic functions on V such that $f_\alpha \rightarrow f$ pointwise on V , and $f_\alpha \rightarrow f$ uniformly on each subgraph Γ of V .*

In particular, each $\Delta_{\overline{V}}(f_\alpha)$ is a discrete measure, and $\Delta_{\overline{V}}(f_\alpha)$ converges weakly to $\Delta_{\overline{V}}(f)$.

See if there's a way to simplify this proof.

PROOF. Let Γ_α be the directed system of all subgraphs of V .

For each α , choose $\varepsilon_\alpha > 0$ in such a way that $\lim_\alpha \varepsilon_\alpha = 0$. For example, one can take

$$\varepsilon_\alpha = \inf_{z \in \Gamma} \frac{1}{\rho(z, \partial V)},$$

where ρ is the path-distance metric on $\mathbb{H}_{\text{Berk}}$. For each α , choose a finite set $\Sigma_\alpha = \{p_{\alpha,1}, \dots, p_{\alpha,M_\alpha}\} \subset \Gamma_\alpha$ containing $\partial\Gamma_\alpha$ such that given $z \in \Gamma_\alpha$, there exists i with $\rho(p_{\alpha,i}, z) \leq \varepsilon_\alpha$. Now for each α , define $F_\alpha \in \text{CPA}(\Gamma_\alpha)$ to be the function on Γ_α whose values at $p_{\alpha,i}$ are given by $F_\alpha(p_{\alpha,i}) = f(p_{\alpha,i})$, and which interpolates linearly on the intervening segments.

Put $f_\alpha(z) = F_\alpha \circ r_{\Gamma_\alpha}(z)$, so $f_\alpha(z)$ is constant on branches off Γ_α . The functions $f_\alpha(x)$ converge pointwise to $f(x)$ on each fixed subgraph $\Gamma \subset V$, since $f|_\Gamma$ is continuous and $\varepsilon_\alpha \rightarrow 0$. By Dini's lemma (Lemma A.7), $f_\alpha \rightarrow f$ uniformly on Γ . Since each $z \in V \cap \mathbb{H}_{\text{Berk}}$ belongs to some subgraph $\Gamma \subset V$, we also see that $f_\alpha(z) \rightarrow f(z)$ pointwise on $V \cap \mathbb{H}_{\text{Berk}}$.

Similarly, if $q \in V \cap \mathbb{P}^1(K)$, then since $r_{\overline{V}, \Gamma_\alpha}(p) \in \partial\Gamma_\alpha \subseteq \Sigma_\alpha$ for all α and $r_{\overline{V}, \Gamma_\alpha}(p) \rightarrow p$, it follows from the continuity of f that $f_\alpha(p) = f(r_{\overline{V}, \Gamma_\alpha}(p))$ converges to $f(p)$. We conclude that f_α converges pointwise to f on V , and the convergence is uniform on every subgraph $\Gamma \subset V$.

We now show that each f_α is subharmonic on V . We have

$$\Delta_{\overline{V}}(f_\alpha) = \Delta_{\Gamma_\alpha}(F_\alpha) = \sum c_{\alpha,i} \delta_{p_{\alpha,i}}$$

for some $c_{\alpha,i} \in \mathbb{R}$. We claim that $c_{\alpha,i} \leq 0$ if $p_{\alpha,i} \notin \partial V$. By Lemma 6.21, for each $p_{\alpha,i} \notin \partial V$

$$- \sum_{\vec{v} \in T_{p_{\alpha,i}}} d_{\vec{v}} f(p_{\alpha,i}) = \Delta_{\Gamma_\alpha} f(p_{\alpha,i}) \leq 0.$$

As $d_{\vec{v}} f_\alpha(p_{\alpha,i}) \geq d_{\vec{v}} f(p_{\alpha,i})$ for each $\vec{v}$, this gives $c_{\alpha,i} = \Delta_{\Gamma_\alpha} F_\alpha(p_{\alpha,i}) \leq 0$. Since $f_\alpha(z)$ is constant on branches off Γ_α ,

$$\Delta_{\overline{V}}(f_\alpha) = \Delta_{\Gamma_\alpha}(F_\alpha) = \sum_{i=1}^{M_\alpha} c_{\alpha,i} \delta_{p_{\alpha,i}}(x).$$

In particular $\Delta_V(f_\alpha) \leq 0$. By construction f_α is continuous. By Lemma 6.9, f_α is subharmonic on V .

The final assertion, that the measures $\Delta_V(f_\alpha)$ converge weakly to $\Delta_V(f)$, follows from Proposition 6.38(A). □

8. The Energy Minimization Principle

Recall that for each $\zeta \in \mathbb{P}_{\text{Berk}}^1$, the Hsia kernel is

$$(6.28) \quad \delta(x, y)_\zeta = \frac{\|x, y\|}{\|x, \zeta\| \|y, \zeta\|} = \frac{\delta(x, y)_{\zeta_{\text{Gauss}}}}{\delta(x, \zeta)_{\zeta_{\text{Gauss}}} \delta(y, \zeta)_{\zeta_{\text{Gauss}}}}.$$

Given a log-continuous probability measure μ (c. f. Definition 8.2), define the *normalized Arakelov-Green's function* $g_\mu(x, y)$ associated to μ by

$$(6.29) \quad g_\mu(x, y) = \int_{\mathbb{P}_{\text{Berk}}^1} -\log_v(\delta(x, y)_\zeta) d\mu(\zeta) + C,$$

where the normalizing constant C is chosen so that

$$(6.30) \quad \iint g_\mu(x, y) d\mu(x) d\mu(y) = 0 .$$

Using (6.28) one sees that

$$g_\mu(x, y) = j_{\zeta_{\text{Gauss}}}(x, y) - u_\mu(x, \zeta_{\text{Gauss}}) - u_\mu(y, \zeta_{\text{Gauss}}) + C$$

and that $C = \iint j_{\zeta_{\text{Gauss}}}(x, y) d\mu(x) d\mu(y)$. Since $u_\mu(z, \zeta_{\text{Gauss}})$ is continuous, $g_\mu(x, y)$ inherits the following properties from $j_{\zeta_{\text{Gauss}}}(x, y)$.

PROPOSITION 6.45. *Let μ be a log-continuous probability measure on $\mathbb{P}_{\text{Berk}}^1$. Then $g_\mu(x, y)$ is symmetric, lower semicontinuous everywhere, continuous off the Type I diagonal, and bounded from below. For each fixed y , the function $G_y(x) = g_\mu(x, y)$ is continuous on $\mathbb{P}_{\text{Berk}}^1 \setminus \{y\}$ and belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$; it is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \{y\}$ and satisfies $\Delta_{\mathbb{P}_{\text{Berk}}^1}(G_y(x)) = \delta_y(x) - \mu$.*

Given a probability measure ν on $\mathbb{P}_{\text{Berk}}^1$, define the generalized potential function

$$(6.31) \quad u_\nu(x, \mu) = \int g_\mu(x, y) d\nu(y) ,$$

and the μ -energy integral

$$(6.32) \quad I_\nu(\mu) = \iint g_\mu(x, y) d\nu(x) d\nu(y) = \int u_\nu(x, \mu) d\nu(x) .$$

Our goal is to prove the following energy minimization principle:

THEOREM 6.46. *Let μ be a log-continuous probability measure on $\mathbb{P}_{\text{Berk}}^1$. Then*

- A) $I_\mu(\nu) \geq 0$ for each probability measure ν on $\mathbb{P}_{\text{Berk}}^1$, and
- B) $I_\mu(\nu) = 0$ if and only if $\nu = \mu$.

The proof rests on establishing analogues of Maria's theorem and Frostman's theorem for the function $f(x) = u_\nu(x, \mu)$. We begin by showing that generalized potential functions have the same properties as usual ones.

PROPOSITION 6.47. *Let μ be a log-continuous probability measure on $\mathbb{P}_{\text{Berk}}^1$, and let ν be an arbitrary probability measure on $\mathbb{P}_{\text{Berk}}^1$. Then $u_\nu(x, \mu)$ is lower semi-continuous everywhere, and is continuous at each $z \notin \text{supp}(\nu)$. For each $p \in \mathbb{P}^1(K)$, as x approaches p along any path $[y, p]$,*

$$\lim_{\substack{x \rightarrow p \\ x \in [y, p]}} f(x) = f(p) .$$

Moreover, $u_\nu(z, \mu)$ belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and $\Delta_{\mathbb{P}_{\text{Berk}}^1}(u_\nu(x, \mu)) = \nu - \mu$.

PROOF. Note that

$$\begin{aligned} u_\nu(x, \mu) &= \iint (j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \zeta) - j_{\zeta_{\text{Gauss}}}(y, \zeta)) \, d\mu(\zeta) d\nu(y) \\ &= u_\nu(x, \zeta_{\text{Gauss}}) - u_\mu(x, \zeta_{\text{Gauss}}) - C' . \end{aligned}$$

Here $C' = \int u_\mu(y, \zeta_{\text{Gauss}}) \, d\nu(y)$ is a finite constant. Since $u_\mu(x, \zeta_{\text{Gauss}})$ is bounded and continuous everywhere, the assertions about continuity, semicontinuity, and path limits follow from the corresponding facts for $u_\nu(x, \zeta_{\text{Gauss}})$ proved in Proposition 4.12.

We have also seen in Example 5.14 that $u_\nu(x, \zeta_{\text{Gauss}})$ and $u_\mu(x, \zeta_{\text{Gauss}})$ belong to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$. By the computations of the Laplacians there, $\Delta(u_\nu(x, \mu)) = (\nu - \delta_{\zeta_{\text{Gauss}}}(x)) - (\mu - \delta_{\zeta_{\text{Gauss}}}(x)) = \nu - \mu$. $\square$

The following lemma plays a key role in the proof of Maria's theorem for the generalized potential function $u_\nu(z, \mu)$. We have already used its analogue for standard potential functions.

LEMMA 6.48. *Let Γ be a subgraph of $\mathbb{P}_{\text{Berk}}^1$, and let ν be an arbitrary probability measure on $\mathbb{P}_{\text{Berk}}^1$. Then for each $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, the restriction of $u_\nu(x, \zeta)$ to Γ is finite and continuous.*

PROOF. First note that for any $\xi \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, there is a constant C such that

$$(6.33) \quad u_\nu(x, \zeta) = u_\nu(x, \xi) + j_{\zeta_{\text{Gauss}}}(x, \xi) - j_{\zeta_{\text{Gauss}}}(x, \zeta) + C .$$

To see this, compute

$$\begin{aligned} u_\nu(x, \zeta) - u_\nu(x, \xi) &= \int j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \zeta) - j_{\zeta_{\text{Gauss}}}(y, \zeta) \, d\nu(y) \\ &\quad - \int j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \xi) - j_{\zeta_{\text{Gauss}}}(y, \xi) \, d\nu(y) \\ &= j_{\zeta_{\text{Gauss}}}(x, \xi) - j_{\zeta_{\text{Gauss}}}(x, \zeta) \\ &\quad + \int j_{\zeta_{\text{Gauss}}}(y, \xi) \, d\nu(y) - \int j_{\zeta_{\text{Gauss}}}(y, \zeta) \, d\nu(y) . \end{aligned}$$

Since ζ_{Gauss} , ξ and ζ all belong to $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, the functions $j_{\zeta_{\text{Gauss}}}(y, \xi)$ and $j_{\zeta_{\text{Gauss}}}(y, \zeta)$ are bounded and continuous on $\mathbb{P}_{\text{Berk}}^1$, and hence the integrals in the last line are finite.

Now take $\xi = r_\Gamma(\zeta)$. Note that there is a constant C_1 such that

$$j_\xi(x, y) + C_1 = j_{\zeta_{\text{Gauss}}}(x, y) - j_{\zeta_{\text{Gauss}}}(x, \xi) - j_{\zeta_{\text{Gauss}}}(y, \xi) .$$

By (6.33), to show that $u_\nu(x, \zeta)$ is continuous on Γ it suffices to show that $u_\nu(x, \xi)$ is continuous on Γ , since $j_{\zeta_{\text{Gauss}}}(x, \xi)$ and $j_{\zeta_{\text{Gauss}}}(x, \zeta)$ are

continuous on all of $\mathbb{P}_{\text{Berk}}^1$. However, for $x \in \Gamma$,

$$\begin{aligned} u_\nu(x, \xi) &= \int j_\xi(x, y) d\nu(y) + C_1 \\ &= \int_\Gamma j_\xi(x, r_\Gamma(y)) d\nu(y) + C_1 = \int_\Gamma j_\xi(x, w) d(r_\Gamma)_*(\nu)(w) + C_1 \end{aligned}$$

Since $j_\xi(x, w)$ is bounded and continuous for $(x, w) \in \Gamma \times \Gamma$, it follows that that $u_\nu(x, \xi)$ is finite and continuous on Γ . $\square$

The following result is an analogue of Maria's theorem from classical potential theory.

PROPOSITION 6.49. *Let μ be a log-continuous probability measure on $\mathbb{P}_{\text{Berk}}^1$, and let ν be a probability measure on $\mathbb{P}_{\text{Berk}}^1$. If there is a constant $M < \infty$ such that $u_\nu(z, \mu) \leq M$ on $\text{supp}(\nu)$, then $u_\nu(z, \mu) \leq M$ for all $z \in \mathbb{P}_{\text{Berk}}^1$.*

PROOF. Put $E = \text{supp}(\nu)$, and let U be a component of $\mathbb{P}_{\text{Berk}}^1 \setminus E$. By Proposition 6.47, $u_\nu(z, \mu)$ is continuous on U and $\Delta_U(u_\nu(z, \mu)) \leq 0$. Hence $u_\nu(z, \mu)$ is subharmonic on U .

Let $f(z) = u_\nu(z, \mu)$. Suppose there exists $p_0 \in U$ with $f(p_0) > M$. As f is continuous on U and points of $\mathbb{H}_{\text{Berk}}$ are dense in U , we may assume without loss of generality that $p_0 \in \mathbb{H}_{\text{Berk}}$. Fix $\sigma > 0$ with $f(p_0) \geq M + \sigma$. By Proposition 6.14, there exists a point $q \in \partial U$ and a path Λ from p_0 to q such that f is non-decreasing along Λ . In particular, $f(z) \geq M + \sigma$ for all $z \in \Lambda^0$ and

$$(6.34) \quad \lim_{\substack{x \rightarrow q \\ x \in \Lambda^0}} f(x) \geq M + \sigma.$$

Now note that the limit on the left-hand side of (6.34) equals $f(q)$. (If $q \in \mathbb{H}_{\text{Berk}}$, this follows from Lemma 6.48, and if $q \in \mathbb{P}^1(K)$, it follows from Proposition 6.47.) Therefore $f(q) \geq M + \sigma$. But $q \in \partial U \subseteq \text{supp}(\nu)$, contradicting the assumption that $f(z) \leq M$ on $\text{supp}(\nu)$. $\square$

Define the ' μ -Robin constant' to be

$$V(\mu) = \inf_{\nu \text{ prob meas}} I_\mu(\nu) = \inf_{\nu \text{ prob meas}} \iint g_\mu(x, y) d\nu(x) d\nu(y),$$

where ν runs over all probability measures supported on $\mathbb{P}_{\text{Berk}}^1$. Trivially $V(\mu) > -\infty$, since $g_\mu(x, y)$ is bounded below, and $V(\mu) \leq 0$, since $I_\mu(\mu) = 0$ by the normalization of $g_\mu(x, y)$. Taking the weak limit of a sequence of measures $\nu_1, \nu_2, \dots$ for which $I_\mu(\nu_n) \rightarrow V(\mu)$ and applying the same argument used to prove the existence of an equilibrium measure, one obtains a probability measure ω for which $I_\mu(\omega) = V(\mu)$.

PROPOSITION 6.50. (Frostman) *Let μ be a log-continuous probability measure on $\mathbb{P}_{\text{Berk}}^1$, and let ω be a probability measure for which $I_\mu(\omega) = V(\mu)$. Then the generalized potential function $u_\omega(z, \mu)$ satisfies*

$$u_\omega(z, \mu) \equiv V(\mu) \quad \text{on } \mathbb{P}_{\text{Berk}}^1.$$

PROOF. First, using a quadraticity argument, we will show that $u_\omega(z, \mu) \geq V(\mu)$ for all $z \in \mathbb{P}_{\text{Berk}}^1$ except possibly on a set f of capacity 0. Then, we will show that $u_\omega(z, \mu) \leq V(\mu)$ on $\text{supp}(\omega)$. By Maria's theorem, $u_\omega(z, \mu) \leq V(\mu)$ for all $z \in \mathbb{P}_{\text{Berk}}^1$. Since a set of capacity 0 is necessarily contained in $\mathbb{P}^1(K)$, it follows that $u_\omega(z, \mu) = V(\mu)$ on $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$. Finally, if $p \in \mathbb{P}^1(K)$, let $z \rightarrow p$ along a path $[y, p]$. Proposition 6.47 gives $u_\omega(p, \mu) = V(\mu)$.

Put

$$\begin{aligned} f &= \{z \in \mathbb{P}_{\text{Berk}}^1 : u_\omega(z, \mu) < V(\mu)\}, \\ f_n &= \{z \in \mathbb{P}_{\text{Berk}}^1 : u_\omega(z, \mu) < V(\mu) - 1/n\}, \quad \text{for } n = 1, 2, 3, \dots \end{aligned}$$

Since $u_\omega(z, \mu)$ is lower semicontinuous, each f_n is closed, hence compact, so f is an F_σ set. By Corollary 4.15, f has capacity 0 if and only if each f_n has capacity 0.

Suppose f_n has positive capacity for some n ; then there is a probability measure σ supported on f_n such that $I_{\zeta_{\text{Gauss}}}(\sigma) := \iint j_{\zeta_{\text{Gauss}}}(x, y) d\sigma(x) d\sigma(y) < \infty$. Since $g_\mu(x, y)$ differs from $j_{\zeta_{\text{Gauss}}}(x, y)$ by a bounded function, $I_\mu(\sigma) < \infty$ as well.

Furthermore, since

$$V(\mu) = \iint g_\mu(x, y) d\omega(x) d\omega(y) = \int u_\omega(x, \mu) d\omega(x),$$

there is a point $q \in \text{supp}(\mu)$ with $u_\omega(q, \mu) \geq V(\mu)$. Since $u_\omega(z, \mu)$ is lower semicontinuous, there is a neighborhood U of q on which $u_\omega(z, \mu) > V(\mu) - 1/(2n)$. After shrinking U if necessary, we can assume its closure $\overline{U}$ is disjoint from f_n . Since $q \in \text{supp}(\omega)$, it follows that $M := \omega(\overline{U}) > 0$. Define a measure σ_1 of total mass 0 by

$$\sigma_1 = \begin{cases} M \cdot \sigma & \text{on } f_n, \\ -\mu & \text{on } \overline{U}, \\ 0 & \text{elsewhere.} \end{cases}$$

We claim that $I_\zeta(\sigma_1)$ is finite. Indeed

$$\begin{aligned} I_\zeta(\sigma_1) &= M^2 \cdot \iint_{f_n \times f_n} g_\mu(x, y) d\sigma(x) d\sigma(y) \\ &\quad - 2M \cdot \iint_{f_n \times \bar{U}} g_\mu(x, y) d\sigma(x) d\omega(y) \\ &\quad + \iint_{\bar{U} \times \bar{U}} g_\mu(x, y) d\omega(x) d\omega(y) . \end{aligned}$$

The first integral is finite by hypothesis. The second is finite because f_n and $\bar{U}$ are disjoint, so $g_\mu(z, w)$ is bounded on $f_n \times \bar{U}$. The third is finite because $I_\mu(\omega)$ is finite and $g_\mu(z, w)$ is bounded below.

For each $0 \leq t \leq 1$, $\omega_t := \omega + t\sigma_1$ is a probability measure. By an expansion like the one above,

$$\begin{aligned} I_\mu(\omega_t) - I_\mu(\omega) &= 2t \cdot \int_E u_\omega(z, \mu) d\sigma_1(z) + t^2 \cdot I_\mu(\sigma_1) \\ &\leq 2t \cdot ((V(\mu) - 1/n) - (V(\mu) - 1/(2n)) \cdot M) + t^2 \cdot I_\mu(\sigma_1) \\ &= (-M/n) \cdot t + I_\mu(\sigma_1) \cdot t^2 . \end{aligned}$$

For sufficiently small $t > 0$, the right side is negative. This contradicts the fact that ω minimizes the energy integral. It follows that f_n has capacity 0, and hence $f = \cup_{n=1}^\infty f_n$ has capacity 0 by Corollary 4.15.

The second part requires showing that $u_\omega(z, \mu) \leq V(\mu)$ for all $z \in \text{supp}(\omega)$. If $u_\omega(z, \mu) > V(\mu)$ for some $q \in \text{supp}(\omega)$, let $\varepsilon > 0$ be small enough that $u_\omega(q, \mu) > V(\mu) + \varepsilon$. The lower semicontinuity of $u_\omega(z, \mu)$ shows there is a neighborhood U of q on which $u_\omega(z, \mu) > V(\mu) + \varepsilon$. Then $T := \omega(U) > 0$, since $q \in \text{supp}(\mu)$. On the other hand, by Lemma 4.14, $\omega(f) = 0$ since $I_\mu(\omega) < \infty$ implies that $I_{\zeta_{\text{Gauss}}}(\omega) < \infty$. Since $u_\omega(z, \mu) \geq V(\mu)$ for all $z \notin f$,

$$\begin{aligned} V(\mu) &= \int_U u_\omega(z, \mu) d\omega(z) + \int_{\mathbb{P}^1_{\text{Berk}} \setminus U} u_\omega(z, \mu) d\omega(z) \\ &\geq T \cdot (V(\mu) + \varepsilon) + (1 - T) \cdot V(\mu) = V(\mu) + T\varepsilon \end{aligned}$$

which is impossible. Hence $u_\omega(z, \mu) \leq V(\mu)$ on $\text{supp}(\omega)$, and Maria's theorem implies that $u_\omega(z, \mu) \leq V(\mu)$ for all z .

As noted at the beginning of the proof, the fact that $u_\omega(p, \mu) = V(\mu)$ for all $p \in \mathbb{P}^1(K)$, now follows from Proposition 6.47. $\square$

We can now complete the proof of Theorem 6.46.

PROOF. (of Theorem 6.46).

Suppose ω minimizes the energy integral $I_\mu(\nu)$. By Proposition 6.50, $u_\omega(z, \mu)$ is constant. It follows that $\Delta(u_\omega(z, \mu)) \equiv 0$. On the other hand, by Proposition 6.47, $\Delta(u_\omega(z, \mu)) = \omega - \mu$. Hence $\omega = \mu$.

However, $I_\mu(\mu) = 0$ by the normalization of $g_\mu(x, y)$, so $V(\mu) = 0$. It follows that $I_\mu(\nu) \geq 0$ for all ν , and if $I_\mu(\nu) = 0$, then $\nu = \mu$. $\square$

Theorem 6.46 is one of the key results needed in [5] for proving a nonarchimedean equidistribution theorem for points of small dynamical height.

9. Notes and further references

CHAPTER 7

Multiplicities

In this chapter, we develop a theory of multiplicities for φ at points in $\mathbb{P}_{\text{Berk}}^1$, with properties analogous to the classical algebraic multiplicities. We then define pullback and pushforward measures under φ , and establish their functoriality properties.

1. Analytic multiplicities

For classical points $a, b \in \mathbb{P}^1(K)$ with $\varphi(a) = b$, write $m_\varphi(a) = m_{\varphi,b}(a)$ for the multiplicity of φ at a . For a, b with $\varphi(a) \neq b$, put $m_{\varphi,b}(a) = 0$. Thus for each $b \in K$

$$\text{div}(\varphi(T) - b) = \sum_{\varphi(a)=b} m_{\varphi,b}(a) \delta_a(x) - \sum_{\varphi(a)=\infty} m_{\varphi,\infty}(a) \delta_a(x) ,$$

and for each $b \in \mathbb{P}^1(K)$

$$\sum_{a \in \mathbb{P}^1(K)} m_{\varphi,b}(a) = d .$$

We seek multiplicities on $\mathbb{P}_{\text{Berk}}^1$ which share these properties.

Our starting point is the following computation:

LEMMA 7.1. *For $b, \zeta \in \mathbb{P}^1(K)$, $b \neq \zeta$, set*

$$h_{b,\zeta}(x) = -\log_v(\delta(\varphi(x), b)_\zeta) .$$

Then

$$\Delta_{\mathbb{P}_{\text{Berk}}^1} h_{b,\zeta} = \sum_{\varphi(a)=b} m_{\varphi,b}(a) \delta_a(x) - \sum_{\varphi(a)=\zeta} m_{\varphi,\zeta}(a) \delta_a(x) .$$

PROOF. It follows from Example 5.13 that

$$\Delta_{\mathbb{P}_{\text{Berk}}^1} (\log_v([\varphi(T) - b]_x)) = \sum_{\varphi(a)=\infty} m_{\varphi,\infty}(a) \delta_a(x) - \sum_{\varphi(a)=b} m_{\varphi,b}(a) \delta_a(x) .$$

On the other hand, by Corollary 3.2, for each $x \in \mathbb{P}_{\text{Berk}}^1$ and $b \in K$, $\delta(x, b)_\infty = [T - b]_x$. Hence

$$\delta(\varphi(x), b)_\infty = [T - b]_{\varphi(x)} = [\varphi(T) - b]_x ,$$

so $-\log_v([\varphi(T) - b]_x) = h_{b,\infty}(x)$. The lemma follows from this in the special case where $\zeta = \infty$.

Now let $\zeta \in \mathbb{P}^1(K)$ be arbitrary. There is a constant C such that

$$\delta(x, b)_\zeta = C \cdot \frac{\delta(x, b)_\infty}{\delta(x, \zeta)_\infty \delta(b, \zeta)_\infty}.$$

Taking logarithms and applying the Laplacian shows that $h_{b,\zeta}(x) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and gives the desired formula when $b \neq \infty$. It is easy to see that the formula holds for $b = \infty$ as well. $\square$

We will now *define* multiplicities on $\mathbb{P}_{\text{Berk}}^1$ in such a way that Lemma 7.1 holds more generally for $b, \zeta \in \mathbb{P}_{\text{Berk}}^1$.

For $z \in \mathbb{P}^1(K)$, define $\varphi^*(\delta_z)$ to be the discrete measure

$$\varphi^*(\delta_z) = \sum_{\varphi(a)=z} m_{\varphi,z}(a) \delta_a.$$

Then Lemma 7.1 can be rephrased as saying that for $b, \zeta \in \mathbb{P}^1(K)$, $b \neq \zeta$, we have

$$\Delta_{\mathbb{P}_{\text{Berk}}^1} h_{b,\zeta} = \varphi^*(\delta_b) - \varphi^*(\delta_\zeta).$$

If $\zeta \in \mathbb{P}^1(K)$ and $b \in \mathbb{H}_{\text{Berk}}$, we now *define* the measure $\varphi^*(\delta_b)$ by the formula

$$(7.1) \quad \Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v(\delta(\varphi(x), b)_\zeta)) = \varphi^*(\delta_b) - \varphi^*(\delta_\zeta).$$

LEMMA 7.2. *The measure $\varphi^*(\delta_b)$ does not depend on the choice of ζ .*

PROOF. We temporarily write $\varphi^*(\delta_b)_\zeta$ for the quantity

$$\Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v(\delta(\varphi(x), b)_\zeta)) + \varphi^*(\delta_\zeta).$$

The key fact is that for any $\zeta, \xi \in \mathbb{P}^1(K)$, there is a constant C such that for all $y \in \mathbb{P}_{\text{Berk}}^1$, we have

$$(7.2) \quad \delta(y, b)_\xi = C \cdot \frac{\delta(y, b)_\zeta}{\delta(y, \xi)_\zeta \delta(b, \xi)_\zeta}.$$

Here $\delta(b, \xi)_\xi$ is a finite nonzero constant since $b \notin \mathbb{P}^1(K)$.

Thus we see that

$$(7.3) \quad \begin{aligned} \Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v(\delta(\varphi(x), b)_\zeta)) - \Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v(\delta(\varphi(x), b)_\xi)) \\ = \Delta_{\mathbb{P}_{\text{Berk}}^1} (-\log_v(\delta(\varphi(x), b)_\zeta)) \\ = \varphi^*(\delta_\xi) - \varphi^*(\delta_\zeta). \end{aligned}$$

On the other hand, the left-hand side of (7.3) is just

$$(\varphi^*(\delta_b)_\zeta - \varphi^*(\delta_\zeta)) - (\varphi^*(\delta_b)_\xi - \varphi^*(\delta_\xi)),$$

so that

$$\varphi^*(\delta_b)_\zeta = \varphi^*(\delta_b)_\xi$$

as desired. $\square$

Although it is not completely obvious, we will see that $\varphi^*(\delta_b)$ is a discrete positive measure. Given this, it makes sense to define the *analytic multiplicity* $m_{\varphi,b}(a)$ for $a, b \in \mathbb{P}_{\text{Berk}}^1$ as the coefficient of δ_a in the expression

$$(7.4) \quad \varphi^*(\delta_b) = m_{\varphi,b}(a) \delta_a .$$

We will see that the real numbers $m_{\varphi,b}(a)$ are nonnegative integers, and that $m_{\varphi,b}(a) > 0$ if and only if $\varphi(b) = a$. So this definition really does give us a reasonable notion of multiplicities on all of $\mathbb{P}_{\text{Berk}}^1$.

The key to understanding the analytic multiplicities is given by the following result.

PROPOSITION 7.3. *If $\langle b_\alpha \rangle$ is a net in $\mathbb{P}^1(K)$ converging to $b \in \mathbb{P}_{\text{Berk}}^1$, then the net of measures $\varphi^*(\delta_{b_\alpha})$ converges weakly to $\varphi^*(\delta_b)$.*

PROOF. Write $\mu_\alpha = \varphi^*(\delta_{b_\alpha})$ and $\mu = \varphi^*(\delta_b)$. Without loss of generality, we may assume that there is a simple domain V containing b and all the points b_α . Fix any point $\zeta \in \mathbb{P}^1(K) \setminus V$, and set $h_\alpha(x) = -\log_v(\delta(\varphi(x), b_\alpha)_\zeta)$ and $h(x) = -\log_v(\delta(\varphi(x), b)_\zeta)$.

By Proposition 6.39, it suffices to show that for any sufficiently large subgraph Γ of $\mathbb{P}_{\text{Berk}}^1$, we have

$$h_\alpha|_\Gamma \rightarrow h|_\Gamma$$

uniformly on Γ .

By the discussion in §5,

$$(7.5) \quad h_\alpha(x) = j_{\zeta_{\text{Gauss}}}(\varphi(x), b_\alpha) - j_{\zeta_{\text{Gauss}}}(\varphi(x), \zeta) - j_{\zeta_{\text{Gauss}}}(b_\alpha, \zeta)$$

and similarly

$$(7.6) \quad h(x) = j_{\zeta_{\text{Gauss}}}(\varphi(x), b) - j_{\zeta_{\text{Gauss}}}(\varphi(x), \zeta) - j_{\zeta_{\text{Gauss}}}(b, \zeta) .$$

Let Γ be a subgraph of $\mathbb{P}_{\text{Berk}}^1$. Without loss of generality, we may assume that Γ contains the Gauss point ζ_{Gauss} . By the retraction formula, we have

$$j_{\zeta_{\text{Gauss}}}(\varphi(x), b_\alpha) = j_{\zeta_{\text{Gauss}}}(r_\Gamma(\varphi(x)), r_\Gamma(b_\alpha))_\Gamma ,$$

and similarly with b instead of b_α .

For $x, y \in \mathbb{P}_{\text{Berk}}^1$, the map

$$g(x, y) = j_{\zeta_{\text{Gauss}}}(r_\Gamma(\varphi(x)), r_\Gamma(y))_\Gamma$$

is the composition of the continuous maps $\varphi \times 1 : \mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1$, $r_\Gamma \times r_\Gamma : \mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1 \rightarrow \Gamma \times \Gamma$, and $j_{\zeta_{\text{Gauss}}} : \Gamma \times \Gamma \rightarrow \mathbb{R}$. It is therefore

Give more precise reference.

Give reference.

Should we prove this? continuous on $\mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1$. In particular, for each fixed y we see that $j_{\zeta_{\text{Gauss}}}(\varphi(x), y)$ is continuous in x . Since Γ is compact, the restriction of $j_{\zeta_{\text{Gauss}}}(\varphi(x), y)$ to Γ is uniformly continuous in x . This implies that

$$j_{\zeta_{\text{Gauss}}}(\varphi(x), b_\alpha) \rightarrow j_{\zeta_{\text{Gauss}}}(\varphi(x), b)$$

uniformly for $x \in \Gamma$.

By the continuity of $j_{\zeta_{\text{Gauss}}}(x, y)$ in y , we have

$$j_{\zeta_{\text{Gauss}}}(b_\alpha, \zeta) \rightarrow j_{\zeta_{\text{Gauss}}}(b, \zeta).$$

Therefore, by (7.5) and (7.6), $h_\alpha(x)$ converges uniformly on Γ to $h(x)$, as desired. $\square$

We now come to our main result about analytic multiplicities.

THEOREM 7.4. *For each $b \in \mathbb{P}_{\text{Berk}}^1$, the measure $\varphi^*(\delta_b)$ defined in (7.1) is positive and discrete. Furthermore, the analytic multiplicities defined by (7.4) satisfy the following properties. For all $a, b \in \mathbb{P}_{\text{Berk}}^1$, we have:*

A) $0 \leq m_{\varphi, b}(a) \in \mathbb{Z}$.

B) $m_{\varphi, b}(a) > 0$ if and only if $\varphi(a) = b$.

C) $\sum_{\varphi(a)=b} m_{\varphi, b}(a) = \deg(\varphi)$.

D) For points a of type I, the analytic multiplicity $m_{\varphi, b}(a)$ coincides with the usual algebraic multiplicity.

PROOF. We already know (D), so it suffices to prove (A), (B), (C).

Let $\mu = \varphi^*(\delta_b)$, and choose a net $\langle b_\alpha \rangle$ in $\mathbb{P}^1(K)$ converging to b . Set $\mu_\alpha = \varphi^*(\delta_{b_\alpha})$. Since $|\mu_\alpha|(\mathbb{P}_{\text{Berk}}^1) = \mu_\alpha(\mathbb{P}_{\text{Berk}}^1) = \deg(\varphi)$ for all α , Proposition 7.3 implies that μ_α converges weakly to μ .

Let $\varepsilon > 0$. For each closed set $B \subset \mathbb{P}_{\text{Berk}}^1$, the Portmanteau theorem (Theorem A.12) shows that there is an α_0 for which

$$\mu_\alpha(B) \leq \mu(B) + \varepsilon$$

whenever $\alpha \geq \alpha_0$. As $\mu_\alpha(B) \geq 0$ for all α , we conclude that $\mu(B) \geq 0$ as well. By regularity, it follows that μ is a positive measure.

To see that μ is discrete, we proceed as follows. Let $z \in \text{supp}(\mu)$, and let $\langle z_\alpha \rangle$ be a net in $\mathbb{P}^1(K)$ converging to z . By assumption, for each open neighborhood U of z we have $\mu(U) > 0$. Choose ε with $0 < \varepsilon < \mu(U)$. The Portmanteau theorem shows that there exists α_0 such that for $\alpha \geq \alpha_0$, we have

$$\mu_\alpha(U) \geq \mu(U) - \varepsilon > 0.$$

Since $\mu_\alpha(U) \in \mathbb{Z}$ for all α , we must in fact have $\mu_\alpha(U) \geq 1$ for all α .

Another application of the Portmanteau theorem shows that for all $\varepsilon > 0$ and α sufficiently large, we have

$$\mu(\overline{U}) \geq \mu_\alpha(U) - \varepsilon \geq 1 - \varepsilon .$$

Therefore $\mu(U) \geq 1$. Since this is true for every open neighborhood U of z , we conclude that $\mu(\{z\}) \geq 1$.

From the weak convergence $\mu_\alpha \rightarrow \mu$, and the fact that each μ_α has total mass $d = \deg(\varphi)$, it follows that μ also has total mass d . Since μ is positive and $\mu(\{z\}) \geq 1$ for every $z \in \text{supp}(\mu)$, it follows that $\text{supp}(\mu)$ is finite. Thus μ is a discrete measure as claimed.

Now that we know that μ is discrete, it follows that for each $z \in \mathbb{P}_{\text{Berk}}^1$ there exists an open neighborhood U of z such that $\mu(\partial U) = 0$. By the Portmanteau theorem, we have $\mu_\alpha(U) \rightarrow \mu(U)$, and therefore $\mu(U) \in \mathbb{Z}$. By definition, we have $\mu(U) = m_{\varphi,b}(z)$, so this proves assertion (A). Since

$$\sum_{a \in \mathbb{P}_{\text{Berk}}^1} m_{\varphi,b}(z) = \mu(\mathbb{P}_{\text{Berk}}^1) = \deg(\varphi) ,$$

we also know that (B) implies (C).

It therefore remains to prove assertion (B), i.e., that for $a, b \in \mathbb{P}_{\text{Berk}}^1$, $\mu(a) > 0$ if and only if $\varphi(a) = b$. For this, choose a net $\langle a_\alpha \rangle$ in $\mathbb{P}^1(K)$ converging to a . Also, note that everything we have said so far applies to *any* net $\langle b_\alpha \rangle$ in $\mathbb{P}^1(K)$ which converges to b .

Suppose that $\varphi(a) = b$. Then by continuity, $\varphi(a_\alpha) \rightarrow b$. Taking $b_\alpha = \varphi(a_\alpha)$, we see that for any sufficiently small neighborhood U of a , we have $\mu_\alpha(U) \rightarrow \mu(U) = m_{\varphi,b}(a)$. But $\mu_\alpha(U)$ is the number of preimages of b_α lying in U , which by construction is at least 1 for α sufficiently large, since $a_\alpha \rightarrow a \in U$. Thus $m_{\varphi,b}(a) \geq 1$.

Suppose conversely that $\varphi(a) \neq b$. Then there exist open neighborhoods U of a and V of b such that $U \cap \varphi^{-1}(V) = \emptyset$. [**Proof:** Since $\varphi(a) \neq b$ and $\mathbb{P}_{\text{Berk}}^1$ is a compact Hausdorff space, there exists an open neighborhood V of b such that $\varphi(a) \notin \overline{V}$. Since a is in the complement of the closed set $\varphi^{-1}(\overline{V})$, there is a whole open neighborhood U of a disjoint from $\varphi^{-1}(\overline{V})$.] For this open neighborhood U of a , and for any net $\langle b_\alpha \rangle$ in $\mathbb{P}^1(K)$ converging to b , there exists α_0 such that $b_\alpha \in V$ for $\alpha \geq \alpha_0$. By construction, there are no preimages of b_α lying in U when $b_\alpha \in V$. It follows that $\mu_\alpha(U) = 0$ for $\alpha \geq \alpha_0$, and thus $\mu(U) = 0$ as well, i.e., $m_{\varphi,b}(a) = 0$. $\square$

As an immediate consequence of Theorem 7.4, we find:

COROLLARY 7.5. *Let $\varphi(T) \in K(T)$ be a nonconstant rational function. Then $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is surjective, and for each $b \in \mathbb{P}_{\text{Berk}}^1$ there are at most $d = \deg(\varphi)$ points a with $\varphi(a) = b$.*

We also obtain the following Berkovich space analogue of the “open mapping theorem” from complex analysis:

COROLLARY 7.6 (Open Mapping Theorem). *Let $\varphi(T) \in K(T)$ be a nonconstant rational function. Then $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is open in the Berkovich topology.*

PROOF. Suppose $U \subset \mathbb{P}_{\text{Berk}}^1$ is an open set for which $\varphi(U)$ is not open. Then there exists $a \in U$ and a net $\langle b_\alpha \rangle$ converging to $b = \varphi(a)$ such that $b_\alpha \notin \varphi(U)$ for all α . By Proposition 7.3, the measures $\mu_\alpha = \varphi^*(\delta_{b_\alpha})$ converge weakly to $\mu = \varphi^*(\delta_b)$. And by Theorem 7.4, $a \in \text{supp}(\mu)$. But each μ_α is supported on the compact set U^c , and therefore μ is supported on U^c as well. This contradicts the fact that $a \in U \cap \text{supp}(\mu)$. $\square$

Justify this.

Another useful result which follows from Theorem 7.4 and Proposition 7.3, is the following.

PROPOSITION 7.7. *If $\psi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R}$ is continuous, then so is the function $\varphi_*\psi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R}$ defined by*

$$\varphi_*\psi(y) = \sum_{\varphi(z)=y} m_{\varphi,y}(z)\psi(z) .$$

PROOF. If $\langle y_\alpha \rangle$ is a net converging to y , we need to show that $\varphi_*\psi(y_\alpha) \rightarrow \varphi_*\psi(y)$. Since

$$\varphi_*\psi(y_\alpha) = \int_{\mathbb{P}_{\text{Berk}}^1} \psi(\varphi^*(\delta_{y_\alpha}))$$

and

$$\varphi_*\psi(y) = \int_{\mathbb{P}_{\text{Berk}}^1} \psi(\varphi^*(\delta_y)) ,$$

the result follows from the fact that $\varphi^*(\delta_{y_\alpha})$ converges weakly to $\varphi^*(\delta_y)$ by Proposition 7.3. $\square$

As a consequence, we can show that φ has a well-defined *degree* on each domain in $\mathbb{P}_{\text{Berk}}^1$.

COROLLARY 7.8. *If U is a domain in $\mathbb{P}_{\text{Berk}}^1$ and $V = \varphi(U)$, then there exists an integer $m = m_{\varphi,U}$ with $1 \leq m \leq \deg(\varphi)$, called the local degree of φ on U such that every point in V has exactly m preimages in U , counting multiplicities.*

PROOF. Define a function $m_U : V \rightarrow \mathbb{R}$ by

$$m_U(b) = \sum_{a \in U} m_{\varphi,b}(a) .$$

Then $m_U = \varphi_* \mathbf{1}$, where $\mathbf{1}$ is the constant function 1 on U . By Proposition 7.7, m_U is continuous. Since m_U takes positive integer values and V is connected, we conclude that m_U is constant on V . □

Note that if $a \in \mathbb{P}_{\text{Berk}}^1$ and $b = \varphi(a)$, then every sufficiently small neighborhood U of a satisfies

$$U \cap \varphi^{-1}(b) = \{a\} .$$

We call such a neighborhood “ φ -small”. For any φ -small neighborhood U of a , the local degree $m_{\varphi,U}$ of φ on U is clearly equal to the analytic multiplicity $m_{\varphi}(a) = m_{\varphi,b}(a)$ of φ at a . We therefore obtain a purely topological interpretation of analytic multiplicities:

COROLLARY 7.9 (Topological characterization of analytic multiplicities). *For every $a \in \mathbb{P}_{\text{Berk}}^1$, and for every sufficiently small neighborhood U of a , the analytic multiplicity $m_{\varphi}(a)$ is equal to the number of preimages (counting multiplicities) of any point in $\varphi(U) \cap \mathbb{P}^1(K)$.*

REMARK 7.10. If K has characteristic zero, then φ has only finitely many critical values in $\mathbb{P}^1(K)$, and therefore all but finitely many type I points of $\varphi(U)$ will have exactly $m_{\varphi}(a)$ distinct preimages in U for any φ -small neighborhood U of a .

Corollary 7.9 shows that our analytic multiplicities coincide with the multiplicities defined by Rivera-Letelier (see for example [54]).

Now show that φ is locally injective iff multiplicity = 1, and perhaps discuss the “ramification function” (p. 143). Then incorporate Example 7.9 concerning ramified points of Berkovich space. Next, discuss the effect of rational maps on path-lengths in $\mathbb{H}_{\text{Berk}}$. For this, it will be useful to discuss annuli and their moduli, and the local degree of a tangent vector. We may as well point out in this context that our results give a proof of the fact that the image of an (open) affinoid is an (open) affinoid. Then use Proposition 7.7 to simplify the discussion of pushforward and pullback measures.

2. The pushforward and pullback measures

Let $\varphi(T) \in K(T)$ be a nonconstant rational function.

If ν is a bounded Borel measure on $\mathbb{P}_{\text{Berk}}^1$, the pushforward measure $\varphi_*(\nu)$ is the Borel measure defined by

$$\varphi_*(\nu)(U) = \nu(\varphi^{-1}(U))$$

for all Borel subsets $U \subset \mathbb{P}_{\text{Berk}}^1$. Here $\varphi^{-1}(U)$ is a Borel set because φ is continuous for the Berkovich topology.

The pullback measure $\varphi^*(\nu)$ is more complicated to define. If $g(x)$ is a continuous function on $\mathbb{P}_{\text{Berk}}^1$, then as in the statement of Proposition 7.7, we define $\varphi_*(g)$ by

$$\varphi_*(g)(y) = \sum_{\varphi(x)=y} m_{\varphi,x}(y) \cdot g(x) .$$

According to Proposition 7.7, the function $\varphi_*(g)$ is continuous.

The definition of $\varphi_*(g)$ shows that $\|\varphi_*(g)\| \leq d \cdot \|g\|$, where $\|g\| = \sup_{x \in \mathbb{P}_{\text{Berk}}^1} |g(x)|$.

The pullback measure $\varphi^*(\nu)$ is the measure representing the bounded linear functional $\Lambda : \mathcal{C}(\mathbb{P}_{\text{Berk}}^1) \rightarrow \mathbb{R}$ defined by $\Lambda(g) = \int \varphi_*(g)(y) d\nu(y)$. Thus

$$(7.7) \quad \int g(x) d\varphi^*(\nu)(x) = \int \varphi_*(g)(y) d\nu(y) .$$

These measures satisfy the usual formal functorial properties.

PROPOSITION 7.11. *Let $\varphi(T) \in K(T)$ be a nonconstant rational function of degree $d = \deg(\varphi)$. Suppose ν is a bounded Borel measure on an open set $U \subset \mathbb{P}_{\text{Berk}}^1$.*

A) Let $\nu_1, \nu_2, \dots$ be a sequence of signed Borel measures on U for which the masses of $|\nu_i|$ are uniformly bounded, and which converge weakly to ν on compact subsets of U . Then $\varphi^(\nu_1), \varphi^*(\nu_2), \dots$ converge weakly to $\varphi^*(\nu)$ on compact subsets of $\varphi^{-1}(U)$.*

If $U = \varphi^{-1}(\varphi(U))$, then $\varphi_(\nu_1), \varphi_*(\nu_2), \dots$ converge weakly to $\varphi_*(\nu)$ on compact subsets of $\varphi(U)$.*

B) $\varphi_(\varphi^*(\nu)) = d \cdot \nu$.*

PROOF. The first assertion in (A) holds because for each $g \in \mathcal{C}(\varphi^{-1}(\overline{U}))$

$$\begin{aligned} \int g(x) d\varphi^*(\nu) &= \int \varphi_*(g)(y) d\nu(y) \\ &= \lim_{n \rightarrow \infty} \int \varphi_*(g)(y) d\nu_n(y) = \lim_{n \rightarrow \infty} \int g(x) d\varphi^*(\nu_n) . \end{aligned}$$

The second holds because for each open subset $V \subset \varphi(U)$,

$$\begin{aligned} \nu_*(V) &= \nu(\varphi^{-1}(V)) \\ &= \lim_{n \rightarrow \infty} \nu_n(\varphi^{-1}(V)) = \lim_{n \rightarrow \infty} \varphi_*(\nu_n)(V), \end{aligned}$$

and equality of two regular Borel measures on open sets implies their equality for all Borel sets.

Part (B) also follows from a simple computation. Formula (7.7) extends to characteristic functions of Borel sets. For each Borel set $E \subset \mathbb{P}_{\text{Berk}}^1$, the identity $\sum_{\varphi(x)=y} m_{\varphi,y}(x) = d$ means that $\varphi_*(\chi_{\varphi^{-1}(E)}) = d \cdot \chi_E$. Hence

$$\begin{aligned} \varphi_*(\varphi^*(\nu))(E) &= \varphi^*(\nu)(\varphi^{-1}(E)) \\ &= \int \chi_{\varphi^{-1}(E)}(x) d\varphi^*(\nu)(x) = \int \varphi_*(\chi_{\varphi^{-1}(E)})(y) d\nu(y) \\ &= \int d \cdot \chi_E(y) d\nu(y) = d \cdot \nu(E). \end{aligned}$$

□

3. The Pullback Formula for subharmonic functions

PROPOSITION 7.12. *Let $\varphi(T) \in K(T)$ be a nonconstant rational map of degree d . Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a domain, and let f be subharmonic in U . Then*

- A) *The function $\varphi^*(f) = f \circ \varphi$ is subharmonic in $\varphi^{-1}(U)$.*
- B) *$\Delta_{\varphi^{-1}(U)}(\varphi^*(f)) = \varphi^*(\Delta_U(f))$.*

Since $\varphi^{-1}(U)$ might not be connected, we need to say what the Laplacian means in this case, or else rephrase the result by assuming that U, U' are domains with $\varphi(U') \subseteq U$, and f is subharmonic in U' , with the conclusion being that $f \circ \varphi$ is subharmonic in U' and $\Delta_{U'}(\varphi^*(f)) = \varphi^*(\Delta_U(f))$. Note that by using an earlier proposition and the fact that every $\Gamma \subset U$ is contained in some simple subdomain V of U , it follows that part (B) holds for all $f \in \text{BDV}(U)$. It might be useful to state it in this generality. Perhaps only state (B) for simple subdomains?

PROOF. (A) To see that $f \circ \varphi$ is subharmonic, we use the Riesz Decomposition Theorem (Proposition 6.35). Let V be a simple subdomain of U , and put $\nu = -\Delta_V(f)$. Let ζ be an arbitrary point in $\mathbb{P}_{\text{Berk}}^1 \setminus \overline{V}$. Proposition 6.35 shows there is a harmonic function h on V

such that for all $x \in V$

$$f(y) = h(y) + \int \log_v(\delta(y, z)_\zeta) d\nu(z) .$$

Composing with $\varphi(x)$ gives

$$f(\varphi(x)) = h(\varphi(x)) + \int_V \log_v(\delta(\varphi(x), z)_\zeta) d\nu(z) .$$

By Corollary 5.38, $h(\varphi(x))$ is harmonic in V . By Lemma 7.1 and Theorem 7.4, for each $z \in V$ the function $\log_v(\delta(\varphi(x), z)_\zeta)$ is subharmonic in $\mathbb{P}_{\text{Berk}}^1 \setminus \varphi^{-1}(\zeta)$, hence in particular in $\varphi^{-1}(V)$. Since $F \circ \varphi - h \circ \varphi$ is upper semicontinuous, it follows from Proposition 6.23 that

$$F(x) = \int_V \log_v(\delta(\varphi(x), z)_\zeta) d\nu(z)$$

is subharmonic on $\varphi^{-1}(V)$. (The function $F(x)$ is not $\equiv -\infty$ on any component of $\varphi^{-1}(V)$, since $F(x) = f(\varphi(x)) - h(\varphi(x))$.) It follows that $f(\varphi(x)) = h(\varphi(x)) + F(x)$ is subharmonic on $\varphi^{-1}(V)$. Since subharmonicity is local, it follows from the open mapping theorem that $f(\varphi(x))$ is subharmonic on $\varphi^{-1}(U)$.

(B) Let V be a simple subdomain of U . By the smoothing theorem (Proposition 6.44), there is a net $\langle f_\alpha \rangle_{\alpha \in A}$ of smooth subharmonic functions on V such that $\lim_\alpha f_\alpha(z) = f(z)$ pointwise for all $z \in V$, and where the convergence is uniform on each subgraph $\Gamma \subset V$. By smoothness, each $\Delta_V(f_\alpha)$ is a discrete measure supported on a finite set of points. We will first prove the pullback formula for the f_α .

Fix $\alpha \in A$ and $\zeta \in \mathbb{P}_{\text{Berk}}^1 \setminus \bar{V}$, and write

$$\nu_\alpha = -\Delta_V(f_\alpha) = \sum_{i=1}^{M_\alpha} c_{\alpha i} \delta_{p_{\alpha i}}(x) .$$

As in the proof of part (A), there is a harmonic function h_α on V such that for all $u \in V$

$$\begin{aligned} f_\alpha(y) &= h_\alpha(y) + \int \log_v(\delta(y, z)_\zeta) d\nu_\alpha(z) \\ &= h_\alpha(y) + \sum_{i=1}^{M_\alpha} c_{\alpha i} \log_v(\delta(y, p_{\alpha i})_\zeta) . \end{aligned}$$

Composing with $\varphi(x)$ gives

$$f_\alpha(\varphi(x)) = h_\alpha(\varphi(x)) + \sum_{i=1}^{M_\alpha} c_{\alpha i} \log_v(\delta(\varphi(x), p_{\alpha i})_\zeta) .$$

As before, $h_\alpha(\varphi(x))$ is harmonic in $\varphi^{-1}(V)$. By Lemma 7.1, each function $h_{\alpha i}(x) = \log_v(\delta(\varphi(x), p_{\alpha i})_\zeta)$ belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(h_{\alpha i}) = \sum_{\varphi(\xi)=\zeta} m_{\varphi,\zeta}(\xi) \cdot \delta_\xi(x) - \sum_{\varphi(a)=p_{\alpha i}} m_{\varphi,p_{\alpha i}}(a) \cdot \delta_a(x)$$

In particular, $h_{\alpha i}(x)$ is subharmonic in $\varphi^{-1}(V)$. We also note that for any p , the measure $\sum_{\varphi(a)=p} m_{\varphi,p}(a) \delta_a(x)$ is precisely the pullback measure $\varphi^*(\delta_p(x))$. Hence

$$\Delta_{\varphi^{-1}(V)}(h_{\alpha i}) = -\varphi^*(\delta_{p_{\alpha i}}(x)) .$$

Summing over all i , we find that

$$\Delta_{\varphi^{-1}(V)}(f_\alpha \circ \varphi) = - \sum_{i=1}^{M_\alpha} c_{\alpha i} \varphi^*(\delta_{p_{\alpha i}}(x)) = \varphi^*(\Delta_V(f_\alpha)) .$$

The functions $f_\alpha \circ \varphi(x)$ converge uniformly to $f \circ \varphi(x)$ on each subgraph $\Gamma \subset V$. Therefore by Proposition 6.38(B), the measures $\Delta_{\varphi^{-1}(V)}(f_\alpha \circ \varphi)$ converge weakly to $\Delta_{\varphi^{-1}(V)}(f \circ \varphi)$. Proposition 6.38(B) also shows the measures $\Delta_V(f_\alpha)$ converge weakly to $\Delta_V(f)$. Hence, by Proposition 7.11, the measures $\varphi^*(\Delta_V(f_\alpha))$ converge weakly to $\varphi^*(\Delta_V(f))$. Thus

$$\Delta_{\varphi^{-1}(V)}(f \circ \varphi) = \varphi^*(\Delta_V(f)) .$$

Exhausting U by simple subdomains, we obtain the desired result. $\square$

4. Notes and further references

Verify that this is the correct reference.

CHAPTER 8

Applications to the dynamics of rational maps

In this chapter we study the dynamics of a rational map φ on $\mathbb{P}_{\text{Berk}}^1$. In particular, we show there is a probability measure μ on $\mathbb{P}_{\text{Berk}}^1$ which satisfies $\varphi_*(\mu) = \mu$ and $\varphi^*(\mu) = d \cdot \mu$. By analogy with the classical case for $\mathbb{P}^1(\mathbb{C})$ (see [46], [33]), we call it the canonical measure. Intuitively, the canonical measure is defined as the negative of the Laplacian of the Call-Silverman local height function (extended to $\mathbb{P}_{\text{Berk}}^1$ in a natural way).

We also show that the potential kernel associated to the canonical measure satisfies a certain energy-minimization principle. These results are utilized in the papers [5] and [1] to investigate properties of algebraic points of small dynamical height.

Chambert-Loir and Favre–Rivera-Letelier have given independent constructions of the canonical measure on $\mathbb{P}_{\text{Berk}}^1$ attached to φ . We refer to their papers [28, 29] for details. Also, we note that Szpiro, Tucker and Pinero ([69]) have constructed a sequence of blowups of a rational map on $\mathbb{P}_1/\text{Spec}(\mathcal{O}_v)$, leading to a sequence of discrete measures supported on the special fibres, and when suitably interpreted in terms of Berkovich space, the weak limit of these measures gives another way of defining the canonical measure.

1. Construction of the canonical measure

Let $\varphi(T) \in K(T)$ be a rational function of degree $d \geq 2$, and choose coprime polynomials $f_1(T), f_2(T) \in K[T]$ with $\max(\deg(f_1), \deg(f_2)) = d$ such that $\varphi(T) = f_2(T)/f_1(T)$. Homogenizing $f_1(T)$ and $f_2(T)$, we obtain homogeneous polynomials $F_1(X, Y), F_2(X, Y) \in K[X, Y]$ of degree d such that $f_1(T) = F_1(1, T)$, $f_2(T) = F_2(1, T)$, and where $\text{Res}(F_1, F_2) \neq 0$.

We are interested in the dynamics of φ on $\mathbb{P}_{\text{Berk}}^1$, so for $n \geq 1$, let $\varphi^{(n)} = \varphi \circ \varphi \circ \cdots \circ \varphi$ (n times). Similarly, define $F^{(n)} = F \circ F \circ \cdots \circ F$ (n times), so that $F^{(n)}(X, Y) = (F_1^{(n)}(X, Y), F_2^{(n)}(X, Y))$ where $F_1^{(n)}(X, Y), F_2^{(n)}(X, Y) \in K[X, Y]$ are homogeneous polynomials of degree d^n .

For the moment, regard $F_1(X, Y)$ and $F_2(X, Y)$ as functions on K^2 and write $F(x, y) = (F_1(x, y), F_2(x, y))$. Let

$$\|(x, y)\| = \max(|x|, |y|)$$

be the usual sup norm on K^2 . We claim there are numbers $0 < B_1 < B_2 < \infty$ such that for all $(x, y) \in K^2$,

$$(8.1) \quad B_1 \cdot \|(x, y)\|^d \leq \|F(x, y)\| \leq B_2 \cdot \|(x, y)\|^d .$$

The upper bound is trivial, since if $B_2 = \max(|c_{1,ij}|, |c_{2,ij}|)$ then by the ultrametric inequality

$$|F_1(x, y)|, |F_2(x, y)| \leq \max_{i,j}(|c_{1,ij}| |x|^i |y|^j) \leq B_2 \|(x, y)\|^d .$$

For the lower bound, note that since $F_1(1, T)$ and $F_2(1, T)$ are coprime, their resultant $b_1 = \text{Res}(F_1(1, T), F_2(1, T)) \in K$ is nonzero. By properties of the resultant, there are polynomials $g_1(T), g_2(T) \in K[T]$ of degree at most $d - 1$ such that

$$g_1(T)F_1(1, T) + g_2(T)F_2(1, T) = b_1 .$$

Homogenizing, we obtain homogenous polynomials $G_1(X, Y), G_2(X, Y)$ of degree $d - 1$ for which

$$(8.2) \quad G_1(X, Y)F_1(X, Y) + G_2(X, Y)F_2(X, Y) = b_1 X^{2d-1} .$$

Similarly, since $F_1(U, 1)$ and $F_2(U, 1)$ are coprime, their resultant b_2 is nonzero, and there are homogeneous polynomials $H_1(X, Y), H_2(X, Y)$ of degree $d - 1$ such that

$$(8.3) \quad H_1(X, Y)F_1(X, Y) + H_2(X, Y)F_2(X, Y) = b_2 Y^{2d-1} .$$

By the upper bound argument applied to $G = (G_1, G_2)$ and $H = (H_1, H_2)$, there is an $A_2 > 0$ such that $\|G(x, y)\|, \|H(x, y)\| \leq A_2 \|(x, y)\|^{d-1}$ for all $(x, y) \in K^2$. By (8.2), (8.3) and the ultrametric inequality,

$$\begin{aligned} |b_1| |x|^{2d-1} &\leq A_2 \|(x, y)\|^{d-1} \|F(x, y)\| , \\ |b_2| |y|^{2d-1} &\leq A_2 \|(x, y)\|^{d-1} \|F(x, y)\| . \end{aligned}$$

Writing $A_1 = \min(|b_1|, |b_2|)$, it follows that

$$A_1 \max(\|x, y\|)^{2d-1} \leq A_2 \|(x, y)\|^{d-1} \|F(x, y)\| .$$

Thus if $B_1 = A_1/A_2$, then $\|F(x, y)\| \geq B_1 \|(x, y)\|^d$.

Taking logarithms, and putting $C_1 = \max(\log_v(B_1), \log_v(B_2))$, we have

$$(8.4) \quad \left| \frac{1}{d} \log_v \|F(x, y)\| - \log_v \|(x, y)\| \right| \leq \frac{C_1}{d} .$$

Give reference.

for all $(x, y) \in K^2 \setminus \{(0, 0)\}$. Inserting $F(x, y)$ for (x, y) in (8.4) and iterating, we find that for each $n \geq 1$,

$$(8.5) \quad \left| \frac{1}{d^n} \log_v \|F^{(n)}(x, y)\| - \frac{1}{d^{n-1}} \log_v \|F^{(n-1)}(x, y)\| \right| \leq \frac{C_1}{d^n}.$$

Put

$$h_{\varphi, v}^{(n)}(x) = \frac{1}{d^n} \log_v \max(|F_1^{(n)}(1, x)|, |F_2^{(n)}(1, x)|).$$

The Call-Silverman local height for φ on $\mathbb{P}^1(K)$ (relative to the point ∞ and the dehomogenization $F_1(1, T), F_2(1, T)$) is defined for $x \in \mathbb{A}^1(K)$ by

$$(8.6) \quad h_{\varphi, v}(x) = \lim_{n \rightarrow \infty} h_{\varphi, v}^{(n)}(x).$$

The fact that the limit exists follows from (8.5), as does the bound

$$(8.7) \quad |h_{\varphi, v}(x) - \log_v(\max(1, |x|))| \leq \sum_{n=1}^{\infty} \frac{C_1}{d^n} = C.$$

Taking logarithms in the identity

$$F^{(n-1)}(1, \varphi(x)) = F^{(n-1)}\left(1, \frac{F_2(1, x)}{F_1(1, x)}\right) = \frac{F^{(n)}(1, x)}{F_1(1, x)^{d^{n-1}}}$$

and letting $n \rightarrow \infty$ gives the functional equation

$$(8.8) \quad h_{\varphi, v}(\varphi(x)) = d \cdot h_{\varphi, v}(x) - \log_v(|F_1(1, x)|),$$

valid on $\mathbb{P}^1(K) \setminus (\{\infty\} \cup \varphi^{-1}(\infty))$. The two properties (8.7), (8.8) characterize the Call-Silverman local height.

We will now “Berkovichize” the local height. For each n , put

$$(8.9) \quad \hat{h}_{\varphi, v}^{(n)}(x) = \frac{1}{d^n} \max(\log_v[F_1^{(n)}(1, T)]_x, \log_v[F_2^{(n)}(1, T)]_x)$$

for $x \in \mathbb{P}_{\text{Berk}}^1$ (we regard it as having the value ∞ at $x = \infty$). Then $\hat{h}_{\varphi, v}^{(n)}(x)$ coincides with $h_{\varphi, v}^{(n)}(x)$ on $\mathbb{A}^1(K)$ and is continuous and strongly subharmonic in $\mathbb{A}_{\text{Berk}}^1$. We claim that for all $x \in \mathbb{A}_{\text{Berk}}^1$

$$(8.10) \quad |\hat{h}_{\varphi, v}^{(n)}(x) - \hat{h}_{\varphi, v}^{(n-1)}(x)| \leq \frac{C_1}{d^n}.$$

Indeed, this holds for all type I points in $\mathbb{A}^1(K)$; such points are dense in $\mathbb{A}_{\text{Berk}}^1$ and the functions involved are continuous, so it holds for all $x \in \mathbb{A}_{\text{Berk}}^1$.

It follows that the functions $\hat{h}_{\varphi, v}^{(n)}(x)$ converge uniformly to a continuous subharmonic function $\hat{h}_{\varphi, v}(x)$ on $\mathbb{A}_{\text{Berk}}^1$ which extends the Call-Silverman local height $h_{\varphi, v}(x)$. By the same arguments as before, for

all $x \in \mathbb{A}_{\text{Berk}}^1$

$$(8.11) \quad |\hat{h}_{\varphi,v}(x) - \log_v(\max(1, [T]_x))| \leq C ,$$

and for all $x \in \mathbb{P}_{\text{Berk}}^1 \setminus (\infty \cup \varphi^{-1}(\infty))$,

$$(8.12) \quad \hat{h}_{\varphi,v}(\varphi(x)) = d \cdot \hat{h}_{\varphi,v}(x) - \log_v([F_1(1, T)]_x) .$$

Actually, this can be viewed as an identity for all $x \in \mathbb{P}_{\text{Berk}}^1$, if one views the right side at $x = \infty$ as given by its limit as $x \rightarrow \infty$.

Let $V_1 = \mathbb{A}_{\text{Berk}}^1 = \mathbb{P}_{\text{Berk}}^1 \setminus \{\infty\}$. Since $\hat{h}_{\varphi,v}$ is subharmonic on V_1 , there is a non-negative measure μ_1 on V_1 such that $\Delta_{V_1}(\hat{h}_{\varphi,v}) = -\mu_1$. On the other hand, each $\hat{h}_{\varphi,v}^{(n)}(x)$ belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ and satisfies

$$\Delta_{\mathbb{P}_{\text{Berk}}^1}(\hat{h}_{\varphi,v}^{(n)}) = \delta_{\infty} - \mu_1^{(n)} ,$$

where $\mu_1^{(n)} \geq 0$ has total mass 1. Since the $\hat{h}_{\varphi,v}^{(n)}$ converge uniformly to $\hat{h}_{\varphi,v}$ on V_1 , it follows that $\Delta_{\mathbb{P}_{\text{Berk}}^1}(\hat{h}_{\varphi,v}^{(n)}) \rightarrow \Delta_{\mathbb{P}_{\text{Berk}}^1}(\hat{h}_{\varphi,v})$. From this, one concludes that the $\mu_1^{(n)}$ converge weakly to μ_1 on simple subdomains of V_1 , that $\hat{h}_{\varphi,v}$ belongs to $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$, and there is a non-negative measure μ on $\mathbb{P}_{\text{Berk}}^1$ of total mass 1 such that

$$(8.13) \quad \Delta_{\mathbb{P}_{\text{Berk}}^1}(\hat{h}_{\varphi,v}) = \delta_{\infty} - \mu .$$

(Note that functions in $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ are allowed take values $\pm\infty$ on points of $\mathbb{P}^1(K)$; the definition of the Laplacian only involves their restriction to $\mathbb{H}_{\text{Berk}}$.)

In the affine patch $V_2 := \mathbb{P}_{\text{Berk}}^1 \setminus \{0\}$, relative to the coordinate function $U = 1/T$, the map φ is given by $F_1(U, 1)/F_2(U, 1)$. By a construction similar to the one above, using the functions $F_1(U, 1)$ and $F_2(U, 1)$, we obtain a function $\hat{g}_{\varphi,v}(x) \in \text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ which is continuous and subharmonic on V_2 and extends the Call-Silverman local height relative to the point 0. For all $x \in V_2$, it satisfies

$$(8.14) \quad |\hat{g}_{\varphi,v}(x) - \log_v(\max(1, [1/T]_x))| \leq C ,$$

$$(8.15) \quad \hat{g}_{\varphi,v}(\varphi(x)) = d \cdot \hat{g}_{\varphi,v}(x) - \log_v([F_2(1/T, 1)]_x) ,$$

where (8.15) holds in the same sense as (8.12). Using the identity $F^{(n)}(U, 1) = F^{(n)}(1, T)/T^{d^n}$, taking logarithms, and letting $n \rightarrow \infty$ gives

$$(8.16) \quad \hat{g}_{\varphi,v}(x) = \hat{h}_{\varphi,v}(x) - \log_v([T]_x) .$$

Applying the Laplacian and using (8.13) shows that

$$(8.17) \quad \Delta_{\mathbb{P}_{\text{Berk}}^1}(\hat{g}_{\varphi,v}) = \delta_0 - \mu .$$

We will refer to the probability measure μ on $\mathbb{P}_{\text{Berk}}^1$ appearing in (8.13) and (8.17) as the *canonical measure* associated to φ .

THEOREM 8.1. *Let $\varphi(T) \in K(T)$ have degree $d \geq 2$. The canonical measure $\mu = \mu_\varphi$ on $\mathbb{P}_{\text{Berk}}^1$ is non-negative and has total mass 1. It satisfies the functional equations $\varphi^*(\mu) = d \cdot \mu$ and $\varphi_*(\mu) = \mu$.*

PROOF. To show $\varphi^*(\mu) = d \cdot \mu$, we use (8.12) and (8.15). Let $x_1, \dots, x_d$ be the zeros (not necessarily distinct) of $F_1(1, T)$, and put $U_1 = \varphi^{-1}(V_1) = \mathbb{P}_{\text{Berk}}^1 \setminus \{x_1, \dots, x_d\}$. By the pullback formula for subharmonic functions (Proposition 7.12), $H(x) := \hat{h}_{\varphi, v}(\varphi(x))$ is subharmonic on U_1 , and

$$\Delta_{U_1}(H) = \varphi^*(\Delta_{V_1}(\hat{h}_{\varphi, v})) .$$

Taking Laplacians in (8.12) gives $\varphi^*(\mu|_{V_1}) = d \cdot \mu|_{U_1}$. Likewise, put $U_2 = \varphi^{-1}(V_2) = \mathbb{P}_{\text{Berk}}^1 \setminus \{y_1, \dots, y_d\}$, where $y_1, \dots, y_d$ are the zeros of $F_2(1/T, 1) = F_2(1, T)/T^d$. Since $F_1(1, T)$ and $F_2(1, T)$ are coprime, the sets $\{x_1, \dots, x_d\}$ and $\{y_1, \dots, y_d\}$ are disjoint. Then $G(x) := \hat{g}_{\varphi, v}(\varphi(x))$ is subharmonic on U_2 , and satisfies

$$\Delta_{U_2}(G) = \varphi^*(\Delta_{V_2}(\hat{g}_{\varphi, v})) .$$

Taking Laplacians in (8.15) gives $\varphi^*(\mu|_{V_2}) = d \cdot \mu|_{U_2}$. Since $V_1 \cup V_2 = U_1 \cup U_2 = \mathbb{P}_{\text{Berk}}^1$, it follows that $\varphi^*(\mu) = d \cdot \mu$.

The identity $\varphi_*(\mu) = \mu$ follows formally from $\varphi^*(\mu) = d \cdot \mu$. By Proposition 7.11(B), $\varphi_*(\varphi^*(\mu)) = d \cdot \mu$. Since $\varphi^*(\mu) = d \cdot \mu$, this gives $\varphi_*(\mu) = \mu$. $\square$

The canonical measure is better behaved than arbitrary measures; in particular, it is “log-continuous” in the following sense:

DEFINITION 8.2. A positive measure ω on $\mathbb{P}_{\text{Berk}}^1$ is *log-continuous* if

$$u_\omega(x) = \int_{\mathbb{P}_{\text{Berk}}^1} -\log_v(\|x, y\|) d\omega(y)$$

is continuous (hence uniformly bounded) on all of $\mathbb{P}_{\text{Berk}}^1$.

In the definition, $-\log_v(\|x, y\|) = -\log_v(\delta(x, y)_{\zeta_{\text{Gauss}}}) = j_{\zeta_{\text{Gauss}}}(x, y)$ where $\zeta_{\text{Gauss}} \in \mathbb{P}_{\text{Berk}}^1$ is the Gauss point, the point corresponding to $B(0, 1)$. Thus $u_\omega(x)$ is the potential function $u_\omega(x, \zeta_{\text{Gauss}})$. In the definition, $\|x, y\|$ could be replaced by any other kernel which differs from it by a bounded continuous function; in particular, it could be replaced by $\delta(x, y)_\zeta$ for any $\zeta \in \mathbb{H}_{\text{Berk}}$.

The log-continuity of the canonical measure follows from:

PROPOSITION 8.3. *Let ω be a positive measure on $\mathbb{P}_{\text{Berk}}^1$ for which $-\omega$ is locally the Laplacian of a continuous subharmonic function. Then ω is log-continuous.*

PROOF. Fix $x \in \mathbb{P}_{\text{Berk}}^1$, and let $V \subset \mathbb{P}_{\text{Berk}}^1$ be a neighborhood on which $-\omega|_V = \Delta_V(f)$ for some continuous subharmonic function f . After shrinking V if necessary, we can that assume V is a simple domain, and that f is bounded and strongly subharmonic on V . Put $\omega_V = \omega|_V$, and consider the potential function

$$u_{\omega_V}(z, \zeta) = \int -\log_v(\delta(x, y)_\zeta) d\omega_V(y) .$$

By Proposition 6.36, $u_{\omega_V}(z, \zeta_{\text{Gauss}})$ is continuous on all of $\mathbb{P}_{\text{Berk}}^1$. It is bounded since $\zeta_{\text{Gauss}} \notin \mathbb{P}^1(K)$.

Since $\mathbb{P}_{\text{Berk}}^1$ is compact, a finite number of such simple domains V_i cover $\mathbb{P}_{\text{Berk}}^1$. It is easy to see that the assertions made for the restriction of ω to the V_i hold also for its restriction to their intersections $V_{i_1} \cap \cdots \cap V_{i_r}$. By the inclusion-exclusion formula, $u_\omega(z, \zeta_{\text{Gauss}})$ is continuous on $\mathbb{P}_{\text{Berk}}^1$. $\square$

EXAMPLE 8.4. A rational function $\varphi(T) \in K(T)$ has *good reduction* if it can be written as $\varphi(T) = f_2(T)/f_1(T)$ where $f_1(T), f_2(T) \in K[T]$ are coprime polynomials whose coefficients belong to the valuation ring $\mathcal{O}$ of K , and where the resultant $\text{Res}(f_1, f_2)$ is a unit in $\mathcal{O}$.

Suppose $\varphi(T)$ has good reduction, and has degree $d \geq 2$. We claim that the canonical measure μ_φ coincides with the Dirac measure $\delta_{\zeta_{\text{Gauss}}}(x)$, where ζ_{Gauss} is the Gauss point of $\mathbb{P}_{\text{Berk}}^1$.

To see this, recall formula (8.1) and note that the proof of that formula shows that under our hypotheses, $B_1 = B_2 = 1$. Thus for all $(x, y) \in K^2$,

$$\|F(x, y)\| = \|(x, y)\|^d .$$

By iteration, for each n , $\|F^{(n)}(x, y)\| = \|(x, y)\|^{d^n}$. Examining the construction of $\hat{h}_{\varphi, v}$, one finds that

$$\hat{h}_{\varphi, v}(x) = \log_v \max(1, [T]_x) = \log_v(\delta(x, \zeta_{\text{Gauss}})_\infty) ,$$

so $\Delta(\hat{h}_{\varphi, v}) = \delta_\infty - \delta_{\zeta_{\text{Gauss}}}$. Hence $\mu_\varphi = \delta_{\zeta_{\text{Gauss}}}$.

This would be a good place to discuss the Jensen-Mahler formula for polynomials.

Replace by homogeneous version.

Give reference.

2. Explicit formula for $g_\mu(x, y)$

Our goal in this section is to give an explicit formula for $g_{\mu_\varphi}(x, y)$ which is quite useful in practice, and to establish an important functional equation satisfied by g_{μ_φ} .

We continue with the notation from §1. In particular, $\varphi \in K(T)$ is a rational function of degree $d \geq 2$, and $F_1(X, Y), F_2(X, Y)$ are homogeneous polynomials of degree d such that $\varphi(T) = F_2(1, T)/F_1(1, T)$ and $\text{Res}(F_1, F_2) \neq 0$. The polynomials F_1, F_2 are well-defined up to multiplication by a common nonzero scalar $c \in K^*$.

With the convention that $-\log \delta_\infty(x, y) = +\infty$ if $x, y \in P^1(K)$ with $x = y$ and $\hat{h}_{\varphi, \infty}(\infty) = +\infty$, we make the following definition.

For $x, y \in \mathbb{P}_{\text{Berk}}^1$, let $R = |\text{Res}(F_1, F_2)|^{-\frac{1}{d(d-1)}}$ and define

$$g_\varphi(x, y) = \begin{cases} -\log_v \delta_\infty(x, y) + \hat{h}_{\varphi, \infty}(x) + \hat{h}_{\varphi, \infty}(y) + \log R & \text{if } x, y \neq \infty \\ \hat{h}_{\varphi, \infty}(x) + \log R & \text{if } y = \infty \\ \hat{h}_{\varphi, \infty}(y) + \log R & \text{if } x = \infty \end{cases}$$

The reason for our particular choice of the constant R will become apparent later on. It is clear from the corresponding properties of the Hsia kernel δ_∞ and the Call-Silverman local height $\hat{h}_{\varphi, \infty}$ that for each fixed $y \in \mathbb{P}_{\text{Berk}}^1$, $g_\varphi(x, y)$ is continuous and in $\text{BDV}(\mathbb{P}_{\text{Berk}}^1)$ as a function of x . It is also clear from the definition that $g_\varphi(x, y)$ is symmetric in x and y .

It is possible to give a more elegant description of $g_\varphi(x, y)$ when $x, y \in \mathbb{P}^1(K)$. To do this, if $x = (x_1, x_2), y = (y_1, y_2) \in K^2$, we define $x \wedge y$ to be

$$x \wedge y = x_1 y_2 - x_2 y_1 ,$$

and let $\|x\| = \max\{|x_1|, |x_2|\}$.

The mapping $F = (F_1, F_2) : K^2 \rightarrow K^2$ is a lifting of φ to K^2 , and we define $F^{(n)} : K^2 \rightarrow K^2$ to be the n th iterate of F . For $z \in K^2 \setminus \{0\}$, define the *homogeneous dynamical height* $H_F : K^2 \setminus \{0\} \rightarrow \mathbb{R}$ by

$$HF(z) := \lim_{n \rightarrow \infty} \frac{1}{d^n} \log \|F^{(n)}(z)\|.$$

(By convention, we define $H_F(0, 0) := -\infty$.)

By a standard telescoping series argument (see [5]), one sees that the limit $\lim_{n \rightarrow \infty} \frac{1}{d^n} \log \|F^{(n)}(z)\|$ exists for all $z \in K^2 \setminus \{0\}$, and $\frac{1}{d^n} \log \|F^{(n)}(z)\|$ converges uniformly on $K^2 \setminus \{0\}$ to $H_F(z)$. The definition of H_F is independent of the norm used to define it; this follows easily from the equivalence of norms on K^2 .

LEMMA 8.5. *Let $x, y \in \mathbb{P}^1(K)$, and let $\tilde{x}, \tilde{y}$ be arbitrary lifts of x, y to $K^2 \setminus \{0\}$. Then*

$$(8.18) \quad g_\varphi(x, y) = -\log |\tilde{x} \wedge \tilde{y}| + H_F(x) + H_F(y) + \log R.$$

Moreover, the right-hand side of 8.18 is independent of the choice of the lifts F_1, F_2 of F .

PROOF. A straightforward computation (see [5] for details). $\square$

We will now show that $g_\varphi(x, y) = g_{\mu_\varphi}(x, y)$ for all $x, y \in \mathbb{P}_{\text{Berk}}^1$. The proof is based on the following observation.

LEMMA 8.6. *For each $y \in \mathbb{P}_{\text{Berk}}^1$, we have $\Delta_x g_\varphi(x, y) = \delta_y - \mu_\varphi$.*

PROOF. This follows from the definition of $g_\varphi(x, y)$, together with the fact that $\Delta_x(-\log_v \delta_\infty(x, y)) = \delta_y - \delta_\infty$ and $\Delta_x(\hat{h}_{\varphi, \infty}) = \delta_\infty - \mu_\varphi$. $\square$

It follows from Proposition 6.45, Lemma 8.6, and the continuity of $g_\varphi(x, y)$ and $g_{\mu_\varphi}(x, y)$ in x that for each $y \in \mathbb{P}_{\text{Berk}}^1$, there exists a constant $C_y \in \mathbb{R}$ such that

$$(8.19) \quad g_\varphi(x, y) = g_{\mu_\varphi}(x, y) + C_y$$

for all $x \in \mathbb{P}_{\text{Berk}}^1$.

In order to show that $C_y = 0$ for all y , it suffices to show that

$$\int g_\varphi(x, y) d\mu_\varphi(x) = \int g_{\mu_\varphi}(x, y) d\mu_\varphi(x) = 0.$$

For g_{μ_φ} , this is straightforward given what we already know.

PROPOSITION 8.7. *For each $y \in \mathbb{P}_{\text{Berk}}^1$, we have*

$$\int_{\mathbb{P}_{\text{Berk}}^1} g_{\mu_\varphi}(x, y) d\mu_\varphi(x) = 0.$$

PROOF. By Proposition 6.47, the potential function

$$u_{\mu_\varphi}(y, \mu_\varphi) = \int g_{\mu_\varphi}(x, y) d\mu_\varphi(x)$$

has Laplacian equal to $\mu_\varphi - \mu_\varphi = 0$. By continuity, $\int g_{\mu_\varphi}(x, y) d\mu_\varphi(x)$ is a constant independent of y . Integrating against μ_φ with respect to y and using the definition of g_{μ_φ} shows that this constant is zero. $\square$

It remains to show that

$$\int_{\mathbb{P}_{\text{Berk}}^1} g_\varphi(x, y) d\mu_\varphi(x) = 0.$$

We will prove this by first establishing a useful functional equation for $g_\varphi(x, y)$.

Let k be a field. By [45, Theorem IX.3.13], if $F = (F_1, F_2)$ and $G = (G_1, G_2)$ where $F_1, F_2 \in k[X, Y]$ are homogeneous of degree d and $G_1, G_2 \in k[X, Y]$ are homogeneous of degree e , then

$$(8.20) \quad \text{Res}(F \circ G) = \text{Res}(F)^e \text{Res}(G)^{d^2}.$$

In particular, it follows by induction on n that

$$(8.21) \quad \text{Res}(F^{(n)}) = \text{Res}(F)^{\frac{d^{n-1}(d^n-1)}{d-1}}.$$

LEMMA 8.8. *If $M \in \text{GL}_2(K)$, let $F' = M^{-1} \circ F \circ M$. Then for all $z, w \in K^2 \setminus \{0\}$,*

$$(8.22) \quad H_F(M(z)) = H_{F'}(z)$$

and

$$(8.23) \quad G_F(M(z), M(w)) = G_{F'}(z, w).$$

Define G_F or change this to g_φ .

PROOF. First of all note that given M , there exist constants $C_1, C_2 > 0$ such that

$$(8.24) \quad C_1 \|z\| \leq \|M(z)\| \leq C_2 \|z\|.$$

Indeed, if we take C_2 to be the maximum of the absolute values of the entries of M then clearly $\|M(z)\| \leq C_2 \|z\|$ for all z . By the same reasoning, if we let C_1^{-1} be the maximum of the absolute values of the entries of M^{-1} , then

$$\|M^{-1}(M(z))\| \leq C_1^{-1} \|M(z)\|,$$

which gives the other inequality.

By the definition of H_F , we have

$$\begin{aligned} H_F(M(z)) &= \lim_{n \rightarrow \infty} \frac{1}{d^n} \log \|F^{(n)}(M(z))\| \\ &= \lim_{n \rightarrow \infty} \frac{1}{d^n} \log \|MF'^{(n)}(z)\| = H_{F'}(z), \end{aligned}$$

where the last equality follows from (8.24). This proves (8.22).

Since $|M(z) \wedge M(w)| = |\det(M)| \cdot |z \wedge w|$, we have

$$(8.25) \quad -\log |M(z) \wedge M(w)| = -\log |\det(M)| - \log |z \wedge w|.$$

On the other hand, by (8.20) and the fact that $|\operatorname{Res}(M)| = |\det(M)|$, it follows that

$$(8.26) \quad -\frac{1}{d(d-1)} \log |\operatorname{Res}(F')| = -\frac{1}{d(d-1)} \log |\operatorname{Res}(F)| - \log |\det(M)| .$$

Putting together (8.22), (8.25), and (8.26) gives (8.23). $\square$

COROLLARY 8.9. *Let $M \in \operatorname{PGL}_2(K)$ be a Möbius transformation. Then for all $z, w \in \mathbb{P}^1(K)$, we have*

$$g_\varphi(M(z), M(w)) = g_{M^{-1} \circ \varphi \circ M}(z, w) .$$

The following result gives the functional equation for $g_\varphi(x, y)$ in the case where $x, y \in \mathbb{P}^1(K)$.

THEOREM 8.10. *For all $x, y \in \mathbb{P}^1(K)$, we have*

$$g_\varphi(\varphi(x), y) = g_\varphi(x, \varphi^*(y)) ,$$

where $y_1, \dots, y_d$ are the preimages of y under φ , counting multiplicity, and

$$g_\varphi(x, \varphi^*(y)) := \sum_{i=1}^d g_\varphi(x, y_i) .$$

PROOF. Using Corollary 8.9, we may assume without loss of generality that $y = \infty$. Let F be a homogeneous lifting of φ , and let $R = |\operatorname{Res}(F)|^{-\frac{1}{d(d-1)}}$. For $z \in K^2 \setminus \{0\}$, let $[z]$ denote the class of z in $\mathbb{P}^1(K)$. Fix $w = (1, 0)$, let $a_1, \dots, a_d$ be the preimages of $[w]$ under φ (counting multiplicities), and let $w_1, \dots, w_d$ be solutions to $F(w_i) = w$ such that $[w_i] = a_i$.

It suffices to show that for all $z \in K^2 \setminus \{0\}$, we have

$$(8.27) \quad G_F(F(z), w) = \sum_{i=1}^d G_F(z, w_i) .$$

Since $H_F(F(z)) = dH_F(z)$ and $H_F(w_i) = \frac{1}{d}H_F(w)$ for all i , (8.27) is equivalent to

$$-\log |F(z) \wedge w| = (d-1) \log R - \sum_{i=1}^d \log |z \wedge w_i| ,$$

which itself is equivalent to

$$(8.28) \quad |F(z) \wedge w| = |\operatorname{Res}(F)|^{1/d} \prod_{i=1}^d |z \wedge w_i| .$$

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ography.

We verify (8.28) by an explicit calculation (compare with [25, Lemma 6.5]).

Write $F = (F_1, F_2) = (\prod_{i=1}^d z \wedge a_i, \prod_{j=1}^d z \wedge b_j)$. Since $F(b_j) = (\prod_{i=1}^d a_i \wedge b_j, 0)$ and $w = (1, 0)$, we may take

$$w_j = \frac{b_j}{(\prod_i a_i \wedge b_j)^{1/d}}$$

for $j = 1, \dots, d$, where for each j we fix some choice of a d th root of $\prod_i a_i \wedge b_j$. Note that $|w_j|$ is independent of which d th root we pick. Thus

$$\prod_{j=1}^d |z \wedge w_j| = \frac{\prod_j |z \wedge b_j|}{\prod_{i,j} |a_i \wedge b_j|^{1/d}} = \frac{|F(z) \wedge w|}{|\text{Res}(F)|^{1/d}},$$

which gives (8.28). $\square$

Recall that for any continuous function $f : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{R}$, the function $\varphi_*(f)$ defined by

$$\varphi_*(f)(y) = \sum_{\varphi(x)=y} m_x f(x)$$

is again continuous. (Here m_x denotes the analytic multiplicity of x as defined in §1.) Using this, we can deduce that Theorem 8.10 remains valid for all $x, y \in \mathbb{P}_{\text{Berk}}^1$:

COROLLARY 8.11. *For all $x, y \in \mathbb{P}_{\text{Berk}}^1$, we have*

$$(8.29) \quad g_\varphi(\varphi(x), y) = g_\varphi(x, \varphi^*(y)) .$$

PROOF. Fix $y \in \mathbb{P}^1(K)$. Then since φ is continuous, both $g_\varphi(\varphi(x), y)$ and $g_\varphi(x, \varphi^*(y))$ are continuous real-valued functions of x for all $x \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$. Since these functions agree on the dense subset $\mathbb{P}^1(K)$ of $\mathbb{P}_{\text{Berk}}^1$ by Theorem 8.10, (8.29) holds for all $x \in \mathbb{P}_{\text{Berk}}^1, y \in \mathbb{P}^1(K)$.

Now fix $x \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$. Then $g_\varphi(\varphi(x), y)$ and $\varphi_* g_\varphi(x, y)$ are continuous and real-valued for all $y \in \mathbb{P}_{\text{Berk}}^1$, and they agree for $y \in \mathbb{P}^1(K)$. Therefore (8.29) holds for all $x \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K), y \in \mathbb{P}_{\text{Berk}}^1$.

Finally, fix $y \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$. Then $g_\varphi(\varphi(x), y)$ and $g_\varphi(x, \varphi^*(y))$ are continuous and real-valued for all $x \in \mathbb{P}_{\text{Berk}}^1$, and they agree on the dense subset $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(K)$, so they agree for all $x \in \mathbb{P}_{\text{Berk}}^1$. $\square$

Using Corollary 8.11, we deduce the following:

COROLLARY 8.12. *For all $y \in \mathbb{P}_{\text{Berk}}^1$, we have*

$$\iint_{\mathbb{P}_{\text{Berk}}^1 \times \mathbb{P}_{\text{Berk}}^1} g_\varphi(x, y) d\mu_\varphi(x) = 0 .$$

PROOF. Fix $y \in \mathbb{P}_{\text{Berk}}^1$. We already know that there is a constant C_y such that $\int g_\varphi(x, y) \mu_\varphi(x) = C_y$, and we need to show that $C_y = 0$.

Since $\varphi_*\mu_\varphi = \mu_\varphi$, it follows from Corollary 8.11 that

$$\begin{aligned} C_y &= \int g_\varphi(x, y) \mu_\varphi(x) = \int g_\varphi(x, y) \varphi_*\mu_\varphi(x) \\ &= \int g_\varphi(\varphi(x), y) \mu_\varphi(x) \\ &= \int g_\varphi(x, \varphi^*(y)) \mu_\varphi(x) \\ &= dC_y, \end{aligned}$$

and thus $C_y = 0$ as desired. $\square$

From Proposition 8.7 and Corollary 8.12, we deduce:

COROLLARY 8.13. $g_\varphi(x, y) = g_{\mu_\varphi}(x, y)$ for all $x, y \in \mathbb{P}_{\text{Berk}}^1$.

3. Introduction to Fatou-Julia theory on $\mathbb{P}_{\text{Berk}}^1$

The modern approach to complex dynamics was initiated by Fatou and Julia in the early part of the twentieth century. The best understood and most classical part of the theory concerns the iteration of rational functions on the complex projective line $\mathbb{P}^1(\mathbb{C})$. The primary goal of the theory is to understand the structure of the Fatou set and its complement, the Julia set.

In p -adic dynamics, there is an analogous, more recent theory for a rational function $\varphi \in \mathbb{C}_p(T)$ acting on $\mathbb{P}^1(\mathbb{C}_p)$. The theory has attained considerable depth (see, e.g. [7], [8], [37], [49], [54]), and one finds both striking similarities and striking differences between rational dynamics over $\mathbb{C}$ and over $\mathbb{C}_p$. (For an overview of the results obtained, see [13].) The differences arise for the most part from the fact that the topology on $\mathbb{C}_p$ is totally disconnected and not locally compact. Even the most basic topological questions about the Fatou and Julia sets have a completely different flavor when working over $\mathbb{C}_p$ versus $\mathbb{C}$. In fact, it is not clear *a priori* what the definition of the Fatou and Julia sets should be in $\mathbb{P}^1(\mathbb{C}_p)$, since (for example) the notions of equicontinuity and normality do not coincide. It took considerable effort to give a satisfactory definition for Fatou components ([55]). Even definitions which initially seem reasonable (such as defining the Fatou set in terms of equicontinuity of the family of iterates $\{\varphi^{(n)}\}$) turn out to behave rather differently than expected. For example, the Julia set of any map with good reduction is empty, in sharp contrast with the situation over

$\mathbb{C}$, where one of the most fundamental facts is that the Julia set is always nonempty.

Our basic premise here is that a rational map $\varphi \in K(T)$ should be thought of as acting not on $\mathbb{P}^1(K)$, but on the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$ over K . The advantages of working in $\mathbb{P}_{\text{Berk}}^1$ are already evident in our construction of the canonical probability measure on $\mathbb{P}_{\text{Berk}}^1$ attached to φ . After giving a definition of the Berkovich Fatou and Julia sets based on equicontinuity (or equivalently, normality), we will show that the canonical measure is supported on the Berkovich Julia set. In particular, it follows that the Berkovich Julia set is always nonempty.

We will also prove an analogue of the Strong Montel theorem for iterates of a rational map defined over a finite extension of $\mathbb{Q}_p$, and apply it to deduce information about the structure of the Berkovich Julia set.

There are (at least) two different results in complex analysis and complex dynamics known as “Montel’s theorem”. We will prove Berkovich space analogues of both of these classical theorems. We will refer to the following result as the *Weak Montel theorem*:

THEOREM (Weak Montel over $\mathbb{C}$). A locally bounded family of analytic functions on a domain $U \subseteq \mathbb{C}$ is normal.

The *Strong Montel theorem*, which is significantly more difficult, is the following result:

THEOREM (Strong Montel over $\mathbb{C}$). A family of meromorphic functions on a domain $U \subseteq \mathbb{C}$ which omits 3 distinct values in $\mathbb{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$ is normal.

For proofs, see Sections 2.1 and 3.2 of [68]. Our Berkovich Strong Montel theorem concerns families of iterates of a rational function defined over a finite extension of $\mathbb{Q}_p$.

In a series of deep and beautiful papers ([54],[53],[55],[56]), Juan Rivera-Letelier has fashioned a powerful set of tools for studying dynamics on $\mathbb{P}_{\text{Berk}}^1$. His work is crucial for our proof of the Strong Montel theorem (Theorem 8.51), upon which all of our subsequent results rely. The major difference between our point of view and Rivera-Letelier’s is that heretofore he has primarily worked with a space $\mathbb{H}_p$ which is naturally isomorphic to $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$, but which is endowed with a stronger topology. In other words, he does not work directly with the space $\mathbb{P}_{\text{Berk}}^1$, endowed with its Berkovich topology, but rather considers separately the action of φ on the spaces $\mathbb{P}^1(\mathbb{C}_p)$ and $\mathbb{H}_p$. (Note:

Rivera-Letelier is currently working on a manuscript [58] in which he reformulates his previous results in the language of Berkovich spaces.)

The Strong Montel theorem for $\mathbb{P}_{\text{Berk}}^1$, like its counterpart over $\mathbb{C}$, appears to be a deep result that cannot be obtained by “elementary” methods. (In contrast, Hsia’s theorem [37] is a fairly elementary fact which provides an even stronger version of Montel’s theorem for $\mathbb{P}^1(\mathbb{C}_p)$.) Once one has the Strong Montel theorem, a number of consequences concerning the topology of the Julia set follow almost immediately. Although the proofs of these consequences are quite similar to their complex counterparts, there are a number of minor differences, so we decided to write out as many of the details as seemed reasonable. We have closely followed Milnor’s excellent monograph ([47]) for most of this material, which can be found in Section 10 below. It is important to note that we only prove the Strong Montel theorem for rational functions defined over a finite extension of $\mathbb{Q}_p$. This assumption is used crucially in our proof of the Weak Montel theorem (from which the strong version is deduced), which is based entirely on the fact that finite extensions of $\mathbb{Q}_p$ are locally compact.

After seeing a draft of our results, Rivera-Letelier found an improvement of our Strong Montel theorem which works for all rational functions defined over $\mathbb{C}_p$. His version of this theorem, together with a discussion of the proof and some consequences, can be found in the Notes at the end of this chapter.

4. Equicontinuity and the Fatou and Julia sets

Add a brief discussion of *uniform structures* to get rid of the dependence on metrizability.

In this section, we define the Berkovich Fatou and Julia sets of a rational map $\varphi \in \mathbb{C}_p(T)$; these are open and closed subsets, respectively, of the Berkovich projective line $\mathbb{P}_{\text{Berk}}^1$. Assuming that the degree of φ is at least 2, we will show that the support of the canonical measure is contained in the Berkovich Julia set of φ . In particular, it will follow that the Berkovich Julia set of a rational map φ of degree at least two is always non-empty.

In order to define the Fatou and Julia sets, we first need to discuss the general notion of topological equicontinuity. The following definition is taken from [59, Section 14.2].

DEFINITION 8.14. Let X and Y be topological spaces, let $\mathcal{F}$ be a family of continuous maps from X to Y , and let $x \in X$ and $y \in Y$ be arbitrary points.

- (a) $\mathcal{F}$ is *topologically equicontinuous at (x, y)* if, given any open neighborhood O of y , there exist open neighborhoods U of x and V of y such that for every $f \in \mathcal{F}$, we have $f(U) \subseteq O$ whenever $f(U) \cap V$ is nonempty.
- (b) $\mathcal{F}$ is *topologically equicontinuous at x* if it is topologically equicontinuous at (x, y) for all $y \in Y$.
- (c) $\mathcal{F}$ is *topologically equicontinuous* if it is topologically equicontinuous at all $x \in X$.

We will need the following lemma from [59, Proposition 14.2.1]:

LEMMA 8.15. *Let $\mathcal{F}$ be a family of continuous maps from X to Y which is topologically equicontinuous at $x \in X$. Then given $K \subseteq O \subseteq Y$ with K compact and O open, there is a neighborhood U of x such that $f(U) \subseteq O$ whenever $f \in \mathcal{F}$ and $f(U) \cap K$ is nonempty.*

DEFINITION 8.16. Suppose that X is a topological space and that Y is a metric space. Let $\mathcal{F}$ be a family of continuous maps from X to Y , and let x be a point of X . We say that $\mathcal{F}$ is *equicontinuous at x* if for every $\varepsilon > 0$, there exists an open neighborhood U of x such that $d(f(x), f(x')) < \varepsilon$ for all $x' \in U$ and all $f \in \mathcal{F}$. We say that $\mathcal{F}$ is *equicontinuous* if it is equicontinuous at every $x \in X$.

Topological equicontinuity is related to equicontinuity as follows (see [59, Exercise 14.2.3]):

PROPOSITION 8.17. *Suppose that X is a topological space and that Y is a metric space. Let $\mathcal{F}$ be a family of continuous maps from X to Y , and let x be a point of X . Then:*

- A) *If $\mathcal{F}$ is equicontinuous at x , then $\mathcal{F}$ is topologically equicontinuous at x .*
- B) *If Y is compact and $\mathcal{F}$ is topologically equicontinuous at x , then $\mathcal{F}$ is equicontinuous at x .*

PROOF. For A), suppose that $\mathcal{F}$ is equicontinuous at x , let $y \in Y$, and let O be an open neighborhood of y in Y . Choose $\varepsilon > 0$ such that $D(y, \varepsilon) \subseteq O$, and put $V = D(y, \varepsilon/3)$. By definition of equicontinuity, there exists an open neighborhood U of x such that $d(f(x), f(u)) < \varepsilon/3$ for all $u \in U$ and all $f \in \mathcal{F}$. Let $f \in \mathcal{F}$, and suppose that $f(U) \cap V$ is nonempty, so that there exists some element $x' \in U$ such that $f(x') \in V$. Then for all $u \in U$, we have

$$\begin{aligned}
 d(f(u), y) &\leq d(f(u), f(x)) + d(f(x), f(x')) + d(f(x'), y) \\
 &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 \\
 &= \varepsilon,
 \end{aligned}$$

so that $f(u) \in O$.

For B), suppose Y is compact and $\mathcal{F}$ is topologically equicontinuous at x . Let $\varepsilon > 0$ be arbitrary. Then for each $y \in Y$, there exist open neighborhoods U_y and $V_y \subseteq D(y, \varepsilon/2)$ of x and y , respectively, such that $f(U_y) \subseteq D(y, \varepsilon/2)$ whenever $f(U_y) \cap V_y \neq \emptyset$.

Since Y is compact, there is a finite subcover $V_1, \dots, V_n := V_{y_1}, \dots, V_{y_n}$ which covers Y . Let $U_1, \dots, U_n$ be the corresponding sets in X , and let $U = \cap U_j$, so that U is an open neighborhood of x .

We claim that $d(f(x), f(x')) < \varepsilon$ whenever $x' \in U$ and $f \in \mathcal{F}$. To see this, note that since $x' \in U$, we have $x' \in U_j$ for all j . Choosing an index k such that $f(x) \in V_k$, we have $f(U_k) \cap V_k \neq \emptyset$, and therefore $f(x') \in D(y_k, \varepsilon/2)$. It follows that

$$\begin{aligned} d(f(x), f(x')) &\leq d(f(x), y_k) + d(y_k, f(x')) \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon \end{aligned}$$

as desired. $\square$

In particular, if X is a topological space and Y is a compact metric space, then the notions of equicontinuity and topological equicontinuity coincide. It follows in this case that the equicontinuity of a family $\mathcal{F}$ of continuous maps from X to Y depends only on the underlying topology on Y , not on the choice of a particular metric on Y .

We now specialize to the case where $X = Y = \mathbb{P}_{\text{Berk}}^1$. Recall that $\mathbb{P}_{\text{Berk}}^1$ is a compact, Hausdorff, and path-connected topological space. Let $\mathcal{F} : U \rightarrow \mathbb{P}_{\text{Berk}}^1$ be a family of continuous maps, where $U \subseteq \mathbb{P}_{\text{Berk}}^1$ is an open set. We say that $\mathcal{F}$ is equicontinuous on U iff it is topologically equicontinuous in the above sense.

REMARK 8.18. Using the fact that the Berkovich subspace topology on $\mathbb{P}^1(\mathbb{C}_p)$ coincides with the topology defined by the spherical metric, it follows that equicontinuity in the Berkovich topology at a point $x \in \mathbb{P}^1(\mathbb{C}_p)$ is equivalent to equicontinuity at x with respect to the spherical metric on $\mathbb{P}^1(\mathbb{C}_p)$. However, equicontinuity at a point of $\mathbb{H}_p = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$ does not seem as easy to describe.

We now define the Berkovich Fatou and Julia sets of a rational function $\varphi(T) \in \mathbb{C}_p(T)$.

DEFINITION 8.19. The *Fatou set* of φ is the open set F_φ consisting of all $x \in \mathbb{P}_{\text{Berk}}^1$ such that the family $\{\varphi^{(n)}\}_{n \geq 1}$ of iterates of φ is equicontinuous at x . The *Julia set* J_φ of φ is the complement of the Fatou set, i.e., $J_\varphi := \mathbb{P}_{\text{Berk}}^1 \setminus F_\varphi$.

By definition, the Fatou set of φ is open, and the Julia set of φ is compact.

The sets $J_\varphi(\mathbb{C}_p) = J_\varphi \cap \mathbb{P}^1(\mathbb{C}_p)$ and $F_\varphi(\mathbb{C}_p) = F_\varphi \cap \mathbb{P}^1(\mathbb{C}_p)$ coincide, respectively, with the Julia sets and Fatou sets of φ in $\mathbb{C}_p$ as defined in [7], [37], and [49]. This follows from Remark 8.18.

Assume that $\deg(\varphi) \geq 2$. Since Benedetto has shown in this case (see [7]) that $F_\varphi(\mathbb{C}_p)$ is always non-empty (in contrast to the situation over the complex numbers), and since $F_\varphi(\mathbb{C}_p) \subseteq F_\varphi$, we conclude:

PROPOSITION 8.20. *If $\deg(\varphi) \geq 2$, then the Berkovich Fatou set F_φ is non-empty.*

Concerning the non-emptiness of the Berkovich Julia set, we have the following result.

THEOREM 8.21. *The support of the canonical probability measure μ_φ attached to φ is contained in the Julia set J_φ .*

PROOF. Let x_0 be a point of the Fatou set. We will show that if U is a sufficiently small open neighborhood of x_0 , then μ_φ restricted to U is the zero measure.

First assume that $x_0 \in \mathbb{A}_{\text{Berk}}^1$. Then it suffices to show that $\hat{h}_{\varphi,v}(x)$ is harmonic on some neighborhood $U \subset \mathbb{A}_{\text{Berk}}^1$ of x_0 . To do this, we follow the general outline of an argument of Fornaess and Sibony [32]. Define

$$O = \{x \in \mathbb{A}_{\text{Berk}}^1 : [T]_x < 2\}$$

$$E = \{x \in \mathbb{A}_{\text{Berk}}^1 : [T]_x \leq \frac{1}{2}\}.$$

By Lemma 8.15, there exists a neighborhood U of x_0 such that $\varphi^{(n)}(U) \subseteq O$ whenever $\varphi^{(n)}(U) \cap E \neq \emptyset$. Therefore for all n we have either $\varphi^{(n)}(U) \subseteq O$ or $\varphi^{(n)}(U) \subseteq \mathbb{P}_{\text{Berk}}^1 \setminus E$. Passing to a subsequence $\varphi^{(n_k)}$ and replacing φ by $1/\varphi$ if necessary, we may therefore assume without loss of generality that $\varphi^{(n_k)}(U) \subseteq O$ for all k , or equivalently that $[\varphi^{(n_k)}(T)]_x < 2$ for all $z \in U$ and all k . In particular, $F_1^{(n_k)} \neq 0$ on U for all k .

We now find that, for $x \in U$, we have

$$\begin{aligned} \hat{h}_{\varphi,v}^{(n_k)}(x) &= \frac{1}{d^{n_k}} \max(\log_v([F_1^{(n_k)}(1, T)]_x), \log_v([F_2^{(n_k)}(1, T)]_x)) \\ &= \frac{1}{d^{n_k}} \log_v([F_1^{(n_k)}(1, T)]_x) + \frac{1}{d^{n_k}} \max(1, \log_v([\varphi^{(n_k)}(T)]_x)). \end{aligned}$$

LaTeX problem: the label is running into the equation.

The last term in (8.30) converges uniformly to zero, since the quantity $[\varphi^{(n_k)}(T)]_x$ is uniformly bounded as k varies. Moreover, the term $\frac{1}{d^{n_k}} \log_v([F_1^{(n_k)}(1, T)]_x)$ is harmonic on U for all k . Since $\hat{h}_{\varphi, v}^{(n_k)}$ converges uniformly to $\hat{h}_{\varphi, v}$, it follows that $\frac{1}{d^{n_k}} \log_v([F_1^{(n_k)}(1, T)]_x)$ converges uniformly to $\hat{h}_{\varphi, v}$ on U . Therefore $\hat{h}_{\varphi, v}$ is harmonic on U as desired.

Finally, for the case $x_0 = \infty$, apply a similar argument to the function $\hat{g}_{\varphi, v}(x)$, using the coordinate function $U = 1/T$ instead of T . $\square$

REMARK 8.22. When φ is defined over a finite extension of $\mathbb{Q}_p$, we will prove in Section 10 that the support of μ_φ is in fact equal to J_φ .

COROLLARY 8.23. *If $\deg(\varphi) \geq 2$, then the Berkovich Julia set J_φ is non-empty.*

REMARK 8.24. By contrast, it is known that $J_\varphi(\mathbb{C}_p)$ (which is the Julia set of φ in $\mathbb{P}^1(\mathbb{C}_p)$) is empty whenever the map φ has good reduction (see [49], [7]).

REMARK 8.25. One can also prove that J_φ is non-empty by using Rivera-Letelier's theorem [54] that every rational map φ has a repelling fixed point in $\mathbb{P}_{\text{Berk}}^1$, together with the fact that repelling fixed points are always in the Berkovich Julia set [54]. (The definition of a repelling fixed point for points in $\mathbb{H}_p = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$ can be found in Section 8 below.)

5. Normality and the topology of locally uniform convergence

Let X, Y be topological spaces. Following Milnor ([47, Section 3]), we introduce the topology of locally uniform convergence on the space $C(X, Y)$ of continuous functions from X to Y .

DEFINITION 8.26. Let $f \in C(X, Y)$. A basis of open neighborhoods for $f \in C(X, Y)$ in the *topology of locally uniform convergence* is given by the open sets of the form

$$N_{K, U}(f) := \{g \in C(X, Y) : (f(x), g(x)) \in U \forall x \in K\},$$

where $K \subseteq X$ is compact and $U \subseteq Y \times Y$ is an open neighborhood of the diagonal.

The following is essentially [47, Lemma 3.1] with some details added for the convenience of the reader.

PROPOSITION 8.27. *If Y is a metric space, then a basis of open neighborhoods for $f \in C(X, Y)$ in the topology of locally uniform convergence is given by the open sets of the form*

$$N_{K, \varepsilon}(f) := \{g \in C(X, Y) : d(f(x), g(x)) < \varepsilon \ \forall x \in K\},$$

where $K \subseteq X$ is compact and $\varepsilon > 0$.

PROOF. We claim that given $K \subseteq X$ compact and $U \subseteq Y \times Y$ an open neighborhood of the diagonal, we can find $\varepsilon > 0$ such that if $x \in K$, $y \in Y$, and $d(f(x), y) < \varepsilon$, then $(f(x), y) \in U$.

To see this, let d denote the metric on Y , and define a metric δ on $Y \times Y$ by $\delta((y_1, y_2), (y'_1, y'_2)) = d(y_1, y'_1) + d(y_2, y'_2)$. It is straightforward to show that the topology induced by δ on $Y \times Y$ is the product topology. Let $\Delta_{f(K)} = \{(y, y) : y \in f(K)\} \subseteq Y \times Y$. For each $z \in \Delta_{f(K)}$, choose an open ball $D(z, \varepsilon_z) = \{z' \in Y \times Y : \delta(z, z') < \varepsilon_z\}$ containing z and contained in U . By compactness of $\Delta_{f(K)}$, a finite subset $\{D(z_j, \varepsilon_j/2)\}_{j=1}^t$ of the collection $\{D(z, \varepsilon_z/2)\}$ will cover $\Delta_{f(K)}$. Let $\varepsilon > 0$ be the minimum of the numbers $\{\varepsilon_j/2\}_{j=1}^t$. Suppose that $x \in K$, $y \in Y$, and $d(f(x), y) < \varepsilon$, and let $w = (f(x), y)$ and $w' = (f(x), f(x))$. Then $\delta(w, w') = d(y, f(x)) < \varepsilon_j/2$ for all j . On the other hand, since $w' \in \Delta_{f(K)}$, there exists an index j such that $\delta(z_j, w') < \varepsilon_j/2$. By the triangle inequality, we have $\delta(w, z_j) < \varepsilon_j$. But then $w \in D(z_j, \varepsilon_j) \subseteq U$, proving the claim.

It follows that $N_{K, \varepsilon}(F) \subseteq N_{K, U}(f)$.

On the other hand, if $U(\varepsilon) := \{(y, y') \in Y \times Y : d(y, y') < \varepsilon\}$, then $N_{K, \varepsilon}(F) = N_{K, U(\varepsilon)}(f)$ by definition. $\square$

From this we easily obtain:

COROLLARY 8.28. *Let X be a topological space and Y a metric space, and let $f_n : X \rightarrow Y$ be a sequence of continuous functions. Let $g \in C(X, Y)$. Then the following are equivalent:*

- (1) f_n converges to g in the topology of locally uniform convergence on $C(X, Y)$.
- (2) $f_n|_K$ converges uniformly to $g|_K$ for every compact subset $K \subseteq X$.

When X is locally compact, these conditions are equivalent to:

- (3) Every point $x \in X$ has an open neighborhood N such that $f_n|_N$ converges uniformly to $g|_N$.

If either of the equivalent conditions (1) or (2) holds, we say that $f_n \rightarrow g$ in $C(X, Y)$, or that f_n converges locally uniformly to g .

Proposition 8.27 shows that if Y is a metric space, then the locally uniform convergence of a sequence $\{f_n\}$ of continuous maps from X to Y depends only on the underlying topology on Y , not on the choice of a particular metric on Y .

DEFINITION 8.29. Let $\{f_n\}$ be a sequence of continuous maps from a topological space X to a topological space Y . We say that $\{f_n\}$ *diverges locally uniformly to infinity in Y* if for all compact sets $K \subseteq X$ and $K' \subseteq Y$, we have $f_n(K) \cap K' = \emptyset$ for n sufficiently large.

DEFINITION 8.30. Let $\mathcal{F}$ be a collection of continuous maps from a topological space X to a topological space Y . We say that $\mathcal{F}$ is *normal* if every infinite sequence $\{f_n\}$ of maps in $\mathcal{F}$ contains a subsequence which either converges locally uniformly to a function $g \in C(X, Y)$, or diverges locally uniformly to infinity in Y . We say that $\mathcal{F}$ is *normal at x* if there is an open neighborhood U of x such that $\mathcal{F}$ is normal on U , i.e., such that the family $\mathcal{F}|_U$ in $C(U, Y)$ is normal.

REMARK 8.31. If Y is compact, then no sequence $\{f_n\}$ can diverge locally uniformly to infinity in Y . Thus $\mathcal{F}$ is normal if and only if every infinite sequence $\{f_n\}$ of maps in $\mathcal{F}$ contains a subsequence which converges locally uniformly.

REMARK 8.32. If X is a separable locally compact metric space, then a family $\mathcal{F}$ is normal on a connected open subset $U \subseteq X$ iff $\mathcal{F}$ is normal at every point of U . (Recall that X is *separable* if it has a countable dense subset.) The proof is the same as the classical one over $\mathbb{C}$ (see e.g. [68, Theorem 2.1.2]). Thus the notion of normality is *local*.

The following lemma gives a useful relationship between two a priori different notions of uniform continuity. In the statement, $\mathcal{A}(V)$ denotes the ring of analytic functions on a strictly affinoid subset V of $\mathbb{P}_{\text{Berk}}^1$, which becomes a Banach algebra when endowed with the spectral norm.

LEMMA 8.33. Let $z \in \mathbb{P}_{\text{Berk}}^1$, and let $V \subset \mathbb{P}_{\text{Berk}}^1$ be a connected strictly affinoid neighborhood of z . Let $\{f_n\}$ be a sequence of analytic functions in $\mathcal{A}(V)$, and let $f \in \mathcal{A}(V)$. Then $f_n \rightarrow f$ in $\mathcal{A}(V)$ if and only if $f_n \rightarrow f$ uniformly on V in the Berkovich topology. In particular, if $f_n \rightarrow f$ in $\mathcal{A}(V)$ then the sequence $\{f_n\}$ is normal at z .

PROOF. If $f_n \rightarrow f$ uniformly on V in the Berkovich topology, then $f_n \rightarrow f$ uniformly on $V(\mathbb{C}_p)$ with respect to the spherical metric topology (which coincides with the subspace topology). Without loss of generality, we may assume that $V \subseteq \mathbb{A}^1(\mathbb{C}_p)$, in which case the spectral norm $\|\cdot\|_V$ on V is just the supremum norm. It thus follows that

$\|f_n - f\|_V \rightarrow 0$. (In fact, we see that the statement $\|f_n - f\|_V \rightarrow 0$ is equivalent to saying that $f_n \rightarrow f$ uniformly on $V(\mathbb{C}_p)$.)

Conversely, suppose that $f_n \rightarrow f$ uniformly with respect to the spherical metric on $V(\mathbb{C}_p)$. Without loss of generality, we may assume that $V \subseteq \mathbb{B}(0, 1)$. By the proof of Urysohn's metrization theorem, the Berkovich topology on V can be given explicitly in the form

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} |\delta(x, \alpha_j)_{\infty} - \delta(y, \alpha_j)_{\infty}|,$$

where $\{\alpha_j\}$ is a countable dense subset of $B(0, 1)$, see §???. Here $\delta(a, b)_{\infty}$ is the Hsia kernel, which agrees with the spherical distance $\|a, b\|$ on $\mathbb{P}^1(\mathbb{C}_p)$. In particular, since

$$||x, z| - |y, z|| \leq \|x, y\|$$

for all $x, y, z \in V(\mathbb{C}_p)$ by the triangle inequality, we have

$$d(x, y) \leq \sum_{j=1}^{\infty} \frac{1}{2^j} \|x, y\| = \|x, y\|$$

for all $x, y \in V(\mathbb{C}_p)$. It follows that $f_n \rightarrow f$ locally uniformly in the Berkovich topology on V as desired. □

It would be nice to prove Lemma 8.33 without appealing to Urysohn's metrization theorem.

REMARK 8.34. Normality for a sequence $\{f_n\}$ of meromorphic functions does not necessarily imply that there is a subsequence f_{n_k} converging to a meromorphic function. For example, if $\{\alpha_n\}$ is a set of coset representatives for the different residue classes in $B(0, 1)$ and $f_n(z) = z + \alpha_n$, then $\{f_n\}$ converges uniformly on the open unit ball $\mathbb{B}(0, 1)^-$ to the constant function $f(z) \equiv \zeta_{\text{Gauss}}$, where $\zeta_{\text{Gauss}} \in \mathbb{P}_{\text{Berk}}^1$ is the Gauss point (the unique boundary point of $\mathbb{B}(0, 1)^-$).

However, if all $\{f_n\}$ are defined over a fixed finite extension of $\mathbb{Q}_p$, then we will see below (Theorem 8.40) that normality *does* imply that there is a subsequence converging to a meromorphic function.

We conclude this section with a version of the Arzela-Ascoli Theorem, which relates the general notions of normality and equicontinuity.

THEOREM 8.35 (Arzela-Ascoli). *Let $\mathcal{F}$ be a collection of continuous maps from a separable, locally compact metric space X to a compact metric space Y , and let $x \in X$. Then $\mathcal{F}$ is equicontinuous at x if and only if $\mathcal{F}$ is normal at x .*

PROOF. (Equicontinuous implies normal): Given $\varepsilon > 0$, let U be an open neighborhood of x such that $z, w \in U$ and $f \in \mathcal{F}$ implies $d_Y(f(z), f(w)) < \varepsilon$. Let $\{f_n\}$ be an infinite sequence of functions from $\mathcal{F}$, and let $\{z_n\}$ be a countable dense subset of U . By (sequential) compactness of Y , for each i the sequence $\{f_n(z_i)\}_{n \geq 1}$ has a convergent subsequence. Taking the corresponding diagonal subsequence $\{g_n\}$ of $\{f_n\}$, we find that $\{g_n\}$ converges at every z_i .

We claim that $\{g_n\}$ converges locally uniformly at x . To see this, let $K \subset U$ be a compact neighborhood of x , and let $0 < \varepsilon' < \varepsilon$. Then there exists a $\delta' > 0$ such that if $d_X(z, w) < \delta'$ then $d_Y(g_n(z), g_n(w)) < \varepsilon'/3$ for all n and all $z, w \in U$. By compactness of K , after relabeling the z_i we may assume that $D(z_1, \delta'), \dots, D(z_k, \delta')$ cover K . As the g_n 's converge at each z_i , there exists N such that if $m, n \geq N$ then $d_Y(g_n(z_i), g_m(z_i)) < \varepsilon'/3$ for all $i = 1, \dots, k$. Finally, if $z \in K$ then $z \in D(z_i, \delta')$ for some $1 \leq i \leq k$ and therefore

$$\begin{aligned} d_Y(g_n(z), g_m(z)) &\leq d_Y(g_n(z), g_n(z_i)) + d_Y(g_n(z_i), g_m(z_i)) \\ &\quad + d_Y(g_m(z_i), g_m(z)) \\ &< \varepsilon'. \end{aligned}$$

It follows that g_n is normal on K , as desired.

(Normal implies equicontinuous): Suppose $\mathcal{F}$ is normal at x . To prove equicontinuity of $\mathcal{F}$ at x , it suffices to show that every sequence of functions in $\mathcal{F}$ has a subsequence which is equicontinuous at x . So let $\{f_n\}$ be a sequence of functions in $\mathcal{F}$. By normality, there is a subsequence $\{h_n\}$ of $\{f_n\}$ converging uniformly to some function h on a compact neighborhood K of x . We claim that $\{h_n\}$ is equicontinuous at x . Indeed, given $\varepsilon > 0$, it follows by uniform convergence that there exists N such that if $z, w \in K$ and $n, m \geq N$ then $\max\{d_Y(h_n(z), h(z)), d_Y(h_m(w), h(w))\} < \varepsilon/3$. As h is uniformly continuous on K , we find that there is an open neighborhood $U \subseteq K$ of x such that if $z, w \in U$ then $d_Y(h(z), h(w)) < \varepsilon/3$. It follows from the triangle inequality that

$$d_Y(h_n(z), h_m(w)) < \varepsilon/3$$

whenever $n, m \geq N$ and $z, w \in U$, proving equicontinuity. $\square$

6. Basic properties of the Fatou and Julia sets

In this section, we discuss some of the basic properties of Berkovich Fatou and Julia sets which can be proved without the use of Montel-type theorems. Later on, we will use the Montel theorems to obtain much more powerful results.

A subset $S \subseteq \mathbb{P}_{\text{Berk}}^1$ is called *completely invariant* under φ if $z \in S$ if and only if $\varphi(z) \in S$.

LEMMA 8.36.

- (a) *The Berkovich Fatou and Julia sets of φ are completely invariant under φ .*
- (b) *For any integer $k \geq 1$, the Berkovich Julia set of $\varphi^{(k)}$ coincides with the Berkovich Julia set of φ .*

PROOF. The proofs are the same as those over the complex numbers, see e.g. Lemmas 4.1 and 4.2 of [47]. $\square$

Each Type I point $z \in \mathbb{P}^1(\mathbb{C}_p)$ of period m has an associated *multiplier*, which is the number $\lambda = (\varphi^{(m)})'(z)$. The multiplier is the same for each of the m points in the forward orbit O of z by the chain rule. We say that the orbit O is *attracting* (resp. *indifferent*, *repelling*) if $|\lambda| < 1$ (resp. $|\lambda| = 1$, $|\lambda| > 1$). In addition, we say that the orbit is *superattracting* if $\lambda = 0$.

If $O \subset \mathbb{P}^1(\mathbb{C}_p)$ is an attracting periodic orbit of period m for φ , the *basin of attraction* of O is the set $\mathbb{A}$ consisting of all $z \in \mathbb{P}_{\text{Berk}}^1$ for which the successive iterates $\varphi^{(m)}(z), \varphi^{(2m)}(z), \dots$ converge towards some point of O . The set $\mathbb{A}(\mathbb{C}_p) = \mathbb{A} \cap \mathbb{P}^1(\mathbb{C}_p) \subset \mathbb{P}^1(\mathbb{C}_p)$ is open and contains the points of O .

LEMMA 8.37.

- A) *The basin of attraction $\mathbb{A}$ of every attracting periodic orbit O of φ is contained in F_φ .*
- B) *Every repelling periodic orbit is contained in J_φ .*
- C) *Every indifferent periodic orbit is contained in F_φ .*

PROOF. The corresponding statements for $\mathbb{P}^1(\mathbb{C}_p)$ are well-known (see e.g. [7]), and follow from the usual arguments over $\mathbb{C}$ (see e.g. Lemma 4.3 of [47]). Thus parts B) and C) of the lemma are true, and $\mathbb{A}(\mathbb{C}_p) \subset F_\varphi(\mathbb{C}_p)$ in part A). To see that $\mathbb{A}$ is contained in F_φ , suppose that $z \in \mathbb{A}$, so that $\varphi^{(km)}(z) \rightarrow z' \in O \cap F_\varphi$ as $k \rightarrow \infty$. If U is an open neighborhood of z' contained in F_φ , then for k large we have $\varphi^{(km)}(z) \in U$, which implies (by the backward invariance of φ) that $z \in F_\varphi$. $\square$

For later reference, we define the *immediate basin of attraction* A_0 of an attracting periodic cycle C to be the analytic component of $\mathbb{A}(\mathbb{C}_p)$ containing C . By definition, the *analytic component* of an open set $X \subset \mathbb{P}^1(\mathbb{C}_p)$ containing a given point $x \in X$ is the union of all connected affinoid domains $Y \subseteq X$ containing x .

7. A Weak Montel Theorem

In this section, we prove that a sequence of rational functions defined over a fixed finite extension of $\mathbb{Q}_p$ which is uniformly bounded on a connected affinoid $V \subseteq \mathbb{P}_{\text{Berk}}^1$ has a subsequence converging to an analytic function on V . Before giving the proof, we first state the analogous result from complex analysis, and then recall some well-known facts from rigid analysis concerning Laurent expansions of functions on V .

7.1. Laurent expansions on affinoids in $\mathbb{P}^1(\mathbb{C}_p)$. Let $V \subseteq \mathbb{P}^1(\mathbb{C}_p)$ be a connected strictly affinoid domain. For convenience, we assume that $\infty \in V$. Let $\mathcal{A}(V)$ denote the ring of analytic functions on V . The complement $\mathbb{P}^1(\mathbb{C}_p) \setminus V$ can be decomposed as a disjoint union of open discs $D_1, \dots, D_s$, where $D_i = B(a_i, |\pi_i|)^-$ with $a_i \in K_v$, $\pi_i \in K_v^*$ for some finite extension K_v of $\mathbb{Q}_p$. For $1 \leq i \leq s$, let $V^{(i)} = \mathbb{P}^1(\mathbb{C}_p) \setminus D_i$ be the corresponding closed disk in $\mathbb{P}^1$ containing ∞ , so that $V = \bigcap_{i=1}^s V^{(i)}$.

The following result is a consequence of Proposition 2.2.6 of [34] and its proof.

PROPOSITION 8.38. *Let $f \in \mathcal{A}(V)$ be an analytic function on V . Then:*

- (i) *f can be written uniquely in the form*

$$f = c + f^{(1)} + \dots + f^{(s)}$$

with $f^{(i)} \in \mathcal{A}(V^{(i)})$, $f^{(i)}(\infty) = 0$ and $c \in \mathbb{C}_p$.

- (ii) *The norm of f is given by*

$$\|f\|_V = \max\{|c|, \max_i \|f_i\|_V\} = \max\{|c|, \max_i \|f_i\|_{V^{(i)}}\}.$$

- (iii) *For $1 \leq i \leq s$ we can represent $f^{(i)}$ by a Laurent series expansion*

$$f^{(i)} = \sum_{m=1}^{\infty} b^{(i)}(m) \left(\frac{\pi_i}{z - a_i} \right)^m$$

with $b^{(i)}(m) \in \mathbb{C}_p$ and $\lim_{m \rightarrow \infty} b^{(i)}(m) = 0$, and

$$\|f^{(i)}\|_{V^{(i)}} = \max_m |b^{(i)}(m)|.$$

- (iv) *Moreover, if $f \in K_v(T)$ is a rational function and $a_i, \pi_i \in K_v$, then $c \in K_v$ and $b^{(i)}(m) \in K_v$ for all i, m .*

REMARK 8.39. We can rephrase part (iv) of Proposition 8.38 as follows. We say that an analytic function f on V is *defined over* K_v if $c \in K_v$ and $b^{(i)}(m) \in K_v$ for all i, m . Then according to (iv), a rational function $f \in K_v(T)$ which is analytic in V is defined over K_v , provided that K_v is large enough so that $a_i, \pi_i \in K_v$ as well.

7.2. Weak Montel. In this section we give the Weak Montel theorem for a sequence of meromorphic functions defined over a local field.

THEOREM 8.40 (Weak Montel).

Let K_v be a finite extension of $\mathbb{Q}_p$, and let f_n be a sequence of meromorphic functions on a connected affinoid subset $V \subset \mathbb{P}^1(\mathbb{C}_p)$, with all f_n defined over K_v . Suppose that there exists a disc $D \subset \mathbb{P}^1(\mathbb{C}_p)$ of positive radius (with respect to the spherical metric) such that $f_n(V) \cap D = \emptyset$ for all n . Then f_n has a subsequence converging to a meromorphic function f on V . Moreover, if all f_n are analytic on V then f is analytic on V as well.

PROOF. By changing coordinates on the source space, we may assume without loss of generality that $\infty \in V$. Furthermore, by changing coordinates on the target space, we may assume without loss of generality that $\infty \in D$, and therefore that each f_n is analytic on V and that there exists a constant $M < \infty$ such that $\|f_n\|_V \leq M$ for all n . By Proposition 8.38, and replacing K_v by a finite extension if necessary, we can write

$$f_n = c_n + \sum_{i=1}^s f_n^{(i)},$$

with

$$f_n^{(i)} = \sum_{m=1}^{\infty} b_n^{(i)}(m) \left(\frac{\pi_i}{z - a_i} \right)^m$$

and $a_i, \pi_i, c_n \in K_v$, $b_n^{(i)}(m) \in K_v$. Since $\|f_n\| \leq M$ for all n , we have $c_n \leq M$ and $b_n^{(i)}(m) \leq M$ for all i, n, m .

As K_v is a finite extension of $\mathbb{Q}_p$, the set $\{x \in K_v : |x| \leq M\}$ is compact. By a standard diagonalization argument, we therefore obtain a subsequence f_{n_k} of functions whose Laurent series coefficients converge. This implies that there exists an analytic function $f \in \mathcal{A}(V)$ such that $f_{n_k} \rightarrow f$ in $\mathcal{A}(V)$, as desired. $\square$

COROLLARY 8.41. *Suppose $\varphi \in K_v(T)$ is a rational function, with K_v a finite extension of $\mathbb{Q}_p$. Let $x \in \mathbb{P}_{\text{Berk}}^1$, and suppose that there exists a neighborhood U of x in $\mathbb{P}_{\text{Berk}}^1$ and a Berkovich disk D such that $\varphi^{(n)}(U) \cap D = \emptyset$ for all n . Then x is in the Berkovich Fatou set of φ .*

PROOF. This follows from Theorem 8.40, Lemma 8.33, and Theorem 8.35. $\square$

REMARK 8.42. It would be interesting to know if Corollary 8.41 remains true without the hypothesis that φ is defined over a finite extension of $\mathbb{Q}_p$. It is worth noting, however, that the Weak Montel Theorem, which was used in our proof of Corollary 8.41, is false without the hypothesis that the functions f_n are all defined over a fixed finite extension K_v of $\mathbb{Q}_p$. For example, let $\{\alpha_n\}$ be a set of coset representatives for the different residue classes in $V = B(0, 1)$, and let $f_n = z + \alpha_n$. Then $f_n(B(0, 1)) = B(0, 1)$ for all n , but f_n clearly has no convergent subsequence in $\mathcal{A}(V) = \mathbb{C}_p\{\{T\}\}$, since $\|f_n - f_m\|_V = 1$ for $n \neq m$.

Similarly, one can verify using the definitions that the sequence $f_n = z + \alpha_n$ is not topologically equicontinuous at the Gauss point of $\mathbb{P}_{\text{Berk}}^1$, so if $f_n \in \mathbb{C}_p(T)$ is a sequence of meromorphic (or even rational) functions and $U \subseteq \mathbb{P}_{\text{Berk}}^1$ is an open set with $f_n(U) \cap D = \emptyset$ for all n and some Berkovich disk D , the family f_n need not be equicontinuous. This is in contrast with the analogous situation over the complex numbers.

With a little more effort, one can give a counterexample to the Weak Montel Theorem in which the f_n are iterates $\varphi^{(N_n)}$ of a polynomial in $\mathbb{C}_p[z]$. More precisely, let

$$\varphi(z) = \varphi_\lambda(z) = \frac{\lambda}{p}z^p + (1 - \frac{\lambda}{p})z^{p+1}.$$

Benedetto ([8]; see also [57]) has shown that there exist parameters $\lambda \in \mathbb{C}_p \setminus \overline{\mathbb{Q}_p}$ for which $\varphi(z)$ has a wandering disc $B = B(x, r)$. That is, the images $\varphi^{(N)}(B)$ are pairwise disjoint, and are not attracted to an attracting fixed point of φ .

Here $x \in \mathbb{C}_p$ is a point with $|x|_p < 1$, and $0 < r < p^{-1/(p-1)}$. The images $\varphi^{(N)}(B)$ remain inside the disc $B(0, 1)$ and outside the disc $B(0, p^{-1/(p-1)})$. There are a sequence of iterates $\varphi^{(N_n)}$ for which the points $x_n := \varphi^{(N_n)}(x)$ satisfy $|x_1|_p > |x_2|_p > \dots > p^{-1/(p-1)}$. Thus for all $i < j$, $|x_i - x_j|_p = \min(|x_i|_p, |x_j|_p) > p^{-1/(p-1)}$, so no subsequence of $\{x_n\}$ is a Cauchy sequence. This shows that the iterates $\{\varphi^{(N_n)}\}$ do not have a subsequence converging to a meromorphic function on $B(x, r)$.

8. The work of Rivera-Letelier

In this section, we summarize some results of Juan Rivera-Letelier from [54],[53],[55],[56], translated into the language of Berkovich spaces,

which will be used in our proof of the Strong Montel Theorem in the next section.

Rivera-Letelier starts with a rational function $\varphi \in \mathbb{C}_p(T)$ and studies the action of φ on $\mathbb{P}^1(\mathbb{C}_p)$ and on the “ p -adic hyperbolic space” $\mathbb{H}_p$, which turns out to be naturally isomorphic to $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$ (see [53, §9.3]), even though at first glance Rivera-Letelier’s definition of $\mathbb{H}_p$ looks rather different. Rivera-Letelier considers a distance function $d(S, S')$ on $\mathbb{H}_p$ which coincides with the path distance ρ as in §???. The resulting metric topology on $\mathbb{H}_p$ is strictly finer than the Berkovich subspace topology coming from the inclusion $\mathbb{H}_p \subset \mathbb{P}_{\text{Berk}}^1$.

One of the most striking aspects of Rivera-Letelier’s work is that one can deduce useful facts about how φ acts on $\mathbb{P}^1(\mathbb{C}_p)$ by studying its action on $\mathbb{H}_p$. (This is perhaps less surprising if one views both of these actions as special cases of the action of φ on $\mathbb{P}_{\text{Berk}}^1$.) In particular, fixed point theorems for the action of φ on $\mathbb{H}_p$ play a prominent role in Rivera-Letelier’s work. In the remainder of this section, we will summarize the results of Rivera-Letelier needed for our proof of the Strong Montel Theorem. There are many interesting results in [54],[53],[55],[56] which we will not touch upon here, for which we encourage the interested reader to consult the original papers.

To every $x \in \mathbb{H}_p$, Rivera-Letelier associates a family of *bouts* (literally: ends). In the context of Berkovich spaces, a bout $\mathcal{P}$ can be thought of as a direction emanating from x , and the collection of bouts at x can therefore be identified with the “tangent space” T_x at x . To each bout $\mathcal{P} \in T_x$, one can associate an open set $B_{\mathcal{P}} \subset \mathbb{P}^1(\mathbb{C}_p)$ in a natural way: $B_{\mathcal{P}}$ is the set of all $y \in \mathbb{P}^1(\mathbb{C}_p)$ such that the path (x, y) lies in the equivalence class $\mathcal{P}$. If x is of Type II (according to Berkovich’s classification), then $B_{\mathcal{P}}$ is an open ball with radius in $|\mathbb{C}_p^*|$; if x is of Type III, then $B_{\mathcal{P}}$ is an irrational ball; and if x is of Type IV, then $B_{\mathcal{P}} = \mathbb{P}^1(\mathbb{C}_p) \setminus \{x\}$. The tangent space T_x itself consists of infinitely many bouts, two bouts, or one bout, according to whether x is Type II, III, or IV, respectively. In the case where x is Type II, there is an important interpretation of the space T_x in terms of residue classes. Let $M \in \text{PGL}(2, \mathbb{C}_p)$ be a Möbius transformation taking the Gauss point ζ_{Gauss} to x . Such a transformation always exists because $\text{PGL}(2, \mathbb{C}_p)$ acts transitively on the subspace $\mathbb{H}_p^{\mathbb{Q}}$ of Type II points. Then M induces in a natural way a bijection between $T_{\zeta_{\text{Gauss}}}$ and T_x . But the tangent space $T_{\zeta_{\text{Gauss}}}$ is canonically isomorphic to $\mathbb{P}^1(\overline{\mathbb{F}}_p)$, since the balls $B_{\mathcal{P}}$ for $\mathcal{P} \in T_x$ are precisely the different residue classes for the reduction map $\pi : \mathbb{P}^1(\mathbb{C}_p) \rightarrow \mathbb{P}^1(\overline{\mathbb{F}}_p)$. We therefore see that the tangent space T_x is (non-canonically) isomorphic to $\mathbb{P}^1(\overline{\mathbb{F}}_p)$.

More generally, the action of any rational function φ on $\mathbb{H}_p$ induces an action on tangent spaces, so that if $\varphi(x) = y$, there is an induced mapping $\varphi_* : T_x \rightarrow T_y$ taking bouts at x to bouts at y . The mapping φ_* , which is the “derivative” of φ , can be explicitly described when x is Type II using the above interpretation of T_x . Let $M \in \mathrm{PGL}(2, \mathbb{C}_p)$ be a Möbius transformation taking the Gauss point ζ_{Gauss} to x , and let M' be a Möbius transformation taking $y = \varphi(x)$ to ζ_{Gauss} . Let $\psi = M' \circ \varphi \circ M$, so that $\psi(\zeta_{\mathrm{Gauss}}) = \zeta_{\mathrm{Gauss}}$. Then ψ has a nontrivial reduction $\tilde{\psi}$ modulo the maximal ideal of $\mathbb{C}_p$, and we can thus view $\tilde{\psi}$ as a nonconstant map from $\mathbb{P}^1(\overline{\mathbb{F}}_p)$ to itself. The key fact is then that there is a finite set $\Sigma \subseteq T_x$ such that $\varphi(B_{\mathcal{P}}) = B_{\varphi_*(\mathcal{P})}$ for all $\mathcal{P} \in T_x \setminus \Sigma$ (see [53, Proposition 3.4] for a proof).

The *local degree* $\deg_x(\varphi)$ of φ at a Type II point x is defined to be the degree of the rational function $\tilde{\psi} \in \overline{\mathbb{F}}_p(T)$. The number $\deg_x(\varphi)$ is independent of the choice of M and M' and satisfies $1 \leq \deg_x(\varphi) \leq \deg(\varphi)$. Rivera-Letelier also defines the local degree of φ at an arbitrary point of $\mathbb{H}_p$, but we don’t need this. It is proved in [53, Corollary 3.5, Proposition 3.10] that $\deg_x(\varphi) = 1$ if and only if φ is a local isometry at x , where “local” and “isometry” both refer to the path distance $d(S, S')$ on $\mathbb{H}_p$, which is invariant under the action of Möbius transformations. By contrast, when $\deg_x(\varphi) > 1$, there is at least one tangent direction along which φ increases distances locally near x by a factor $m \geq 2$. (In all cases, φ is a local isometry along all but finitely many bouts.)

We now discuss fixed points in $\mathbb{H}_p$. If φ has a fixed point in $\mathbb{H}_p$, then by [55, Lemma 8.4], φ must have a Type II (or *rational*) fixed point. There are two types of rational fixed points in $\mathbb{H}_p$, which Rivera-Letelier calls *indifferent* and *repelling*, respectively. A fixed point $x \in \mathbb{H}_p^{\mathbb{Q}}$ is called *indifferent* if $\deg_x(\varphi) = 1$, and *repelling* if $\deg_x(\varphi) > 1$. More generally, a periodic point $x \in \mathbb{H}_p^{\mathbb{Q}}$ with minimal period $n \geq 1$ is a fixed point of $\varphi^{(n)}$, and is classified as indifferent or repelling accordingly.

The following is an immediate consequence of the “Fixed Point Property” proved in [55]:

THEOREM 8.43 (Fixed Point Theorem). *A rational map φ of degree at least 2 must have either a rational fixed point in $\mathbb{H}_p$ or an attracting fixed point in $\mathbb{P}^1(\mathbb{C}_p)$ (or both).*

As a consequence of his study of the analytic components of the “domain of quasiperiodicity”, and in particular, the fact (see [54, Theorem 3]) that every such component must be a connected open affinoid, Rivera-Letelier also proves (see [53, Theorem C]):

THEOREM 8.44. *If φ has a periodic point in $\mathbb{H}_p$, then it has a repelling periodic point in $\mathbb{H}_p^{\mathbb{Q}}$.*

Combining these two theorems, we deduce:

COROLLARY 8.45. *φ must have either a repelling periodic point in $\mathbb{H}_p^{\mathbb{Q}}$ or an attracting periodic point in $\mathbb{P}^1(\mathbb{C}_p)$.*

A related result (Theorem B of [53]) is:

THEOREM 8.46. *φ always has a repelling fixed point, either in $\mathbb{P}^1(\mathbb{C}_p)$ or in $\mathbb{H}_p^{\mathbb{Q}}$.*

Since one can show that repelling fixed points (in $\mathbb{P}^1(\mathbb{C}_p)$ or in $\mathbb{H}_p^{\mathbb{Q}}$) are always in the Berkovich Julia set J_φ , this theorem can be used to give another proof of the fact that the Berkovich Julia set is non-empty (Corollary 8.23; see also Remark 8.25).

We also note the following property of rational fixed points, which follows from the description of the derivative mapping φ_* in terms of the reduction of φ ([55], Lemma 4.1):

PROPOSITION 8.47. *If $x \in \mathbb{H}_p^{\mathbb{Q}}$ is fixed by φ , then every bout $\mathcal{P} \in T_x$ is preperiodic under φ_* , and there are periodic bouts of arbitrarily large prime order.*

Note that the periodicity depends upon the fact that the residue field $\overline{\mathbb{F}}_p$ of $\mathbb{C}_p$ is a direct limit of finite fields. The proposition would be false if we replaced $\mathbb{C}_p$ by an arbitrary complete and algebraically closed field containing $\mathbb{Q}_p$.

A point $x \in \mathbb{P}_{\text{Berk}}^1$ is *grand orbit finite* (or *exceptional*) if the set of all forward and backward iterates of x is finite. In other words, if we define the grand orbit of x to be

$$GO(x) = \{y \in \mathbb{P}_{\text{Berk}}^1 : \varphi^{(m)}(x) = \varphi^{(n)}(y) \text{ for some } m, n \geq 0\},$$

then x is grand orbit finite if and only if $GO(x)$ is finite. By analogy with the situation over $\mathbb{C}$, we define the *exceptional locus* E_φ of φ to be the set of all grand orbit finite points of $\mathbb{P}_{\text{Berk}}^1$.

LEMMA 8.48.

- A) *The exceptional locus E_φ of φ contains at most 3 points. At most 2 exceptional points are in $\mathbb{P}^1(\mathbb{C}_p)$, and at most 1 is in $\mathbb{H}_p = \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{P}^1(\mathbb{C}_p)$.*
- B) *$E_\varphi(\mathbb{C}_p) \subset F_\varphi(\mathbb{C}_p)$, and there is an exceptional point in $\mathbb{H}_p$ (necessarily of Type II) if and only if φ has simple reduction (i.e., good reduction in some coordinate).*

PROOF. Since $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is surjective by §???, φ must map each grand orbit onto itself. So φ maps each finite grand orbit bijectively onto itself, and thus each finite grand orbit is a periodic cycle. The fact that there are at most 2 exceptional points in $\mathbb{P}^1(\mathbb{C}_p)$ then follows from the Riemann-Hurwitz formula, since φ has at most $2d - 2$ critical points (counting multiplicity), and each backward invariant point must be a critical point with multiplicity $d - 1$. This argument also shows that every finite grand orbit is superattracting, and hence contained in the Fatou set $F_\varphi(\mathbb{C}_p)$.

The fact that there is at most 1 exceptional point in $\mathbb{H}_p$, together with the remaining statements in A) and B), are consequences of [53, Theorem 3]. $\square$

Finally, we discuss two useful results from [55], both of which say that, under suitable hypotheses, the union of all forward images under φ of an open set $U \subset \mathbb{P}^1(\mathbb{C}_p)$ must be quite large. The first result follows from Proposition 5.1 of [55], and the second from Proposition 6.1 of *loc. cit.* (note that one of the X 's in the statement of [55, Prop. 6.1] should be a C).

PROPOSITION 8.49. *Let $U \subset \mathbb{P}_{\text{Berk}}^1$ be a connected open affinoid containing a repelling periodic point of $\mathbb{H}_p^\mathbb{Q}$, and let $U(\mathbb{C}_p) = U \cap \mathbb{P}^1(\mathbb{C}_p)$ be the corresponding open affinoid subset of $\mathbb{P}^1(\mathbb{C}_p)$. Then*

$$\bigcup_{n \geq 0} \varphi^{(n)}(U(\mathbb{C}_p)) \supseteq \mathbb{P}^1(\mathbb{C}_p) \setminus E_\varphi(\mathbb{C}_p).$$

PROPOSITION 8.50. *Let $A_0 \subset \mathbb{P}^1(\mathbb{C}_p)$ be the immediate basin of attraction for some attracting periodic orbit of φ , and let $X \subset \mathbb{P}^1(\mathbb{C}_p)$ be a connected open affinoid which intersects both C and its complement. Then*

$$\bigcup_{n \geq 0} \varphi^{(n)}(X) \supseteq \mathbb{P}^1(\mathbb{C}_p) \setminus E_\varphi(\mathbb{C}_p).$$

9. The Strong Montel Theorem

We now prove a version of the Strong Montel theorem for the Berkovich projective line. In doing so, we make use of the results of Rivera-Letelier discussed in the previous section.

Let $\varphi \in \mathbb{C}_p(T)$ be a rational function of degree at least 2. Define the *Rivera-Letelier Fatou set* F_φ^{RL} to be

$$F_\varphi^{\text{RL}} = \{x \in \mathbb{P}_{\text{Berk}}^1 : \text{there exists a neighborhood } V \text{ of } x \\ \text{such that } \bigcup_{n=1}^\infty \varphi^{(n)}(V) \text{ omits at least 3 points of } \mathbb{P}^1(\mathbb{C}_p)\}.$$

We also let $F_\varphi^{\text{RL}}(\mathbb{C}_p) = F_\varphi^{\text{RL}} \cap \mathbb{P}^1(\mathbb{C}_p)$. Note that $F_\varphi \subseteq F_\varphi^{\text{RL}}$, as follows easily from the definitions.

The Strong Montel theorem over $\mathbb{C}$ implies that the complex analogue of F_φ^{RL} coincides with the Fatou set of φ (defined in terms of equicontinuity/normality). Under the assumption that φ is defined over a finite extension K_v of $\mathbb{Q}_p$, we will show later on that $F_\varphi^{\text{RL}} = F_\varphi$ in $\mathbb{P}_{\text{Berk}}^1$ as well.

Using the results of Section 8 and the Weak Montel Theorem, we now prove:

THEOREM 8.51 (Strong Montel Theorem).

Suppose $\varphi \in K_v(T)$ is a rational function of degree at least 2, with K_v a finite extension of $\mathbb{Q}_p$. Let $x \in \mathbb{P}_{\text{Berk}}^1$, and suppose that there exists a neighborhood V of x in $\mathbb{P}_{\text{Berk}}^1$ such that $\bigcup_{n=1}^\infty \varphi^{(n)}(V)$ omits at least three points of $\mathbb{P}^1(\mathbb{C}_p)$. Then

- A) each subsequence $\{\varphi^{(n_k)}\}$ has a further subsequence converging to a K_v -rational meromorphic function on V ;*
- B) x is in the Berkovich Fatou set of φ .*

PROOF. Since the connected affinoid sets form a fundamental system of compact neighborhoods for the Berkovich topology, we can assume without loss of generality that V is a connected affinoid. Also, by Corollary 8.41, it is enough to prove that every subsequence of $\{\varphi^{(n_k)}\}$ has a further subsequence $\{\varphi^{(n'_k)}\}$ such that $W := \bigcup_{n=1}^\infty \varphi^{(n'_k)}(V(\mathbb{C}_p))$ omits a disk of positive radius in $\mathbb{P}^1(\mathbb{C}_p)$. (In more concrete terms, we are saying that it suffices to prove the implication “Weak Montel implies Strong Montel”.) Finally, note that since V is a connected affinoid, V is contained in a connected component U of F_φ^{RL} and $V(\mathbb{C}_p)$ is contained in some analytic component X of $F_\varphi^{\text{RL}}(\mathbb{C}_p)$. With these preliminary observations made, the proof now breaks into two cases.

Suppose first that there exists a repelling periodic point $z \in \mathbb{H}_p^\mathbb{Q}$ of period m . Then that z is a fixed point of $\varphi^{(m)}$. By Proposition 8.47, there exist periodic bouts $\mathcal{P}, \mathcal{P}' \in T_x \setminus \Sigma$ whose orbits under $\varphi_*^{(m)}$ are disjoint. If $B_{\mathcal{P}}, B_{\mathcal{P}'}$ are the corresponding open balls, then since $\mathcal{P}, \mathcal{P}' \notin \Sigma$, it follows that $B_{\mathcal{P}}$ and $B_{\mathcal{P}'}$ are periodic under φ with *disjoint* orbits. We claim that the corresponding Berkovich open balls $\mathbb{B}_{\mathcal{P}}$ and $\mathbb{B}_{\mathcal{P}'}$ are connected components of F_φ^{RL} . Indeed, let $\mathbb{B}$ be one of these balls; then $\mathbb{B}$ is contained in F_φ^{RL} by periodicity, and no strictly larger open set can be contained in F_φ^{RL} because the unique boundary point z of $\mathbb{B}$ is not contained in F_φ^{RL} by Proposition 8.49. This proves the claim.

Let C, C' be the orbits of $\mathbb{B}, \mathbb{B}'$ under φ ; each is a finite disjoint union of Berkovich open balls. Now consider the component U of F_φ^{RL} containing V . Then exactly one of the following is true:

- (i) $\varphi^{(n)}(U) \in C$ for some $n \geq 1$;
- (ii) $\varphi^{(n)}(U) \in C'$ for some $n \geq 1$;
- (iii) $\varphi^{(n)}(U) \notin C \cup C'$ for any $n \geq 1$.

In all three cases, we see that $\cup_{n \geq 1} \varphi^{(n)}(U)$ omits at least one of the open sets C or C' , which proves the result under the assumption that there is a repelling periodic point $z \in \mathbb{H}_p^{\mathbb{Q}}$.

Now suppose that there does not exist a repelling periodic point $z \in \mathbb{H}_p^{\mathbb{Q}}$. Then by Corollary 8.45, there is an attracting fixed point $z_0 \in \mathbb{P}^1(\mathbb{C}_p)$. Let $A_0 \subset \mathbb{P}^1(\mathbb{C}_p)$ be the immediate attracting basin of z_0 , and let X be the analytic component of $F_{\varphi}^{\text{RL}}(\mathbb{C}_p)$ containing x . Since φ acts on the set of analytic components of $F_{\varphi}^{\text{RL}}(\mathbb{C}_p)$, exactly one of the following alternatives holds:

- (i) $\varphi^{(n)}(X) = A_0$ for some $n \geq 1$;
- (ii) $\varphi^{(n)}(X) \cap A_0 = \emptyset$ for all $n \geq 1$.

In Case (ii), we are clearly done since A_0 is open in $\mathbb{P}^1(\mathbb{C}_p)$. Similarly, in Case (i) we find upon passing to a subsequence that there exists a small open disk D around z_0 such that $\cup_k \varphi^{(n_k)}(X) \subseteq D$ (see [54, Corollary 4.6]). But then $\cup_k \varphi^{(n_k)}(X)$ omits all of $\mathbb{P}^1(\mathbb{C}_p) \setminus D$, which has nonempty interior, and we are done. $\square$

We note that Theorem 8.51 implies that the Berkovich and Rivera-Letelier Fatou sets coincide:

COROLLARY 8.52. *Let $\varphi \in K_v(T)$ be a rational function of degree at least 2, with K_v a finite extension of $\mathbb{Q}_p$. Then $F_{\varphi} = F_{\varphi}^{\text{RL}}$.*

PROOF. Theorem 8.51 implies that $F_{\varphi} \supseteq F_{\varphi}^{\text{RL}}$. The other containment follows easily from the definitions. $\square$

Theorem 8.51 should be compared with the following result, which is the Berkovich space interpretation of a theorem of Hsia [37]:

THEOREM 8.53 (Hsia's Theorem). *Let $D \subset \mathbb{P}_{\text{Berk}}^1$ be a closed disk, and let $\mathcal{F}$ be a family of meromorphic functions on D . Suppose that $\bigcup_{f \in \mathcal{F}} f(D)$ omits two distinct points of $\mathbb{P}^1(\mathbb{C}_p)$. Then $\mathcal{F}$ is equicontinuous at all $x \in D \cap \mathbb{P}^1(\mathbb{C}_p)$.*

There is an even stronger version of the Strong Montel theorem when a point of $\mathbb{H}_p$ is omitted:

THEOREM 8.54. *Suppose $\varphi \in K_v(T)$ is a rational function of degree at least 2, with K_v a finite extension of $\mathbb{Q}_p$. Let $x \in \mathbb{P}_{\text{Berk}}^1$, and suppose that there exists a connected affinoid neighborhood V of x in $\mathbb{P}_{\text{Berk}}^1$ such that $\cup_{n=1}^{\infty} \varphi^{(n)}(V)$ omits at least one point of $\mathbb{H}_p$. Then $x \in F_{\varphi}$.*

PROOF. Let y denote the omitted point of $\mathbb{H}_p$. If y is of Type II or III, then the complement $\mathbb{P}_{\text{Berk}}^1 \setminus \{y\}$ contains at least 2 connected components, each of which is open in $\mathbb{P}_{\text{Berk}}^1$. Since each $\varphi^{(n)}(V)$ is connected, by passing to a subsequence we find that $\cup \varphi^{(n_k)}(V)$ omits an entire open set in $\mathbb{P}_{\text{Berk}}^1$. Thus x is in F_φ by the Weak Montel Theorem.

Finally, suppose that y is of Type IV. We will say that a closed affinoid domain $W \subset \mathbb{P}_{\text{Berk}}^1$ is K_v -rational if after a K_v -rational change of coordinates, it can be written in the form $W = \mathbb{P}_{\text{Berk}}^1 \setminus \bigcup_{i=1}^m \mathbb{B}(a_i, r_i)^-$ where each $a_i \in K_v$ and each r_i is in the value group of $K_v^\times$. After enlarging K_v if necessary, we can assume that V is K_v -rational. It is not hard to see that for any nonconstant $f \in K_v(T)$ and any K_v -rational affinoid W , $f(W)$ is either a K_v -rational closed affinoid or all of $\mathbb{P}_{\text{Berk}}^1$. In particular, since $y \notin \varphi^{(n)}(V)$ for all n , it follows that each $\varphi^{(n)}(V)$ is a K_v -rational closed affinoid.

Recall that since y is of type IV, it corresponds to a sequence of nested discs $B(a_1, r_1) \supset B(a_2, r_2) \supset \cdots$ in $\mathbb{C}_p$ with empty intersection, such that $r = \lim_{i \rightarrow \infty} r_i > 0$. It is easy to see that if an open Berkovich disc $\mathbb{B}(a, s)^-$ contains y , then $s > r$. Let s_0 be the smallest element of the value group of $K_v^\times$ with $s_0 > r$. There is a minimal disc $\mathbb{B}(b_0, s_0)^-$ which contains y . Similarly a codisc $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{B}(a, s)$ can contain y only if $s < r$; there is a maximal s_1 in the value group of $K_v^\times$ with $s_1 < r$, and there is a minimal codisc such that $y \in \mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{B}(b_1, s_1)$.

We can decompose each $\varphi^{(n)}(V)$ as $\mathbb{P}_{\text{Berk}}^1 \setminus \bigcup_{i=1}^{m_n} \mathbb{B}(a_i^{(n)}, r_i^{(n)})^-$ or as $\mathbb{B}(a_0^{(n)}, r_0^{(n)}) \setminus \bigcup_{i=1}^{m_n} \mathbb{B}(a_i^{(n)}, r_i^{(n)})^-$ for certain $a_i^{(n)} \in K_v$ and $r_i^{(n)}$ in the value group of $K_v^\times$. Since $y \notin \varphi^{(n)}(V)$, it must belong to $\mathbb{B}(a_i^{(n)}, r_i^{(n)})^-$ for some i , or to $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{B}(a_0^{(n)}, r_0^{(n)})$. By the discussion above, $\varphi^{(n)}(V)$ must omit either $\mathbb{B}(b_0, s_0)^-$ or $\mathbb{P}_{\text{Berk}}^1 \setminus \mathbb{B}(b_1, s_1)$. Upon passing to a subsequence if necessary we can assume that all the $\varphi^{(n_k)}(V)$ omit the same open set. We can then finish the argument exactly as before. $\square$

As an immediate corollary of Theorems 8.51 and 8.54, we obtain:

COROLLARY 8.55. *Suppose $\varphi \in K_v(T)$ is a rational function of degree at least 2, with K_v a finite extension of $\mathbb{Q}_p$. Let $x \in \mathbb{P}_{\text{Berk}}^1$, and suppose that there exists a connected affinoid neighborhood V of x in $\mathbb{P}_{\text{Berk}}^1$ such that $\cup_{n=1}^\infty \varphi^{(n)}(V)$ omits at least three points of $\mathbb{P}_{\text{Berk}}^1$. Then $x \in F_\varphi$.*

This result is sharp, since if V is any connected affinoid neighborhood of the Gauss point of $\mathbb{P}_{\text{Berk}}^1$ not containing 0 and ∞ and

$\varphi(T) = T^2$, then $\cup_{n=1}^{\infty} \varphi^{(n)}(V)$ omits the two points 0 and ∞ , but $V \not\subseteq F_{\varphi}$, see Example 8.67 below.

10. Consequences of the Strong Montel Theorem

As in complex dynamics, one can obtain a number of structural results concerning the Fatou and Julia sets using the strong version of Montel's Theorem. We describe some of those results in this section, closely following the exposition of [47, Chapter 4].

Throughout this section, φ denotes a rational map of degree $d \geq 2$ defined over a finite extension K_v of $\mathbb{Q}_p$.

We can use Lemma 8.48 to characterize maps with simple reduction in terms of their Julia sets (recall that φ has simple reduction if after a change of coordinates, it has good reduction.)

LEMMA 8.56. *The Julia set J_{φ} consists of a single point (necessarily a Type II point of $\mathbb{H}_p$) if and only if φ has simple reduction.*

PROOF. If $J_{\varphi} = \{\zeta\}$ consists of a single point, then the result follows from Lemma 8.48, since ζ is exceptional by the complete invariance of the Julia set. On the other hand, suppose φ has simple reduction. Conjugating by a Mobius transformation (i.e., changing coordinates), we may assume without loss of generality that φ has good reduction. By [53, Section 4.1], this is equivalent to the statement that φ fixes the Gauss point $\zeta_{\text{Gauss}} \in \mathbb{H}_p$. Let $U = \mathbb{P}_{\text{Berk}}^1 \setminus \{\zeta_{\text{Gauss}}\}$. Then U is a disjoint union of Berkovich open disks D_{α} (residue classes) parametrized by the points $\alpha \in \mathbb{P}^1(\overline{\mathbb{F}}_p)$. Furthermore, for each $n \geq 1$, $\varphi^{(n)}$ acts on the set D_{α} via its reduction acting on $\mathbb{P}^1(\overline{\mathbb{F}}_p)$ (see Proposition 4.6 of [54]). In particular, for each $\alpha \in \mathbb{P}^1(\overline{\mathbb{F}}_p)$, the union $\cup_{n=1}^{\infty} \varphi^{(n)}(D_{\alpha})$ is a *finite* disjoint union of residue classes. In particular, $D_{\alpha} \subseteq F_{\varphi}$ by the Weak Montel theorem. Hence $U \subseteq F_{\varphi}$, which together with Corollary 8.23 implies that $J_{\varphi} = \{\zeta\}$. $\square$

The following result is analogous to Theorem 4.7 of [47].

THEOREM 8.57 (Transitivity Theorem). *Let $x \in J_{\varphi}$ and let N be an open neighborhood of x . Let U be the union of the forward images $\varphi^{(n)}(N)$ for $n \geq 1$. Then:*

- A) U contains all of $\mathbb{H}_p$, and all but at most two points of $\mathbb{P}^1(\mathbb{C}_p)$.
- B) U contains all of J_{φ} .
- C) If N is sufficiently small, then $U = \mathbb{P}_{\text{Berk}}^1 \setminus E_{\varphi}(\mathbb{C}_p)$.

PROOF. A) The set $\mathbb{P}^1(\mathbb{C}_p) \setminus U(\mathbb{C}_p)$ can contain at most two points; otherwise, since $\varphi(U) \subseteq U$, it would follow from Theorem 8.51 that U is contained in the Fatou set, contradicting the fact that $x \in U \cap$

J_φ . The fact that $\mathbb{H}_p \setminus E_\varphi$ is contained in U follows similarly from Theorem 8.54. However, if $\mathbb{H}_p \cap E_\varphi$ is nonempty, then by Lemma 8.56, φ has simple reduction and $J_\varphi = \{P\}$ is a single point; thus $x = P \in N$ by assumption.

B) Since $\varphi(U) \subseteq U$, any preimage of a point $z \in \mathbb{P}^1(\mathbb{C}_p) \setminus U(\mathbb{C}_p)$ must also belong to the finite set $\mathbb{P}^1(\mathbb{C}_p) \setminus U(\mathbb{C}_p)$. It follows easily that z is grand orbit finite, and thus (by Lemma 8.48) belongs to the Fatou set of φ .

C) If $N \subseteq \mathbb{P}_{\text{Berk}}^1 \setminus E_\varphi$, it follows easily from the proofs of A) and B) that $U = \mathbb{P}_{\text{Berk}}^1 \setminus E_\varphi$. $\square$

The following corollary is stronger than what one obtains classically over $\mathbb{C}$, due to the absence in the non-archimedean setting of rational maps with empty Fatou set (e.g. Lattès maps).

COROLLARY 8.58. *The Berkovich Julia set J_φ always has empty interior.*

PROOF. If $J = J_\varphi$ has an interior point z_1 , then choosing a neighborhood $N \subset J$ of z_1 , the union U of forward images of N under φ is both contained in J (by the forward invariance of J) and everywhere dense in $\mathbb{P}_{\text{Berk}}^1$ (since $\mathbb{P}_{\text{Berk}}^1 \setminus U$ contains at most 3 points by the Transitivity Theorem). Since J is closed, this implies that $J = \mathbb{P}_{\text{Berk}}^1$, and thus that F_φ is empty, contradicting Proposition 8.20. $\square$

COROLLARY 8.59. *If $\mathbb{A} \subset \mathbb{P}_{\text{Berk}}^1$ is the basin of attraction for some periodic orbit, then the topological boundary $\partial\mathbb{A}$ of $\mathbb{A}$ is equal to the Julia set J_φ .*

PROOF. If N is any neighborhood of $x \in J$, then $\varphi^{(n)}(N) \cap \mathbb{A} \neq \emptyset$ for some $n \geq 1$ by Theorem 8.57. Thus N itself intersects $\mathbb{A}$, which shows that $J \subseteq \overline{\mathbb{A}}$. Since $\mathbb{A}$ is contained in the Fatou set of φ by Lemma 8.37, it follows that $J \subseteq \partial\mathbb{A}$. On the other hand, it is easily deduced from the definitions that $\partial\mathbb{A} \subseteq J$. $\square$

COROLLARY 8.60. *The closure of the backward orbit of any non-exceptional point of $\mathbb{P}_{\text{Berk}}^1$ contains J_φ . In particular, the backward orbit of any point in J_φ is dense in J_φ .*

PROOF. Let x be a non-exceptional point (which means that the backward orbit of x is infinite), and let x_1, x_2, x_3 be in the backward orbit $O^-(x)$ of x . If U is any open set which intersects J_φ , then since $\{\varphi^{(n)}\}$ is not normal on U , we must have $x_i \in \varphi^{(n)}(U)$ for some $n \geq 1$ and some $i \in \{1, 2, 3\}$ by Corollary 8.55. Thus $x \in \varphi^{(m)}(U)$ for some $m \geq 1$, which means that U intersects $O^-(x)$. $\square$

COROLLARY 8.61. *If φ does not have simple reduction (in which case J_φ consists of a single point in $\mathbb{H}_p$), then J_φ is perfect (and in particular is uncountable).*

PROOF. If φ has simple reduction, then by Lemma 8.56, J_φ consists of a single point, which is an exceptional point of Type II in $\mathbb{H}_p$.

Conversely, suppose that φ does not have simple reduction. Then by Lemma 8.48, we see that $E_\varphi \cap \mathbb{H}_p = \emptyset$. Furthermore, J_φ is infinite, since if it were finite it would consist of grand orbit finite points in $\mathbb{P}^1(\mathbb{C}_p)$, which are necessarily in the Fatou set by Lemma 8.48. So J_φ contains at least one limit point z_0 , and z_0 is non-exceptional. But then the iterated preimages of z_0 form a dense set of non-isolated points in J_φ , and thus J_φ is perfect (i.e., contains no isolated points). $\square$

COROLLARY 8.62. *The Julia set J_φ is either connected, or else has uncountably many connected components.*

PROOF. In the case where J_φ is perfect, the proof is identical to its classical counterpart (see [47, Corollary 4.12]). The only other possibility is that J_φ consists of a single point, in which case it is trivially connected. $\square$

COROLLARY 8.63. *If K is any closed subset of $\mathbb{P}_{\text{Berk}}^1$ containing at least three points which is completely invariant under φ , then K contains J_φ .*

PROOF. If U denotes the complement of K , then U is also completely invariant under ϕ . In particular, $\varphi^{(n)}(U) \subseteq U$ for all $n \geq 1$. Since U omits at least three points of $\mathbb{P}_{\text{Berk}}^1$, it follows from Corollary 8.55 that $U \subseteq F_\varphi$, and thus $J_\varphi \subseteq K$. $\square$

The last corollary allows us to characterize the support of the canonical measure μ_φ .

THEOREM 8.64. *The support of the canonical measure μ_φ is precisely J_φ .*

PROOF. Recall that $\text{supp}(\mu_\varphi) \subseteq J_\varphi$ by Theorem 8.21. To show the opposite inequality, it suffices by Corollary 8.63 to show that $\text{supp}(\mu_\varphi)$ is completely invariant under φ . This follows formally from the fact that $\varphi_*\mu_\varphi = \mu_\varphi$ by §???. Indeed, let $\mu = \mu_\varphi$, and let K be the complement of $\text{supp}(\mu)$. If $x \notin \text{supp}(\mu)$ (i.e. $x \in K$), then there exists an open neighborhood U of x such that $\mu(U) = 0$. By definition of φ_* , we have $\mu(\varphi^{-1}U) = (\varphi_*\mu)(U) = \mu(U) = 0$, so that $\varphi^{-1}(x) \subset \varphi^{-1}(U) \subseteq K$. It follows that K is backward invariant under

φ . But since $\varphi : \mathbb{P}_{\text{Berk}}^1 \rightarrow \mathbb{P}_{\text{Berk}}^1$ is surjective, every backward invariant set is completely invariant under φ . Thus K , and therefore its complement μ_φ , is completely invariant under φ . $\square$

11. Polynomial dynamics

In this section, we assume that $\varphi \in K[T]$ is a *polynomial* of degree $d \geq 2$ defined over a finite extension K of $\mathbb{Q}_p$.

DEFINITION 8.65. The *filled Julia set* K_φ of φ is defined to be

$$K_\varphi := \bigcup_{M>0} \{x \in \mathbb{P}_{\text{Berk}}^1 : |\varphi^{(n)}(T)|_x \leq M \ \forall n \geq 1\}.$$

It is easy to see that K_φ is actually the set of all $x \in \mathbb{P}_{\text{Berk}}^1$ such that the sequence $|\varphi^{(n)}(T)|_x$ stays bounded as $n \rightarrow \infty$. We may choose M so that $K_\varphi \subseteq \mathbb{B}(0, M)$; in particular, K_φ is a compact and bounded subset of $\mathbb{A}_{\text{Berk}}^1$. It is non-empty, because the preperiodic points of φ in $\mathbb{A}^1(\mathbb{C}_p)$ are contained in K_φ . It is not hard to see that there are always infinitely many preperiodic points of φ in $\mathbb{A}^1(\mathbb{C}_p)$, so that in fact K_φ is infinite. Also, it follows by continuity that K_φ is the closure of $K_\varphi(\mathbb{C}_p)$, where $K_\varphi(\mathbb{C}_p) := K_\varphi \cap \mathbb{P}^1(\mathbb{C}_p)$.

THEOREM 8.66. *The Julia set of φ is the topological boundary of the filled Julia set, i.e., $J_\varphi = \partial K_\varphi$.*

PROOF. This follows from Corollary 8.59, since in the present situation the complement of K_φ is the attracting basin for the (super)attracting fixed point ∞ . $\square$

EXAMPLE 8.67. Consider the polynomial $\varphi(T) = T^2$. The filled Julia set is

$$K_\varphi = \mathbb{B}(0, 1) = \{z \in \mathbb{A}_{\text{Berk}}^1 : |T|_z \leq 1\}.$$

The Julia set of φ is the boundary of K_φ , which consists of a single point, the Gauss point $\zeta_{\text{Gauss}} \in \mathbb{B}(0, 1) \subset \mathbb{P}_{\text{Berk}}^1$. The canonical measure μ_φ in this case is a point mass supported at ζ_{Gauss} .

The fact that $J_\varphi = \{\zeta_{\text{Gauss}}\}$ also follows from Lemma 8.56.

EXAMPLE 8.68. If $\varphi(T) = \frac{T^p - T}{p}$, then it is well-known that $K_\varphi(\mathbb{C}_p) = \mathbb{Z}_p \subset \mathbb{P}^1(\mathbb{C}_p)$, which is compact as a subset of both $\mathbb{P}^1(\mathbb{C}_p)$ and $\mathbb{P}_{\text{Berk}}^1$. Thus K_φ , which is the closure of $K_\varphi(\mathbb{C}_p)$, equals $\mathbb{Z}_p$ as well. Since $\mathbb{Z}_p$ has empty interior, it also follows that $J_\varphi = J_\varphi(\mathbb{C}_p) = \mathbb{Z}_p$. The canonical measure μ_φ in this case is the normalized Haar measure on $\mathbb{Z}_p$.

EXAMPLE 8.69. Let p be an odd prime, and choose $a \in \mathbb{C}_p$ such that $|a| = p^\epsilon > 1$, where $0 < \epsilon \leq \frac{p}{2p+2}$. If $\varphi(T) = T^2(T - a)^p$, then by Example 3 of [7], φ does not have simple reduction but $J_\varphi(\mathbb{C}_p)$ is nonetheless empty. From Lemma 8.56 and Corollary 8.61, we conclude that J_φ is an uncountable subset of $\mathbb{H}_p$.

Note that for a polynomial $\varphi(T) \in \mathbb{C}_p[T]$ of degree at most $p + 1$, Benedetto has shown that $J_\varphi(\mathbb{C}_p)$ is empty if and only if φ has simple reduction (Proposition 4.9 of [7]).

12. A more general Strong Montel theorem

We give a more general version of the Strong Montel theorem (Theorem 8.51) due to J. Rivera-Letelier, which he showed to us shortly after we sent him a draft of this manuscript. We will give a sketch of the proof, and refer to [58] for further details. We then discuss some consequences of Rivera-Letelier's result.

Rivera-Letelier's Strong Montel theorem is the following result:

THEOREM 8.70 (Strong Montel Theorem). *Suppose $\varphi \in \mathbb{C}_p(T)$ is a rational function of degree at least 2. Let $x \in \mathbb{P}_{\text{Berk}}^1$, and suppose that there exists a neighborhood V of x in $\mathbb{P}_{\text{Berk}}^1$ such that $\bigcup_{n=1}^{\infty} \varphi^{(n)}(V)$ omits at least three points of $\mathbb{P}^1(\mathbb{C}_p)$, or at least one point of $\mathbb{H}_p$. Then x is in the Berkovich Fatou set of φ .*

The proof is based on Rivera-Letelier's classification of the periodic components of F_φ^{RL} , together with a useful normality lemma for wandering Fatou components.

In order to give the idea of the proof of Theorem 8.70, we briefly recall Rivera-Letelier's classification theorem. One defines the following types of components for a rational map $\varphi \in \mathbb{C}_p(T)$ of degree $d \geq 2$:

Immediate basins of attraction: If $z_0 \in \mathbb{P}^1(\mathbb{C}_p)$ is an attracting periodic point of period n , its *basin of attraction* is the set

$$\hat{A}_\varphi(z_0) = \{z \in \mathbb{P}_{\text{Berk}}^1 : \varphi^{(nk)}(z) \rightarrow z_0 \text{ when } k \rightarrow \infty\}.$$

One easily shows that $\hat{A}_\varphi(z_0)$ is open and invariant under $\varphi^{(n)}$. The connected component of $\hat{A}_\varphi(z_0)$ containing z_0 is called the *immediate basin of attraction* of z_0 ; it is also open and invariant under $\varphi^{(n)}$. Rivera-Letelier has shown that $\hat{A}_\varphi(z_0)$ is contained in F_φ (and hence in F_φ^{RL}), and that the immediate basin of attraction of z_0 is a connected component of F_φ^{RL} .

Domain of quasi-periodicity: The *domain of quasi-periodicity* of φ , denoted $\hat{E}_\varphi$, is defined to be the interior of the set of points in $\mathbb{P}_{\text{Berk}}^1$

which are recurrent under φ . Recall that a point is called *recurrent* if it lies in the closure of its forward orbit. By definition, $\hat{E}_\varphi$ is open and invariant under φ , and is disjoint from each basin of attraction. Rivera-Letelier has shown that $\hat{E}_\varphi$ is contained in F_φ (and hence in F_φ^{RL}), and that each connected component of $\hat{E}_\varphi$ is in fact a connected component of F_φ^{RL} .

Rivera-Letelier's classification of periodic Fatou components says that these are the only possibilities:

THEOREM 8.71 (Rivera-Letelier). *Every periodic connected component of F_φ^{RL} is either an immediate basin of attraction or a connected component of the domain of quasi-periodicity of φ . Moreover, immediate basins of attraction and connected components of the domain of quasi-periodicity are contained in F_φ , i.e., the iterates of φ form a normal family on such domains.*

REMARK 8.72. The proof of Theorem 8.71 uses the fact that the residue field of $\mathbb{C}_p$ is a union of finite fields, and in particular the proof does not extend to rational functions defined over an arbitrary complete and algebraically closed non-archimedean field.

A component of F_φ^{RL} which is not preperiodic is called a *wandering component*. The following lemma allows one to prove normality of the iterates of φ on wandering Fatou components:

LEMMA 8.73 (Rivera-Letelier). *If $\mathcal{F}$ is a family of analytic maps defined on a connected open subset W of $\mathbb{P}_{\text{Berk}}^1$, and if the images of W under the elements of $\mathcal{F}$ are pairwise disjoint, then $\mathcal{F}$ is a normal family.*

PROOF. Fix $w_0 \in W$, and assume that $f_j(w_0) \rightarrow \zeta$ for some infinite sequence f_j in $\mathcal{F}$. We claim that f_j converges uniformly on W to the constant function $f(z) \equiv \zeta$. To see this, V be an affinoid neighborhood of ζ in $\mathbb{P}_{\text{Berk}}^1$. Then for $j \gg 0$, $f_j(W)$ must be contained in V , because $f_j(W) \cap V \neq \emptyset$, V has only finitely many boundary points in $\mathbb{P}_{\text{Berk}}^1$, and each boundary point is in $f_j(W)$ for at most one value of j by the hypothesis of pairwise disjointness. This implies the claim and proves that $\mathcal{F}$ is a normal family. $\square$

Using Lemma 8.73, one obtains the following corollary:

COROLLARY 8.74. *Every wandering component of F_φ^{RL} is contained in F_φ .*

Using Theorem 8.71 and Corollary 8.74, one proves Theorem 8.70 as follows:

PROOF. We know that $F_\varphi \subseteq F_\varphi^{\text{RL}}$, and for the opposite containment it suffices to show that every connected component of F_φ^{RL} is contained in F_φ . One easily reduces the case of a preperiodic component to the corresponding result for periodic components. The fact that $F_\varphi = F_\varphi^{\text{RL}}$ therefore follows from Theorem 8.71 and Corollary 8.74. One immediately deduces the Strong Montel theorem in the case where $\bigcup_{n=1}^\infty \varphi^{(n)}(V)$ omits at least three points of $\mathbb{P}^1(\mathbb{C}_p)$. When $\bigcup_{n=1}^\infty \varphi^{(n)}(V)$ omits at least one point of $\mathbb{H}_p$, one shows as in the proof of Theorem 8.54 that $\bigcup_{n=1}^\infty \varphi^{(n)}(V)$ must in fact omit an entire disk, so the result follows from the previous case. $\square$

COROLLARY 8.75. *Let $\varphi \in \mathbb{C}_p(T)$ be a rational map of degree at least 2, and let $U \subseteq \mathbb{P}_{\text{Berk}}^1$ be a connected open set. Then the following are equivalent:*

- (1) *The iterates $\varphi^{(n)}$ form a normal family on U .*
- (2) *The iterates $\varphi^{(n)}$ form an equicontinuous family on U .*
- (3) *Every $x \in U$ has a neighborhood $U' \subseteq U$ such that the union $\bigcup_{n=1}^\infty \varphi^{(n)}(U')$ of all forward iterates of U' under φ omits at least three points of $\mathbb{P}_{\text{Berk}}^1$.*

So one can characterize the Berkovich Fatou set of φ as the largest open set U on which any of the conditions (1)-(3) in Corollary 8.75 holds, and the Berkovich Julia set J_φ as the complement of the Berkovich Fatou set. Using Theorem 8.70, and imitating some classical arguments over $\mathbb{C}$ as in §10, Rivera-Letelier has proved the following result concerning the Berkovich Julia set:

THEOREM 8.76. *Let $\varphi \in \mathbb{C}_p(T)$ be a rational map of degree at least 2. Then the following are equal:*

- (1) *The Berkovich Julia set J_φ .*
- (2) *The closure of the set of repelling periodic points in $\mathbb{P}_{\text{Berk}}^1$ for φ .*
- (3) *The support of the canonical measure μ_φ .*

REMARK 8.77. In condition (2) of the theorem, we are including the repelling periodic points in $\mathbb{H}_p$. It is still an open problem to determine whether the closure in $\mathbb{P}^1(\mathbb{C}_p)$ of the repelling periodic points contained in $\mathbb{P}^1(\mathbb{C}_p)$ is dense in $J_\varphi(\mathbb{C}_p)$ when $J_\varphi(\mathbb{C}_p) \neq \emptyset$.

Using Theorem 8.70, it is easy to check that all of the results discussed in §10 and §11, which were stated for $\varphi \in K_v(T)$ with $[K_v : \mathbb{Q}_p]$ finite, hold more generally for any rational map $\varphi \in \mathbb{C}_p(T)$ of degree $d \geq 2$.

13. Notes and further references

APPENDIX A

Some results from analysis and topology

In this appendix, we recall some results from analysis and topology which are used in the text.

1. Convex functions

(See [52, §2.6]) Let $-\infty \leq a < b \leq \infty$. A function $\psi : (a, b) \rightarrow \mathbb{R}$ is called *convex* if for every $t_1, t_2 \in (a, b)$,

$$\psi((1 - \lambda)t_1 + \lambda t_2) \leq (1 - \lambda)\psi(t_1) + \lambda\psi(t_2)$$

for all $0 \leq \lambda \leq 1$.

A convex function on (a, b) is automatically continuous. Also, if $\psi \in \mathcal{C}^2(a, b)$, then ψ is convex if and only if $\psi'' \geq 0$ on (a, b) .

In the proof of Proposition C.13(C), we will use the following result:

LEMMA A.1. *Let $g : (a, b) \rightarrow \mathbb{R}$ be a function such that for each $t \in (a, b)$, both one-sided derivatives $g'_-(t) = \lim_{h \rightarrow 0^-} (g(t+h) - g(t))/h$ and $g'_+(t) = \lim_{h \rightarrow 0^+} (g(t+h) - g(t))/h$ exist. Suppose that*

A) $g'_-(x) \leq g'_+(x)$ for each $x \in (a, b)$, and

B) $g'_+(x) \leq g'_-(y)$ for each $x, y \in (a, b)$ with $x < y$.

Then $g(t)$ is convex on (a, b) .

PROOF. Note that $g(t)$ is continuous, since the left and right derivatives exist at each point. Fix $x, y \in (a, b)$ with $x < y$, and let

$$L_{x,y}(t) = \frac{g(y) - g(x)}{y - x} \cdot (t - x) + g(x)$$

be the line through $(x, g(x))$ and $(y, g(y))$. We claim that $g(t) \leq L_{x,y}(t)$ for all $x < t < y$. Suppose to the contrary that there is a point $z \in (x, y)$ where $g(z) > L_{x,y}(z)$. Put $\alpha = (g(z) - g(x))/(z - x)$, $\beta = (g(y) - g(z))/(y - z)$. Then $\alpha > \beta$.

We claim there are a point $u \in (x, z)$ where $g'_+(u) \geq \alpha$, and a point $v \in (z, y)$ where $g'_-(v) \leq \beta$. This contradicts $g'_+(u) \leq g'_-(v)$.

First consider the interval (x, z) . Let $L_{x,z}(t) = \alpha \cdot (t - x) + g(x)$ be the line through $(x, g(x))$ and $(z, g(z))$, and put $f(t) = g(t) - L_{x,z}(t)$. Then $f(x) = f(z) = 0$. We will now apply the argument in Rolle's theorem. Since $f(t)$ is continuous, there is a point $u \in (x, y)$ where f

achieves a maximum or a minimum. If u is a minimum, then $f'_-(u) \leq 0$ and $f'_+(u) \geq 0$, so $g'_-(u) \leq \alpha$ and $g'_+(u) \geq \alpha$. If u is a maximum, then $f'_-(u) \geq 0$ and $f'_+(u) \leq 0$, so $g'_-(u) \geq \alpha$ and $g'_+(u) \leq \alpha$. Since $g'_+(z) \geq g'_-(u)$ it must be that $g'_-(u) = g'_+(u) = \alpha$. In either case we have a point u where $g'_-(u) \leq \alpha$ and $g'_+(u) \geq \alpha$. This second inequality is the one we want.

Next consider the interval (z, y) . By the same reasoning as before, there is a point $v \in (z, y)$ where $g'_-(v) \leq \beta$ and $g'_+(v) \geq \beta$. This time it is the first inequality that we want. $\square$

2. Upper and lower semicontinuous functions

Let X be a topological space. We denote by $\mathcal{C}(X)$ the space of continuous functions $f : X \rightarrow \mathbb{R}$, and by $\mathcal{C}_c(X)$ the space of compactly supported continuous functions $f : X \rightarrow \mathbb{R}$.

Recall that a real-valued function $f : X \rightarrow [-\infty, \infty)$ is *upper semicontinuous* if for each $x_0 \in X$,

$$\limsup_{x \rightarrow x_0} f(x) \leq f(x_0) .$$

This is equivalent to requiring that $f^{-1}([-\infty, b))$ be open for each $b \in \mathbb{R}$. According to [19, §6.2, Proposition 3], an upper semicontinuous function is automatically *strongly upper semicontinuous*, in the sense that

$$\limsup_{x \rightarrow x_0} f(x) = f(x_0) .$$

Think about what happens if f is allowed to take the value $-\infty$. Bourbaki as stated doesn't allow this.

If X is *compact*, then according to [52, Theorem 2.1.2], an upper semicontinuous function f on X is bounded above and achieves its maximum value on X .

If $f : X \rightarrow [-\infty, \infty)$ is locally bounded above, we define the *upper semicontinuous regularization* f^* of f by

$$f^*(x) = \limsup_{y \rightarrow x} f(y) .$$

See [52, Definition 3.4.1]. The function f^* is upper semicontinuous and $f^* \geq f$ on X . Moreover, f^* is the *smallest* function with these properties, in the sense that if g is upper semicontinuous and $g \geq f$, then $g \geq f^*$. In particular, if f is upper semicontinuous then $f^* = f$.

Similarly, $f : X \rightarrow (-\infty, \infty]$ is *lower semicontinuous* if for each $x_0 \in X$,

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0) ,$$

or equivalently,

$$\liminf_{x \rightarrow x_0} f(x) = f(x_0) .$$

According to [31, Proposition 7.11(c)], we have the following result:

LEMMA A.2. *If $\mathcal{G}$ is a family of lower semicontinuous functions on X , and if $f(x) = \sup\{g(x) : g \in \mathcal{G}\}$, then f is lower semicontinuous.*

Similarly, if $\mathcal{G}$ is a family of upper semicontinuous functions on X , and if $f(x) = \inf\{g(x) : g \in \mathcal{G}\}$, then f is upper semicontinuous.

Furthermore, if X is a locally compact Hausdorff space and f is a nonnegative lower semicontinuous function on X , then by [31, Proposition 7.11(e)], we have

$$f(x) = \sup\{g(x) : g \in \mathcal{C}_c(X), 0 \leq g \leq f\}.$$

From this and the monotone convergence theorem, one deduces (see [31, Corollary 7.13]):

PROPOSITION A.3. *Let X be a locally compact Hausdorff space. If μ is a positive Radon measure and f is lower semicontinuous and bounded below on X , then*

$$(A.1) \quad \int f \, d\mu = \sup \left\{ \int g \, d\mu : g \in \mathcal{C}_c(X), 0 \leq g \leq f \right\}.$$

Note that if X is compact and Hausdorff, then every lower semicontinuous function on X is bounded below.

3. Nets

Our primary reference for this section is [31, §4.3], see also §29 of [50].

In the theory of metric spaces, convergent sequences play a major role. For example, a real-valued function f on a metric space is continuous if and only if $x \rightarrow x_n$ implies that $f(x_n) \rightarrow f(x)$. But for a general topological space, this is not true: even if X is compact Hausdorff, there can be discontinuous functions on X for which $f(x_n) \rightarrow f(x)$ whenever $x_n \rightarrow x$. The problem is that the indexing set $\mathbb{N}$ for a sequence is too small to capture the complexities of an arbitrary topological space. To rectify the situation, we introduce the notion of a net, in which the index set $\mathbb{N}$ for sequences is replaced by an arbitrary directed set.

A *directed set* is a set A equipped with a reflexive and transitive binary relation $\geq$, and such that for any $\alpha, \beta \in I$, there exists $\gamma \in A$ with $\gamma \geq \alpha$ and $\gamma \geq \beta$. For example, if X is a topological space, the set of all open neighborhoods of a point $x \in X$ is a directed set, with $V \geq U$ iff $V \subseteq U$.

A subset B of a directed set A is called *cofinal* in A if for each $\alpha \in A$, there exists $\beta \in B$ such that $\beta \geq \alpha$. More generally, a function $h : B \rightarrow A$ is called *cofinal* if for each $\alpha_0 \in A$, there exists $\beta_0 \in B$ such that $\beta \geq \beta_0$ implies $h(\beta) \geq \alpha_0$.

If X is a set, a *net* in X is a mapping $\alpha \mapsto x_\alpha$ from A to X . We write $\langle x_\alpha \rangle_{\alpha \in A}$, or just $\langle x_\alpha \rangle$, to denote a net indexed by A . It is important to note that a sequence is the same thing as a net indexed by $\mathbb{N}$.

Let X be a topological space, and let $x \in X$. A net $\langle x_\alpha \rangle$ in X *converges* to x (written $x_\alpha \rightarrow x$) if for every neighborhood U of x , there exists $\alpha_0 \in A$ such that $x \in U$ for all $\alpha \geq \alpha_0$.

A point $x \in X$ is a *cluster point* of $\langle x_\alpha \rangle$ if for every neighborhood U of x and every $\alpha \in A$, there exists $\beta \geq \alpha$ such that $x_\beta \in U$.

LEMMA A.4. *If $E \subseteq X$, then $x \in X$ belongs to the closure of E iff there is a net in E which converges to x .*

The following result (see [31, Proposition 4.19]) was alluded to above:

PROPOSITION A.5. *If X, Y are topological spaces and $f : X \rightarrow Y$, then f is continuous at $x \in X$ iff $f(x_\alpha) \rightarrow f(x)$ for every net $\langle x_\alpha \rangle$ converging to x .*

A topological space X is Hausdorff iff every net in X converges to at most one point.

A *subnet* of a net $\langle x_\alpha \rangle_{\alpha \in A}$ is a net $\langle y_\beta \rangle_{\beta \in B}$ together with a cofinal map $\beta \mapsto \alpha_\beta$ from B to A such that $y_\beta = x_{\alpha_\beta}$. Note that the map $\beta \mapsto \alpha_\beta$ is not required to be injective.

The following result gives an analogue of the Bolzano-Weierstrass theorem which holds in a general topological space.

- PROPOSITION A.6. (a) *If $\langle x_\alpha \rangle_{\alpha \in A}$ is a net in X , then $x \in X$ is a cluster point of $\langle x_\alpha \rangle$ iff $\langle x_\alpha \rangle$ has a subnet which converges to x .*
- (b) *A topological space X is compact iff every net in X has a convergent subnet (which happens iff every net in X has a cluster point).*

There is also a version of Dini's lemma for nets of functions:

LEMMA A.7. *Let X be a compact topological space, and suppose $\langle f_\alpha \rangle_{\alpha \in A}$ is a monotonically increasing or decreasing net in $\mathcal{C}(X)$. If f_α converges pointwise to a continuous function $f \in \mathcal{C}(X)$, then $f_\alpha \rightarrow f$ uniformly on X .*

PROOF. We may suppose that $\langle f_\alpha \rangle$ is monotonically increasing. Let $\varepsilon > 0$. For each $\alpha \in A$, set

$$U_\alpha = \{x \in X : f_\alpha(x) > f(x) - \varepsilon\}.$$

Then $U_\alpha \supseteq U_{\alpha'}$ if $\alpha \geq \alpha'$, and $\cup_\alpha U_\alpha = X$. By compactness, there is an $\alpha_0 \in A$ such that $U_{\alpha_0} = X$. Thus $f(x) - \varepsilon < f_{\alpha_0}(x) \leq f(x)$ for all $x \in X$ and all $\alpha \geq \beta$, so that $f_\alpha \rightarrow f$ uniformly on X . $\square$

4. Measure theoretic terminology

Let X be a topological space, and let Σ be a σ -algebra on X . A *signed measure* on (X, Σ) is a function $\nu : \Sigma \rightarrow [-\infty, \infty]$ assuming at most one of the values $\pm\infty$ such that $\nu(\emptyset) = 0$ and if E_j is a sequence of disjoint subsets in Σ , then

$$\nu(\cup_{j=1}^{\infty} E_j) = \sum_{j=1}^{\infty} \nu(E_j),$$

with $\sum_{j=1}^{\infty} \nu(E_j)$ converging absolutely when $\nu(\cup_{j=1}^{\infty} E_j)$ is finite.

A *positive measure* is one for which $\nu : \Sigma \rightarrow [0, \infty]$. A positive measure is *finite* if $\nu(X)$ is finite, and is a *probability measure* if $\nu(X) = 1$.

Every signed measure on X can be written as a difference of two positive measures, at least one of which is finite. More precisely, one has the *Jordan Decomposition Theorem*: If ν is a signed measure, there exist unique positive measures ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$. The latter condition means that there exists a partition of X into disjoint subsets $E^+, E^- \in \Sigma$ such that $\nu(E^+) = \nu(E^-) = 0$. The *total variation* $|\nu|$ of ν is the positive measure defined as $|\nu| = \nu^+ + \nu^-$.

A signed measure is *finite* if $|\nu|$ is finite.

If X is a topological space, then unless otherwise specified, a *measure* on X will mean a finite, signed Borel measure on X .

5. Radon measures

Our primary reference in this section is [31].

Let X be a locally compact Hausdorff space, and let μ be a positive Borel measure on X . If E is a Borel subset of X , then μ is called *outer regular* on E if

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}$$

and *inner regular* on E if

$$\mu(E) = \sup\{\mu(E') : E' \subseteq E, E' \text{ compact}\}.$$

We say that μ is *regular* if it is both inner and outer regular on all Borel sets. We say that μ is a *Radon measure* if it is finite on all compact sets, outer regular on all Borel sets, and inner regular on all open sets.

If X is *compact* (or more generally if X is a countable union of compact sets), then μ is Radon if and only if it is regular and finite on all compact sets.

More generally, a signed Borel measure is said to be a *signed Radon measure* iff its positive and negative parts are both Radon measures.

For many topological spaces which one encounters in practice, every Borel measure is Radon. This is the case, for example, for any locally compact Hausdorff space in which every open set is a countable union of compact sets [31, Theorem 7.8]. However, there are compact Hausdorff spaces which admit finite, non-Radon Borel measures (see [31, Exercise 7.15]).

Let $M(X)$ denote the normed vector space consisting of all signed Radon measures on X , together with the norm given by $\|\mu\| = |\mu|(X)$, and let $\mathcal{C}_0(X)$ (resp. $\mathcal{C}(X)$) denote the space of compactly supported (resp. all) continuous functions $f : X \rightarrow \mathbb{R}$.

THEOREM A.8 (Riesz Representation Theorem). *The map $\mu \mapsto \int_X f d\mu$ is an isometric isomorphism from $M(X)$ to $\mathcal{C}_0(X)^*$.*

COROLLARY A.9. *If X is a compact Hausdorff space, then $\mathcal{C}(X)^*$ is isometrically isomorphic to $M(X)$.*

Let μ_α be a net in $M(X)$. We say that μ_α *converges vaguely* to $\mu \in M(X)$ iff $\int f d\mu_\alpha \rightarrow \int f d\mu$ for all $f \in \mathcal{C}_0(X)$. The corresponding topology on $M(X)$ is called the *vague topology*. (This is the same thing as the weak* topology on $M(X)$, and by abuse of language is also referred to sometimes as the weak topology.)

Together with Alaoglu's theorem ([31, Theorem 5.18]), the Riesz Representation Theorem implies:

THEOREM A.10. *If X is a compact Hausdorff space, the unit ball $\{\mu \in M(X) : \|\mu\| \leq 1\}$ in $M(X)$ is compact in the vague topology.*

As a corollary, we have:

THEOREM A.11 (Prohorov's theorem for nets). *If X is a compact Hausdorff space, then every net μ_α in $M(X)$ has a vaguely convergent subnet.*

Of probability
measures?

Change weak convergence to vague convergence throughout, or vice-versa.

6. Portmanteau theorem

We will make use in the text of the following variant of the well-known “Portmanteau theorem” (Theorem 2.1 of [14]). We include a proof for lack of a suitable reference for the exact statement which we need.

THEOREM A.12. *Let X be a locally compact Hausdorff space, and let $\langle \mu_\alpha \rangle_{\alpha \in I}$ be a net of positive Radon measures, each of total mass $M > 0$, converging vaguely to a positive Radon measure μ on X . Then:*

- (a) $\mu_\alpha(A) \rightarrow \mu(A)$ for every Borel set $A \subseteq X$ with $\mu(\partial A) = 0$.
- (b) More generally, for every Borel set $A \subseteq X$ and every $\varepsilon > 0$, there exists $\alpha_0 \in I$ such that

$$(A.2) \quad \mu(A^\circ) \leq \mu_\alpha(A) \leq \mu(\bar{A}) + \varepsilon$$

for all $\alpha \geq \alpha_0$.

PROOF. It suffices to prove (b), since if (b) holds and $\mu(\partial A) = 0$, then for every $\varepsilon > 0$, there exists α_0 such that for $\alpha \geq \alpha_0$ we have:

$$\begin{aligned} \mu_\alpha(A) &\leq \mu_\alpha(\bar{A}) \leq \mu(\bar{A}) + \varepsilon = \mu(A) + \varepsilon \\ \mu_\alpha(A) &\geq \mu_\alpha(A^\circ) \geq \mu(A^\circ) - \varepsilon = \mu(A) - \varepsilon. \end{aligned}$$

Thus $|\mu_\alpha(A) - \mu(A)| \leq \varepsilon$ for $\alpha \geq \alpha_0$, i.e., $\mu_\alpha(A) \rightarrow \mu(A)$.

To prove (b), we recall from Proposition A.3 that if f is lower semicontinuous and bounded below on X , then for all α we have

$$\begin{aligned} \int f d\mu_\alpha &= \sup \left\{ \int g d\mu_\alpha : g \in \mathcal{C}_c(X), 0 \leq g \leq f \right\} \\ \int f d\mu &= \sup \left\{ \int g d\mu : g \in \mathcal{C}_c(X), 0 \leq g \leq f \right\}. \end{aligned}$$

So for any $\varepsilon > 0$ there exists $g \in \mathcal{C}_c(X)$ with $0 \leq g \leq f$ such that

$$\int g d\mu \geq \int f d\mu - \varepsilon/2.$$

By vague convergence, there exists $\alpha_0 \in I$ such that for $\alpha \geq \alpha_0$ we have

$$\left| \int g d\mu_\alpha - \int g d\mu \right| \leq \varepsilon/2.$$

In particular, since $0 \leq g \leq f$, for $\alpha \geq \alpha_0$ we have

$$(A.3) \quad \int f d\mu_\alpha \geq \int g d\mu_\alpha \geq \int f d\mu - \varepsilon.$$

If $U \subseteq X$ is open, then the characteristic function χ_U of U is lower semicontinuous, and applying (A.3) gives

$$\mu_\alpha(U) = \int \chi_U d\mu_\alpha \geq \int \chi_U d\mu - \varepsilon = \mu(U) - \varepsilon.$$

for all $\alpha \geq \alpha_0$.

If $B \subseteq X$ is closed, then since $\mu(X) = \lim \mu_\alpha(X) = M$, we have

$$\mu_\alpha(B) = \mu_\alpha(X) - \mu_\alpha(B^c) \leq M - \mu(B^c) + \varepsilon = \mu(B) + \varepsilon$$

for all $\alpha \geq \alpha_0$. □

7. Baire measures

A subset E of X is called a *Baire set* if it belongs to the σ -algebra generated by the sets

$$\{x \in X : f(x) \geq \alpha\}$$

for $\alpha \in \mathbb{R}$ and $f \in \mathcal{C}_c(X)$. (This σ -algebra is called the *Baire σ -algebra* on X .) On a locally compact Hausdorff space, every Baire set is also a Borel set, but there are compact Hausdorff spaces where the class of Borel sets is strictly larger than the class of Baire sets [59, §13.1, Problem 6]. A *Baire measure* on X is a measure defined on the σ -algebra of all Baire sets which is finite for each compact Baire set.

It is clear that one can replace $\{x \in X : f(x) \geq \alpha\}$ in the definition of the Baire σ -algebra by any or all of $\{x \in X : f(x) \geq \alpha\}$, $\{x \in X : f(x) > \alpha\}$, $\{x \in X : f(x) \leq \alpha\}$, and $\{x \in X : f(x) < \alpha\}$.

A useful fact about Baire sets is that if X is compact (or more generally if X is a countable union of compact sets), then every Baire measure on X is regular [59, Corollary 12]. In particular:

PROPOSITION A.13. *If X is a compact Hausdorff space for which the Baire and Borel σ -algebras coincide, then every finite Borel measure on X is a Radon measure.*

8. The one-point compactification

A reference for this section is §29 of [50].

A basic result in topology is that any locally compact Hausdorff space has a *minimal compactification*. More precisely, we have the following result.

THEOREM A.14. *Let X be a locally compact Hausdorff space. Then there exists a compact Hausdorff space Y containing X such that $Y \setminus X$ consists of a single point. Moreover, if Y' is another such space, then*

there is a homeomorphism between Y and Y' which is the identity on X .

The space Y given by the theorem is called the *one-point compactification* of X . As a set, we have $Y = X \cup \{\infty\}$ (disjoint union), where the symbol ∞ denotes an object which does not belong to X . The open subsets of Y are defined to be the open subsets of X , together with all sets of the form $Y \setminus C$, where $C \subset X$ is compact.

The converse of Theorem A.14 is also true, since an open subspace of a compact Hausdorff space is locally compact.

If X is already compact, then ∞ is an isolated point of Y . If X is not compact, then Y is dense in X .

For example, the one-point compactification of a locally compact field F is homeomorphic to $\mathbb{P}^1(F)$, the projective line over F . (In particular, the one-point compactification of $\mathbb{R}$ is homeomorphic to S^1 , and the one-point compactification of $\mathbb{C}$ is homeomorphic to S^2 .)

APPENDIX B

$\mathbb{R}$ -trees and Gromov hyperbolicity

In this appendix, we present a self-contained discussion of $\mathbb{R}$ -trees, including the connection with Gromov 0-hyperbolicity. Although most of the results here are known, our organization and presentation of this material is somewhat novel. We have been strongly influenced, however, by [51, §2.2], [20, Chapter III.H], and [27, Chapter 3].

1. Definitions

A *geodesic segment* in a metric space (X, d) is the image $|\alpha| = \alpha([a, b])$ of an isometric embedding $\alpha : [a, b] \rightarrow X$ of a (possibly degenerate) real interval $[a, b]$ into X . We say that α is a geodesic from x to y , or that α *joins* x and y , if $\alpha(a) = x$ and $\alpha(b) = y$. The points x and y are called the *endpoints* of α . By abuse of terminology, we sometimes identify a geodesic segment $|\alpha|$ with the parametrizing map α .

A *geodesic space* is a metric space in which any two points x, y can be joined by a geodesic segment. In general, there could be many different geodesic segments joining two given points of X .

An $\mathbb{R}$ -*tree* is a metric space (T, d) such that for any two points $x, y \in T$, there is a *unique* geodesic segment in T from x to y . We will denote this geodesic segment by $[x, y]$.

A *rooted $\mathbb{R}$ -tree* is a triple (T, d, ζ) consisting of an $\mathbb{R}$ -tree (T, d) together with the choice of a reference point $\zeta \in T$ called the *root*.

A point x in an $\mathbb{R}$ -tree T is said to be *ordinary* if the complement $T \setminus \{x\}$ has two connected components. A point which is not ordinary will be called a *branch point* of T . If $T \setminus \{x\}$ has just one connected component, x is also called an *endpoint* of T . A *finite $\mathbb{R}$ -tree* is an $\mathbb{R}$ -tree which has only finitely many branch points.

2. Geodesic triangles

Let T be an $\mathbb{R}$ -tree. For every $x, y, z \in T$, let $\Delta = \Delta_{xyz} \subseteq T$ be the union of the three geodesic segments joining these three points:

$$\Delta = [x, y] \cup [y, z] \cup [x, z] .$$

It is easily verified that Δ is a *tripod*, i.e., a finite $\mathbb{R}$ -tree consisting of three (possibly degenerate) edges joined at a common vertex v .

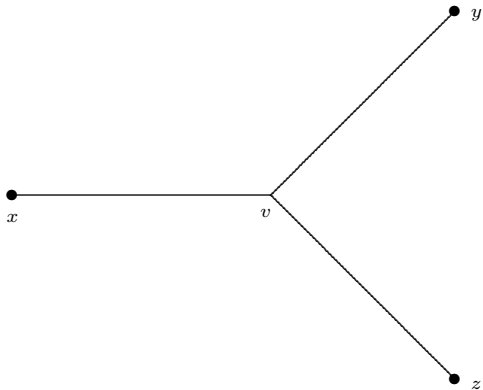


FIGURE 1. A tripod with vertices x, y, z and center v .

Moreover, x, y , and z are the endpoints of the tripod. The point v of the tripod where the three edges meet is called the *center*.

Following Gromov, we now establish a sort of converse to the observation that every triangle in an $\mathbb{R}$ -tree is a tripod, thereby obtaining a useful characterization of $\mathbb{R}$ -trees.

Let X be a geodesic space. A *geodesic triangle* in X consists of a set $\{x, y, z\}$ of points of X (the “vertices” of the triangle), together with geodesic segments $\alpha_{xy}, \alpha_{yz}, \alpha_{xz}$ (the “edges” of the triangle) joining these points pairwise. We usually identify a geodesic triangle with the underlying subspace $\Delta = \Delta_{xyz}$ of X given by

$$\Delta = \alpha_{xy} \cup \alpha_{yz} \cup \alpha_{xz} ,$$

and call x, y, z the *vertices* of Δ . For example, a tripod T is a geodesic triangle whose vertices are the endpoints of T .

If Δ_i is a geodesic triangle in the metric space X_i for $i = 1, 2$, a *simplicial map* $f : \Delta_1 \rightarrow \Delta_2$ is a continuous map taking the vertices of Δ_1 to the vertices of Δ_2 and edges of Δ_1 to edges of Δ_2 .

DEFINITION B.1. A geodesic space X is called *strongly hyperbolic* if every geodesic triangle Δ in X is isometric to a tripod T_Δ via a simplicial map $\chi_\Delta : \Delta \rightarrow T_\Delta$.

PROPOSITION B.2. A geodesic space X is strongly hyperbolic if and only if it is an $\mathbb{R}$ -tree.

PROOF. By what we have already said, it suffices to prove that if a geodesic space X is strongly hyperbolic, then there is a *unique* geodesic

segment joining any two given points x and y . Suppose $\alpha, \beta : [a, b] \rightarrow X$ are geodesic segments joining x and y . Take any point $z \in |\alpha|$, let $c = \alpha^{-1}(z) \in [a, b]$, and consider the geodesic triangle Δ with vertices x, y, z formed by the geodesic segments $\alpha_1 = \alpha|_{[a, c]}$, $\alpha_2 = \alpha|_{[c, b]}$, and β . Choose an isometry $\chi : \Delta \rightarrow T$ to a tripod T_Δ , and let d_T denote the metric on T_Δ . Let v be the point of X mapping to the center of the tripod T_Δ . If $\ell(\gamma)$ denotes the length of a geodesic segment γ , then we have

$$\begin{aligned} \ell(\alpha) &= \ell(\alpha_1) + \ell(\alpha_2) = d_T(\chi(x), \chi(z)) + d_T(\chi(z), \chi(y)) \\ &\geq d_T(\chi(x), \chi(y)) = \ell(\beta), \end{aligned}$$

with equality if and only if $\chi(z) = \chi(v)$, i.e., if and only if $z = v$. Thus $\ell(\alpha) \geq \ell(\beta)$. By symmetry, we conclude that $\ell(\alpha) = \ell(\beta)$, and therefore $z = v$. But then $z \in |\beta|$, and since z was arbitrary, it follows that $|\alpha| \subseteq |\beta|$. By symmetry, we see that $|\alpha| = |\beta|$ as desired. $\square$

It is sometimes useful to think of strong hyperbolicity in terms of quadrilaterals rather than triangles. If X is a geodesic space, $x_1, \dots, x_4 \in X$, and α_{ij} is a geodesic segment joining x_i and x_j for each unordered pair (i, j) with $i \neq j$, we call the union of the segments α_{ij} a *geodesic quadrilateral* with vertices x_i . Thus a geodesic quadrilateral has four vertices and six sides (“four sides plus two diagonals”).

There is also an obvious 4-endpoint analogue of a tripod, which we call a *quadrupod*: it is a finite $\mathbb{R}$ -tree obtained by gluing two tripods along a common edge:

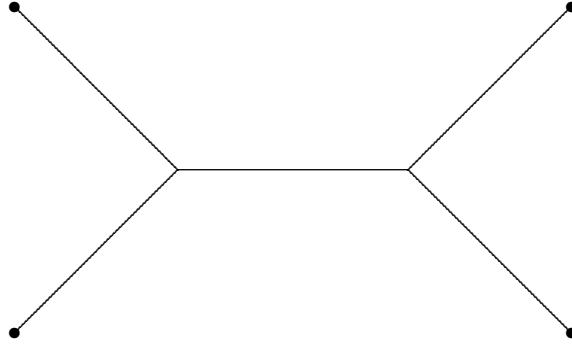


FIGURE 2. A quadrupod.

LEMMA B.3. *A geodesic space X is strongly hyperbolic if and only if every geodesic quadrilateral Q in X is isometric to a quadrupod T_Q via a simplicial isometry.*

PROOF. The “if” direction is clear. For the reverse implication, suppose X is strongly hyperbolic. By Proposition B.2, there is a *unique* geodesic segment joining any two points of X . For any points $x, y, z, w \in X$, let Δ_{xyz} be the geodesic triangle corresponding to x, y, z , and choose an isometry $\chi_{xyz} : \Delta_{xyz} \rightarrow T_{xyz}$ to a tripod T_{xyz} . Let v be the point of X mapping to the center of T_{xyz} , and choose an isometry $\chi_{xvw} : \Delta_{xvw} \rightarrow T_{xvw}$. It follows easily from the uniqueness of geodesics in X that the isometries χ_{xyz} and χ_{xvw} can be “glued” to give an isometry from the geodesic quadrilateral with vertices x, y, z, w to a quadripod. $\square$

A useful consequence of Lemma B.3 is that, in the definition of strongly hyperbolic, one may restrict to geodesic triangles containing a fixed vertex x .

COROLLARY B.4. *Let X be a geodesic space, and fix a “base point” $x \in X$. Then X is strongly hyperbolic if and only if for every pair of points $y, z \in X$ and every geodesic triangle Δ with vertices x, y, z , there is a simplicial isometry from Δ to a tripod T_Δ .*

PROOF. The “only if” direction is clear. For the reverse implication, let $y, z, w \in X$, and let Q be any geodesic quadrilateral with vertices x, y, z, w . By assumption, Q is isometric to a quadripod T_Q . Restricting this isometry to the geodesic triangle Δ_{yzw} with vertices y, z, w , we see that Δ_{yzw} is isometric to a tripod as required. $\square$

3. The Gromov product

Let (X, d) be a metric space, and for $x, y, z \in X$, define the *Gromov product* $(x|y)_z$ of x and y relative to z by the formula

$$(B.1) \quad (x|y)_z = \frac{1}{2} (d(x, z) + d(y, z) - d(x, y)) .$$

As motivation for this definition, note that if (T, d_T) is a tripod with vertices x, y, z and center v , then $d_T(x, v) = (y|z)_x$, $d_T(y, v) = (x|z)_y$, and $d_T(z, v) = (x|y)_z$.

DEFINITION B.5. A metric space X is called *0-hyperbolic* (in the sense of Gromov) if for every $x, y, z, w \in X$, we have

$$(B.2) \quad (x|y)_w \geq \min\{(x|z)_w, (y|z)_w\} .$$

REMARK B.6. More generally, if $\delta \geq 0$ is a real number, a metric space X is called *δ -hyperbolic* if for every $x, y, z, w \in X$, we have

$$(B.3) \quad (x|y)_w \geq \min\{(x|z)_w, (y|z)_w\} - \delta ,$$

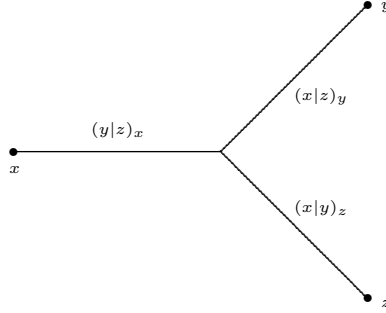


FIGURE 3. The Gromov product on a tripod.

and X is called *Gromov hyperbolic* if it is δ -hyperbolic for some δ . See [20] for a detailed discussion of Gromov hyperbolicity, including a number of equivalent definitions.

The proof of the following proposition is adapted from [51, §2.2] (see also [20, Chapter III.H]). Before giving the proof, we note the following easily verified facts. If Δ is a geodesic triangle in a metric space X , there is a unique pair (T_Δ, χ_Δ) consisting of a tripod T_Δ and a surjective simplicial map $\chi_\Delta : \Delta \rightarrow T_\Delta$ whose restriction to each edge of Δ is an isometry. Moreover, χ_Δ is an *isometry* if and only if χ_Δ is *injective*.

PROPOSITION B.7. *If X is a geodesic space, then X is strongly hyperbolic if and only if X is 0-hyperbolic.*

PROOF. Suppose X is strongly hyperbolic, and let Q be any geodesic quadrilateral with vertices x, y, z, w . Since Q is isometric to a quadripod, it suffices to note upon inspection that a quadripod is 0-hyperbolic.

Conversely, suppose X is 0-hyperbolic. Let Δ be a geodesic triangle in T with vertices x, y, z , and let $\chi = \chi_\Delta : \Delta \rightarrow T_\Delta$ be the canonical piecewise isometry to the tripod T_Δ . Suppose $w, w' \in \Delta$ are points for which $\chi(w) = \chi(w')$. Without loss of generality, we can assume that $w \in \overline{xy}$ and $w' \in \overline{xz}$, where $\overline{xy}$ (resp. $\overline{xz}$) denotes the edge of Δ connecting x and y (resp. x and z). Since $\chi(w) = \chi(w')$, we have $d(x, w) = d(x, w') = r$ for some real number $r \geq 0$. Let v be the center of T_Δ ; since $\chi(w) = \chi(w')$ is situated between $\chi(x)$ and v , and since $d_T(\chi(x), v) = (y|z)_x$, we have $(y|z)_x \geq r$. Also, we have $(w|y)_x = (w'|z)_x = r$. As X is 0-hyperbolic, we see that

$$(w|w')_x \geq \min\{(w|y)_x, (y|w')_x\} \geq \min\{(w|y)_x, (y|z)_x, (z|w')_x\} \geq r.$$

On the other hand, by the definition of the Gromov product we have $(w|w')_x = r - \frac{1}{2}d(w, w')$, from which we obtain $d(w, w') \leq 0$. It follows that $w = w'$, and therefore χ is injective. As previously noted, this implies that χ is an isometry. $\square$

Fix a base point ζ in the metric space X . X is called *0-hyperbolic relative to ζ* if for every $x, y, z \in X$, we have

$$(B.4) \quad (x|y)_\zeta \geq \min\{(x|z)_\zeta, (y|z)_\zeta\}.$$

By definition, X is 0-hyperbolic if and only if it is 0-hyperbolic relative to ζ for *all* $\zeta \in X$. We now prove the *a priori* stronger fact that X is 0-hyperbolic if and only if it is 0-hyperbolic relative to ζ for *some* $\zeta \in X$. First, we need the following lemma.

LEMMA B.8. *The metric space X is 0-hyperbolic with respect to the basepoint $\zeta \in X$ if and only if for any points $x, y, z, w \in X$, we have*

$$(B.5) \quad (x|y)_\zeta + (z|w)_\zeta \geq \min\{(x|z)_\zeta + (y|w)_\zeta, (x|w)_\zeta + (y|z)_\zeta\}.$$

PROOF. If (B.5) holds, then since $(z|\zeta)_\zeta = 0$, taking $w = \zeta$ shows that X is 0-hyperbolic with respect to ζ .

Conversely, suppose X is 0-hyperbolic with respect to ζ . The desired inequality (B.5) is symmetric with respect to x and y , and with respect to w and z . Let

$$M = \max\{(x|z)_\zeta, (x|w)_\zeta, (y|z)_\zeta, (y|w)_\zeta\}.$$

If $M = (x|w)_\zeta$, interchange the roles of w and z . If $M = (y|z)_\zeta$, interchange the roles of x and y . If $M = (y|w)_\zeta$, interchange the roles both of x and y and of w and z . In this way, we may assume without loss of generality that $M = (x|z)_\zeta$.

By 0-hyperbolicity, we have

$$(x|y)_\zeta \geq \min\{(x|z)_\zeta, (y|z)_\zeta\} = (y|z)_\zeta$$

and

$$(z|w)_\zeta \geq \min\{(x|z)_\zeta, (x|w)_\zeta\} = (x|w)_\zeta.$$

Thus

$$(x|y)_\zeta + (z|w)_\zeta \geq (x|w)_\zeta + (y|z)_\zeta,$$

which proves (B.5). $\square$

For $x, y, z, w \in X$, define the “cross ratio” of (x, y, z, w) relative to ζ to be

$$(x; y; z; w)_\zeta = (x|y)_\zeta + (z|w)_\zeta - (x|z)_\zeta - (y|w)_\zeta.$$

LEMMA B.9. *The cross ratio $(x; y; z; w)_\zeta$ is independent of ζ .*

PROOF. This is a direct calculation using the definition of the Gromov product. $\square$

By Lemma B.9, we may write $(x; y; z; w)$ instead of $(x; y; z; w)_\zeta$. As a consequence of Lemmas B.9 and B.8, we obtain:

COROLLARY B.10. *If a metric space X is 0-hyperbolic with respect to some basepoint $\zeta \in X$, then it is 0-hyperbolic with respect to every basepoint $\zeta \in X$ (i.e., it is 0-hyperbolic).*

PROOF. By Lemma B.8, X is 0-hyperbolic with respect to ζ if and only if

$$(B.6) \quad \min\{(x; y; z; w)_\zeta, (x; y; w; z)_\zeta\} \geq 0.$$

But by Lemma B.9, condition (B.6) is independent of ζ . $\square$

4. $\mathbb{R}$ -trees and partial orders

Let $(T, \leq)$ be a partially ordered set (poset) satisfying the following two axioms:

- (P1) T has a unique maximal element ζ , called the *root* of T .
- (P2) For each $x \in T$, the set $S_x = \{z \in T : z \geq x\}$ is totally ordered.

We say that T is a *parametrized rooted tree* if there is a function $\alpha : T \rightarrow \mathbb{R}_{\geq 0}$ with values in the nonnegative reals such that:

- (P3) $\alpha(\zeta) = 0$.
- (P4) α is *order-reversing*, in the sense that $x \leq y$ implies $\alpha(x) \geq \alpha(y)$.
- (P5) The restriction of α to any full totally ordered subset of T gives a bijection onto a real interval.

(A totally ordered subset S of T is called *full* if $x, y \in S$, $z \in T$, and $x \leq z \leq y$ implies $z \in S$.)

Although inspired by it, this definition differs slightly from the one given in [27, Chapter 3].

Given two elements x and y of a parametrized rooted tree T , one can use the parametrization α to define the *least upper bound* $x \vee y$ of x and y . Indeed, since the set $S_x = \{z \in T : z \geq x\}$ is full and totally ordered, axiom (P5) provides a bijection from S_x to a real interval $[0, R_x]$. The intersection of S_x and $S_y = \{z \in T : z \geq y\}$ contains the root ζ , so it is a non-empty subset of S_x . Now define

$$x \vee y = \alpha^{-1} \left(\sup_{z \in S_x \cap S_y} \alpha(z) \right).$$

It is straightforward to show that $x \leq x \vee y$, $y \leq x \vee y$, and that if $z \in T$ is any point with $x \leq z$ and $y \leq z$, then $x \vee y \leq z$.

We now describe a one-to-one correspondence between rooted $\mathbb{R}$ -trees and parametrized rooted trees.

Given a rooted $\mathbb{R}$ -tree (T, d, ζ) , one defines a partial order on T by declaring that $x \leq y$ iff y belongs to the unique path between x and ζ , and a parametrization α on T by setting $\alpha(x) = d(x, \zeta)$. It is easy to check that the resulting structure satisfies the axioms (P1)-(P5) for a parametrized rooted tree.

Conversely, given a parametrized rooted tree $(T, \leq)$ with root ζ and parametrization α , define a distance function d on T by

$$d(x, y) = \alpha(x) + \alpha(y) - 2\alpha(x \vee y) .$$

It is easy to check that d defines a metric on T , and that there exists a geodesic segment joining any two points $x, y \in T$, namely the concatenation of the segments $\sigma_x = \{z : x \leq z \leq x \vee y\}$ and $\sigma_y = \{z : y \leq z \leq x \vee y\}$.

PROPOSITION B.11. *(T, d) is an $\mathbb{R}$ -tree.*

PROOF. First note that the Gromov product on (T, d) with respect to ζ satisfies the identity

$$(x|y)_\zeta = \alpha(x \vee y) .$$

We claim that T is 0-hyperbolic with respect to ζ . This is equivalent to the assertion that for every $x, y, z \in T$, we have

$$(B.7) \quad \alpha(x \vee y) \geq \min\{\alpha(x \vee z), \alpha(y \vee z)\} .$$

By axiom (P2), the set $\{x \vee y, x \vee z, y \vee z\}$ is totally ordered. Since (B.7) is symmetric in x and y , we may assume without loss of generality that $y \vee z \leq x \vee z$. But then

$$x \vee y \leq (x \vee y) \vee z = x \vee (y \vee z) \leq x \vee (x \vee z) = x \vee z .$$

Since α is order-reversing, it follows that $\alpha(x \vee y) \geq \alpha(x \vee z)$, proving (B.7), and thus that T is 0-hyperbolic with respect to ζ . By Corollary B.10, T is 0-hyperbolic. The result now follows from Proposition B.2. $\square$

One verifies easily that the maps we have just defined

$$\{\text{rooted } \mathbb{R}\text{-trees}\} \rightleftarrows \{\text{parametrized rooted trees}\}$$

are inverse to one another, and therefore there is an equivalence between the notion of a rooted $\mathbb{R}$ -tree and that of a parametrized rooted tree.

5. The weak and strong topologies

In this section, we mostly follow the exposition of [27, Chapter 5].

Let (X, d) be an $\mathbb{R}$ -tree. The *strong topology* on X is defined to be the topology induced by the metric d . There is another natural and very useful topology on X , called the *weak topology*. In order to define it, we need some preliminary definitions.

Let $p \in X$. We define an equivalence relation on $X \setminus \{p\}$ by declaring that $x \sim y$ if and only if the unique geodesic segments from p to x and p to y share a common initial segment. An equivalence class is called a *tangent vector* at p , and the set of tangent vectors at p is called the (projectivized) *tangent space* of p , denoted T_p or $T_p(X)$. A point x in the equivalence class $\vec{v}$ is said to *represent* the tangent vector $\vec{v}$.

If $\vec{v} \in T_p(X)$ is a tangent vector at a point $p \in X$, define

$$U(p; \vec{v}) = \{x \in X \setminus \{p\} : x \text{ represents } \vec{v}\}.$$

By definition, the weak topology is the topology generated by all sets of the form $U(p; \vec{v})$ (i.e., such sets form a sub-base of open sets for the weak topology).

In [24], the weak topology is called the *observer's topology*, since one can think of $U(p; \vec{v})$ as the set of all points which can be seen by an observer standing at p and looking in the direction $\vec{v}$. A fundamental system of open neighborhoods of a point $p \in X$ is given by sets of the form

$$(B.8) \quad U(q_1; \vec{v}_1) \cap \cdots \cap U(q_k; \vec{v}_k)$$

with $p \in U(q_i; \vec{v}_i)$ for all i . A set of the form (B.8) can be thought of as the set of all points which can simultaneously be seen by k observers situated at points of $X \setminus \{p\}$ and looking in the direction of p .

It is easy to see that the weak topology on a metric tree is always Hausdorff. As the names suggest, there is a relation between the weak and strong topologies. The following lemma and its proof are taken from [27, Proposition 5.5]. In the statement, we write X_{strong} for X endowed with the strong topology and X_{weak} for X endowed with the weak topology.

LEMMA B.12. *The strong topology on an $\mathbb{R}$ -tree (X, d) is stronger (i.e., finer) than the weak topology. Equivalently, the natural map $\psi : X_{\text{strong}} \rightarrow X_{\text{weak}}$ which is the identity on the underlying sets is continuous.*

PROOF. Let x_n be a sequence in X which converges in the strong topology to x . Consider X as a rooted tree with x as the root, and let $\leq$ be the corresponding partial order. If $x_n \not\rightarrow x$ in the weak topology,

then there must be a point $y < x$ such that $x_n \leq y$ for infinitely many n . But $d(x_n, x) \geq d(y, x)$ for such n , a contradiction. $\square$

COROLLARY B.13. *An $\mathbb{R}$ -tree X is arcwise connected in its weak topology.*

REMARK B.14. By abuse of terminology, we use the terms “arcwise connected” and “path-connected” interchangeably throughout this monograph.

There is also a relationship between connectedness in the strong and weak topologies. The following lemma and its proof are taken from [24, Proposition 1.7].

LEMMA B.15. *X_{strong} and X_{weak} have the same connected subsets.*

PROOF. A connected subset of X_{strong} is arcwise connected, and therefore is arcwise connected in the weak topology by Lemma B.12.

Conversely, let C be a connected subset of X_{weak} , and assume that it is not connected in the strong topology. Then it is not convex, so that there exist points $x, y \in C$ and $z \in X \setminus C$ with $z \in [x, y]$. Let $\vec{v}$ be the tangent vector at z representing the path $[z, x]$. Then $U = U(z; \vec{v})$ and $V = X \setminus (U \cup \{z\})$ are disjoint open sets covering C . Moreover, both U and V are non-empty, as $x \in U$ and $y \in V$. This contradicts the connectedness of C . $\square$

As a consequence, it is easy to see that elements of $T_p(X)$ are in one-to-one correspondence with the connected components of $X \setminus \{p\}$ in either the strong or weak topology.

We now show that arcwise connectedness in the weak and strong topologies coincide. (This is not explicitly pointed out in [27] or [24].)

PROPOSITION B.16. *An injective mapping $i : [0, 1] \rightarrow X$ is continuous in the weak topology if and only if it is continuous in the strong topology.*

PROOF. Since the canonical map $\psi : X_{\text{strong}} \rightarrow X_{\text{weak}}$ is continuous, it is clear that if i is strongly continuous then it is weakly continuous. To show the converse, suppose i is weakly continuous, and let Λ be the image of i . Let $x = i(0)$, $y = i(1)$, and let σ be the unique geodesic segment from x to y . We claim that $\Lambda \subseteq \sigma$. Indeed, if not then there exists $z \in \Lambda \setminus \sigma$. Since $z \notin \sigma$, it is easy to see that x and y are contained in the same connected component of $X \setminus \{z\}$. Let t be the unique point of $[0, 1]$ for which $i(t) = z$. Since $i : [0, 1] \rightarrow \Lambda$ is a continuous bijection between compact Hausdorff spaces (where Λ is endowed with the weak subspace topology), it must be a homeomorphism. Thus 0

and 1 lie in the same connected component of $i^{-1}(z) \subseteq [0, 1] \setminus \{t\}$. This contradiction proves our claim that $\Lambda \subseteq \sigma$. It follows that $\psi^{-1}(\Lambda)$ is compact in the *strong* topology. The inverse of the map $\psi^{-1} \circ i : [0, 1] \rightarrow \psi^{-1}(\Lambda)$ is $i^{-1} \circ \psi$, which is a continuous bijection between compact Hausdorff spaces, and therefore is a homeomorphism. Thus $\psi^{-1} \circ i$ is continuous, i.e., i is continuous in the strong topology. $\square$

COROLLARY B.17. *An $\mathbb{R}$ -tree X is uniquely arcwise connected in its weak topology.*

COROLLARY B.18. *A subset of X is arcwise connected in the weak topology if and only if it is arcwise connected in the strong topology.*

APPENDIX C

Metrized Graphs

In this appendix, we recall some basic facts about metrized graphs from [4]. For the reader's convenience, we have extracted (with minor modifications) certain portions of [4], [3], and [23] to make our presentation of the theory as self-contained as possible. All metrized graphs are assumed to be *finite*.

Ask the Canadian Journal of Math for permission to reproduce portions of [4] here.

1. Definitions

Intuitively, a metrized graph is a finite graph whose edges are thought of as line segments having a well-defined length. In particular, Γ is a one-dimensional manifold except at finitely many “branch points”, where it looks locally like an n -pointed star. The path-length function along each edge extends to a metric on all of Γ , making it a compact metric space.

A rigorous definition of a metrized graph (following [74]) is as follows.

DEFINITION C.1. A metrized graph Γ is a compact, connected metric space such that for each $p \in \Gamma$, there exist a radius $r_p > 0$ and an integer $n_p \geq 1$ such that p has a neighborhood $V_p(r_p)$ isometric to the star-shaped set

$$S(n_p, r_p) = \{z \in \mathbb{C} : z = te^{k \cdot 2\pi i / n_p} \text{ for some } 0 \leq t < r_p \text{ and some } k \in \mathbb{Z}\},$$

equipped with the path metric.

A finite $\mathbb{R}$ -tree, as defined in §4 of Chapter 1, is the same thing as a metrized graph without cycles. Although $\mathbb{R}$ -trees suffice for the purpose of studying $\mathbb{P}_{\text{Berk}}^1$, the more general concept of a metrized graph is needed to study Berkovich curves of higher genus. Since the theory of Laplacians is not much more difficult to set up on an arbitrary metrized graph than on a finite $\mathbb{R}$ -tree, we describe the theory in this more general context.

By a *vertex set* for Γ , we mean a finite set of points S such that $\Gamma \setminus S$ is a union of open intervals whose closures have distinct endpoints. A vertex set necessarily contains all endpoints and branch points of Γ . If Γ has loops, it also contains at least one interior point from each loop.

There is a close connection between metrized graphs and finite weighted graphs. Given a metrized graph Γ , any choice of a vertex set S for Γ gives rise to a weighted graph G which one may call a *model* for Γ ; a different choice of vertex set leads to an equivalent weighted graph. Conversely, every weighted graph G determines a metrized graph in the obvious way, so there is a one-to-one correspondence between metrized graphs and equivalence classes of finite weighted graphs (see [4] and [3] for further details).

Often, when given a metrized graph Γ , one chooses without explicit comment a vertex set S , together with distinguished parametrizations of the edges of the corresponding model G . The definition of the Laplacian given below is independent of these implicit choices.

By an *isometric path* (or simply a *path*) in Γ , we will mean an injective length-preserving continuous map from the real interval $[0, L]$ into Γ . We will say that an isometric path $\gamma : [0, L] \rightarrow \Gamma$ *emanates from* p , and *terminates at* q , if $\gamma(0) = p$ and $\gamma(L) = q$. We call two isometric paths emanating from p *equivalent* if they share a common initial segment. For each $p \in \Gamma$, we let $T_p(\Gamma)$ (the projectivized “tangent space” to p) denote the set of equivalence classes of isometric paths emanating from p . It is easy to see that $|T_p(\Gamma)| = n_p$, i.e., there is a bijection between elements of T_p and the “edges” of Γ bordering p .

It is useful to associate to each element of $T_p(\Gamma)$ a formal “unit tangent vector” $\vec{v}$, and to write $p + t\vec{v}$ instead of $\gamma(t)$, where $\gamma : [0, L] \rightarrow \Gamma$ is a representative path. If $f : \Gamma \rightarrow \mathbb{R}$ is a function, and $\vec{v}$ is a formal unit tangent vector at p , then we will define the “(one-sided) derivative of f in the direction $\vec{v}$ ” to be

$$d_{\vec{v}}f(p) = \lim_{t \rightarrow 0^+} \frac{f(p + t\vec{v}) - f(p)}{t} = \lim_{t \rightarrow 0^+} \frac{f(\gamma(t)) - f(p)}{t},$$

provided the limit exists as a finite number.

2. The space $\text{CPA}(\Gamma)$

Let $\text{CPA}(\Gamma)$ be the space of continuous, piecewise-affine, real-valued functions on Γ . (By a *piecewise affine function* f , we mean that there is a vertex set S_f for Γ such that f is affine on each edge in $\Gamma \setminus S_f$ with respect to an arclength parametrization of that edge.) If $f \in \text{CPA}(\Gamma)$,

then clearly the directional derivatives $d_{\vec{v}}(p)$ are defined for all $p \in \Gamma$ and all $\vec{v} \in T_p(\Gamma)$.

Chinburg and Rumely ([23]) introduced a Laplacian operator on CPA(Γ). Their Laplacian is a map from CPA(Γ) to the space of discrete signed measures on Γ . We will take the Laplacian to be the negative of theirs, and put

$$\Delta(f) = \sum_{p \in \Gamma} \left(- \sum_{\vec{v} \in T_p(\Gamma)} d_{\vec{v}} f(p) \right) \delta_p(x) ,$$

where $\delta_p(x)$ is the Dirac measure at p . The operator Δ coincides, in a precise sense, with the usual combinatorial Laplacian on a finite weighted graph – see [3] for details. By abuse of notation, we will write $\Delta(f)(p)$ for $\Delta(f)(\{p\}) = - \sum_{\vec{v} \in T_p(\Gamma)} d_{\vec{v}} f(p)$. Here are some elementary properties of Δ :

PROPOSITION C.2. *Let $f, g \in \text{CPA}(\Gamma)$. Then*

- A) $\Delta(f) \equiv 0$ if and only if $f = C$ for some constant C .
- B) $\Delta(f) = \Delta(g)$ if and only if $f = g + C$ for some constant C .
- C) If f is nonconstant, then $f(x)$ achieves its maximum at a point p where $\Delta(f)(p) > 0$, and its minimum at a point q where $\Delta(f)(q) < 0$.
- D) $\int_{\Gamma} f \Delta(g) = \int_{\Gamma} g \Delta(f)$.
- E) The total mass $\Delta(f)(\Gamma) = 0$.

PROOF. For (A), if $f = C$, clearly $\Delta(f) \equiv 0$. Conversely, suppose $\Delta(f) \equiv 0$. Put $M = \max_{x \in \Gamma} f(x)$, and let $G = \{x \in \Gamma : f(x) = M\}$. Since f is continuous, G is closed. To see that G is also open, note that if $p \in G$, then $d_{\vec{v}} f(p) \leq 0$ for each $\vec{v}$ at p , since M is the maximum value of f . Since f is piecewise affine, if f were not locally constant at p then we would have $d_{\vec{v}} f(p) < 0$ for some $\vec{v}$, but then $\Delta(f)(p) > 0$, contrary to our assumption. Thus, G is both open and closed, and since Γ is connected, $G = \Gamma$.

Trivially (A) implies (B).

For (C), suppose f is nonconstant, and let M and G be as in (A). If $p \in \partial G$, then the argument above shows that $\Delta(f)(p) > 0$. Similarly, f achieves its minimum at a point where $\Delta(f)(q) < 0$.

For (D), let S be a vertex set for Γ such that f and g are both affine on each edge of $\Gamma \setminus S$. Given an edge e_i of $\Gamma \setminus S$, fix an orientation of it, and let $\gamma_i : [0, L_i] \rightarrow e_i$ be an arclength parametrization. By abuse of notation, write $f(x) = f(\gamma_i(x))$ and $g(x) = g(\gamma_i(x))$ on e_i . Integration by parts shows that

$$\int_{\Gamma} f(x) \Delta(g)(x) = \sum_i \int_{e_i} f'(x) g'(x) dx = \int_{\Gamma} g(x) \Delta(f)(x) .$$

Part (E) follows from (D), taking $g(x) \equiv 1$. $\square$

3. The potential kernel $j_z(x, y)$

There is a *fundamental potential kernel* $j_z(x, y)$ on Γ which (for fixed y, z) belongs to $\text{CPA}(\Gamma)$ and “inverts the Laplacian”. It is defined by requiring that for each fixed $y, z \in \Gamma$,

$$(C.1) \quad \begin{cases} \Delta_x(j_z(x, y)) = \delta_y(x) - \delta_z(x) , \\ j_z(z, y) = 0 . \end{cases}$$

The uniqueness of $j_z(x, y)$ follows from Proposition C.2 (B). Its existence can be shown in general using circuit theory (see [23]), or by linear algebra (see [74] or [3]).

When Γ is a finite $\mathbb{R}$ -tree, the existence of a function satisfying (C.1) is immediate: given $x, y, z \in \Gamma$, let w be the point where the path from x to z and the path from y to z first meet, and put $j_z(x, y) = \rho(z, w)$, the path length from z to w . Then along the path from z to y , $j_z(x, y) = \rho(z, x)$; on branches off that path, $j_z(x, y)$ is constant. One easily checks that the function thus defined satisfies the equations (C.1). In other words (see Appendix B), the function $j_z(x, y)$ on a finite $\mathbb{R}$ -tree Γ is just the Gromov product $(x|y)_z$:

$$(C.2) \quad j_z(x, y) = (x|y)_z .$$

The potential kernel has the following physical interpretation. View Γ as an electric circuit with terminals at y and z , and with the resistance of each edge given by its length. Then $j_z(x, y)$ is the voltage at x when current 1 enters at y and exits at z , with reference voltage 0 at z . By its construction, $j_z(x, y)$ belongs to $\text{CPA}(\Gamma)$, and its slope along any edge of Γ has absolute value at most 1.

PROPOSITION C.3.

A) $j_z(x, y)$ is non-negative, bounded, symmetric in x and y , and jointly continuous in x, y, z .

B) If $f \in \text{CPA}(\Gamma)$ satisfies $\Delta(f) = \sum_{i=1}^n c_i \delta_{p_i}(x)$ then there is a constant such that

$$f(x) = \sum_{i=1}^n c_i j_z(x, p_i) + C .$$

C) For each $z, \zeta \in \Gamma$,

$$(C.3) \quad j_\zeta(x, y) = j_z(x, y) - j_z(x, \zeta) - j_z(y, \zeta) + j_z(\zeta, \zeta) .$$

D) Fix $x, z \in \Gamma$, let $h(y) = j_z(x, y)$, and let $\vec{v} \in T_x(\Gamma)$. Then the directional derivative of h in the direction $\vec{v}$ satisfies

$$|D_{\vec{v}}(h)(x)| \leq 1 .$$

PROOF. (A) The non-negativity of $j_z(x, y)$ follows from its defining properties (C.1) and Proposition C.2 (C).

For symmetry, fix a, b, z and apply Proposition C.2 (D) to $f(x) = j_z(x, b)$ and $g(x) = j_z(x, a)$. Then $\Delta(f) = \delta_b(x) - \delta_z(x)$ and $\Delta(g) = \delta_a(x) - \delta_z(x)$. Since $f(z) = g(z) = 0$,

$$j_z(a, b) = \int_{\Gamma} f(x) \Delta(g)(x) = \int_{\Gamma} g(x) \Delta(f)(x) = j_z(b, a) .$$

Clearly $j_z(x, y)$ is continuous in x for each fixed y, z , with $|j_z(x, y) - j_z(p, y)| \leq \rho(x, p)$ since the absolute value of the slope of $j_z(x, y)$ along each edge is at most 1. From this, one deduces that it is jointly continuous in x, y for each fixed z , and indeed that for given p, q

$$\begin{aligned} |j_z(x, y) - j_z(p, q)| &\leq |j_z(x, q) - j_z(p, q)| + |j_z(x, q) - j_z(x, y)| \\ (C.4) \qquad \qquad &\leq \rho(x, p) + \rho(y, q) \end{aligned}$$

using the symmetry shown above. Finally, for any fixed z_0 , we claim that

$$(C.5) \qquad j_z(x, y) = j_{z_0}(x, y) - j_{z_0}(x, z) - j_{z_0}(z, y) + j_{z_0}(z, z) .$$

The joint continuity of $j_z(x, y)$ in x, y , and z follows from this and (C.4). Its boundedness follows from continuity and the compactness of Γ .

To prove (C.5), note that for fixed y, z

$$\Delta_x(j_z(x, y)) = \delta_y(x) - \delta_z(x) = \Delta_x(j_{z_0}(x, y) - j_{z_0}(x, z)) ,$$

so by Proposition C.2 (B) there is a constant $C_{y,z}$ such that $j_z(x, y) = j_{z_0}(x, y) - j_{z_0}(x, z) + C_{y,z}$. Taking $x = z_0$ shows $C_{y,z} = j_z(z_0, y)$. In other words, we have

$$(C.6) \qquad j_z(x, y) = j_{z_0}(x, y) - j_{z_0}(x, z) + j_z(z_0, y) .$$

Similarly, applying Δ_y to $j_z(z_0, y)$ and $-j_{z_0}(z, y)$ (and using symmetry), one finds there is a constant C_z such that $j_z(z_0, y) = -j_{z_0}(z, y) + C_z$. Taking $y = z$ shows $C_z = j_{z_0}(z, z)$. Combining these gives (C.5). This also proves (C).

For part (B), first note that if $\Delta(f) = \sum_{i=1}^n c_i \delta_{p_i}(x)$ then $\sum_{i=1}^n c_i = 0$ by Proposition C.2 (E). Using this, one sees that

$$\Delta\left(\sum_{i=1}^n c_i j_{z_0}(x, p_i)\right) = \sum_{i=1}^n c_i \delta_{p_i}(x) ,$$

and the result follows from Proposition C.2 (B).

Finally, to prove (D), one notes using the physical interpretation of $j_z(x, y)$ that if $\vec{v}$ corresponds to an edge e in a suitable model for

Γ , then $D_{\vec{v}}(h)(x)$ represents the current flowing through e when one unit of current enters the circuit at y and exits at z . (The sign of $D_{\vec{v}}(h)(x)$ determines which way the current is flowing.) It is therefore clear from physical considerations that $|D_{\vec{v}}(h)(x)| \leq 1$. (For a more rigorous mathematical proof, see, e.g., [3].) $\square$

4. The space $\text{Zh}(\Gamma)$

Recall that if $f \in \text{CPA}(\Gamma)$ then

$$(C.7) \quad \Delta(f) = \sum_{p \in \Gamma} \left(- \sum_{\vec{v} \text{ at } p} d_{\vec{v}} f(p) \right) \delta_p(x) .$$

It is possible to define a Laplacian on larger classes of functions. For example, Zhang ([74]) defined a Laplacian on the space of continuous, piecewise $\mathcal{C}^2$ functions whose one-sided directional derivatives $d_{\vec{v}} f(p)$ exist for all $p \in \Gamma$. We will write $\text{Zh}(\Gamma)$ for this space. Zhang put

$$(C.8) \quad \Delta_{\text{Zh}}(f) = -f''(x) dx + \sum_{p \in \Gamma} \left(- \sum_{\vec{v} \text{ at } p} d_{\vec{v}} f(p) \right) \delta_p(x) ,$$

where $f''(x)$ is taken relative to the arclength parametrization on each segment in the complement of an appropriate vertex set X_f for Γ . In other words, $f''(x) = \frac{d^2}{dt^2} f(p + t\vec{v})$ for $x = p + t\vec{v} \in \Gamma \setminus X_f$.

Again, $\Delta_{\text{Zh}}(f)$ is a measure on Γ . The condition that the directional derivatives exist for all p is easily seen to be equivalent to requiring that $f'' \in L^1(\Gamma)$. For a function $f \in \text{CPA}(\Gamma)$, $f'' = 0$ in the complement of a vertex set, so Zhang's Laplacian is compatible with the one defined by (C.7).

Zhang's Laplacian is a hybrid of the usual (positive definite) Laplacian $-f''$ on $\mathbb{R}$ and the combinatorial Laplacian on a weighted graph. It satisfies appropriate analogues of the properties given in Proposition C.2. (See Proposition C.13 for a more general statement.) The particular form of definition (C.8) is justified by the observation that $\int f \Delta(g) = \int g \Delta(f)$ for all $f, g \in \text{Zh}(\Gamma)$, a fact which one easily verifies using integration by parts.

It turns out that one can extend the Laplacian to a still broader class of functions. This observation is fundamental for the applications given in this book. To motivate the definition of the more general Laplacian operator, note that Zhang's Laplacian has the following property.

LEMMA C.4. (Mass Formula) *Let $E \subset \Gamma$ be a set which is a finite union of points and closed intervals. Then for each $f \in \text{Zh}(\Gamma)$,*

$$(C.9) \quad \Delta_{\text{Zh}}(f)(E) = - \sum_{p \in \partial E} \sum_{\substack{\vec{v} \in T_p(\Gamma) \\ \text{outward}}} d_{\vec{v}} f(p)$$

where ∂E is the boundary of E , and a direction $\vec{v}$ at p is called “outward” if the edge it corresponds to leads away from E .

PROOF. Choose a vertex set S for Γ which includes the endpoints of all segments in E , all isolated points of E , and all points where $f''(x)$ is not defined. Put $S_E = S \cap E$; then $E \setminus S_E$ is a finite union of open intervals (a_i, b_i) . By definition

$$(C.10) \quad \Delta_{\text{Zh}}(f)(E) = - \sum_i \int_{a_i}^{b_i} f''(x) dx + \sum_{p \in S_E} \left(- \sum_{\vec{v} \in T_p(\Gamma)} d_{\vec{v}} f(p) \right) \delta_p(x) .$$

On the other hand, for each interval (a_i, b_i)

$$\int_{a_i}^{b_i} f''(x) dx = f'(b_i) - f'(a_i)$$

where $f'(a_i)$ and $f'(b_i)$ are one-sided derivatives at the endpoints. If a_i, b_i correspond to $p, q \in S_E$, then for the direction vectors $\vec{v}, \vec{w}$ at p, q leading into (a_i, b_i) we have $d_{\vec{v}} f(p) = f'(a_i)$, $d_{\vec{w}} f(q) = -f'(b_i)$. It follows that

$$- \int_{a_i}^{b_i} f''(x) dx = d_{\vec{w}} f(q) + d_{\vec{v}} f(p) .$$

Thus, in (C.10) the sum over the integrals cancels all the directional derivatives for directions leading into E . What remains is the sum over directional derivatives in directions outward from E . $\square$

The Mass Formula says that for a closed set E which is a finite union of segments and isolated points, $\Delta_{\text{Zh}}(f)(E)$ is determined simply by knowing the directional derivatives of f at the points of ∂E . In particular, taking $E = \Gamma$, we see that $\Delta_{\text{Zh}}(f)(\Gamma) = 0$, since $\partial \Gamma$ is empty. As the complement of an open interval U is a closed set E of the type we have been discussing, it follows that

$$(C.11) \quad \Delta_{\text{Zh}}(f)(U) = -\delta(f)(E) = \sum_{p \in \partial U} \sum_{\substack{\vec{v} \in T_p(\Gamma) \\ \text{inward}}} d_{\vec{v}} f(p) .$$

Taking unions of open and closed sets, we see that for any $T \subset \Gamma$ which is a finite union of points and intervals (open, closed, or half-open) there

is a formula for $\Delta_{\text{Zh}}(f)(T)$ in terms of the directional derivatives $d_{\vec{v}}(f)$ at points of ∂T . (This formula is easy to write out, but messy; see (C.12) below.) The collection of such sets forms a Boolean algebra $\mathcal{A}(\Gamma)$.

Conversely, suppose $f : \Gamma \rightarrow \mathbb{R}$ is any function whose directional derivatives $d_{\vec{v}}f(p)$ exist for all p and $\vec{v}$. Taking Lemma C.4 as the starting point, one could hope to *define* the Laplacian of f by (C.9) and (C.11). This does indeed give a finitely additive set function $\Delta(f)$ on the Boolean algebra $\mathcal{A}(\Gamma)$. However, $\Delta(f)$ does not necessarily extend to a measure; there are pathological examples (similar to Weierstrass's continuous but nowhere differentiable functions) which oscillate so much that $\Delta(f)$ is not countably additive.

An additional condition is necessary: we need to assume that f is of *bounded differential variation* – see (C.13) below for a precise definition. We will write $\text{BDV}(\Gamma)$ for the space of functions of bounded differential variation. It is easy to see that $\text{CPA}(\Gamma) \subset \text{Zh}(\Gamma) \subset \text{BDV}(\Gamma)$.

In Theorem C.6 of the next section, we will show that if $f \in \text{BDV}(\Gamma)$, then the finitely additive function $\Delta(f)$, defined initially on $\mathcal{A}(\Gamma)$, extends to a finite signed Borel measure of total mass 0 on Γ . Conversely, we will see in Corollary C.11 that if μ is a finite signed Borel measure on Γ with total mass 0, then there is an $f \in \text{BDV}(\Gamma)$, unique up to addition of an arbitrary real constant, for which $\Delta(f) = \mu$. In other words, there is a natural bijection between finite signed Borel measures of total mass zero on Γ and functions in $\text{BDV}(\Gamma)$ modulo constant functions. This gives a precise meaning to the catchphrase “ $\text{BDV}(\Gamma)$ is the largest space of (continuous) functions on which $\Delta(f)$ can be defined as a measure”.

We will see in Proposition C.13 that the properties of the Laplacian proved in Proposition C.2 for functions in $\text{CPA}(\Gamma)$ extend to the more general case of functions in $\text{BDV}(\Gamma)$.

5. The space $\text{BDV}(\Gamma)$

For the convenience of the reader, we have reproduced portions of [4] in this section.

Let $\mathcal{D}(\Gamma)$ be the class of all functions on Γ whose one-sided derivatives exist everywhere, i.e.,

$$\mathcal{D}(\Gamma) = \{f : \Gamma \rightarrow \mathbb{R} : d_{\vec{v}}f(p) \text{ exists for each } p \in \Gamma \text{ and } \vec{v} \in T_p(\Gamma)\} .$$

It is easy to see that each $f \in \mathcal{D}(\Gamma)$ is continuous.

As in the previous section, let $\mathcal{A} = \mathcal{A}(\Gamma)$ be the Boolean algebra of subsets of Γ generated by the connected open sets. Each $S \in \mathcal{A}$ is a finite disjoint union of sets isometric to open, half-open, or closed

intervals (we consider isolated points to be degenerate closed intervals); conversely, all such sets belong to $\mathcal{A}$. Define the set $b(S)$ of boundary points of S to be the collection of points belonging to the closures of both S and $\Gamma \setminus S$. It is easy to see that each $S \in \mathcal{A}$ has only finitely many boundary points. Note that under this definition, if $\Gamma = [0, 1]$ and $S = [0, \frac{1}{2}]$, for example, then the left endpoint 0 is not a boundary point of S .

For each $p \in b(S)$, let $\text{In}(p, S)$ be the set of ‘inward-directed unit vectors at p ’: the set of all $\vec{v} \in T_p(\Gamma)$ for which $p + t\vec{v}$ belongs to S for all sufficiently small $t > 0$. Let $\text{Out}(p, S) = T_p(\Gamma) \setminus \text{In}(p, S)$ be the collection of ‘outward-directed unit vectors at p ’. If p is an isolated point of S , then $\text{In}(p, S) = \emptyset$ and $\text{Out}(p, S) = T_p(\Gamma)$.

If $f \in \mathcal{D}(\Gamma)$, then we can define a finitely additive set function m_f on $\mathcal{A}$ by requiring that for each $S \in \mathcal{A}$

$$(C.12) \quad m_f(S) = \sum_{\substack{b \in b(S) \\ b \notin S}} \sum_{\vec{v} \in \text{In}(b, S)} d_{\vec{v}} f(p) - \sum_{\substack{b \in b(S) \\ b \in S}} \sum_{\vec{v} \in \text{Out}(b, S)} d_{\vec{v}} f(p) .$$

Thus, for an open set $S \in \mathcal{A}$,

$$m_f(S) = \sum_{b \in b(S)} \sum_{\vec{v} \in \text{In}(b, S)} d_{\vec{v}} f(p) ,$$

for a closed set $S \in \mathcal{A}$,

$$m_f(S) = - \sum_{b \in b(S)} \sum_{\vec{v} \in \text{Out}(b, S)} d_{\vec{v}} f(p) ,$$

and for a set $S = \{p\}$ consisting of a single point,

$$m_f(\{p\}) = - \sum_{\vec{v} \in T_p(\Gamma)} d_{\vec{v}} f(p) .$$

The finite additivity is clear if one notes that each $S \in \mathcal{A}$ can be written as a finite disjoint union of open intervals and points, and that for each boundary point p of S , the set $\text{In}(p, S)$ coincides with $\text{Out}(p, \Gamma \setminus S)$.

Note that

$$m_f(\emptyset) = m_f(\Gamma) = 0 ,$$

since by our definition, both the empty set and the entire graph Γ have no boundary points. It follows for any $S \in \mathcal{A}$

$$m_f(\Gamma \setminus S) = -m_f(S) .$$

If $f_1, f_2 \in \mathcal{D}$ and $c_1, c_2 \in \mathbb{R}$, then it is easy to see that the set function corresponding to $c_1 f_1 + c_2 f_2$ is

$$m_{c_1 f_1 + c_2 f_2} = c_1 m_{f_1} + c_2 m_{f_2} .$$

We will say that a continuous function $f \in \mathcal{D}(\Gamma)$ is of *bounded differential variation*, and write $f \in \text{BDV}(\Gamma)$, if there is a constant $B > 0$ such that for any countable collection $\mathcal{F}$ of pairwise disjoint sets in $\mathcal{A}$,

$$(C.13) \quad \sum_{S_i \in \mathcal{F}} |m_f(S_i)| \leq B .$$

It is easy to see that $\text{BDV}(\Gamma)$ is a linear subspace of $\mathcal{D}(\Gamma)$.

PROPOSITION C.5. *Let Γ be a metrized graph.*

A) *If $f \in \text{BDV}(\Gamma)$, then there are only countably many points $p_i \in \Gamma$ for which $m_f(p_i) \neq 0$, and $\sum_i |m_f(p_i)|$ converges.*

B) *In the definition of $\text{BDV}(\Gamma)$ one can restrict to families of pairwise disjoint connected open sets, or connected closed sets. More precisely, a function $f \in \mathcal{D}(\Gamma)$ belongs to $\text{BDV}(\Gamma)$ if and only if*

(1) *there is a constant B_1 such that for any countable family $\mathcal{F}$ of pairwise disjoint connected open sets $V_i \in \mathcal{A}$,*

$$\sum_{\substack{V_i \in \mathcal{F} \\ V_i \text{ connected, open}}} |m_f(V_i)| \leq B_1 ; \quad \text{or}$$

(2) *there is a constant B_2 such that for any countable family $\mathcal{F}$ of pairwise disjoint connected closed sets $E_i \in \mathcal{A}$,*

$$\sum_{\substack{E_i \in \mathcal{F} \\ E_i \text{ connected, closed}}} |m_f(E_i)| \leq B_2 .$$

PROOF. For A), note that for each $0 < n \in \mathbb{Z}$ there can be at most $nB = B/(1/n)$ points $p \in \Gamma$ with $|m_f(\{p\})| \geq 1/n$, so there are at most countably many points with $m_v(\{p\}) \neq 0$. Moreover,

$$\sum_{\substack{p \in \Gamma \\ m_f(\{p\}) \neq 0}} |m_f(\{p\})| < B .$$

For B), note that if $f \in \text{BDV}(\Gamma)$ then $B(1)$ and $B(2)$ hold trivially. Conversely, first suppose $B(1)$ holds. We begin by showing that there are only countably many points $p \in \Gamma$ for which $m_f(p) \neq 0$, and that

$$(C.14) \quad \sum_{m_f(\{p\}) \neq 0} |m_f(\{p\})| < 2B_1 .$$

To see this, suppose that for some $0 < n \in \mathbb{Z}$, there were more than $2nB_1$ points p with $|m_v(\{p\})| \geq 1/n$. Then either there would be more than nB_1 points with $m_v(\{p\}) \geq 1/n$, or more than nB_1 points with $m_v(\{p\}) \leq -1/n$. Suppose for example that $p_1, \dots, p_r$ satisfy

$m_v(\{p_i\}) \leq -1/n$, where $r > nB_1$. Put $K = \{p_1, \dots, p_r\}$, and let $V_1, \dots, V_s$ be the connected components of $\Gamma \setminus K$. Then

$$\begin{aligned} B_1 &< -(m_f(\{p_1\}) + \dots + m_f(\{p_r\})) = -m_f(K) \\ &= m_f(V_1 \cup \dots \cup V_s) = m_v(V_1) + \dots + m_v(V_s) \\ &\leq |m_f(V_1)| + \dots + |m_f(V_r)| \leq B_1. \end{aligned}$$

Hence there can be only countably many points p_i with $m_f(\{p_i\}) \neq 0$, and (C.14) holds.

Now let $\mathcal{F}$ be any countable collection of pairwise disjoint sets $S_i \in \mathcal{A}$. Each S_i can be decomposed as a finite disjoint union of connected open sets and sets consisting of isolated points, and if

$$S_i = V_{i1} \cup \dots \cup V_{i,r_i} \cup \{p_{i1}\} \cup \dots \cup \{p_{i,s_i}\}$$

is such a decomposition, then

$$\sum_{S_i \in \mathcal{F}} |m_f(S_i)| \leq \sum_{i,j} |m_f(V_{ij})| + \sum_{i,k} |m_f(\{p_{ik}\})| \leq 3B_1.$$

It follows that $f \in \text{BDV}(\Gamma)$.

Next suppose that $B(2)$ holds; we will show that $B(1)$ holds as well. Consider a countable collection $\mathcal{F}$ of pairwise disjoint connected open sets V_i . Decompose $\mathcal{F} = \mathcal{F}_+ \cup \mathcal{F}_-$, where $V_i \in \mathcal{F}_+$ iff $m_f(V_i) \geq 0$, and $V_i \in \mathcal{F}_-$ iff $m_f(V_i) < 0$. Relabel the sets so that $\mathcal{F}_+ = \{V_1, V_3, V_5, \dots\}$. For each n , put $K_n = \Gamma \setminus (V_1 \cup V_3 \cup \dots \cup V_{2n+1})$, and decompose K_n as a finite disjoint union of connected closed sets $E_1, \dots, E_r$. Then

$$\begin{aligned} \sum_{i=0}^n m_f(V_{2i+1}) &= m_f(\Gamma \setminus K_n) = |m_f(K_n)| \\ &\leq \sum_{j=1}^r |m_f(E_j)| \leq B_2. \end{aligned}$$

Letting $n \rightarrow \infty$ we see that $\sum_{V_i \in \mathcal{F}_+} m_f(V_i) \leq B_2$. Similarly $\sum_{V_i \in \mathcal{F}_-} |m_f(V_i)| \leq B_2$, so $\sum_{V_i \in \mathcal{F}} |m_f(V_i)| \leq 2B_2$, and we are done. $\square$

We now come to the main result of this section.

THEOREM C.6. *If $f \in \text{BDV}(\Gamma)$, then the finitely additive set function m_f extends to a finite, signed Borel measure m_f^* of total mass 0 on Γ .*

PROOF. We begin with a reduction. It suffices to show that the restriction of m_f to each edge e extends to a (finite, signed Borel) measure on e . Identifying e with its parametrizing interval, we can

assume without loss of generality that $\Gamma = [a, b]$ is a closed interval and that $f : [a, b] \rightarrow \mathbb{R}$ is in $\text{BDV}([a, b])$.

We next decompose m_f into an atomic and an atomless part.

Let $\{p_1, p_2, \dots\}$ be the points in $[a, b]$ for which $m_f(\{p_i\}) \neq 0$. For brevity, write $c_i = m_f(\{p_i\})$; by hypothesis, $\sum_i |c_i|$ converges. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(t) = -\frac{1}{2} \sum_i c_i |t - p_i| ,$$

and let $m_g^* = \sum_i c_i \delta_{p_i}(x)$ be the Borel measure giving the point mass c_i to each p_i .

By a direct computation, one checks that for each $x \in \mathbb{R}$, both one-sided derivatives $g'_\pm(x)$ exist, with

$$g'_-(x) = \frac{1}{2} \left(\sum_{x \leq p_i} c_i - \sum_{x > p_i} c_i \right), \quad g'_+(x) = \frac{1}{2} \left(\sum_{x < p_i} c_i - \sum_{x \geq p_i} c_i \right) .$$

For any closed subinterval $[c, d] \subset [a, b]$,

$$\begin{aligned} m_g([c, d]) &= g'_-(c) - g'_+(d) \\ &= \sum_{c \leq p_i \leq d} c_i = m_g^*([c, d]) . \end{aligned}$$

In particular the set function m_g has precisely the same point masses as m_f . Similar computations apply to open and half-open intervals. Thus the measure m_g^* extends m_g .

Replacing $f(t)$ by $h(t) = f(t) - g(t)$, we are reduced to the situation where $h \in \text{BDV}([a, b])$ has no point masses. This means that for each $p \in (a, b)$,

$$0 = m_h(\{p\}) = h'_-(p) - h'_+(p),$$

or in other words that $h'(p)$ exists. By hypothesis, both $h'_+(a)$ and $h'_-(b)$ exist, so $h(t)$ is differentiable on $[a, b]$. The fact that $h \in \text{BDV}([a, b])$ means that $h'(t)$ is of bounded total variation.

Find a direct reference for this.

We claim that $h'(t)$ is continuous on $[a, b]$. Suppose it were discontinuous at some point p . By [66], p.109, the existence of $h'(t)$ for all t means that $h'(t)$ cannot have a jump discontinuity, so the discontinuity must be due to oscillation. Hence there would be an $\varepsilon > 0$, and a sequence of points $t_1, t_2, \dots$, either monotonically increasing or monotonically decreasing to p , such that $|h'(t_i) - h'(t_{i+1})| > \varepsilon$ for each i . Assume for convenience that the t_i are monotonically increasing. Considering the intervals $[t_1, t_2], [t_3, t_4], \dots$ we see that

$$\sum_{i=1}^{\infty} |m_h([t_{2i-1}, t_{2i}])| = \infty ,$$

contradicting the fact that $h \in \text{BDV}([a, b])$.

For $a \leq t \leq b$, let $T(t)$ be the cumulative total variation function of $h'(t)$: letting $\mathcal{Q}$ vary over all finite partitions $a = q_0 < q_1 < \cdots < q_m = t$ of $[a, t]$,

$$T(t) = \sup_{\mathcal{Q}} \sum_{j=1}^m |h'(q_j) - h'(q_{j-1})| .$$

Then $T : [a, b] \rightarrow \mathbb{R}$ is monotone increasing, and is continuous since $h'(t)$ is continuous. By Proposition 12 on p.301 of [59], there is a unique finite signed Borel measure ν on $[a, b]$ such that $\nu((c, d]) = T(d) - T(c)$ for each half-open interval $(c, d] \subset [a, b]$. Since $T(t)$ is continuous, ν has no point masses.

Next put

$$T_1(t) = T(t) - h'(t) .$$

Then $T_1(t)$ is also monotone increasing and continuous, so by the same Proposition there is a bounded Borel measure ν_1 on $[a, b]$ such that for each half-open interval $(c, d] \subset [a, b]$,

$$\nu_1((c, d]) = T_1(d) - T_1(c) .$$

As before, ν_1 has no point masses.

Now define the Borel measure $m_h^* = \nu_1 - \nu$. By construction, m_h^* has finite total mass. For any half-open interval $(c, d] \subset [a, b]$,

$$m_h^*((c, d]) = h'(c) - h'(d) = m_h((c, d]) .$$

Since the measures ν and ν_1 have no point masses,

$$m_h^*([c, d]) = m_h^*((c, d)) = m_h^*([c, d)) = m_h^*((c, d]) .$$

Thus m_h^* extends the finitely additive set function m_h .

Finally, adding the measures m_g^* extending m_g and m_h^* extending m_h , we obtain the desired measure $m_f^* = m_g^* + m_h^*$ extending m_f . Since $m_f(\Gamma) = 0$, we have $m_f^*(\Gamma) = 0$ as well. $\square$

6. The Laplacian on a metrized graph

If $f \in \text{BDV}(\Gamma)$, we define the Laplacian $\Delta(f)$ to be the measure given by Theorem C.6:

$$\Delta(f) = m_f^* .$$

Recall from §4 that there is another notion of Laplacian defined by Zhang on the space $\text{Zh}(\Gamma)$.

LEMMA C.7. *The space $\text{Zh}(\Gamma)$ is a subset of $\text{BDV}(\Gamma)$.*

PROOF. We first prove that the directional derivatives $d_{\vec{v}}f(p)$ exist for each $p \in X_f$ and each $\vec{v} \in T_p(\Gamma)$. Fix such a p and $\vec{v}$, and let t_0 be small enough that $p + t\vec{v} \in \Gamma \setminus X_f$ for all $t \in (0, t_0)$. By abuse of notation, we will write $f(t)$ for $f(p + t\vec{v})$. By hypothesis, $f \in \mathcal{C}^2((0, \delta))$. By the Mean Value Theorem, $d_{\vec{v}}f(p)$ exists if and only if $\lim_{t \rightarrow 0^+} f'(t)$ exists (in which case the two are equal). Given $\varepsilon > 0$, let $0 < \delta < t_0$ be small enough that $\int_{(0, \delta)} |f''(t)| dt < \varepsilon$; this is possible since $f'' \in L^1(\Gamma)$. Then for all $t_1, t_2 \in (0, \delta)$,

$$(C.15) \quad |f'(t_2) - f'(t_1)| \leq \int_{t_1}^{t_2} |f''(t)| dt < \varepsilon,$$

which proves that $\lim_{t \rightarrow 0^+} f'(t) = d_{\vec{v}}f(p)$ exists.

Equation (C.15) also implies (in the notation of §4) that for every countable family $\mathcal{F}$ of pairwise disjoint connected closed sets $E_i \in \mathcal{A}$, we have

$$\sum_{E_i \in \mathcal{F}} |m_f(E_i)| \leq \sum_{p \in X_f} |m_f(\{p\})| + \int_{\Gamma} |f''(t)| dt < \infty,$$

so that $f \in \text{BDV}(\Gamma)$ as desired. $\square$

It follows easily from the Mass Formula (Lemma C.4) that Zhang's Laplacian, defined in (C.8), coincides for $f \in \text{Zh}(\Gamma)$ with the Laplacian on the larger space $\text{BDV}(\Gamma)$. More precisely, we have the following result.

PROPOSITION C.8.

A) If $f \in \text{Zh}(\Gamma)$, then $\Delta_{\text{Zh}}(f) = \Delta(f)$.

B) If $f \in \text{BDV}(\Gamma)$, and $\Delta(f)$ has the form $\Delta(f) = g(x)dx + \sum_{p \in X} c_p \delta_{p_i}(x)$ for a piecewise continuous function $g \in L^1(\Gamma)$ and a finite set X , then $f \in \text{Zh}(\Gamma)$. Furthermore, let X_g be a vertex set for Γ containing X and the finitely many points where $g(x)$ is not continuous; put $c_p = 0$ for each $p \in X_g \setminus \{p_1, \dots, p_m\}$. Then $f''(x) = -g(x)$ for each $x \in \Gamma \setminus X_g$, $f'(x)$ is continuous on the closure of each segment of $\Gamma \setminus X_g$ (interpreting $f'(x)$ as a one-sided derivative at each endpoint), and $\Delta_p(f) = -c_p$ for each $p \in X_g$.

PROOF. A) Suppose $f \in \text{Zh}(\Gamma)$. For each $p \in X_f$, we have $\Delta(f)(\{p\}) = \Delta_{\text{Zh}}(f)(\{p\})$. To show that $\Delta(f) = \Delta_{\text{Zh}}(f)$ it suffices to show that $\Delta(f)((c, d)) = \Delta_{\text{Zh}}(f)((c, d))$ for each open interval (c, d) contained in an edge of $\Gamma \setminus X_f$. But

$$\begin{aligned} \Delta(f)((c, d)) &= m_f((c, d)) = f'(c) - f'(d) \\ &= - \int_c^d f''(x) dx = \Delta_{\text{Zh}}((c, d)). \end{aligned}$$

B) Now suppose $f \in \text{BDV}(\Gamma)$, and $\Delta(f)$ has the given form. Consider an edge in $\Gamma \setminus X_g$, and identify it with an interval (a, b) by means of the distinguished parametrization. For each $x \in (a, b)$, we have $\Delta(f)(x) = 0$, so $f'(x)$ exists. If $h > 0$ is sufficiently small, then

$$f'(x+h) - f'(x) = -\Delta(f)([x, x+h]) = -\int_x^{x+h} g(t) dt ,$$

while if $h < 0$ then

$$f'(x+h) - f'(x) = \Delta(f)([x+h, x]) = \int_{x+h}^x g(t) dt = -\int_x^{x+h} g(t) dt .$$

Hence

$$f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h \rightarrow 0} \left(-\frac{1}{h} \int_x^{x+h} g(t) dt \right) = -g(x) .$$

The assertion that $\Delta_p(f) = -c_p$ for each $p \in X_g$ is clear. $\square$

COROLLARY C.9. *If $f \in \text{BDV}(\Gamma)$ and $\Delta(f) = \sum_{i=1}^k c_i \delta_{p_i}$ is a discrete measure, then $f \in \text{CPA}(\Gamma)$.*

PROOF. Since $\Delta(f)$ is discrete, it follows by Proposition C.8 that $f(x) \in \text{Zh}(\Gamma)$ and $\Delta(f) = \Delta_{\text{Zh}}(f)$. Fixing a vertex set X for Γ , we see that $f''(x) = 0$ on $\Gamma \setminus X$, so $f(x)$ is affine on each segment of $\Gamma \setminus X$, which means that $f \in \text{CPA}(\Gamma)$. $\square$

The following proposition is the key technical result needed to see that every measure of total mass zero is the Laplacian of some function in $\text{BDV}(\Gamma)$.

PROPOSITION C.10. *If ν is a finite signed Borel measure on Γ , and if $h(x) = \int_{\Gamma} j_z(x, y) d\nu(y)$, then $h \in \text{BDV}(\Gamma)$ and $\Delta(h) = \nu - \nu(\Gamma)\delta_z$.*

PROOF. Fix $z \in \Gamma$, and let $h(x) = \int j_z(x, y) \nu(y)$.

We first show that $h(x) \in \mathcal{D}(\Gamma)$, i.e., that all directional derivatives of h exist. More precisely, if $\vec{v} \in T_x(\Gamma)$, we claim that the partial derivative $\partial_{x, \vec{v}} h(x)$ in the direction $\vec{v}$ at x exists and satisfies

$$(C.16) \quad \partial_{x, \vec{v}} \left(\int j_z(x, y) \nu(y) \right) = \int (\partial_{x, \vec{v}} j_z(x, y)) \nu(y) ,$$

i.e., we may “differentiate under the integral sign”. Note that the right-hand side of (C.16) is well-defined, because for fixed x, z , and $\vec{v}$, the function $\partial_{x, \vec{v}} j_z(x, y)$ is piecewise continuous in y .

To verify (C.16), note that for each fixed y , and for $\epsilon > 0$ sufficiently small, we have

$$j_z(x + \epsilon \vec{v}, y) - j_z(x, y) = \epsilon \cdot \partial_{x, \vec{v}} j_z(x, y)$$

Give proof or reference – it’s even piecewise *affine*.

by the fact that $j_z(x, y)$ is piecewise affine in x . From this one readily obtains (C.16).

Next, we claim that if ν_n is any sequence of measures with $|\nu_n - \nu|(\Gamma) \rightarrow 0$, and if $h_n(x) = \int j_z(x, y) \nu_n(y)$. then

$$(C.17) \quad \partial_{x,\bar{v}} h_n(x) \rightarrow \partial_{x,\bar{v}} h(x) .$$

This follows from the estimate

$$|\partial_{x,\bar{v}} h_n(x) - \partial_{x,\bar{v}} h(x)| \leq \sup_{y \in \Gamma} (|\partial_{x,\bar{v}} j_z(x, y)|) \cdot |\nu_n - \nu|(\Gamma) ,$$

since $|\partial_{x,\bar{v}} j_z(x, y)| \leq 1$ for all y by Part (D) of Proposition C.3.

Let ν_n^+ (resp. ν_n^-) be a sequence of discrete measures which converge weakly to ν^+ (resp. ν^-), and set $\nu_n = \nu_n^+ - \nu_n^-$. Then $(\nu_n - \nu)(\Gamma) \rightarrow 0$. If $\nu_n = \sum a_i \delta_{q_i}$ and D is a closed segment in Γ , then

$$(\Delta h_n)(D) = \sum a_i \Delta_x(j_z(x, q_i)) = \sum a_i (\delta_{q_i} - \delta_z) = \nu_n(D) - \nu_n(\Gamma) \delta_z(D) .$$

Passing to the limit as n tends to infinity, (C.17) implies that $m_h(D) = \nu(D) - \nu(\Gamma) \delta_z(D)$.

For any countable collection $\mathcal{F}$ of pairwise disjoint sets D_i of the type above, it follows that

$$\sum_{i=1}^{\infty} |m_h(D_i)| \leq 2|\nu|(\Gamma) ,$$

so by Proposition C.5, $h \in \text{BDV}(\Gamma)$. The measure $\Delta(h) = m_h^*$ attached to h is determined by its value on closed intervals, so it must coincide with $\nu - \nu(\Gamma) \delta_z$. $\square$

As an immediate consequence of Proposition C.10, we obtain:

COROLLARY C.11. *If ν is a finite signed Borel measure of total mass zero on Γ , then there exists a function $h \in \text{BDV}(\Gamma)$ such that $\Delta h = \nu$.*

By part (B) of Proposition C.13 below, h is unique up to addition of an arbitrary real constant.

We can deduce another useful result from Proposition C.10. For any finite signed Borel measure ν on Γ , put

$$j_\nu(x, y) = \int_{\Gamma} j_\zeta(x, y) d\nu(\zeta) .$$

Clearly $j_\nu(x, y)$ is symmetric, and is jointly continuous in x and y . It also has the following useful property:

PROPOSITION C.12. *For each $y \in \Gamma$, the function $F_y(x) = j_\nu(x, y)$ belongs to the space $\text{BDV}(\Gamma)$, and satisfies*

$$\Delta_x(F_y) = \nu(\Gamma) \cdot \delta_y - \nu .$$

PROOF. From (C.3), we see that for any fixed $z \in \Gamma$, we have

$$F_y(x) = \int j_\zeta(x, y) \nu(\zeta) = \nu(\Gamma) j_z(x, y) - \int j_z(x, \zeta) \nu(\zeta) - C ,$$

where C is independent of x . By Proposition C.10, we find that

$$\Delta_x F_y(x) = \nu(\Gamma) (\delta_y - \delta_z) - (\nu - \nu(\Gamma) \delta_z) = \nu(\Gamma) \delta_y - \nu$$

as desired. $\square$

It is clear from the definitions that the Laplacian on $\text{BDV}(\Gamma)$ is a linear operator:

$$\Delta(\alpha f + \beta g) = \alpha \Delta(f) + \beta \Delta(g)$$

for $f, g \in \text{BDV}(\Gamma)$ and $\alpha, \beta \in \mathbb{R}$.

We also note the following additional properties of the Laplacian on $\text{BDV}(\Gamma)$, which extend those proved in Proposition C.2 for functions in $\text{CPA}(\Gamma)$.

PROPOSITION C.13. *If $f, g \in \text{BDV}(\Gamma)$, then*

- A) $\Delta(f) \equiv 0$ if and only if $f = C$ for some constant C .
- B) $\Delta(f) = \Delta(g)$ if and only if $f = g + C$ for some constant C .
- C) If f is nonconstant, then $f(x)$ achieves its maximum at a point p in the support of $\Delta(f)^+$, and its minimum at a point q in the support of $\Delta(f)^-$.
- D) $\int_\Gamma f(x) \Delta(g)(x) = \int_\Gamma g(x) \Delta(f)(x)$.
- E) The total mass $\Delta(f)(\Gamma) = 0$.

PROOF. A) Trivially $f = C$ implies $\Delta(f) = 0$. Conversely, if $\Delta(f) = 0$, then by part Corollary C.9, we have $f \in \text{CPA}(\Gamma)$. The result now follows from part (A) of Proposition C.2.

B) This follows from (A) by the linearity of Δ .

C) Let $x_1 \in \Gamma$ be a point where the continuous function f achieves its maximum, and let Γ_1 be the connected component of $\{x \in \Gamma : f(x) = f(x_1)\}$ containing x_1 . Since f is non-constant, we have $\Gamma_1 \neq \Gamma$.

If x_0 is a boundary point of Γ_1 , we claim that $x_0 \in \text{supp}(\Delta_\Gamma(f)^+)$. Suppose not. Letting $\rho(x, y)$ denote the metric on Γ , there is a neighborhood $\Gamma_{x_0}(\varepsilon) = \{x \in \Gamma : \rho(x, x_0) < \varepsilon\}$ with $\Delta_\Gamma(f)|_{\Gamma_{x_0}(\varepsilon)} \leq 0$. After shrinking ε if necessary, we can assume that $\Gamma_{x_0}(\varepsilon)$ is a star, a union of

half-open segments $[x_0, x_0 + \varepsilon \vec{v}_i)$ where the $\vec{v}_i$ are the unit tangent vectors at x_0 . Since $f(x_0)$ is the maximum value of $f(x)$ on Γ , necessarily $d_{\vec{v}_i} f(x_0) \leq 0$ for each i . If $d_{\vec{v}_i} f(x_0) < 0$ for some i , then

$$\Delta_\Gamma(f)(x_0) = - \sum_i d_{\vec{v}_i} f(x_0) > 0$$

contradicting $x_0 \notin \text{supp}(\Delta_\Gamma(f)^+)$. Hence $d_{\vec{v}_i} f(x_0) = 0$ for each i .

Since x_0 is a boundary point of Γ_1 , there is a point $y_0 \in \Gamma_{x_0}(\varepsilon)$ where $f(y_0) < f(x_0)$. Let i be such that $y_0 \in e_i = (x_0, x_0 + \varepsilon \vec{v}_i)$. As $\Delta_\Gamma(f)|_{\Gamma_{x_0}(\varepsilon)} \leq 0$, it follows that $\Delta_\Gamma(f)|_{e_i} \leq 0$. For each $y = x_0 + t\vec{v}_i \in e_i$, there are two unit tangent vectors at y : write $\vec{v}_+$ for the one that leads away from x_0 , and $\vec{v}_-$ for the one that leads towards x_0 .

By the definition of $\Delta_\Gamma(f)$, for each $y \in e_i$

$$0 \geq \Delta_\Gamma(f)(y) = -d_{\vec{v}_-} f(y) - d_{\vec{v}_+} f(y)$$

so $d_{\vec{v}_+} f(y) \geq -d_{\vec{v}_-} f(y)$. Similarly, for each open subsegment $(y_1, y_2) \subset e_i$,

$$0 \geq \Delta_\Gamma(f)((y_1, y_2)) = d_{\vec{v}_+} f(y_1) + d_{\vec{v}_-} f(y_2)$$

so $-d_{\vec{v}_-} f(y_2) \geq d_{\vec{v}_+} f(y_1)$.

For the function $g(t) = f(x_0 + t\vec{v}_i)$, we have $g'_-(t) = -d_{\vec{v}_-} f(y)$ and $g'_+(t) = d_{\vec{v}_+} f(y)$. By Lemma A.1 f is convex on e_i . Since $d_{\vec{v}_i} f(x_0) = 0$, $f(x_0 + t\vec{v}_i)$ is nondecreasing in t . Hence $f(y_0) \geq f(x_0)$, a contradiction. Thus $x_0 \in \text{supp}(\Delta_\Gamma(f)^+)$.

The case of a minimum is similar.

D) Set $\nu = \Delta(f)$, so that $\nu(\Gamma) = 0$ by part (E). For fixed y, z , we know from Proposition C.10 that $h(y) = \int j_z(x, y) d\nu(x)$ belongs to $\text{BDV}(\Gamma)$ and $\Delta h = \nu$. It follows from (B) that $f = h + C$ for some constant C . By Fubini's theorem, and the fact that $\Delta(g)(\Gamma) = 0$, we have

$$\int_\Gamma f \Delta(g) = \int_\Gamma h \Delta(g) = \iint_{\Gamma \times \Gamma} j_z(x, y) \Delta(f)(x) \Delta(g)(y) .$$

The result now follows by symmetry.

E) This has already been noted, and follows immediately from the definition of the Laplacian on $\text{BDV}(\Gamma)$.

□

APPENDIX D

Brief overview of Berkovich's theory

In this appendix, we recall some basic definitions and results from Berkovich's general theory of analytic spaces. The primary references for this material are Berkovich's book [9] and his paper [10]. For the most part, we do not give proofs, although we do provide a self-contained proof of the compactness of Berkovich affinoids which is different from the one in [9].

All rings considered in this appendix will be commutative rings with an identity element 1.

We plan to expand this Appendix and make it a little more self-contained.

1. Seminorms and norms

A *seminorm* on a ring $\mathcal{A}$ is a function $\|\cdot\| : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ with values in the set of nonnegative reals such that for every $f, g \in \mathcal{A}$, we have

- (S1) $\|0\| = 0, \|1\| = 1$.
- (S2) $\|f + g\| \leq \|f\| + \|g\|$.
- (S3) $\|f \cdot g\| \leq \|f\| \cdot \|g\|$.

A seminorm $\|\cdot\|$ defines a topology on $\mathcal{A}$ in the usual way, and this topology is Hausdorff if and only if $\|\cdot\|$ is a *norm*, meaning that $\|f\| = 0$ if and only if $f = 0$.

A *normed ring* is pair $(\mathcal{A}, \|\cdot\|)$ consisting of a ring $\mathcal{A}$ and a norm $\|\cdot\|$. It is called a *Banach ring* if $\mathcal{A}$ is complete with respect to this norm. Any ring may be regarded as a Banach ring with respect to the *trivial* norm for which $\|0\| = 0$ and $\|f\| = 1$ for $f \neq 0$.

A seminorm $\|\cdot\|$ on a ring $\mathcal{A}$ is called *multiplicative* if for all $f, g \in \mathcal{A}$, we have

$$(S3)' \quad \|f \cdot g\| = \|f\| \cdot \|g\| \quad ,$$

and it is called *non-archimedean* if

$$(S2)' \quad \|f + g\| \leq \max\{\|f\|, \|g\|\} \quad .$$

A seminorm $\|\cdot\|$ on a normed ring $(\mathcal{A}, \|\cdot\|)$ is called *bounded* if

(S4) there exists a constant $C > 0$ such that $|f| \leq C\|f\|$ for all $f \in \mathcal{A}$.

For a Banach ring, it is well-known (see for example [31, Proposition 5.2]) that boundedness is equivalent to continuity.

LEMMA D.1. *If $|\cdot|$ is multiplicative, then condition (S4) is equivalent to:*

(S4)' $|f| \leq \|f\|$ for all $f \in \mathcal{A}$.

PROOF. Since $|f^n| \leq C\|f^n\|$, we have $|f| \leq \sqrt[n]{C}\|f\|$ for all $n \geq 1$. Passing to the limit as n tends to infinity yields the desired result. $\square$

2. The spectrum of a normed ring

Let $(\mathcal{A}, \|\cdot\|)$ be a (commutative) Banach ring. We define a topological space $\mathcal{M}(\mathcal{A})$, called the *spectrum* of $\mathcal{A}$, as follows. As a set, $\mathcal{M}(\mathcal{A})$ consists of all bounded multiplicative seminorms on $\mathcal{A}$. The topology on $\mathcal{M}(\mathcal{A})$ is defined to be the weakest one for which all functions of the form $|\cdot| \rightarrow |f|$ for $f \in \mathcal{A}$ are continuous.

It is useful from a notational standpoint to denote points of $X = \mathcal{M}(\mathcal{A})$ by a letter such as x , and the corresponding bounded multiplicative seminorm by $[\cdot]_x$. With this terminology, a subbase of open sets for the topology on X is given by the collection

$$\begin{aligned} U(f, \alpha) &= \{x \in X : [f]_x < \alpha\} \\ V(f, \alpha) &= \{x \in X : [f]_x > \alpha\} \end{aligned}$$

for all $f \in \mathcal{A}$ and all $\alpha \in \mathbb{R}$.

Equivalently, one may define the topology on $\mathcal{M}(\mathcal{A})$ as the *topology of pointwise convergence*: a net $\langle x_\alpha \rangle$ in $\mathcal{M}(\mathcal{A})$ converges to $x \in \mathcal{M}(\mathcal{A})$ if and only if $[f]_{x_\alpha}$ converges to $[f]_x$ in $\mathbb{R}$ for all $f \in \mathcal{A}$.

THEOREM D.2. *If $\mathcal{A}$ is a Banach ring, then the spectrum $\mathcal{M}(\mathcal{A})$ is a non-empty compact Hausdorff space.*

PROOF. This is proved in Theorem 1.2.1 of [9]. The fact that $\mathcal{M}(\mathcal{A})$ is Hausdorff is an easy exercise. The proof that $\mathcal{M}(\mathcal{A})$ is always non-empty is rather subtle, though. (In many cases of interest, however, such as when the norm on $\mathcal{A}$ is multiplicative, the fact that $\mathcal{A}$ is non-empty is obvious.)

We give a different proof from the one in [9] of the compactness of $\mathcal{M}(\mathcal{A})$. It suffices to prove that every net in $X = \mathcal{M}(\mathcal{A})$ has a convergent subnet. Let T be the space $\prod_{f \in \mathcal{A}} [0, \|f\|]$ endowed with the product topology. By Tychonoff's theorem, T is compact. By

Lemma D.1, there is a natural map $\iota : X \rightarrow T$ sending $x \in X$ to $([f]_x)_{f \in \mathcal{A}}$, and ι is clearly injective.

Let $\langle x_\alpha \rangle$ be a net in X . Since T is compact, $\langle \iota(x_\alpha) \rangle$ has a subnet $\langle \iota(y_\beta) \rangle$ converging to an element $(\alpha_f)_{f \in \mathcal{A}} \in T$. Define a function $[\cdot]_y : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ by $[f]_y = \alpha_f$. It is easily verified that $[\cdot]_y$ is a bounded multiplicative seminorm on $\mathcal{A}$, and thus defines a point $y \in X$. By construction, we have $\iota(y) = \lim_\beta \iota(y_\beta)$. This implies that $\lim_\beta [f]_{y_\beta} = [f]_y$ for all $f \in \mathcal{A}$, i.e., $y_\beta \rightarrow y$. Thus $\langle x_\alpha \rangle$ has a convergent subnet as desired. $\square$

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